Praxia Update

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Volume 54, Issue 7 · June 2026 · 30 new articles

Medical Tourism for Regenerative Medicine Treatments: Legitimate Therapy or Snake Oil?

Rodeo SA

PubMed · DOI

Return to Play, Career Longevity, and Reruptures After Primary Anterior Cruciate Ligament Reconstruction With Lateral Extra-articular Tenodesis in Professional Soccer and Rugby Players: A Comparison of Hamstring Tendon Versus Bone-Patellar Tendon-Bone Autografts.

Lynskey SJ, Ambrosanio A, Motesharei A, Jones M, Ball S, Church JS, Williams A

BACKGROUND: Elite pivoting athletes are at a high risk of sustaining anterior cruciate ligament (ACL) injuries and reruptures. While bone-patellar tendon-bone (BPTB) and hamstring tendon (HT) autografts are both widely used for primary ACL reconstruction (ACLR), comparative data in the setting of routine lateral extra-articular tenodesis (LET) remain limited.

PURPOSE: To compare return-to-play (RTP) rates, the rerupture risk, and career outcomes in elite rugby and soccer players undergoing ACLR with LET using BPTB versus HT autografts.

STUDY DESIGN: Cohort study; Level of evidence, 4.

METHODS: Elite rugby and soccer players with an isolated ACL rupture underwent primary ACLR with LET using either a BPTB or HT autograft. Graft choice was based on preferences of the surgeon, athlete, and athlete's medical team. Professional match participation and rerupture rates were tracked for a mean of 5.6 years (range, 2.0-12.7 years). Primary outcomes were RTP rate, time to RTP, and postoperative career length. Secondary outcomes included graft reruptures, match exposure, and level of play. Survival analysis and multivariable Cox regression were used to evaluate the predictors of career duration.

RESULTS: A total of 223 knees (217 athletes) were included (BPTB: n = 126; HT: n = 97). RTP rates were high and comparable between those with BPTB grafts (94.4%) and those with HT grafts (95.9% […]

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Association Between Posterior Tibial Slope and Graft Survival in High-Risk Anterior Cruciate Ligament Reconstruction With Lateral Extra-articular Tenodesis.

Pineda T, Mazy D, Ramos-Rojas J, Cance N, Dan MJ, Demey G, Dejour DH

BACKGROUND: Posterior tibial slope (PTS) is a well-established anatomic risk factor for anterior cruciate ligament (ACL) graft failure. Lateral extra-articular tenodesis (LET) is increasingly used as an adjunctive procedure in high-risk patients to reduce rerupture rates; however, how its protective effect varies across the continuum of slope values remains insufficiently characterized.

PURPOSE: To determine how PTS modifies the protective effect of LET in high-risk patients and to evaluate the influence of PTS on graft failure across different risk profiles.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: A retrospective cohort of 585 patients who underwent primary ACL reconstruction (ACLR) with hamstring tendon autografts between 2014 and 2017 was analyzed at a minimum follow-up of 6 years. Patients were nonrandomly allocated according to predefined clinical risk criteria into a low-risk group (isolated ACLR) and

a high-risk group (ACLR + LET). A graft rerupture was defined clinically and confirmed by imaging. Multivariable logistic regression was used to assess the association between PTS and graft failure. Restricted cubic spline regression was also used to explore nonlinear relationships and identify slope ranges associated with low and high failure probabilities.

RESULTS: PTS independently predicted a graft rerupture in both groups, with a stronger effect i […]

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Lateral Extra-articular Tenodesis Restores Knee Rotational and Translational Stability Regardless of Fixation Type: A Matched-Pair Cadaveric Study.

Ihn H, Quintana D, Puzzitiello R, Sakelaris P, Rawson H, Barber N, Ernat JJ, Aoki SK et al.

BACKGROUND: Lateral extra-articular tenodesis (LET) is used as an adjunct to anterior cruciate ligament reconstruction (ACLR) to enhance stability. However, understanding of the optimal fixation method is limited.

PURPOSE/HYPOTHESIS: The purpose was to compare biomechanical knee stability using 3 femoral LET fixation methods before and after cyclic loading. It was hypothesized that the fixation achieved with a suture anchor or an interference screw would demonstrate greater stability than that achieved with a metallic staple.

STUDY DESIGN: Controlled laboratory study.

METHODS: Ten matched pairs of cadaveric knees (mean age 64 years) underwent modified Lemaire LET. Each pair received suture anchor fixation on 1 knee and either interference screw (arm A) or metallic staple (arm B) fixation on the contralateral knee. Specimens were tested in 4 states: intact (native), anterolateral knee lesion, modified Lemaire (LET) procedure, and modified Lemaire (LET) procedure after cyclic loading. Cyclic loading consisted of 2000 cycles of 5 N·m internal rotation at 1 Hz. Anterior translation was applied at 90 N of force using visual marker motion capture. Internal rotation was measured using a 5 N·m moment and a digital inclinometer.

RESULTS: No significant differences in anterior translation were observed between the different states (anterolateral lesion, LET, and post-cyclic loading) […]

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Intraoperative Manual Resistance and Direct Intra-articular Visualization Maneuvers for Anterior Cruciate Ligament Reconstruction: A Retrospective Cohort Study.

Cantrell WA, Hale ME, Scariano G, O'Toole RA, Serna A, Scarcella M, Farrow LD

BACKGROUND: The use of femoral suspensory fixation devices is a popular technique in anterior cruciate ligament reconstruction (ACLR), but there is no consensus on the best intraoperative method for confirming proper button deployment. Recent literature suggests that manual resistance alone is at a high risk of causing button malpositioning or floating without utilizing intraoperative fluoroscopic imaging.

PURPOSE: To investigate the incidence of femoral suspensory button malpositioning in patients undergoing ACLR when intraoperative manual resistance and direct intra-articular visualization maneuvers without imaging are used to confirm fixation as well as to compare the rates of button malpositioning between fixed- and adjustable-loop fixation devices.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: All patients who underwent ACLR by 3 board-certified, sports medicine fellowship-trained orthopaedic surgeons at a single institution were included. All 3 surgeons used intraoperative manual resistance and direct intra-articular visualization maneuvers to verify button device positioning. Patients were identified using prospectively collected administrative data from an institutional orthopaedic database. Patients were excluded if they did not have suspensory fixation devices used for reconstruction or if they lacked postoperative radiographs. Improper device deployme […]

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Secondary Muscle Injuries and Performance Decline After Anterior Cruciate Ligament Reconstruction in Professional Soccer: A Retrospective Matched Cohort Study.

Mazza D, De Carli A, De Carli F, Gagliardo S, Suraci F, Liberati Petrucci G, Santoriello V, Zeppilli P

BACKGROUND: Professional soccer players who return to play after anterior cruciate ligament reconstruction (ACLR) often experience secondary muscle injuries, but the true incidence, timing, and contributing factors remain incompletely defined.

HYPOTHESIS/PURPOSE: It was hypothesized that soccer players returning after ACLR would show a higher incidence of secondary muscle injuries than matched controls, with an increased risk associated with premature return to play (RTP). The purpose of this study was to identify the incidence and risk factors for muscle injuries after ACLR and to evaluate their impact on athletic performance.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: A total of 40 professional male soccer players who underwent ACLR between 2020 and 2023 were followed for 24 months after RTP. Players were matched 1:1 to controls based on age, position, league, and preinjury minutes played. Collected variables included RTP timing, muscle injury incidence and timing, injury characteristics, performance metrics (minutes played, goals/assists, defensive actions), and market value changes. Analyses included paired t tests, multivariable logistic regression, Kaplan-Meier survival curves, and Cox proportional hazards modeling.

RESULTS: Within 12 months of RTP, 13 of 40 players (32.5%) in the ACLR group sustained at least one secondary muscle injury compared with […]

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Surgical Management of Horizontal Cleavage Tears of the Medial Meniscus: A Biomechanical Analysis During Simulated Gait.

Gould HP, Chastain K, Wright-Chisem J, Letendre S, Parides M, Maher SA, Ranawat A

BACKGROUND: Horizontal cleavage tears (HCTs) comprise one-quarter of all diagnosed meniscal pathologies. Despite several static/quasi-static cadaveric studies showing that HCTs and their repairs do not change contact mechanics while leaflet resections do, it is unclear if these conclusions hold during activities of daily living.

HYPOTHESIS/PURPOSE: The purpose of this study was to quantify the comparative biomechanical effects of meniscal HCT, HCT repair, and leaflet resection on joint contact forces throughout simulated gait. It was hypothesized that HCT repair would not change contact mechanics relative to the intact meniscus, while inferior meniscal leaflet resection would lead to increased contact stress and decrease contact areas throughout simulated gait.

STUDY DESIGN: Controlled laboratory study.

METHODS: Six human cadaveric knees were placed on a multidirectional dynamic simulator. Electronic pressure sensors were inserted across the medial tibial plateau, and contact forces throughout simulated gait were quantified. Contact stress, contact area, and percent meniscal loading were quantified for intact, HCT, HCT repair, and inferior leaflet conditions.

RESULTS: No significant differences in contact mechanics occurred between the intact meniscus and HCT or between the intact meniscus and HCT repair. Inferior leaflet resection resulted in a statistically significant de […]

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Medial Meniscus Posterior Root Repair in Knees With ≥5° of Varus Alignment Is Associated With Greater Osteoarthritis Progression Compared With <5° Varus Alignment.

Dzidzishvili L, Casanova F, López-Torres II, Moews LD, Perez Lloveras GO, Thamrongskulsiri N, Allende F, Chahla J

BACKGROUND: Despite advances in medial meniscus posterior root (MMPR) repair techniques over the past decade, concerns persist about subsequent osteoarthritis (OA) progression.

PURPOSE: To assess postoperative progression to OA after isolated MMPR repair in patients with normal alignment versus varus alignment.

STUDY DESIGN: Case-control study; Level of evidence, 3.

METHODS: A total of 128 patients who underwent isolated MMPR repair with available pre- and postoperative radiographs were included. Patients were stratified by mechanical hip-knee-ankle (HKA) angle into normal alignment (<5°; n = 83) and varus alignment (≥5°; n = 45) groups. Radiographic assessments included HKA angle, OA grading using the Kellgren-Lawrence (KL) classification, joint space width (JSW), meniscal extrusion, and repair healing on magnetic resonance imaging (MRI). Postoperative patient-reported outcomes were assessed using the International Knee Documentation Committee (IKDC) score. Logistic regression identified independent predictors of OA progression, and Pearson correlations were used to examine relationships among patient, radiographic, and clinical variables.

RESULTS: After a minimum follow-up of 2 years, OA progression occurred in 32.5% of normal alignment knees and 53.3% of varus-aligned knees (P = .02), with varus alignment associated with 2.37-fold higher odds of progression (95% CI, 1.13 […]

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Isolated Medial Patellofemoral Ligament Reconstruction for Recurrent Patellar Dislocations With Increased Femoral Anteversion and an Absent/Mild J-sign.

Zhang Z, Wang D, Liu Y, Wang X, Zhang H

BACKGROUND: The influence of greater femoral anteversion on outcomes after medial patellofemoral ligament (MPFL) reconstruction is increasingly recognized, leading to the more frequent use of combined derotational femoral osteotomy. However, the indications for this additional invasive procedure remain poorly defined, particularly in patients without severe patellar maltracking.

PURPOSE: To evaluate the clinical outcomes of isolated MPFL reconstruction in patients with recurrent patellar dislocations (RPDs), increased femoral anteversion (>30°), and an absent or mild J-sign.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: A retrospective cohort study was conducted on 52 consecutive patients (mean age, 19.1 ± 5.9 years) with RPDs, femoral anteversion >30°, and an absent/mild J-sign who underwent isolated MPFL reconstruction between 2018 and 2022. Rotational deformities of the lower extremities, including femoral anteversion, tibiofemoral rotation, and tibial external torsion, were assessed using computed tomography. Clinical and radiological outcomes were assessed preoperatively and at a minimum 2-year follow-up. Additionally, outcomes were compared between patients with and without severe trochlear dysplasia (Dejour type B or D) to evaluate its potential confounding effect.

RESULTS: This study evaluated the outcomes of isolated MPFL reconstruction in 52 patients […]

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Bone Marrow Aspirate Concentrate Improves the Early Osseous Integration of Fresh Osteochondral Allografts in the Knee: A Randomized Controlled Trial.

Gelber PE, Ramírez-Bermejo E, Caviasso G, Juncosa-Chacón J, Fariñas O

BACKGROUND: Fresh osteochondral allograft (FOCA) transplantation is an increasingly used technique for treating symptomatic cartilage defects in young and active patients. However, insufficient osseous integration of the graft remains a primary cause of failure. Bone marrow aspirate concentrate (BMAC), rich in mesenchymal stem cells, may enhance graft integration.

PURPOSE: To determine whether the use of autologous BMAC in FOCA transplantation of the knee improves osseous integration on computed tomography (CT) during the first postoperative year and yields superior clinical outcomes compared to non-BMAC-treated grafts at 2-year follow-up.

STUDY DESIGN: Randomized clinical trial; Level of evidence, 1.

METHODS: We conducted a single-center, prospective, randomized controlled trial in 36 patients undergoing FOCA transplantation. Patients who met the inclusion criteria were randomly assigned to either a BMAC group or non-BMAC group. CT was performed at 3, 6, and 12 months, and imaging findings were evaluated using the semiquantitative assessment CT osteochondral allograft (ACTOCA) scoring system. Clinical outcomes (International Knee Documentation Committee, Kujala, Western Ontario Meniscal Evaluation Tool, and Tegner scores) were assessed preoperatively and at 6, 12, and 24 months.

RESULTS: Osseous integration at the host-graft junction on CT was superior in the BMAC group at […]

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Knee Chondral Shear Injury Repair: A Biomechanical Laboratory Comparison of Suture Bridge to Chondral Pin Fixation.

Duru NO, Stoner AJ, Frey C, Pham NS, Ellis HB, Schmitz MR, Yen YM, Tompkins MA et al.

BACKGROUND: Delamination of chondral fragments in pediatric patients is common and is usually caused by osteochondritis dissecans (OCD) or patellar dislocations. For fragments with minimal to no bone, fixation with screws may not be ideal due to hardware prominence on the cartilage or suboptimal screw purchase, which can result in hardware migration and loss of fixation. Lower-profile chondral fixation devices may be ideal in these circumstances, such as the suture bridge fixation construct and bioabsorbable chondral pin fixation.

HYPOTHESIS: Suture bridge constructs would provide superior fixation over chondral pins for chondral shear injury fragments.

STUDY DESIGN: Controlled laboratory study.

METHODS: Seven pediatric cadaveric femurs were utilized. Circular lesions, 15 mm in diameter, were created on both femoral condyles. One lesion was randomized to chondral pin repair, while the other received suture bridge repair using 2-0 suture fixated with suture anchors. Each specimen was then potted before undergoing biomechanical testing on a materials testing frame. Each construct underwent precyclic rotational shear testing, cyclic loading, and postcyclic rotational shear testing. The stiffness (N·m/deg) of each repair during pre- and postcyclic rotational shear testing was recorded and compared using paired t tests.

RESULTS: Suture bridge fixations, compared with chondral pi […]

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Establishment of a Rat Pull-Out Repair Model Demonstrating Fibrous Meniscus-Bone Healing and Suppression of Meniscal Extrusion After Medial Meniscus Posterior Root Tear.

Katsumata T, Nakagawa Y, Yoshida R, Matsuda J, Nakamura T, Miyatake K, Katagiri H, Seki R et al.

BACKGROUND: Medial meniscus posterior root tear (MMPRT) disrupts meniscal hoop tension and is associated with osteoarthritis progression. Although pull-out repair is a standard surgical technique for MMPRT, its precise healing process remains unclear.

PURPOSE/HYPOTHESIS: This study aimed to establish a rat model for MMPRT pull-out repair, investigating its biomechanical and histological healing and its effect on medial meniscus extrusion (MME). It was hypothesized that pull-out repair would show reduced MME and restore biomechanical property toward normal.

STUDY DESIGN: Controlled laboratory study.

METHODS: A total of 100 male Wistar rats were assigned to the repair (MMPRT pull-out repair with external fixation) or nonrepair (unrepaired MMPRT with external fixation) group. Contralateral knees served as controls. Operated knees were immobilized with an external fixator for 2 weeks. Evaluations at 4 and 12 weeks included micro-computed tomography to quantify MME using the meniscal extrusion ratio (MER), biomechanical testing for tensile strength, and macroscopic/microscopic assessments.

RESULTS: The repair group showed a significantly lower MER than the nonrepair group at all time points. No difference in maximum load to failure was found between 4 and 12 weeks. At both time points, there was no significant difference between the repair group and the healthy leg. Histological […]

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Femoroacetabular Impingement Rabbit Model Induced by Exercise and Mechanical Stimulation.

Du C, Luo Y, Zhou G, Cao Y, Li S, Gao G, He TC, Xu Y

BACKGROUND: Femoroacetabular impingement (FAI) is one of the major causes of hip pain. The pathogenesis of FAI remains unclear, and few animal models are available to study its mechanisms.

PURPOSE: To evaluate bone deformity formation, labrum, and articular cartilage injury induced by mechanical and exercise stimulation, and to establish an FAI model in New Zealand White rabbits.

STUDY DESIGN: Controlled laboratory study.

METHODS: 6-month-old (6M) and 1-month-old (1M) male rabbits were used. Three-dimensional printed polylactic acid scaffolds were surgically fixed to the acetabular rim, followed by postoperative treadmill training. Specimens were collected at 8 and 12 weeks. Bone deformities were assessed by micro-computed tomography. Histological evaluation included hematoxylin and eosin staining for morphology, Safranin O/Fast Green (SO/FG), and toluidine blue staining for proteoglycan and glycosaminoglycan deposition, and immunofluorescence for SRY-box transcription factor 9 (SOX9), aggrecan (ACAN), runt-related transcription factor 2 (RUNX2), and cluster of differentiation 31 (CD31) expression. Mechanical properties of the labrum and cartilage were evaluated by nanoindentation, with the Young's modulus as the stiffness parameter.

RESULTS: Gross evaluation (control [Ctrl] vs FAI, 6M: $P < .0001$, 1M: $P < 0001$) and micro-CT revealed significant bone deformities and cartilag […]

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Knotless Versus Knotted Suture Bridge for Rotator Cuff Tears: A Biomechanical and Histological Study in an Animal Model.

Du H, Guo X, Liu J, Xu Y, Wan M, Xiao W, Zhang P, Jiang X et al.

BACKGROUND: Knotted suture bridge repair (KSB) has been widely demonstrated as an effective method for rotator cuff tears. However, it is associated with high retear rates, primarily attributed to medial row failure resulting from knot-induced stress concentration and compromised tendon perfusion. Knotless suture bridge repair (KLSB) has recently gained considerable attention as a promising alternative technique. Currently, no consensus exists regarding the necessity of medial row knot tying in suture bridge repair for rotator cuff tears. This study aimed to compare the histological and biomechanical outcomes between KLSB and KSB for rotator cuff repair.

HYPOTHESIS/PURPOSE: KLSB will yield superior histological tendon-to-bone healing as compared with KSB. The purpose was to compare the biomechanical and histological outcomes of KLSB and KSB for rotator cuff tears in a rabbit model.

STUDY DESIGN: Controlled laboratory study.

METHODS: Acute bilateral supraspinatus tears were created in the shoulders of 56 New Zealand White rabbits. KLSB and KSB procedures were randomly assigned to the left or right shoulder. Sixteen animals were euthanized at 2, 4, and 8 weeks postoperatively, with 8 rabbits allocated to histological evaluation and the remaining 8 to biomechanical testing. An additional 8 animals were utilized exclusively for initial biomechanical evaluation at week 0.

RESULT […]

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Mechanically Preconditioned Biomimetic Gradient Scaffold for Tendon-to-Bone Regeneration in a Rabbit Rotator Cuff Tear Model.

Chen Y, Li Y, Aili D, Li J, Zhang C, Zhao C, Liu Q

BACKGROUND: Healing of the tendon-bone interface (TBI) after rotator cuff injury is limited, often leading to scar-mediated repair. Decellularized tendon fibrocartilage-bone composite (dTFBC) and bone marrow mesenchymal stem cell sheets (BMSCS) improve repair in preclinical models, but full scaffold encapsulation with mechanical preconditioning remains unexplored.

PURPOSE: To evaluate TBI regeneration using mechanically preconditioned, BMSCS-encapsulated dTFBC scaffolds in a rabbit model.

STUDY DESIGN: Controlled laboratory study.

METHODS: A total of 48 male New Zealand White rabbits were randomized into 4 groups ($n = 12$ each): (1) standard repair (control); (2) repair with dTFBC; (3) repair with dTFBC-BMSCS; and (4) repair with mechanically preconditioned dTFBC-BMSCS (mdTFBC-BMSCS). Healing at 12 weeks was assessed via gross observation, histomorphological and immunohistochemical analyses, and biomechanical testing.

RESULTS: The mdTFBC-BMSCS group demonstrated a higher ultimate failure load (mean, 203.9 ± 65.6 N) than both the control (118 ± 37.4 N; $P = .003$) and dTFBC groups (127 ± 44.1 N; $P = .008$), with no significant difference between the 2 BMSCS-treated groups. Histological analysis revealed enhanced fibrocartilage formation, improved collagen fiber organization, and reduced inflammation infiltration at the TBI in the mdTFBC-BMSCS group. Immunohistochemical analysis […]

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Obesity Amplifies the Detrimental Effects of Delayed Repair on Tendon-to-Bone Healing After Rotator Cuff Injury in a Rat Model.

Wu Y, Liu Q, Li X, Ai Y, Zhao S, Ji X, Li M, Xiu J et al.

BACKGROUND: Rotator cuff repair outcomes are influenced by systemic metabolic status and tear chronicity. Although obesity and delayed surgery are each linked to poorer healing, whether obesity potentiates the detrimental effects of repair delay on tendon-to-bone healing remains incompletely defined.

HYPOTHESIS: Delayed repair would impair tendon-to-bone healing individually, and obesity would exacerbate the consequences of repair delay, resulting in worse structural, functional, and molecular outcomes.

STUDY DESIGN: Controlled laboratory study.

METHODS: A total of 48 Sprague-Dawley rats were randomly assigned to 4 groups defined by repair timing (early vs delayed) and metabolic status (healthy vs obese). Outcomes were assessed at 4 and 8 weeks postrepair using micro-magnetic resonance imaging (MRI), gait analysis, and histology/immunofluorescence; biomechanical testing and lipidomics were performed at 8 weeks.

RESULTS: Across assessments, delayed repair was associated with impaired recovery, and obesity exacerbated these effects, yielding worse outcomes. Specifically, the combined condition exhibited a more unfavorable MRI appearance and gait metrics, collagen architecture, collagen type 3-rich scarring, proteoglycan staining, and cartilage-related marker expression. Notably, lipid infiltration markers FABP4 and perilipin 2 increased by approximately 6-fold ($P < .0001$) and […]

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Can a Clinician Accurately Diagnose Iliopsoas Tendinitis on a Physical Examination?

Johnstone T, Pierre K, Roh E, Fredericson M, Safran MR

BACKGROUND: Iliopsoas (IP) tendinitis is an underrecognized but important cause of anterior hip pain in active patients. Some studies have reported that a physical examination (PE) has limited reliability in making a diagnosis of IP tendinitis. Differentiating anterior hip pain due to IP tendinitis from intra-articular sources is critical to appropriately and expediently treat patients.

HYPOTHESIS/PURPOSE: PE techniques can detect IP tendinitis, and we sought to quantify the diagnostic utility of these techniques using responses to an ultrasound-guided anesthetic injection as the reference standard.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: Charts from a single surgeon's practice were reviewed. Patients with suspected IP tendinitis who received an ultrasound-guided IP bursa injection were included. A pain reduction ≥50% immediately or at first follow-up defined the positive reference standard. Recorded PE variables were absolute and relative seated hip flexion (SHF) weakness, pain with SHF, and absolute and relative tenderness to palpation (TTP) of the IP tendon. Radiographs were systematically reviewed for relevant parameters and concomitant abnormalities, and magnetic resonance imaging (MRI) scans obtained within 12 months of the diagnosis were examined for changes consistent with IP tendinitis. Diagnostic accuracy statistics were calculated.

RESULTS: A t […]

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Predictors of Achieving the Patient Acceptable Symptom State at 5 Years After Primary Hip Arthroscopy in High-Level Adult Athletes.

Owens JS, Lee MS, Jimenez AE, Harris WT, Domb BG

BACKGROUND: Variables predictive of achieving clinically meaningful outcomes in high-level adult athletes after primary hip arthroscopy at midterm follow-up remain incompletely defined.

PURPOSE: To identify variables predictive of achieving the patient acceptable symptom state (PASS) for the Hip Outcome Score-Sports-Specific Subscale (HOS-SSS) at a minimum 5-year follow-up after primary hip arthroscopy in high-level adult athletes.

STUDY DESIGN: Case-control study; Level of evidence, 3.

METHODS: Data were prospectively collected and retrospectively reviewed for adult (≥18 years old) athletes who underwent primary hip arthroscopy for femoroacetabular impingement syndrome between February 2010 and August 2016. Inclusion criteria consisted of participation in high school, collegiate, or professional sports within 1 year before surgery as well as the availability of preoperative and minimum 5-year patient-reported outcome scores for the modified Harris Hip Score, Nonarthritic Hip Score, HOS-SSS, and visual analog scale for pain. Exclusion criteria were age <18 or >50 years, workers' compensation status, previous ipsilateral hip surgery/conditions, Tönnis grade >1 osteoarthritis, or unwillingness to participate. Patients were stratified based on achievement of the PASS for the HOS-SSS at 5-year follow-up. Univariate and multivariate logistic regression analyses were performed to […]

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Return to Sport in Professional Athletes With Borderline Hip Dysplasia After Hip Arthroscopy for Femoroacetabular Impingement.

Uppstrom T, Felan NA, Lind DRG, Dornan GJ, Speshock A, Philippon MJ

BACKGROUND: Hip arthroscopy for femoroacetabular impingement (FAI) in professional athletes is associated with significant improvements in postoperative pain and function and a return-to-sport (RTS) rate >80%. However, RTS rate after hip arthroscopy for FAI in professional athletes with borderline dysplasia is unknown.

PURPOSE: To assess RTS rates exclusively in professional athletes with borderline dysplasia after hip arthroscopy for treatment of FAI.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: A total of 40 professional and Olympic athletes (42 hips) with borderline dysplasia (lateral center edge angle [LCEA] 18°-25°) underwent hip arthroscopy for FAI between 2005 and 2022. RTS was defined as competing in a single professional game at an equal level after surgery. Data were retrospectively obtained for each athlete from publicly available sport-specific data sources.

RESULTS: The study included 42 hips. Patients undergoing primary hip arthroscopy demonstrated a 79% RTS rate, whereas patients undergoing revision hip arthroscopy demonstrated a 25% RTS rate (P = .006). Among athletes who successfully returned, RTS occurred at a median of 7.5 months (range, 3.1-24.3 months) after surgery. The mean age at the time of surgery was 28.9 years (range, 18-53.5 years). In total, 21 hips (50%) had an Outerbridge grade 3 or 4 defect at the time of surgery, and 10 (24%) […]

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Return-to-Work and Clinical Outcomes After Anterior Latissimus Dorsi and Teres Major Tendon Transfer for Irreparable Subscapularis Tears.

Baek CH, Kim BT, Kim JG, Lim C, Kim SJ

BACKGROUND: Irreparable subscapularis tears represent a significant clinical challenge, especially in active patients aiming to preserve shoulder function. Anterior latissimus dorsi and teres major (LDTM) tendon transfer has been proposed to restore internal rotation and anterior shoulder stability; however, data on return-to-work (RTW) outcomes remain limited.

PURPOSE/HYPOTHESIS: The study aimed to evaluate clinical outcomes and RTW rates after anterior LDTM transfer in patients with irreparable subscapularis tears, with the hypothesis that the procedure would provide favorable clinical results and a high RTW rate.

STUDY DESIGN: Case series; Level of evidence, 4.

METHODS: A retrospective review was conducted on patients who underwent LDTM transfer for irreparable subscapularis tears, with or without concomitant supraspinatus involvement. Inclusion criteria were irreparable subscapularis tears with medial tendon retraction and grade 3 or 4 fatty infiltration, without advanced glenohumeral arthritis. Exclusion criteria included prior bone surgery, incomplete follow-up, or loss to follow-up. Clinical assessment included patient-reported outcome measures, such as visual analog scale, American Shoulder and Elbow Surgeons form, Constant score, and Single Assessment Numeric Evaluation, as well as range of motion and strength. Tendon integrity was evaluated via magnetic resonance i […]

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Summative Report on Time Out of Play and Injury Rates for Major and Minor League Baseball: An Analysis of 83,215 Injuries From 2011 to 2023.

Ward GH, Meta F, Cabarcas B, Hebert E, Griffith TB, Meister K, Camp CL

BACKGROUND: Recent reports suggest increasing trends in injuries, primarily among pitchers and upper extremity injuries, among Major League Baseball (MLB) and Minor League Baseball (MiLB) players. Accordingly, an updated comprehensive report on injury epidemiology in professional baseball is needed.

PURPOSE: To (1) update the literature with a comprehensive epidemiological report of all injuries in MLB and MiLB; (2) provide a detailed breakdown of characteristics of the top 50 injuries; and (3) analyze long-term trends in injuries and days missed (DM) over time.

STUDY DESIGN: Case Series; Level of evidence, 4.

METHODS: The MLB Health and Injury Tracking System was used to identify injuries in MLB and MiLB players from 2011 to 2023. The inclusion criteria consisted of injuries occurring during spring training, regular season, or postseason activities that resulted in ≥1 DM from play. Basic and advanced player and injury characteristics were analyzed-including a list of the top 50 most common injuries.

RESULTS: From 2011 to 2023, a total of 83,215 musculoskeletal injuries occurred among MLB and MiLB players, resulting in 1,280,887 total DM, with a mean (median) DM per injury of 18 (11). The upper extremity accounted for 46% of injuries, and the lower extremity for 30%. The most common injury was hamstring strains (n = 6603; 7.9%); hamate fractures had the highest rate of surg […]

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Lateral Extra-articular Tenodesis With Anterior Cruciate Ligament Reconstruction in Pediatric and Skeletally Immature Patients: A Systematic Review and Meta-analysis.

Kotipalli S, Haidl T, Vivekanantha P, de Sa D, Kay J

BACKGROUND: Lateral extra-articular tenodesis (LET), alongside anterior cruciate ligament reconstruction (ACLR), has been shown to improve rerupture and rotational laxity in patients <25 years. However, safety and efficacy in both general pediatric (<18 years) and skeletally immature patients are important to identify.

PURPOSE: To assess clinical outcomes and complications after the LET procedure with ACLR in the pediatric and skeletally immature population.

STUDY DESIGN: Meta-analysis; Level of evidence, 4.

METHODS: Three databases were searched on December 5, 2024. Data were collected on study characteristics, demographics, surgical details, LET indications, patient-reported outcome measures, return to sport (RTS), rerupture rates, and complications. A meta-analysis of graft rerupture and RTS was performed using a Mantel-Haenszel and fixed-effects model (pooled effect measure: odds ratio [OR] with 95% CI).

RESULTS: Nine studies comprising 317 patients (318 knees) were included, of whom 204 patients (205 knees) were skeletally immature. The mean age of all patients and skeletally immature patients was 14.6 years (range, 8-18 years) and 13.6 years (range, 8-16.1 years), respectively. Common indications for LET included a grade 2+ pivot shift and intention to return to a high level of sport. The pooled RTS rate of ACLR+LET was 96% (92%-99%; I2 = 48%) and 98% (94%-100%; I2 = […]

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Return to Sport After Revision Anterior Cruciate Ligament Reconstruction: A Systematic Review and Meta-analysis.

Gopinatth V, Touhey DC, Barksdale EM, Knapik DM

BACKGROUND: Revision anterior cruciate ligament (ACL) reconstruction (ACLR) is a well-established procedure to restore knee stability and improve function after a failed primary ACLR. In active individuals, patient, injury, and operative variables influencing successful return to sport (RTS) after revision ACLR remain poorly understood.

PURPOSE: To evaluate RTS outcomes in patients undergoing revision ACLR.

STUDY DESIGN: Meta-analysis, Level of evidence, 4Methods:A systematic review was conducted in accordance with the 2020 PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines. A literature search was conducted by querying 5 databases from inception through January 2025 to identify studies reporting on RTS outcomes in athletes undergoing revision ACLR. Meta-analysis was performed using random-effects models at 95% confidence intervals, with odds ratios used for comparative studies.

RESULTS: A total of 52 studies, consisting of 3814 patients, met inclusion criteria. The mean patient age was 27.6 ± 8.4 years, with 66.3% (2340/3532) of the patients being male. Soccer was the most commonly reported sport (24.6%; 390/1584), followed by basketball (17.6%; 278/1584) and football (7.8%; 124/1584). The overall pooled RTS rate was 77.8% (95% CI, 0.732-0.824), with the RTS rate to the previous level of competition being 48.2% (95% CI, 0.410-0.553). The […]

PubMed · DOI

Comparative Outcomes of Autograft and Allograft in Hip Labral Reconstruction: A Systematic Review and Meta-analysis.

Kurapatti M, Yuro M, Tao BS, Ramey MD, Swaminathan S, Locke AR, Koehne NH, Parisien RL

BACKGROUND: There is currently no consensus regarding the superiority of allografts or autografts in labral hip reconstruction.

PURPOSE: To compare patient outcomes after arthroscopic labral reconstruction using autografts versus allografts.

STUDY DESIGN: Meta-analysis; Level of evidence, 4.

METHODS: A systematic review and meta-analysis were conducted according to PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines. The PubMed, Embase, and Scopus databases were queried on February 2, 2025. Original outcome studies on arthroscopic hip labral reconstruction with clearly delineated graft type and a mean follow-up of at least 1 year were included. Studies with a mean follow-up <1 year, nonarthroscopic labral reconstruction, or unclear graft-specific outcomes were excluded. Data were extracted from 30 studies (1968 hips: 1071 autograft, 897 allograft). A random-effects meta-analysis was performed using RStudio to assess patient-reported outcomes, revision surgery, and complication rates.

RESULTS: There were no significant differences in most patient-reported outcomes: Harris Hip Score (allograft: 27.5 vs autograft: 24.6; P = .17), Hip Outcome Score (HOS) Activities of Daily Living (20.9 vs 22.8; P = .59), HOS Sports (33.2 vs 34.0; P = .65), SF-12 Physical score (10.0 vs 7.0; P = .19), and visual analog scale pain score (-3.5 vs -4.2; P = .34). […]

PubMed · DOI

Risk Factors for Shoulder Stiffness After Rotator Cuff Repair: A Systematic Review and Meta-analysis.

Zhan H, Yang Z, Liu P, Wang Y, Zhao Z, Zhang B, Liang Q, Zheng J et al.

BACKGROUND: Shoulder stiffness after rotator cuff repair (RCR) represents a primary determinant of patient-reported quality of life. The implementation of predictive strategies for the early identification of risk factors is critical for mitigating this complication.

PURPOSE: To evaluate the incidence of shoulder stiffness after RCR and to identify associated risk factors.

STUDY DESIGN: Systematic review and meta-analysis; Level of evidence, 4.

METHODS: A comprehensive literature search was conducted in PubMed, Embase, Web of Science, and the Cochrane Library from database inception to February 28, 2025. Factors reported in ≥2 studies with extractable data underwent meta-analysis; otherwise, a qualitative synthesis was performed. Sensitivity analyses were executed for risk factors exhibiting substantial heterogeneity or potential reporting bias. Effect sizes for risk factors were calculated using odds ratios (ORs) or mean differences (MDs) with 95% confidence intervals (CIs).

RESULTS: This meta-analysis incorporated 20 studies comprising 34,565 patients undergoing RCR. The pooled incidence of postoperative stiffness was 12.9% (95% CI, 8.5%-18.0%), with significant variations based on diagnostic criteria (15.8% for range of motion vs 2.2% for intervention-based definitions; $P < .001$) and follow-up duration (18.9% at <6 months vs 9.2% at ≥6 months; $P < .05$). Overall, 33 poten […]

PubMed · DOI

Clinical Outcomes and Complications After Sternoclavicular Joint Reconstruction for Chronic Instability: A Systematic Review.

Kapoor U, Khoo KJ, Hsu JE, Matsen FA, Schiffman CJ

BACKGROUND: Chronic sternoclavicular (SC) joint instability is a relatively rare orthopaedic pathology that is treated with surgical reconstruction when symptoms persist despite adequate nonsurgical treatment. There are a variety of surgical techniques; however, the literature on the topic is limited to small case series.

PURPOSE: To provide a comprehensive systematic review of the surgical variables, clinical outcomes, and complications after SC joint reconstruction.

STUDY DESIGN: Systematic review; Level of evidence, 4.

METHODS: A systematic review of the literature was conducted following the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines. PubMed and MEDLINE were searched using a comprehensive search strategy for articles involving SC joint reconstructions. A descriptive and critical analysis of the results was performed.

RESULTS: Twelve studies comprising 164 patients (169 SC joint reconstructions) were identified. Bicortical reconstruction was the most frequently used technique overall (63% of cases) and was used more often in the setting of posterior instability (87%). All studies demonstrated significant improvements in patient-reported outcome measures (PROMs) regardless of graft type, technique, or direction of instability. Complication rates were similar between techniques (9% bicortical vs 15% unicortical; $P = .23$), althoug […]

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Patient-Reported Outcomes and Revision Rates After ACL Reconstruction With Quadriceps Versus Hamstring and Patellar Tendon Autografts: Letter to the Editor.

Kocak S

PubMed · DOI

Patient-Reported Outcomes and Revision Rates After ACL Reconstruction With Quadriceps Versus Hamstring and Patellar Tendon Autografts: Response.

Rizvanovic D, Waldén M, Forssblad M, Cristiani R, von Essen C, Stålman A

PubMed · DOI

Using a Residual Pivot Shift as the Indication to Perform a Lateral Extra-articular Tenodesis During ACL Reconstruction Using Autologous Hamstring Grafts Is Associated With Improved Surgical Outcomes: Response.

Porter MD

PubMed · DOI

Using a Residual Pivot Shift as the Indication to Perform a Lateral Extra-articular Tenodesis During ACL Reconstruction Using Autologous Hamstring Grafts Is Associated With Improved Surgical Outcomes: Letter to the Editor.

Dheeraj Y, Sankhwar DK

PubMed · DOI

BJJ

The Bone & Joint Journal

Volume 108-B, Issue 6 · June 2026 · 14 new articles

Line-to-line cementation in hip arthroplasty : what is the current role of the 'French Paradox'?

Andriollo L, Vermue H, Haddad FS, Lustig S

AIMS: The 'French Paradox' cementing technique challenges conventional principles of femoral component fixation by advocating line-to-line femoral component insertion and a thin cement mantle of variable thickness. This contradicts the widely accepted recommendation of a 2 mm to 4 mm cement mantle thickness. However, this approach has shown good clinical results. This review aims to summarize the current evidence regarding the clinical outcomes, biomechanical analysis, and ongoing controversies surrounding the French Paradox cementing technique in total hip arthroplasty (THA).

METHODS: A systematic literature search was conducted to identify studies investigating the concept of line-to-line cementing, or the French Paradox, in the context of primary THA. Studies were identified through database searches (PubMed/MEDLINE, Embase, Web of Science, and Cochrane Library) up to March 2025 and selected based on predefined criteria. Data extraction and quality assessment were performed independently. Due to methodological heterogeneity, findings were synthesized narratively, following PRISMA guidelines.

RESULTS: Biomechanical data support the effectiveness of the line-to-line cementing technique, showing superior rotational stability and adequate fixation even in osteoporotic bone. Clinical studies report excellent survivorship rates, with ten- and 17-year implant survival exceeding 9 […]

PubMed · DOI

Increasing mobility in hospital after hip fracture.

Kearney RS, Costa ML

Each year there are 70,000 new hip fractures in the UK, with an ongoing annual cost to health and social care services of £4.3 billion. Following surgery, 30% of patients are not returned to their usual home by 120 days. Current UK national standards for rehabilitation recommend two hours of physiotherapy in the first postoperative week. It has been shown that increasing the frequency and duration of physiotherapy could lead to more rapid recovery and earlier discharge from hospital, with more patients returning to their preinjury place of residence. However, the National Institute for Health and Care Excellence found this evidence to be insufficient to recommend a change in national guidelines. The World Hip Trauma Evaluation (WHiTE) 15: INITIATE is a two-arm, multicentre, cluster, randomized controlled trial with embedded process and economic evaluations. Half of the centres where these patients are treated will provide routine care, and the other half will provide routine care with increased duration and frequency of physiotherapy combined with strategies to increase mobility-focused activities.

PubMed · DOI

Diagnosis of aseptic loosening after hip arthroplasty : international Delphi consensus.

Hopman AGM, Blankevoort L, Lustig S, Amanatullah DF, Schafroth MU, Kievit AJ, Hip Arthroplasty Loosening (HAL) Study Group, Abdel MP et al.

AIMS: This Delphi consensus study aimed to achieve international expert agreement on clinical and radiological criteria for diagnosing aseptic component loosening in total hip arthroplasty (THA). The consensus involved revision hip surgeons from diverse healthcare settings worldwide, aiming to standardize diagnostic criteria and enhance clinical decision-making and future research comparability.

METHODS: A three-round Delphi study was conducted in accordance with Conducting and Reporting of Delphi Studies guidelines. A panel of 125 orthopaedic surgeons from 32 countries was approached based on geographical representation and experience in revision THA. In Round 1, experts listed clinical and imaging criteria; in Rounds 2 and 3, derived statements were rated on a Likert scale. Consensus was a priori defined as ≥ 70% agreement. A total of 43 experts completed Round 1, and 37 completed all three Delphi rounds.

RESULTS: Consensus was achieved on 25 clinical and radiological statements. Investigations for periprosthetic joint infection was unanimously considered the first step. Limping and pain were general indicators of loosening. Pain during weightbearing, hip rotation, start-up, or localized to the thigh was linked to femoral loosening. Pain at start-up, in the groin, or during activities was associated with acetabular loosening. Radiographs were the primary imaging modality, s […]

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Decreased evertor strength following peroneus longus tendon harvest for anterior cruciate ligament reconstruction : an isokinetic muscle strength testing study.

Kotian RN, Ajoy SM, Panduranga R, Visweswara RD, Kumar P, Galagali DA, Patil SC, Rafi Vaderi BM

AIMS: There are conflicting results in the literature regarding the safety of the peroneus longus tendon (PLT) as a potential graft for anterior cruciate ligament (ACL) reconstruction, and no long-term studies with objective assessment exist. Therefore, we assessed the donor ankle following PLT harvest for ACL reconstruction using isokinetic muscle strength testing (IMST) of the evertors and invertors of both feet.

METHODS: The study included 31 patients with a mean age of 32.61 years (SD 10.60) and a mean follow-up period of 18.34 months (SD 6.25). IMST of evertors and invertors of both feet was conducted. Subjective outcomes were assessed using the Foot and Ankle Ability Measure (FAAM) score and the Foot and Ankle Outcome Score (FAOS). To compare the different parameters between timepoints, repeated measures of analysis of variance/Friedman test were used.

RESULTS: The mean length and diameter of the PLT graft were 29.32 cm (SD 2.26) and 9.32 mm (SD 0.92), respectively. A statistically significant decrease in peak torque at 60°/second was observed at six months for both evertors ($p < 0.001$) and invertors ($p = 0.010$). No statistically significant difference in peak torque was observed at one year at 60°/second (evertors: $p = 0.386$; invertors: $p = 0.685$) or 120°/second (evertors: $p = 0.071$; invertors: $p = 0.156$). The difference in FAAM ($p = 0.621$) and FAOS ($p = 0.456$) scores […]

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National patient-reported outcome measures data and the National Joint Registry : a report on the data completeness and quality.

Singh Benning M, Penfold CM, Matharu GS, Sayers A, Blom AW, Whitehouse MR, Wilkinson JM, Judge A

AIMS: Since April 2009, patients undergoing NHS-funded primary elective total knee or hip arthroplasties (TKAs or THAs) should be invited to complete patient-reported outcome measures (PROMs). The aim of this study was to use National Joint Registry (NJR) data to identify patient and demographic differences between those with complete versus those with missing PROMs.

METHODS: Patients undergoing publicly funded elective primary TKA or THA between April 2009 and December 2018 recorded in the NJR were eligible. NJR data were linked to the English Hospital Episode Statistics (HES), and PROMs data. Oxford Knee/Hip scores (OKS/OHS) were eligible for inclusion if recorded up to 18 weeks preoperatively (Q1) and between six and 12 months postoperatively (Q2). Proportions were used to describe the completeness of PROMs data. The following variables were assessed to determine associations with completeness of PROMs data: age, sex, American Society of Anesthesiologists (ASA) grade, BMI, socioeconomic deprivation, baseline (Q1) OKS/OHS, and hospital.

RESULTS: Of 570,449 eligible TKAs and 507,962 eligible THAs, complete pre- and postoperative PROMs were available for 229,794 TKAs (40.3%) and 210,929 THAs (41.5%). There were no differences in the patient and demographic factors between those with and without complete PROMs. Patients with complete PROMs had higher preoperative PROMs (less p […]

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Infection following foot and ankle surgery : a subanalysis of data captured from the UK Foot and Ankle Thromboembolism (FATE) audit.

Shepherd J, Mason L, Houchen-Wolloff L, Malhotra K, Mangwani J, UK-FATE Collaborative, Ahmad I, Zeb J et al.

AIMS: The UK Foot and Ankle Thromboembolism (UK-FATE) Collaborative evaluated incidence of venous thromboembolism (VTE) in foot and ankle surgery and Achilles tendon (AT) rupture. As part of the data collection, infection within 90 days was a secondary outcome. This study aimed to evaluate overall rate of superficial and deep infection following foot and ankle surgery in trauma, elective, and acute diabetic surgery.

METHODS: Data were collected prospectively across 68 centres in the UK for all patients who underwent foot and ankle surgery between 1 June and 30 November 2022. Over 500,000 data points were collected, including development of superficial and deep infection within 90 days of procedure and presence of comorbidities.

RESULTS: UK-FATE captured 90-day infection data for a total of 9,723 cases. When nonoperatively managed AT ruptures were excluded, infection data were available for 9,591 trauma, elective, and acute diabetic procedures. Overall superficial and deep infection rates in all procedures were 4.1% (n = 397/9,591) and 1.4% (n = 133/9,591), respectively. Both superficial (7.4% (n = 29); p < 0.001) and deep infection (8.7% (n = 34); p < 0.001) rates were highest in acute diabetic procedures. Deep infection differed between trauma (1.4% (n = 65/4,351)) and elective (0.7% (n = 34/4,785)) procedures (p < 0.001), but superficial infection rates were similar (3.9% ([…]

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The experience of stemmed pyrocarbon shoulder hemiarthroplasty in New Zealand : ten years of registry data.

Konarski AJ, Mohammed KD

AIMS: Pyrocarbon has been used as a bearing surface in hemiarthroplasty (HA) of the shoulder for more than a decade. The use of pyrocarbon is thought to reduce the rate of glenoid erosion compared with metal HA, and reduce the risk of glenoid complications compared with anatomical total shoulder arthroplasty (aTSA). However, there is little information about the long-term outcomes. The aim of this study was to investigate the use of pyrocarbon HA (PyCHA) in New Zealand, and the clinical outcomes and revision rates.

METHODS: The New Zealand National Joint Registry was used to obtain data on all patients undergoing HA of the shoulder with a stemmed PyCHA. Data which were assessed included the indication for surgery, patient-reported outcome scores, and revision rates.

RESULTS: Between January 2014 and December 2024, 288 stemmed PyCHAs were recorded in 277 patients, with a mean age of 55 years (20 to 86). The mean follow-up was 50 months (3 to 128). The use increased 2.3-fold during the study period. The most common indication for surgery was osteoarthritis (OA) (154 (53.5%)), followed by avascular necrosis (AVN) (52 (18.1%)), the sequelae of trauma (48 (16.7%)) and after recurrent dislocation (38 (13.2%)). The mean Oxford Shoulder Score (OSS) was 36.4 (SD 8.2) six months postoperatively, increasing to 40.8 (SD 7.9) at five years. The revision rate was 0.75 per 100 component-yea […]

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Revision after anatomical total shoulder arthroplasty : the influence of age on outcomes in a large nationwide registry study.

Smeitink N, Schröder FF, Dorrestijn O, Spekenbrink-Spooren A, Dahmen J, van Noort A, van den Bekerom MPJ, Veen EJD

AIMS: Primary anatomical total shoulder arthroplasty (aTSA) is generally the preferred treatment for active young patients with increased functional demands who need a shoulder arthroplasty. There is little information, however, about the survival of this implant in patients aged < 65 years. The aim of this study was to investigate the differences in revision rates after primary aTSA in patients aged < 65 years compared with those aged ≥ 65 years, within ten years, hypothesizing that younger patients would have higher revision rates due to increased functional demands and activity levels.

METHODS: Data on 3,622 primary aTSAs performed between January 2014 and January 2024 were extracted from the Dutch Arthroplasty Register. A total of 1,486 were performed in patients aged < 65 years, and 2,136 in those aged ≥ 65 years. Kaplan-Meier survival and multivariate Cox regression analyses were performed.

RESULTS: Within ten years of follow-up, 159 revisions (4%) were required. Significantly more revisions were performed in the younger group (89 (6%)) than in the older group (70 (3%); p < 0.001). The cumulative revision percentages were 11% (95% CI 8 to 14) and 6% (95% CI 4 to 7; p < 0.001), respectively. The most common indications for revision in both groups were cuff rupture, instability, and loosening of the glenoid component. In patients aged ≥ 65 years, both revisions for peripr […]

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Computer navigation in total shoulder arthroplasty : analysis from the Australian Orthopaedic Association National Joint Replacement Registry.

Wilcox BM, Campbell RJ, Gill DRJ, Harries D, Yeoh TS, Low AK

AIMS: Computer navigation is capable of improving the accuracy of glenoid baseplate and screw positioning in shoulder arthroplasty, but there is very limited evidence of its effect on clinical outcomes. In Australia, there is one commonly used total shoulder arthroplasty prosthesis with the option for intraoperative navigation. The aim of this study was to compare revision rates for a single prosthesis design, performed with or without computer navigation.

METHODS: Data were obtained from a large national arthroplasty registry for all primary stemmed anatomical total shoulder arthroplasty (aTSA) and reverse total shoulder arthroplasty (rTSA) procedures using a single-design platform prosthesis (SDPP) from 1 June 2011 to 31 December 2022. Procedures were grouped by the use of computer navigation versus no technology assistance. Stemless procedures were excluded. A subgroup analysis included only procedures from 31 January 2017, when computer navigation for this device was first recorded, to enable a time-matched comparison. Further subanalyses assessed outcomes by glenoid morphology (from 2017) and mortality, limited to the first eligible procedure per patient. Kaplan-Meier estimates of survivorship described the time to first revision. Hazard ratios (HRs) from Cox proportional hazard models, adjusted for age and sex, were used to compare revision rates.

RESULTS: There were 5, […]

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The association between the morphological patterns of medium to large rotator cuff tears and the retear rate after arthroscopic repair.

Lee S, Lee HA, Shin SJ

AIMS: The aim of this study was to evaluate factors associated with retear of the rotator cuff after arthroscopic repair of cuff tears and to investigate whether these factors vary according to the pattern of the tear.

METHODS: Between March 2014 and June 2021, patients who underwent arthroscopic complete rotator cuff r n epair for symptomatic 2 cm to 4 cm full-thickness cuff tears with a minimum follow-up of two years were retrospectively reviewed. The patterns of the tears were classified intraoperatively after debridement of degenerative tissues, as crescent-shaped, U-shaped, anterior L-shaped, and posterior L-shaped tears. Based on MRI or ultrasound performed six months postoperatively, patients were divided into healed and retear groups. Clinical and radiological parameters relating to retear were compared between the groups, and multiple regression analysis was performed to identify factors associated with retear. The interaction between the patterns of tear and each independent retear-related factor was assessed using logistic regression.

RESULTS: Of 713 patients, the tendon healed in 605 (84.9%) and 108 (15.1%) had a retear. In the multiple regression analysis, an anterior L-shaped tear (odds ratio (OR) 3.26 (95% CI 1.58 to 6.71); p = 0.001) was the most powerful independent factor associated with retear, followed by age (OR 1.04 (95% CI 1.01 to 1.06); p = 0.008), ant […]

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Pain Rehabilitation to Optimize Major Orthopaedic Trauma REcovery (PROMOTE) compared with routine care : a multicentre, parallel-group, randomized feasibility trial with an embedded qualitative study.

Fordham B, Quarmby LM, Costa ML, Painton R, Draper K, Knight R, Achten J, Appelbe D et al.

AIMS: The aim of this study was to assess the feasibility of conducting a randomized controlled trial (RCT) to test the effectiveness of an online, biopsychosocial intervention for the management of recovery in patients with complex lower-limb orthopaedic trauma.

METHODS: This was a multicentre, parallel-group, randomized feasibility trial with an embedded qualitative study. Patients were recruited from four UK NHS major trauma centres if they were aged ≥ 16 years and had undergone surgery for complex lower-limb orthopaedic trauma. They were randomized to gain access to a biopsychosocial support website to target psychological predictive factors of poor outcomes for three months in addition to routine care, plus four one-to-one sessions with healthcare professionals to increase adherence, compared with routine care only. The primary outcome was patient acceptance, and secondary outcomes were the patients' adherence and intervention delivery fidelity.

RESULTS: A total of 57 of 112 eligible patients (51%) participated (≥ 50% study feasibility criterion) and a mean of 2.7 were recruited per centre per month (≥ 2 criterion). They attended a median of three out of four sessions (3 to 4 criterion). However, the intervention delivery fidelity criterion (≥ 90% of intervention providers deliver the intervention as per manual) was not met (58% observed). The retention criterion (< 20% […]

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Implant survival and limb salvage rates after combined proximal tibial and distal femoral endoprosthetic replacement.

Baird CAE, Archer JE, Green NM, Burahee A, Riaz M, Ahmed Kamel S, Stevenson J, Kurisunkal V et al.

AIMS: Combined endoprosthetic replacement of the ipsilateral distal femur and proximal tibia is rarely performed. It is usually undertaken to achieve limb salvage after resection of primary bone tumours around the knee. This study aims to determine the viability of 'composite' endoprosthetic reconstruction around the knee, and report implant survival and limb salvage rates.

METHODS: A retrospective review of a prospectively maintained departmental database was undertaken to identify patients who had undergone composite endoprosthetic reconstruction around the knee. The primary outcome of interest was revision-free implant survival. Secondary outcomes were amputation-free survival, the Musculoskeletal Tumor Society (MSTS) score, and extensor lag. A total of 34 patients (21 male and 13 female) with a mean age of 32 years (14 to 66) underwent surgery between January 1997 and November 2024, and were followed up for a median interval of 12.6 years (IQR 3 to 14).

RESULTS: The oncological diagnoses were osteosarcoma (n = 26), Ewing's sarcoma (n = 5), and one each of giant cell tumour of bone, spindle cell sarcoma, chondrosarcoma, and synovial sarcoma. Of the eight primary and 26 revision cases, the pooled five- and ten-year revision-free implant survival rates were 77% (95% CI 63 to 94%) and 61% (95% CI 45 to 83), respectively (Kaplan-Meier). Resection length and construct:stem rati […]

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Natural history of radiological coronal alignment of lower limbs in children and adolescents with achondroplasia.

Sawamura K, Matsushita M, Mishima K, Imagama S, Kitoh H

AIMS: Achondroplasia (ACH) is the most common skeletal dysplasia worldwide, caused by mutations in the fibroblast growth factor receptor 3 gene and characterized by a disproportionately short stature, and many orthopaedic complications. Genu varum affects between 80% and 90% of children with ACH, but the longitudinal natural history of lower limb alignment remains unclear. The aim of this study was to describe age-related changes in lower limb alignment in children with ACH by analyzing longitudinal radiological parameters from infancy to skeletal maturity.

METHODS: This retrospective cohort study included 42 children with ACH, who had at least two radiological assessments at our institution between January 2003 and December 2024. Whole-leg supine anteroposterior radiographs were obtained before the age of two years and in the standing position thereafter. There were 22 females and 20 males, and a total of 254 limbs were analyzed. The parameters which were analyzed included the femorotibial angle (FTA), mechanical axis angle (MAA), mechanical lateral proximal femoral angle (mLPFA), mechanical lateral distal femoral angle (mLDFA), medial proximal tibial angle (MPTA), and lateral distal tibial angle (LDTA). Radiographs were grouped into nine age groups between six months and 18 years. Longitudinal changes were analyzed using one-way analysis of variance with Bonferroni post hoc […]

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Femoral nerve palsy in brace treatment for developmental dysplasia of the hip : incidence and outcomes in a prospective international cohort.

Schaeffer EK, Wang AWT, Hu J, Nguyen V, Sankar WN, Williams N, Global Hip Dysplasia Study Group, Mulpuri K et al.

AIMS: Femoral nerve palsy is a potential complication of brace treatment for children with developmental dysplasia of the hip (DDH). Little is known about its causes, and previous studies have been limited by their small sample size, retrospective design, and/or being single-centre series. The aim of this study was to examine the risk factors for femoral nerve palsy in the largest prospective cohort of children to date with DDH treated using an orthosis.

METHODS: A global multicentre prospective database of children with DDH was analyzed. Those treated primarily using an orthosis were included. Mixed-effects logistic regression was used to identify risk factors for the development of femoral nerve palsy, including sex, age at the time of diagnosis and application of an orthosis, the type of orthosis, the location of the femoral head, femoral head cover, and α angle. Both univariate and multivariate analyses were conducted.

RESULTS: The study included 3,008 children (5,012 affected hips) who were enrolled from 21 centres in seven countries; 99 hips (2.0%) in 94 children (3.1%) developed a femoral nerve palsy, which occurred at a median of seven days (IQR 6 to 21) after the application of a brace. A significantly increased proportion of children who developed a femoral nerve palsy were treated in a Pavlik harness compared with those who did not develop a femoral nerve palsy (84 […]

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Clinical Orthopaedics & Related Research

Volume 484, Issue 6 · June 2026 · 39 new articles

Guest Editorial: The Health Benefits of Bayesian Reasoning.

Rezk E, Ring D

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Editor's Spotlight/Take 5: Few FDA Approved AI/ML Orthopaedic Devices Have EU MDR Equivalents or Peer-Reviewed Validation.

Manner PA

PubMed · DOI

Few FDA Approved AI/ML Orthopaedic Devices Have EU MDR Equivalents or Peer-Reviewed Validation.

Bracken A, Whelehan S, Merghani K, Sheehan E, Feeley I

BACKGROUND: Artificial intelligence (AI) and machine-learning (ML) technologies are increasingly being incorporated into orthopaedic medical devices, offering the potential to enhance surgical planning, intraoperative precision, and personalized care. The incorporation of AI into medical devices introduces unique risks and challenges that traditional regulatory pathways in the United States and European Union (EU) were not designed to address. Although the EU Artificial Intelligence Act is currently being implemented to provide a specific framework for AI medical device assessment, no equivalent directive exists in the United States.

QUESTIONS/PURPOSES: In consideration of these evolving regulatory contexts, this study sought to address the following questions related to FDA-approved, AI/ML-enabled orthopaedic devices: (1) How often are FDA-approved, AI/ML-enabled orthopaedic devices also approved under the EU Medical Device Regulation (MDR)? (2) How often are AI/ML-enabled orthopaedic devices approved through the FDA 510(k) pathway using non-AI predicate devices? (3) How often do approved AI/ML-enabled orthopaedic devices have independent, peer-reviewed evidence supporting their use? (4) How often are the details of the training and validation datasets of these devices disclosed in publicly available regulatory documents?

METHODS: A total of 37 AI/ML-enabled orthopaedic devi […]

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Not the Last Word: One of These Predictions, ± 1, Will Come True.

Bernstein J

PubMed · DOI

Your Best Life: Practicing Detachment From Negativity.

Kelly JD

PubMed · DOI

Medicolegal Sidebar: What to Do When a Patient Threatens to Harm Someone.

Bal BS

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Virtue Ethics in a Value-driven World: When Cost Shapes Choice-Ethical Decisions in Rural Fracture Care.

Salphale YS, Babhulkar S

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Intersection of Gender and Orthopaedic Culture: Great and Courageous Leadership.

Zillmer DA

PubMed · DOI

ArtiFacts: Smith-Petersen's Accidental Prosthesis.

Hawk AJ

PubMed · DOI

CORR Synthesis: How Should PROM Thresholds Be Determined and Interpreted to Reflect Clinically Meaningful Change in Orthopaedic Surgery?

Vallurupalli N, Padon B, Yao JJ

PubMed · DOI

Classifications in Brief: The AO Spine Upper Cervical Injury Classification System.

Khela M, Gendelberg D

PubMed · DOI

Severe Damage to the Ligamentous-Fossa-Foveolar Complex Is Common in Patients Undergoing Surgical Hip Dislocation for Femoroacetabular Impingement.

Stetzelberger VM, Popa V, Schwab JM, Tannast M

BACKGROUND: Ligamentum teres lesions are increasingly linked to hip pain and poor surgical outcomes. However, the extent of damage to the full ligamentous-fossa-foveolar complex (LFFC) and the factors associated with this finding remain unclear in young patients undergoing joint-preserving surgery.

QUESTIONS/PURPOSES: In this study, we sought (1) to determine the percentage of patients undergoing surgical hip dislocation for femoroacetabular impingement who have diagnosable LFFC lesions using a previously validated grading system and (2) to identify clinical and radiographic factors associated with more severe LFFC damage.

METHODS: Between July 2016 and December 2023 the senior author treated 402 hips with surgical hip dislocation for symptoms and radiological signs of femoroacetabular impingement (cam morphology, decreased acetabular coverage, or femoral version abnormalities) without advanced osteoarthritis (Tönnis > II). Based on that, 34% (136 of 402) were excluded because of previous surgery, posttraumatic FAI etiologies, incomplete documentation, tumor associated, pediatric hip disorders (Perthes disease and slipped capitis femoris epiphysis), or avascular necrosis. This left 66% (266 of 402) for final analysis in this retrospective, institutional review board-approved study. Mean age was 29 ± 8 years, and 46% (122 of 266) were male. Standardized AP and axial radiograph […]

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CORR Insights®: Severe Damage to the Ligamentous-Fossa-Foveolar Complex Is Common in Patients Undergoing Surgical Hip Dislocation for Femoroacetabular Impingement.

Faure PA

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Does Insertional Reattachment for Acute Sleeve Avulsion Fracture of the Achilles Tendon Provide Sustained Function and Sports Participation?

Xiong S, Li Y, Xie X, Ma K, Pi Y, Yang Y, Zhao F, Chen L et al.

BACKGROUND: Insertional reattachment of the Achilles tendon is the recommended surgical treatment for acute sleeve avulsion fractures at the tendon insertion in patients fit for surgery and who are comfortable taking the risks associated with the procedure. However, there is a paucity of evidence regarding the longitudinal clinical outcomes and sustained sports participation at minimum 2- and 5-year follow-up after this procedure.

QUESTIONS/PURPOSES: (1) What are the 2- and 5-year patient-reported outcomes after insertional reattachment for acute sleeve avulsion fractures of the Achilles tendon? (2) What proportion of patients return to sports after insertional reattachment for acute sleeve avulsion fractures of the Achilles tendon? (3) What patient factors are associated with inferior patient-reported outcomes, including the VAS, the American Orthopaedic Foot and Ankle Society (AOFAS) Ankle-Hindfoot score, the Foot Function Index (FFI), and Tegner scores, at minimum 5-year follow-up?

METHODS: A retrospective study was performed that evaluated patients who underwent insertional reattachment for acute sleeve avulsion fractures of the Achilles tendon between December 2011 and December 2019 at our institution. During this period, we treated 55 patients for this injury. Surgery was generally offered to patients who presented with acute posterior heel pain and functional loss cons […]

PubMed · DOI

CORR Insights®: Does Insertional Reattachment for Acute Sleeve Avulsion Fracture of Achilles Tendon Provide Sustained Function and Sports Participation?

Humbyrd CJ

PubMed · DOI

Does First Metatarsal Head Lowering During Minimally Invasive Chevron Akin Osteotomy Yield No Significant Difference in Patient-reported Outcomes Compared With Second Distal Metatarsal Minimally Invasive Osteotomy for Treating Intractable Plantar Keratosis?

Choi JY, Yi Y, Choi SM, Suh JS

BACKGROUND: Hallux valgus is a common forefoot deformity that alters forefoot loading and often leads to secondary complications such as intractable plantar keratosis beneath the second metatarsal head. Although the link between hallux valgus and intractable plantar keratosis is recognized, whether two procedures that correct intractable plantar keratosis through different mechanisms-minimally invasive chevron Akin osteotomy (MICA) with first metatarsal head lowering and conventional MICA combined with second distal metatarsal minimally invasive osteotomy (DMMO)-have differential effects on symptom improvement has not, to our knowledge, been assessed.

QUESTIONS/PURPOSES: (1) Do patients with hallux valgus and intractable plantar keratosis beneath the second metatarsal head have distinctive radiographic features compared with those with hallux valgus alone? (2) How do the two different surgical strategies (MICA with first metatarsal head lowering versus MICA with second DMMO) differ in terms of lesser metatarsal head height and intractable plantar keratosis resolution? (3) What are the differences in the Foot and Ankle Ability Measure (FAAM), including the subscales for activities of daily living (ADL) and sports, and other complications between these techniques?

METHODS: Between January 2017 and June 2024, two surgeons treated 194 feet with hallux valgus using MICA after pred […]

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CORR Insights®: Does First Metatarsal Head Lowering During Minimally Invasive Chevron Akin Osteotomy Yield No Significant Difference in Patient-reported Outcomes Compared With Second Distal Metatarsal Minimally Invasive Osteotomy for Treating Intractable Plantar Keratosis?

Zide JR

PubMed · DOI

Body Fat Percentage But Not BMI Was Related to Clinical Function in Patients With Adhesive Capsulitis After Manipulation Under Anesthesia.

Liu H, Luo J, Wang X, Cai H, Wang C, Dai X, Shen W

BACKGROUND: Adhesive capsulitis can substantially impair shoulder function, and manipulation under anesthesia is commonly used for patients with persistent symptoms, yet outcomes vary widely. Previous research has suggested that obesity, commonly assessed through BMI, influences recovery in musculoskeletal conditions. However, BMI does not differentiate between fat and lean mass, which may be more relevant to tissue healing and functional recovery.

QUESTIONS/PURPOSES: (1) Does body fat percentage, rather than BMI, predict clinical outcomes after manipulation under anesthesia for adhesive capsulitis? (2) Do patients with different body fat percentage levels experience distinct postoperative clinical outcomes? (3) Are common metabolic comorbidities independent predictors of poor surgical recovery in this population?

METHODS: Between December 2022 and December 2023, we treated 280 patients with adhesive capsulitis. Of these, 28% (78 of 280) of patients experienced satisfactory outcomes after nonsurgical treatment, 2% (6 of 280) of patients had concurrent shoulder disorders, and 4% (10 of 280) of patients lacked detailed clinical evaluations. The remaining 66% (186 of 280) of patients underwent manipulation under anesthesia and were the focus of this prospective, observational study. Among these patients, we excluded 0.5% (1 of 186) who received an additional nerve block treatmen […]

PubMed · DOI

CORR Insights®: Body Fat Percentage But Not BMI Was Related to Clinical Function in Patients With Adhesive Capsulitis After Manipulation Under Anesthesia.

Rojas Lievano J

PubMed · DOI

Could the Scapular Spike Sign Be Used as a Radiographic Proxy for Surgical Indications?

Brahme IS, Shaikh H, Bober K, Mavrommatis S, LaRoque MC, Winter JD, Cole PA

BACKGROUND: Scapula fractures are rare injuries (< 1% of fractures), and there remains substantial variability regarding decision-making for the treatment of displaced scapula fractures. Scapula fractures are often underdiagnosed and therefore undertreated. Despite the existence of surgical interventions, there remains a paucity of understanding with regard to the appropriate triaging of displaced scapula fractures, which can lead to significant long-term detriment to patients who sustain these injuries. With a better understanding of surgical indications, surgeons can more efficiently counsel and treat patients with scapula fractures.

QUESTIONS/PURPOSES: (1) In patients with displaced extraarticular scapular fractures meeting surgical indications (lateral border offset > 2 cm, glenopolar angle ≤ 22°, angulation ≥ 45°, double disruption of the shoulder suspensory complex, or open fractures), what was the median lateral border displacement beyond the edge of the glenoid (spike sign)? (2) What amount of displacement (spike sign) indicates surgery based on previously established criteria, and what are the sensitivity, specificity, positive predictive value, and negative predictive value of the displacement threshold for indicating surgery? (3) What are the interobserver and intraobserver reliability of the spike sign?

METHODS: This was a retrospective, diagnostic study evaluatin […]

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CORR Insights®: Could the Scapular Spike Sign Be Used as a Radiographic Proxy for Surgical Indications?

Schiffman C

PubMed · DOI

Gait Function at 20 Years or More After Rotationplasty Shows Pseudo Knee Motion, Decreased Walking Speed, and Increased Energy Cost of Walking.

Krebbekx GGJ, Waterval NFJ, Brehm MA, Nollet F, Ham JSJ, Kerkhoffs GMMJ, Schaap GR, Bramer JAM et al.

BACKGROUND: Rotationplasty is a surgical procedure primarily used for malignant bone tumors around the knee when conventional limb salvage is not feasible because of tumor extent or patient preference. The procedure repurposes the ankle to function as a pseudo knee. While early outcomes are generally good, long-term gait performance in adulthood has not been well described.

QUESTIONS/PURPOSES: This study addresses the following questions about patient outcomes 20 years or more after rotationplasty: (1) What are the walking speed and energy cost as well as the spatiotemporal and gait parameters compared with a control group without lower limb diseases? (2) How are walking speed and energy cost related to age, follow-up duration, and gait parameters? (3) Does thigh-shank length discrepancy correlate with the energy cost of walking and gait parameters?

METHODS: Between 1980 and 2002, a total of 70 patients underwent rotationplasty (all Winkelmann Type A1) at two centers in Amsterdam. Of these, 37% (26) died, 4% (3) underwent amputation, and 9% (6) could not be traced. Of the remaining 35 patients, 6% (2) lived abroad, 9% (3) declined participation, and 3% (1) had a nonfitting prosthesis, leaving 83% (29 of 35) of patients available for evaluation at a median (IQR) follow-up time of 33 years (29 to 35). Rotationplasty was performed for osteosarcoma in 76% (22 of 29) of patients, […]

PubMed · DOI

CORR Insights®: Gait Function at 20 Years or More After Rotationplasty Shows Pseudo Knee Motion, Decreased Walking Speed, and Increased Energy Cost of Walking.

Leithner A

PubMed · DOI

What Predicts a Subsequent Skeletal-related Event, and How Do Factors Associated With Mortality Differ After an Initial Versus a Subsequent Event?

Yen HH, Ogink PT, Yen HK, Chen JH, Huang WL, Lin TC, Lin WH, Hsieh HC et al.

BACKGROUND: Metastatic bone disease presents challenges from debilitating symptoms and poor prognosis. Given the extended life expectancy attributed to improved treatment modalities, such as molecular treatments, patients are at risk of experiencing subsequent skeletal-related events over a prolonged period. While previous studies have identified prognostic factors for initial skeletal-related events, it remains unclear whether these findings can be applied to patients who develop subsequent events. To address this knowledge gap and help clinicians proactively manage this growing patient population, our study aimed to identify the factors that are associated with developing subsequent skeletal-related events in patients who have already experienced bone metastasis.

QUESTIONS/PURPOSES: (1) What factors are associated with a higher risk of developing subsequent skeletal-related events with death as a competing risk? (2) How do factors associated with mortality differ between patients treated for an initial skeletal-related event and those treated for subsequent skeletal-related events?

METHODS: We conducted a retrospective study of 4159 adult patients treated for bone metastasis from January 2010 to December 2018 in Taiwan. The data were drawn from a tertiary referral center and a local hospital. Patients were included if they had a surgical procedure and/or radiotherapy for an […]

PubMed · DOI

CORR Insights®: What Predicts a Subsequent Skeletal-related Event, and How Do Factors Associated With Mortality Differ After an Initial Versus a Subsequent Event?

Nystrom LM

PubMed · DOI

Can Orthopaedic Surgery Go Green? Environmental Footprint of Disposable Versus Reusable Instruments in TKA.

Monadjemi S, Villatte G, Erivan R, Descamps S

BACKGROUND: The environmental impact of surgical instruments and procedures is attracting increasing attention. Single-use instruments are gaining adoption across various healthcare settings, including in orthopaedic surgery, because of their convenience and immediate availability. However, the replacement of reusable instruments, whether kept in stock (RI-S) or procured on loan (RI-L), with single-use instruments has raised obvious ecologic concerns, such as waste management, greenhouse gas emissions, and resource depletion.

QUESTIONS/PURPOSES: Our study aimed (1) to assess the environmental impact (as estimated by acidification, climate change, freshwater eutrophication, water depletion, and resource depletion) of both single-use instruments and RI-S or RI-L employed in TKA procedures and (2) to identify whether one option is more ecologically sustainable (as determined by a life cycle assessment [LCA]).

METHODS: A comprehensive LCA was conducted according to ISO 14040/44 guidelines to calculate the environmental impact of single-use instruments and reusable instruments by considering raw material extraction, manufacturing, distribution, reprocessing (where applicable), and end-of-life waste management. Data were collected both from a 5-year retrospective audit of the manufacturer and from our hospital's sterile facility. Reusable instruments were assessed under two scenari […]

PubMed · DOI

CORR Insights®: Can Orthopaedic Surgery Go Green? Environmental Footprint of Disposable Versus Reusable Instruments in TKA.

Lutnick E

PubMed · DOI

More Frequent Electrical Stimulation Impairs Nerve Regeneration After Median Nerve Repair in a Rat Model.

Kosinski L, Lama C, Rebello E, Ge J, Katarincic J, Gil J

BACKGROUND: Peripheral nerve regeneration via electrical stimulation on nerve repair has been shown to potentiate axonal growth and functional recovery. However, it remains unknown whether increasing the frequency of electrical stimulation improves, diminishes, or has no effect on the quality of nerve regeneration after repair. Additionally, few animal studies have investigated the potential beneficial effects of serial stimulations on accelerating and improving end-organ functionality over time. We therefore sought to evaluate the additive effect of weekly versus biweekly serial electrical stimulation on median nerve regeneration after transection and repair in a rat model using a stimulated grip strength testing method.

QUESTIONS/PURPOSES: (1) Does biweekly versus weekly electrical stimulation lead to greater grip strength forces and therefore, accelerated nerve regeneration? (2) Does frequency of electrical stimulation affect muscle morphology, compound muscle action potential amplitude, muscle weight, or axon count?

METHODS: Twenty-one male Sprague Dawley rats (7 to 8 weeks old, weighing 225 to 250 g at study onset) were randomized into three groups (7 rats per group): (1) median nerve transection and repair with biweekly electrical stimulation, (2) median nerve transection and repair with weekly electrical stimulation, and (3) control group receiving weekly electrical st […]

PubMed · DOI

CORR Insights®: More Frequent Electrical Stimulation Impairs Nerve Regeneration After Median Nerve Repair in a Rat Model.

Schroen CA

PubMed · DOI

Letter to the Editor: What Is the Probability of Radial Nerve Recovery After Surgical Repair of Humerus Fractures Accounting for Time Since Injury?

Schroen CA

PubMed · DOI

Reply to the Letter to the Editor: What Is the Probability of Radial Nerve Recovery After Surgical Repair of Humerus Fractures Accounting for Time Since Injury?

Krijnen NA, Teunis T

PubMed · DOI

Letter to the Editor: Do Functional Patient-reported Outcome Measures Reflect the Activities That Matter Most to Patients After Hip or Knee Arthroplasty?

Riddle DL

PubMed · DOI

Letter to the Editor: Do Functional Patient-reported Outcome Measures Reflect the Activities That Matter Most to Patients After Hip or Knee Arthroplasty?

Zhang L

PubMed · DOI

Reply to the Letter to the Editor: Do Functional Patient-reported Outcome Measures Reflect the Activities That Matter Most to Patients After Hip or Knee Arthroplasty?

Karimijashni M, Poitras S

PubMed · DOI

Letter to the Editor: Editorial: How to Make the Most of Open Notes.

Lee HJ

PubMed · DOI

Letter to the Editor: What Factors Were Associated With Implant Failure After Posterior Vertebral Column Resection for Severe Thoracolumbar Pott Deformity?

Shi B, Li L, Xie W

PubMed · DOI

Reply to the Letter to the Editor: What Factors Were Associated With Implant Failure After Posterior Vertebral Column Resection for Severe Thoracolumbar Pott Deformity?

Zhong J, Cao K

PubMed · DOI

Letter to the Editor: Total Arthroplasty Versus Trapeziectomy With Ligamentoplasty for Trapeziometacarpal Osteoarthritis: 5-year Outcomes.

Karjalainen T

PubMed · DOI

Reply to the Letter to the Editor: Total Arthroplasty Versus Trapeziectomy With Ligamentoplasty for Trapeziometacarpal Osteoarthritis: 5-year Outcomes.

Simón-Pérez C, Frutos-Reoyo EJ, Martín-Ferrero MÁ

PubMed · DOI

JBJS

JBJS (American)

Volume 108, Issue 11 · June 2026 · 20 new articles

Sports Unite All of Us.

Chen AF, Piuzzi NS, Moatshe G

PubMed · DOI

Coronal Malalignment in the Skeletally Immature: The Next Frontier in Understanding the Risk of ACL Injury?: Commentary on an article by Joshua T. Bram, MD, et al.: " Greater Valgus Alignment in Pediatric and Adolescent Patients with a Primary ACL Tear Compared with Healthy Controls ".

Boublik M, Judkins BL

PubMed · DOI

Cost Analysis Is Foundational to Measuring Value: Commentary on an article by Jeffrey S. Mun, BA, et al.: "Drivers of Labor and Supply Cost Variation in Anterior Cruciate Ligament Reconstruction. A Multicenter Time-Driven Activity-Based Costing Analysis".

Day CS, Yousef AH

PubMed · DOI

Navigating the Next Decade of Hip Preservation: Commentary on an article by Jeffrey J. Nepple, MD, MS, et al.: " Survivorship of Femoroacetabular Impingement Surgery at Mean 10-Year Follow-up. A Prospective, Multicenter Cohort Study ".

Wettstein M, Christofilopoulos P

PubMed · DOI

What's New in Sports Medicine.

Al-Hourani K, Khan S, Boufadel P, Uppstrom TJ, Tanaka MJ, Parisien RL, Ma R, Li X

PubMed · DOI

Enhancing Surgeon Longevity and Performance: Six Lanes from Elite Athlete Optimization Applied to Surgical Leadership.

Hasan SS, Kelly BT, Ramkumar PN

PubMed · DOI

ACL Surgery in 2026: Is There a Reason Not to Add a Lateral Extra-Articular Tenodesis?

Frank RM

PubMed · DOI

Beyond Elite Performance: Lessons on Health and Identity from Pablo Aimar: Insights from the Former Elite Soccer Player and Current World Cup-Winning Assistant Coach of the Argentine National Team.

Aimar PA, Martinez D, Chahla J, Piuzzi NS

PubMed · DOI

From Sideline to Specialty: The Birth of Sports Medicine in the United States of America.

Bergfeld JA

PubMed · DOI

Malformation in Medicine.

Stidham S

PubMed · DOI

Cell Therapy.

Lee CA

PubMed · DOI

A Novel Hybrid Training Model for Open Fracture Management in Rwanda.

Kadiyala S, Powis E, Mirahmadi A, Mercado C, Yu S, Velichala SR, Colannino A, Hung I et al.

➢ Open fractures are a critical global health challenge that disproportionately affect individuals in low- and middle-income countries (LMICs), primarily due to road traffic collisions. Surgical management of open fractures is 1 of the 3 essential bellwether procedures identified by The Lancet Commission on Global Surgery. ➢ We developed and evaluated a novel hybrid course on open fracture management for surgical trainees and practicing surgeons in Rwanda, combining a self-directed, virtual, pre-course curriculum with a live, in-person workshop in Kigali in June 2025 that was simultaneously live-streamed for virtual attendees. Prerecorded multilingual lectures (English and French) and curated peer-reviewed articles provided foundational knowledge in advance and prepared learners for in-person didactics, case discussions, and skills training. ➢ The in-person workshop included didactic sessions and discussions of local clinical cases from Rwanda related to open fracture management and other orthopaedic emergencies, along with hands-on practice in fracture external fixation and negative pressure wound therapy using affordable devices designed for resource-constrained practice. ➢ The workshop engaged 160 active learners (37 in-person, 123 virtual) and demonstrated high overall satisfaction among 84 survey respondents, with an average rating of 4.6 out of 5. ➢ Self-reported confiden […]

PubMed · DOI

Evaluation and Management of Meniscal Tears.

Morgan JT, Nishioka T, Casanova F, Moatshe G, LaPrade RF, Chahla J

➢ Meniscal preservation has become the central management principle of meniscal tears. Biomechanical evidence has demonstrated that meniscal resection increases joint contact stress, accelerates osteoarthritis progression, and worsens long-term outcomes compared with repair and nonoperative management. ➢ Treatment decisions should be individualized based on tear morphology, tissue quality, and patient-specific factors. ➢ Repair technique selection (all-inside, inside-out, or outside-in) should be dictated by the tear location and pattern. ➢ Adjunct treatment strategies, such as biologic augmentation, may be used selectively to increase the potential for meniscal healing, although these strategies have inconsistent outcomes.

PubMed · DOI

Multiligament Knee Injuries.

Super JT, Chahla J, Geeslin AG, Moatshe G, LaPrade RF

➢ Multiligament knee injuries (MLKIs) encompass a heterogeneous spectrum of severe knee trauma, presenting ongoing challenges in their diagnosis, classification, management, and postoperative rehabilitation. This review synthesizes the current evidence with expert clinical perspectives to summarize key principles in evaluation and management. ➢ Thorough clinical examination, stress radiography, and magnetic resonance imaging can improve injury characterization and objective quantification of pathologic laxity to guide surgical planning. ➢ Contemporary reconstruction strategies emphasize the detection of posteromedial corner, posterolateral corner, and meniscal pathologies, while recognizing that appropriate management of these associated injuries protects cruciate reconstruction grafts. ➢ Treatment timing remains controversial, with increasing evidence and consensus for early, comprehensive single-stage surgery when feasible in selected patients. ➢ Modern approaches to MLKI management should prioritize restoration of anatomy, biomechanical stability, meticulous planning to avoid tunnel convergence, and rehabilitation strategies.

PubMed · DOI

Comparison of Autograft Types in Anterior Cruciate Ligament Reconstruction: A Systematic Review and Bayesian Network Meta-Analysis of Randomized Clinical Trials.

Vosoughi F, Younesian S, Mousavi SM, Shaker F, Menbari Oskouie I

BACKGROUND: The literature regarding optimal autograft choice for anterior cruciate ligament (ACL) reconstruction (ACLR) remains inconclusive. This network meta-analysis (NMA) compares common autografts for primary ACLR.

METHODS: PubMed, Scopus, Web of Science, and Embase were searched up to May 3, 2025, for randomized clinical trials (RCTs) on primary ACLR in adults that compared ≥2 of the following tendon autografts: 4-strand semitendinosus (4SST), 4-strand semitendinosus-gracilis (4SSTG), its 5-strand variant (5SSTG), bone-patellar tendon-bone (BPTB), quadriceps tendon with bone (QTB), and free quadriceps tendon (FQT). Outcomes analyzed in the NMA were the International Knee Documentation Committee (IKDC) subjective score, Lysholm score, Tegner Activity Scale, anteroposterior (instrumented) and rotational (pivot-shift) stability, and rerupture or revision ACLR rate. Autografts were ranked using surface under the cumulative ranking (SUCRA) values.

RESULTS: A total of 44 RCTs with 3,491 patients were included in the NMA. With respect to the IKDC, QTB was statistically superior to BPTB (mean difference = 3.46, 95% credible interval [CrI]: 0.29 to 6.77), although the difference was likely not clinically meaningful. QTB ranked highest for the IKDC (SUCRA = 90.1%) and Tegner (SUCRA = 85.3%), while BPTB ranked lowest for the IKDC and Lysholm. With respect to knee laxity, QTB rank […]

PubMed · DOI

Knee Injectables in Young Athletes: Evidence, Recommendations, and Clinical Application.

Hantouly A, Bedi A, Khan M

Knee pain resulting from acute trauma and overuse injury is common among athletes and represents a major cause of reduced performance, time loss from sport, and long-term sequelae including osteoarthritis. Injectable therapies are frequently used as a nonoperative treatment modality to alleviate symptoms and facilitate early return to sport. This review evaluates the current evidence on commonly used knee injectables in the younger athletic population with pre-arthritic knee pain, including corticosteroids, hyaluronic acid, platelet-rich plasma (PRP), and other biologics. Relevant literature was identified without restriction on study design and with a focus on athlete-specific outcomes and clinical applicability. Overall, the available evidence on knee injectables for athletes remains limited, heterogeneous, and largely extrapolated from older, nonathletic cohorts. In the absence of available athlete-specific guidelines, most injectables carry weak and/or conditional recommendations, highlighting the need for individualized treatment and shared decision-making. High-quality, sport-specific clinical trials are required to establish clear guidelines and optimize outcomes in this population.

PubMed · DOI

Professional Athlete Redefined: How to Think About the Changing Youth Sports Landscape.

Paredes-Barbeito A, Jain C, Anthony C, Humbyrd CJ

PubMed · DOI

Greater Valgus Alignment in Pediatric and Adolescent Patients with a Primary ACL Tear Compared with Healthy Controls.

Bram JT, Beber SA, Lu S, Pascual-Leone N, Groff KD, Green DW, Fabricant PD

BACKGROUND: Coronal plane angular deformity remains under-investigated in the context of pediatric anterior cruciate ligament (ACL) tears. We hypothesized that baseline coronal alignment in pediatric and adolescent patients with a first-time ACL injury would differ from that in a matched healthy comparison population of patients without knee pathology.

METHODS: Patients ≤18 years of age who underwent primary ACL reconstruction and had preoperative lower-extremity hip-to-ankle alignment radiographs (cases) and individuals without lower-extremity conditions that would influence alignment (controls) were matched 1:1 on the basis of age (±1 year) and sex. Coronal plane parameters included the hip-knee-ankle angle (HKA), mechanical axis deviation (MAD), mechanical lateral distal femoral angle (mLDFA), and medial proximal tibial angle (MPTA). Decision stump analyses were used to identify clinically relevant alignment threshold values.

RESULTS: A total of 200 patients were included in the analysis (100 per group). The mean age was 12.7 ± 2.1 years in the ACL group (58% White/Caucasian, 50% female) and 13.1 ± 2.4 years in the control group (49% White/Caucasian, 50% female). Compared with controls, patients with an ACL tear demonstrated increased valgus alignment across all 4 parameters: MAD (-4.1 ± 7.8 versus -0.3 ± 7.6 mm; $p < 0.001$), HKA (-1.4° ± 2.6° versus -0.5° ± 2.3°; $p = 0.006$ […]

PubMed · DOI

Drivers of Labor and Supply Cost Variation in Anterior Cruciate Ligament Reconstruction: A Multicenter Time-Driven Activity-Based Costing Analysis.

Mun JS, Dean MC, Gillinov SM, Poutre RL, Chenna SS, Allen BJ, Beck da Silva Etges AP, Treloar JA et al.

BACKGROUND: Understanding drivers of supply and labor cost variation in orthopaedic surgery is crucial to provide value-based care. Time-driven activity-based costing (TDABC) is a more accurate methodology for capturing costs of care than traditional methods. Anterior cruciate ligament reconstruction (ACLR) is one of the most performed outpatient procedures within orthopaedic surgery. The purpose of this study was to characterize the cost composition of ACLR and identify factors that drive cost variation.

METHODS: Cost data for supplies and time-based personnel usage were extracted from electronic health records and were used to calculate costs using TDABC. TDABC methodology was applied to calculate the cost of personnel usage by multiplying the duration and associated cost per minute. Descriptive statistics and mixed-effects modeling were used to determine cost drivers.

RESULTS: This study included 861 patients who underwent ACLR at 8 hospitals. The mean patient age (and standard deviation) was 31.1 ± 11.6 years. Of the 861 patients, 350 were male and 511 were female; 85.6% of patients were White, 8.1% were Asian, and 3.4% were Black. There was 3.2-fold variation in supply costs ($2,950) and 1.6-fold variation in labor costs ($940) between the 10th and 90th percentiles. Overall, supply costs accounted for 58.2% of total costs, whereas labor costs comprised the remaining 41.8 […]

PubMed · DOI

Survivorship of Femoroacetabular Impingement Surgery at Mean 10-Year Follow-up: A Prospective, Multicenter Cohort Study.

Nepple JJ, Hood H, Kim YJ, Beaulé P, Sierra R, Millis M, Robben Z, Drain C et al.

BACKGROUND: Long-term outcomes of femoroacetabular impingement (FAI) surgery, particularly survivorship, are critical to guide treatment decision-making and patient counseling, yet only a limited number of studies have reported mid- to long-term survivorship. The purpose of this study was to report survivorship rates at a mean 10-year follow-up in a large, multicenter FAI surgery cohort and to identify clinical predictors of survivorship.

METHODS: A prospective, multicenter cohort study assessed patients treated for FAI with hip arthroscopy or surgical dislocation from 2008 to 2012. At a minimum of 8 years, 362 hips (80.1%) had follow-up that permitted assessment of total hip arthroplasty (THA)-free survivorship. A Cox proportional-hazards model was developed to identify risk factors for THA.

RESULTS: tThe cohort included 362 hips with a mean patient age of 32.1 years; 53% were in females, and 95.6% were in Caucasian patients. The THA-free survivorship of the cohort was 90.6% at a mean of 10.4 ± 1.6 years postoperatively. Risk factors for THA were older age at surgery ($p = 0.01$), male sex ($p = 0.02$), body mass index of ≥30 kg/m 2 ($p = 0.009$), and femoral head chondromalacia ($p < 0.001$).

CONCLUSIONS: This study demonstrates that FAI surgery yielded durable 10-year THA-free survivorship of 90.6%. Older age at surgery, obesity, male sex, and femoral head chondromalacia were key […]

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The Journal of Arthroplasty

Volume 41, Issue 7 · July 2026 · 48 new articles

Letter Regarding "Rotational Alignment in Total Knee Arthroplasty: A Randomized Clinical Trial of Akagi Versus Sahin Tibial Referencing Techniques".

Sun X, Jian J

PubMed · DOI

Reply to Letter Regarding: "Rotational Alignment in Total Knee Arthroplasty: A Randomized Clinical Trial of Akagi Versus Sahin Tibial Referencing Techniques".

Nunes RPS, Franciozi CE, Aihara AY, Marques FS, Kubota MS, Luzo MVM

PubMed · DOI

Letter Regarding "Impact of Obesity on Daily Activity Following Total Knee Arthroplasty".

Güne■ Z

PubMed · DOI

Response to the Letter to the Editor Commenting on "Impact of Obesity on Daily Activity Following Total Knee Arthroplasty".

Wu KA, Kugelman DN, Shah SN, Ryan SP, Bolognesi MP, Seyler TM, Wellman SS

PubMed · DOI

Letter Regarding "Is It Safe to Give Meloxicam or Celecoxib to Chronically Anticoagulated Patients Following Primary Total Knee Arthroplasty?".

Smitterberg CW, Mont MA

PubMed · DOI

Reply to Letter Regarding "Is It Safe to Give Meloxicam or Celecoxib to Chronically Anticoagulated Patients Following Primary Total Knee Arthroplasty?".

Liu J, Lee GC

PubMed · DOI

Letter regarding 'A Randomized, Double-Blind, Placebo-Controlled Trial on the Efficacy of Dexamethasone Combined with Neuraxial Anesthesia in Reducing Pain and Opioid Consumption after Primary Cementless Total Hip Arthroplasty Using the Direct Anterior Approach'.

Uysal A■, Ya■ar E

PubMed · DOI

Reply to Letter Regarding "A Randomized, Double-Blind, Placebo-Controlled Trial on the Efficacy of Dexamethasone Combined with Neuraxial Anesthesia in Reducing Pain and Opioid Consumption after Primary Cementless Total Hip Arthroplasty Using the Direct Anterior Approach".

Schroven W, Verrewaere D, Deckmyn T, Vandekerckhove M, Van Eemeren A, Vanlommel J

PubMed · DOI

Reply to Letter to the Editor on "Predictive Factors of Pain and Functional Outcome Five Years Following Total Hip Arthroplasty: A Prospective Function and Outcomes Research for Comparative Effectiveness in Total Joint Replacement Cohort Study".

Ayers DC

PubMed · DOI

Fear and the Adoption of Technology in Total Hip Arthroplasty.

Blumenfeld TJ, Fitzsimmons LE, Goodman SB, Amanatullah DF

PubMed · DOI

Perioperative Management of Diabetic Medications in Patients Undergoing Elective Total Joint Arthroplasty: A Review of Current Guidelines and Recommendations.

Kumar J, Fawcett MA, Sherman CE, Springer BD

BACKGROUND: The prevalence of diabetes in patients undergoing total joint arthroplasty (TJA) is steadily increasing. Diabetes predisposes patients to major complications such as prosthetic joint infection, which poses major challenges to the patient and surgeon. Therefore, careful perioperative glucose management in these patients is critical. Medications have been the mainstay in the management of diabetes; however, these medications are each associated with their unique risk profile in the perioperative period.

METHODS: This review evaluated the available literature regarding the 10 main diabetic medication classes, their mechanism of action, adverse effects, and current perioperative management recommendations.

RESULTS: Available literature addressing perioperative diabetic medication management in TJA is still lacking; however, there are still many guidelines that have been synthesized for these medications in both an elective surgery setting and within TJA.

CONCLUSIONS: Based on this literature review, we provided a summation of the perioperative guidelines for the ten main diabetic drug classes and what the current available perioperative guidelines are. Many of these medication classes have not been studied yet in the setting of TJA. However, we hope this summation offers orthopaedic providers a practical clinical guideline and insight into managing and treating their […]

PubMed · DOI

Intravenous Meloxicam Versus Ketorolac for Pain Control Following Total Joint Arthroplasty: A Double-Blind Randomized Controlled Trial.

Rizzo MG, Costello JP, Constantinescu DS, Samineni AV, Moore MR, Shittu AT, Tabibi O, Osman BM et al.

BACKGROUND: Although joint arthroplasty surgery is considered safe and effective in reducing osteoarthritis-related pain long-term, the surgery is known to be painful in the immediate postoperative period. Therefore, optimal pain management has become a crucial goal for orthopedic surgeons. Nonsteroidal anti-inflammatory drugs are often used but can cause adverse effects due to their mechanism of action. Meloxicam, however, has a unique mechanism of action, allowing it to better spare the gut lining, translating to reduced complications. This double-blind randomized controlled trial sought to compare intravenous (IV) meloxicam and IV ketorolac for total joint arthroplasty regarding pain scores, opioid consumption, and lengths of stay.

METHODS: All other institutional protocols and medications remained unchanged. A total of 223 patients were randomized. The meloxicam group received 30 mg IV meloxicam preoperatively two hours before primary incision, while the ketorolac group received 15 mg IV ketorolac intraoperatively followed by 15 mg IV every six hours, scheduled for two doses. Pain, opioid consumption, and nausea were recorded at two and 24 hours postoperatively. An a priori power analysis demonstrated a need for 170 total patients. The study was registered on clinicaltrials.gov (NCT05291598) and received Institutional Review Board approval.

RESULTS: The IV meloxicam group […]

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Unicompartmental Knee Arthroplasty for Isolated Compartment Osteonecrosis: A 20-Year Experience.

Meissner N, Bukowski BR, Larson DR, Hannon CP, Bedard NA, Abdel MP

BACKGROUND: Osteonecrosis of the knee, whether primary (spontaneous) or secondary, can often necessitate surgical intervention. If advanced osteonecrosis is confined to a single compartment, unicompartmental knee arthroplasty (UKA) is often considered. The purpose of this study was to update and extend our prior series, with specific emphasis on implant survivorship, radiographic results, clinical outcomes, and complications in a concise cohort of UKAs for osteonecrosis.

METHODS: We retrospectively identified 70 consecutive UKAs for osteonecrosis performed at a single tertiary care academic institution from 2002 to 2022. Of these, 64 UKAs (91%) were performed for primary osteonecrosis, and six UKAs (9%) were performed for secondary osteonecrosis. Osteonecrosis was in the medial femoral condyle in 61 UKAs (87%), medial tibial plateau in six UKAs (9%), and lateral femoral condyle in three UKAs (4%). All UKAs were cemented, with 77% utilizing a fixed-bearing design and 23% utilizing a mobile-bearing design. The mean patient age was 66 years, with 59% being women. The mean follow-up was eight years (range, two to 20).

RESULTS: The 10-year survivorship free of any revision for the entire cohort was 91%. The predominant indication for revision was the development of osteoarthritis in the lateral compartment in three knees. The 10-year survivorship free of any reoperation was 88%. T […]

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Outcomes and Implant Survival in Lateral Unicompartmental Knee Arthroplasty: The Path Less Traveled.

Nwankwo TN, Li KK, Parks NL, Fricka KB

BACKGROUND: Unicompartmental knee arthroplasty (UKA) is an effective treatment for isolated lateral compartment osteoarthritis (OA). Surgeons have been reticent to adopt this procedure due to the difficulty of exposure and historically higher rates of revision when compared to total knee arthroplasty (TKA). The purpose of this study was to evaluate our series of lateral UKAs.

METHODS: We retrospectively reviewed 548 fixed-bearing, cemented lateral UKAs performed at a single institution between January 2001 and May 2023. The diagnosis was OA in 98.5% of cases, and 71% of patients were women. The average age and body mass index at the time of surgery were 67 years (range, 28 to 93) and 27.8 (range, 18.0 to 49.5), respectively. The primary outcomes included implant survivorship and revision rate with associated causes. The secondary outcomes included complication rates and analysis of preoperative and postoperative tibio-femoral angle.

RESULTS: The 10-years survivorship was 95.7%. There were 19 cases (3.5%) that required revision after a mean time in situ of 6.5 years, most commonly due to progression of OA (n = 14), followed by aseptic loosening (n = three), periprosthetic fracture (n = one), and infection (n = one). An additional 11 cases (2.0%) required reoperation without revision. The early (≤ 90 days) complication rate was 2.2%. Postoperative limb alignment did not differ […]

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Spin Is Prevalent in the Majority of Abstracts of Patello-femoral Arthroplasty Studies.

Feingold CL, Lin EH, Young BA, Kumaran P, Barcenas AB, Podosin MA, Stone AV, Liu JN

BACKGROUND: The purpose of our study was to investigate the prevalence and types of reporting biases in patello-femoral arthroplasty (PFA) studies.

METHODS: PubMed, Scopus, and Web of Science were searched for the term "patellofemoral arthroplasty" using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. Articles were included if they were (1) primary research articles, (2) investigating patello-femoral arthroplasties, (3) level of evidence (LOE) 1 to 3, (4) published after 2013, and (5) published in English in a peer-reviewed journal. Articles were excluded if they were systematic reviews or meta-analyses, case reports, not published in English, or if their full text was inaccessible. Abstracts were designated as "positive," "neutral," or "negative" outcomes based on whether they favored PFA. The included abstracts were assessed for the 13 most common types of spin. Fisher's exact tests were used to analyze the relationship between the different spin types and study characteristics. There were 37 articles that met the inclusion criteria; the articles were published between 2015 and 2024.

RESULTS: Spin was identified in 22 (59.5%) studies, and nine (69.2%) spin types were identified at least once. Spin types three and 14 were identified most frequently in 10 (27.0%) studies each. The amount of spin in each abstract ranged from none to […]

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Rotational Alignment in Total Knee Arthroplasty: A Randomized Clinical Trial of Akagi Versus Sahin Tibial Referencing Techniques.

Nunes RPS, Franciozi CE, Aihara AY, Marques FS, Kubota MS, Luzo MVM

BACKGROUND: Rotational malalignment of the tibial component in total knee arthroplasty (TKA) contributes to patellofemoral complications and functional impairment. While several anatomical landmarks have been proposed for intraoperative referencing, no single method has been universally accepted. This study compared the rotational accuracy and clinical outcomes of Akagi's line and Sahin's combined axis referencing techniques.

METHODS: A total of 78 patients undergoing primary TKA were randomized and prospectively allocated to two groups: Akagi (n = 40) and Sahin (n = 38). Rotational alignment was assessed postoperatively via computed tomography using the method proposed by the "Société française de la hanche et du genou" by two independent examiners. Clinical outcomes were evaluated preoperatively and at one and two years postoperatively using range of motion, Knee Injury and Osteoarthritis Outcome Score, the 12-Item Short Form Survey, and Forgotten Joint Score. Statistical comparisons were performed using appropriate parametric and nonparametric tests based on the normality of data distribution.

RESULTS: There were no statistically significant differences observed between groups in tibial, femoral, or combined rotational alignment ($P > 0.05$ for all comparisons), with mean combined rotation values remaining within ± 3°, the commonly accepted safe zone. Both groups demonstrate […]

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Diagnosing Flexion Laxity After Knee Arthroplasty Using the "Hanging" Clunk Test and the Standard-To-Hanging Lateral Difference.

Blumenfeld TJ, Bargar WL, Amanatullah DF

BACKGROUND: Flexion laxity after total knee arthroplasty (TKA) may be difficult to diagnose. We describe simple clinical and radiographic tests that may aid in the identification of this problem.

METHODS: There were 99 consecutive nonselected patients who had 128 TKAs presenting to our clinic for scheduled or unscheduled follow-up at a minimum of one year postprocedure, who were enrolled in an institutional review board-approved study. Patients were divided into two groups (cases = self-report of pain, controls = no pain). All patients underwent routine clinical and radiographic evaluation, as well as physical examination for a hanging clunk sign, and one additional radiographic examination (hanging lateral radiograph).

RESULTS: For the cases group (n = 51), a positive hanging clunk sign was identified in six patients. Clinical flexion laxity was identified in two patients. For the control group (n = 76), a positive hanging clunk sign was identified in 15 patients. Clinical flexion laxity was identified in three patients. At a standard-to-hanging lateral difference ≥ four mm, 11 patients complained of pain, and seven did not (61%, specificity: 91%, sensitivity: 23%, accuracy: 64%). The overall incidence of a positive hanging clunk sign was 16.5%.

CONCLUSION: Isolated flexion laxity may be difficult to diagnose. Flexion laxity after TKA may be difficult to diagnose. The clini […]

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Joint Line Obliquity Affects Survivorship Following Total Knee Arthroplasty: 12.7-Year Follow-Up of 800 Cases.

Yang HY, Song EK, Seo JH, Kang SJ, Seon JK

BACKGROUND: The purpose of this study was to evaluate the influence of joint line obliquity (JLO) on survivorship following total knee arthroplasty (TKA) over a mean follow-up of 12.7 years METHODS: We reviewed the records for 800 patients who had undergone TKA for primary knee osteoarthritis between January 2004 and December 2010. Based on postoperative alignment, patients were categorized into two groups: the mechanically aligned group (mechanical axis of 0 ± 3°) and the outlier group (mechanical axis beyond 0 ± 3°). In addition, patients were stratified into four groups according to joint line obliquity (JLO), as measured on full-length standing weight-bearing radiographs using both the original classification (sum of lateral distal femoral angle and medial proximal tibial angle) and a modified classification (joint surface angle relative to the hip-knee-ankle axis). Kaplan-Meier survival analysis was used to estimate implant survival, with aseptic revision for mechanical failure defined as the endpoint.

RESULTS: Mechanical failure occurred in 3.5% of cases (28 of 800) over a mean follow-up of 12.7 years. Although the original JLO classification was not associated with aseptic revision, the modified classification revealed significant differences within the outlier group. Notably, lateral joint line inclination was associated with the highest failure rate (5.2% [10 of 193]; […]

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Is a Combination of Diosmin and Hesperidin Effective on Swelling, Pain, and Range of Motion After Total Knee Arthroplasty?

Girgin AB, Duman E

BACKGROUND: Swelling is one of the hindrances to functional recovery after total knee arthroplasty (TKA). The aim of this study was to investigate the effect of a diosmin and hesperidin combination on early rehabilitation of patients after TKA.

METHODS: This randomized controlled study included patients who underwent unilateral primary TKA between October 2024 and June 2025. There were two groups that were determined as the diosmin and hesperidin group and the control group. The control group did not receive any placebo and only received standard postoperative treatment. The patients' extremity circumferences from the suprapatellar, patellar, and infrapatellar regions; visual analog scale scores at rest and during movement; knee joint range of motion; inflammatory markers in the blood at 24 and 48 hours, and 14 days postoperatively; and complications were compared.

RESULTS: Flexion degrees were 78.8 on the first postoperative day, 86.7 on the second postoperative day, and 104.9 on the 14th postoperative day in the diosmin and hesperidin group, while they were 53.9, 65, and 91 degrees, respectively, in the control group ($P < 0.001$). Although there was a significant difference in suprapatellar and patellar circumference increases on postoperative days one and two between the two groups, there was a significant difference in suprapatellar, patellar, and infrapatellar circumferen […]

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Preoperative Testosterone Replacement Therapy Is Associated With Increased Complication Risk After Total Knee Arthroplasty: A Propensity-Matched Analysis of 13,250 Patients.

Omurzakov A, Omurzakov AM, Bhatti P, Debbi EM, Gausden EB, Chalmers BP

BACKGROUND: While testosterone replacement therapy (TRT) is known to affect cardiovascular physiology, its impact on outcomes following total knee arthroplasty (TKA) remains unclear. This study aimed to assess whether preoperative TRT use is associated with increased complications following TKA.

METHODS: A retrospective cohort study using a large national database was performed. Patients undergoing primary TKA before February 2020 with 5-year follow-up were stratified based on preoperative TRT use within one year of surgery. Patients who had a history of septic arthritis, osteonecrosis, or pathologic fractures were excluded. Propensity score matching (1:1) was used to balance cohorts. Outcomes included medical complications at 90 days and one year postoperatively and periprosthetic complications up to five years postoperatively. After matching, 6,625 patients were included in each cohort.

RESULTS: At 90 days, TRT use was associated with higher rates of pulmonary embolism (odds ratio (OR) 1.4, 95% confidence interval (CI): 1.0 to 1.8, $P = 0.041$), pneumonia (OR 1.8, 95% CI: 1.4 to 2.3, $P < 0.001$), acute kidney injury (OR 1.5, 95% CI: 1.2 to 1.8, $P < 0.001$), and sepsis (OR 1.8, 95% CI: 1.3 to 2.4, $P < 0.001$). At one year, TRT patients had elevated deep vein thrombosis (OR 1.4, 95% CI: 1.1 to 1.6, $P < 0.001$), cardiac event (OR 1.3, 95% CI: 1.0 to 1.6, $P = 0.018$), and pneumonia (O […]

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Sleep Management Combined With Rehabilitation Exercise to Alleviate Chronic Pain Following Total Knee Arthroplasty: A Randomized Controlled Trial.

Han X, Zhao X, Ge Y, Wang B, Shang X, Lu Z, Ma Y, Li F

BACKGROUND: Chronic pain following total knee arthroplasty (TKA) remains a frequent and debilitating complication that impairs recovery and quality of life. Current pharmacological treatments provide limited relief for persistent pain, especially when complicated by sleep or mood disorders. This study investigated whether combining cognitive behavioral therapy for insomnia (CBT-I) with structured rehabilitation exercise could offer a nonpharmacologic solution.

METHODS: A single-center, randomized controlled trial was conducted with 80 patients experiencing persistent pain ≥ three months after TKA. Participants were randomized into two groups (n = 40 each): an intervention group receiving CBT-I plus tailored physical rehabilitation and a control group receiving conventional analgesics. Assessments were performed at baseline and at weeks one, two, four, and 12 postintervention. Primary outcomes included Visual Analog Scale (VAS), Hospital for Special Surgery (HSS) score, range of motion (ROM), and knee joint swelling (measured by limb circumference). The secondary outcomes included the Pittsburgh Sleep Quality Index (PSQI) and the Hospital Anxiety and Depression Scale (HADS). Safety indicators were monitored.

RESULTS: All follow-up assessments (week one, two, four, and 12) were calculated from the start of the intervention (baseline, T0). All participants were enrolled three to […]

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Triple-Taper Collared Stem Morphology Influences Periprosthetic Femoral Fracture Risk: A Comparison of Single-, Dual-, Triple-, and Quadrangular-Taper Stems.

LeBrun DG, Ruiz V, Gausden EB, Westrich GH

BACKGROUND: Stem morphology is an important contributor to periprosthetic femoral fracture (PFF) risk following total hip arthroplasty (THA). New-generation triple-taper collared stems may be associated with a lower risk of PFF. Our objective was to compare the incidence of PFFs in triple-taper collared stems to three other cementless stem morphologies.

METHODS: A single-institution retrospective cohort study was performed. All primary THA cases (n = 23,448) from February 2016 to December 2023 using cementless single-taper (n = 8,529), dual-taper (n = 8,514), triple-taper collared (n = 4,024), and quadrangular-taper stems (n = 2,381) from four major implant companies were evaluated. Outcomes were any PFF, intraoperative PFF, and early (90 days) postoperative PFF.

RESULTS: Any PFF occurred in 60 of 8,529 (0.70%) single-taper, 114 of 8,514 (1.34%) dual-taper, 15 of 4,024 (0.37%) triple-taper collared, and 26 of 2,381 (1.09%) quadrangular-taper stems (overall difference, $P < 0.01$). Intraoperative fractures occurred in 30 single-taper (0.35%), 69 dual-taper (0.81%), 10 triple-taper collared (0.25%), and 22 (0.92%) quadrangular-taper stems ($P < 0.01$). Postoperative fractures occurred in 30 single-taper (0.35%), 47 dual-taper (0.55%), five triple-taper collared (0.12%), and four (0.17%) quadrangular-taper stems ($P < 0.01$). Collared stems (including quadrangular-taper collared stems […]

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Influence of Coronal and Sagittal Femoral Stem Alignment on Mid-Term Functional Outcomes in Direct Anterior Approach Total Hip Arthroplasty Utilizing a Cementless, Trochanteric-Sparing Short Stem.

Poursalehian M, Hajiaghajani S, Tabatabaei Irani P, Mortazavi SMJ

BACKGROUND: Neutral femoral-stem alignment is thought to promote durability and function after total hip arthroplasty (THA), but the clinical relevance of modest malalignment, particularly in modern, cementless short stems inserted through a direct anterior approach (DAA), remains uncertain. This study aimed to evaluate whether coronal or sagittal femoral stem malalignment affects pain and function after DAA THA with a cementless short stem.

METHODS: We retrospectively reviewed 348 hips (306 patients) that underwent primary DAA THA. Stem orientation was measured on standardized anteroposterior (AP) and lateral radiographs obtained one year postoperatively. Patient-reported outcome measures (PROMs), including Visual Analog Scale (VAS) pain, Harris

Hip Score (HHS), and Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC), were collected at a mean of 4.3 ± 1.2 years. Multivariable linear regression examined factors linked to malalignment and its independent association with PROMs.

RESULTS: Coronal malalignment occurred in 25 of 348 hips (7.2%) and sagittal malalignment in 77 of 348 (22.1%). Coronal malalignment did not significantly affect final PROMs (VAS, HHS, WOMAC; $P > 0.05$). Sagittal malalignment was associated with lower HHS (91.9 ± 7.6 versus 93.6 ± 6.8; = 0.010) and higher WOMAC (11.9 ± 10.2 versus 9.7 ± 9.1; $P = 0.005$). The absolute differences (1.7 HH […]

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The Elixhauser Comorbidity Index as a Predictor of Disruptive Bleeding in Total Hip Arthroplasty: Impact on Outcomes and Costs.

Ng MK, Mont MA, Afolabi M, V PN, Kumar A, Johnston SS

BACKGROUND: With an estimated one million cases per year anticipated by 2030, total hip arthroplasty (THA) is a popular and successful treatment for end-stage hip osteoarthritis. The objectives of this study were to: (1) measure the incidence of disruptive bleeding; (2) determine the demographics of patients and the burden of Elixhauser comorbidity related to disruptive bleeding; and (3) assess the effect of disruptive bleeding on health care utilization (inpatient costs, hospital lengths of stay, and 90-day readmission rates).

METHODS: In this cohort study, patients who had primary THA in 2019 were identified using a national health care database. Multivariate logistic regressions were used to examine possible associations with disruptive bleeding after patients were grouped by Elixhauser Comorbidity Index (ECI), which ranged from 0 to ≥ 6. Both 95% confidence intervals (CIs) and odds ratios (ORs) were presented for every ECI. Generalized linear models were used to assess the effects of disruptive bleeding on hospital lengths of stay, 90-day readmission rates, and overall expenses.

RESULTS: Disruptive hemorrhage occurred in 6,586 (6.0%) of the 109,482 patients receiving THA. Higher bleeding risk was linked to women and Medicare coverage ($P < 0.01$). For instance, the risks of disruptive bleeding rose from OR = 1.48 (95% CI 1.23 to 1.78, $P < 0.001$) for ECI = 2 to OR = 3.81 (95 […]

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Defining Preoperative Anemia Thresholds for Revision Total Hip Arthroplasty.

Sutton R, Leipman JH, Linton A, Sherman MB, Martinazzi BJ, Fellheimer HS, Fillingham YA

BACKGROUND: In an era of value-based medical management, anemia optimization before revision total hip arthroplasty (THA) could improve outcomes. Simulation models in orthopedic literature help define the likelihood of successful surgery based on varying thresholds of preoperative parameters. We evaluated outcomes for revision THA if patients were either approved/denied surgery based on incremental preoperative anemia thresholds.

METHODS: We retrospectively reviewed consecutive revision THA patients from 2019 to 2024 at a single institution. Exclusion criteria were primary THA, no preoperative hematocrit, and incomplete 1-year follow-up. Simulated preoperative hematocrit cutoffs of 39 through 33% were used. Preoperative hematocrit above each threshold was "approved" for surgery, while below each threshold was "denied" surgery. Intraoperative transfusions, postoperative transfusions, discharge dispositions, 90-day readmissions, and complications were evaluated. Area under the curve (AUC) determined anemia thresholds needed for an event not to occur.

RESULTS: If the strictest 39% hematocrit threshold were used, 626 of 1,118 (56%) patients would be denied surgery, with denied patients having higher readmissions (16.9 versus 13.2%), intraoperative transfusions (45.7 versus 11.6%), postoperative transfusions (37.2 versus 14%), nonhome discharge (66 versus 35.8%), and complications […]

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Cementless Titanium Fibermesh Cup in Cup-Cage Construct for Severe Acetabular Defect in Revision Total Hip Arthroplasty.

Chen IH, Tzeng SC, Wang CT

BACKGROUND: Addressing bony defects in cases of severe acetabular bone loss remains challenging, and the cup-cage construct presents a potential solution. While tantalum cups have demonstrated favorable outcomes within the cup-cage construct, the efficacy of employing cup-cage constructs with titanium fibermesh remains to be elucidated.

METHODS: We conducted a retrospective, single-center study of patients who had Paprosky type 3A or 3B acetabular bone loss and underwent cup-cage reconstruction with a cementless titanium fibermesh cup between 2011 and 2019, with a minimum 5-year follow-up period. Implant survivorship was assessed using Kaplan-Meier analyses, while clinical outcomes were evaluated, and sequential radiographic evaluations were performed to identify bone restoration processes. There were 29 patients included who had a mean age of 63 years (range, 41 to 82) and a mean follow-up duration of 125.7 months (range, 76 to 163).

RESULTS: Cumulative implant survivorship was 97% at two years (95% confidence interval: 78 to 100) and 90% (95% confidence interval: 71 to 98) at five and 10 years. Radiographically, all cases (29 of 29) demonstrated graft incorporation and osseointegration, with medial acetabular wall restoration observed in instances of pelvic discontinuity, and the hip center was effectively restored. Postoperatively, there were significant improvements in fu […]

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Trends in Antibiotic-Resistant Bacteria in Surgically Treated Periprosthetic Joint Infection: A Decade of Experience at a Regional Referral Center.

Chandler CC, Graham SD, Hietpas KT, Fehring TK, Otero JE

BACKGROUND: Periprosthetic joint infection (PJI) represents a devastating complication following total joint arthroplasty. Understanding trends in causative organisms and antibiotic resistance patterns is necessary for optimizing treatment strategies. We investigated (1) whether there has been a change in prevalence of drug-resistant organisms in surgically treated PJI over the past decade and (2) if there are risk factors associated with drug-resistant PJI.

METHODS: A retrospective review of our institutional database was performed for surgically treated PJI between January 1, 2010, and December 31, 2021. A total of 915 patients met Musculoskeletal Infection Society criteria for PJI. The prevalence of patients who had drug-resistant organisms was recorded and compared over the study duration. Of 915 procedures, 25.1% (N = 230) were caused by resistant organisms. The most common resistant organisms were methicillin-resistant Staphylococcus aureus (63.0%), methicillin-resistant Staphylococcus epidermidis (24.3%), and resistant coagulase-negative Staphylococci (16.1%).

RESULTS: Total PJI cases increased from 53 in 2010 to 109 in 2021, while resistant organism cases increased from 22 to 30. The percentage of resistant PJI decreased from 41.5% (22 of 53) in 2010 to 25.1% (30 of 109) in 2021, with no significant trend (P = 0.85). Patients who had higher comorbidity burden had sign […]

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Results of Two-Stage Exchange for Periprosthetic Joint Infection Following Total Knee Arthroplasty: Low Rates of Failure in 463 Knees at 10 Years.

Terhune EB, Carstens MF, Fruth KM, Hannon CP, Bedard NA, Berry DJ, Abdel MP

BACKGROUND: Some recent data suggest similar reinfection rates between two- and one-stage exchange arthroplasties for infected total knee arthroplasties (TKAs). However, those data are limited in follow up and analysis of aseptic failure rates. The purpose of this study was to assess the 10-years results of TKA two-stage exchange in one of the largest series to date.

METHODS: We identified 463 infected TKAs treated with two-stage exchange from 1991 to 2021 at a single institution. The mean age at reimplantation was 68 years, 49% were women; mean body mass index was 32 (range, 16 to 64). Diagnosis of periprosthetic joint infection was based on the 2011 Musculoskeletal Infection Society criteria. Cumulative incidences of reinfection and revision with death as a competing risk were calculated. The mean follow-up was eight years (range, two to 30).

RESULTS: The cumulative incidence of reinfection was 6% at one year, 14% at five years, and 19% at 10 years. Factors predictive of reinfection included elevated body mass index (hazard ratio (HR) 1.3 for every 5-point increase), McPherson host grade C (HR: 2.3), previous revision surgery (HR: 2.1) and Anderson Orthopaedic

Research Institute type-3 bone loss (HR: 4.6). The cumulative incidence of aseptic revision was 9% at 10 years. The cumulative incidence of any revision was 26% at 10 years. The most common indications for revision we […]

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High Satisfaction Despite 32% Persistent Hip Flexor Weakness After Endoscopic Tenotomy for Iliopsoas Impingement Following Total Hip Arthroplasty.

Twardy V, Rösch MBJ, Pohlig F, von Eisenhart-Rothe R, Banke IJ

BACKGROUND: Iliopsoas impingement (IPI) is a serious complication after total hip arthroplasty (THA), causing chronic pain and functional impairment. Endoscopic iliopsoas tenotomy (EIT) is considered among the least invasive options to prevent implant revision. This study aimed to evaluate postoperative outcomes with a focus on pain relief, objective hip flexor strength, and patient satisfaction. We hypothesized that patients would report high satisfaction despite potential persistent strength deficits.

METHODS: This retrospective, consecutive, monocentric, single-surgeon level III study with prospective outcome evaluation included 31 THA patients (mean age 63 years, 18 women/13 men, mean body mass index (BMI) 28) treated with EIT for IPI between 2015 and 2022. The mean follow-up was 37.0 months (range, 24 to 82). The IPI was based on clinical findings and 3-Tesla metal artifact reduction sequences (MARS)-magnetic resonance imaging (MRI), and other causes of painful THA were excluded. Outcome measures included pain (visual analog scale (VAS)), Harris Hip Score (HHS), Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC), and standardized isometric hip flexion strength testing using a dynamometer, performed on the operated and contralateral limbs in all patients.

RESULTS: Pain improved significantly (VAS 8.3 ± 1.3 to 2.9 ± 2.3, $P < 0.001$) with a postoperative […]

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Efficacy of Commercially Available Irrigation Solutions on Removal of Biofilms Grown on Porous Titanium Implants: An In Vitro Study.

Miller A, Nasir M, Bou-Akl T, Pawlitz P, Markel DC

BACKGROUND: Periprosthetic joint infection (PJI) remains a major complication in orthopaedic surgery. A challenge in treating PJI is bacterial biofilm formation on implants, which confers a barrier to conventional antibiotic therapies and often necessitates implant removal and revision surgery. The goal of this study was to compare the efficacy of six commercially available irrigation solutions in decreasing bacterial load and disrupting biofilm formed on three dimensionally printed porous titanium discs that are similar to the surface characteristics of TJA implants.

METHODS: An in vitro model with three dimensionally printed titanium discs was used. Staphylococcus aureus was cultured to induce biofilm formation on the discs. There were six irrigation solutions applied according to the manufacturer's instructions. The effectiveness of each solution in reducing bacterial load was measured by sonication and colony-forming unit counts. Scanning electron microscopy was employed to observe biofilm disruption and structural changes.

RESULTS: When comparing groups, all had significantly fewer total colony-forming units after irrigation than after use of the chlorhexidine gluconate solution ($P < 0.001$). There were no other significant differences in the average total bacteria post-irrigation between the other groups. The scanning electron microscopy imaging revealed that the biofilm […]

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Recovery Plateaus and Ceiling Effects of Commonly Used Patient-Reported Outcome Measures Following Primary Total Knee Arthroplasty.

Bayram JM, Clement ND, Deehan DJ, London NJ, Pandit HG, Holloway NJ, Clarke JV

BACKGROUND: Various patient-reported outcome measures (PROMs) are used following total knee arthroplasty (TKA), but the timing of recovery plateaus and the presence of ceiling effects remain unclear. This study aimed to describe these characteristics of commonly used PROMs following TKA.

METHODS: This retrospective analysis of prospective data included 229 patients (mean age, 64 years; range, 43 to 75) who underwent primary TKA. Outcomes were collected preoperatively, at six weeks, six months, and annually up to four years using the Oxford Knee Score (OKS), Forgotten Joint Score, Knee Injury and Osteoarthritis Outcome Score subscales, EuroQol 5-Dimension, EuroQol Visual Analog Scale, Objective Knee Society Score, and range of motion. Recovery trajectories were modeled using linear mixed-effects models, and plateaus were identified through pairwise comparisons. Ceiling effects and proportions of patients achieving the minimal important change (MIC), patient acceptable symptom state, and maximum scores were tracked over time.

RESULTS: Knee-specific PROMs plateaued by two years, while health-related quality of life measures plateaued earlier (EuroQol 5-Dimension at one year, EuroQol Visual Analog Scale at six months), as did physical outcome measures (one year). For knee-specific PROMs, the proportions achieving the MIC and patient acceptable symptom state stabilized by one and […]

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Morbidity and Mortality Analysis of Primary Total Knee Arthroplasty in Patients Aged > 90 Years: Insights From the German Arthroplasty Registry.

Beckers G, Simon D, Grimberg A, Steinbrück A, Schröder L, Holzapfel BM

BACKGROUND: This study aimed to evaluate the safety of primary total knee arthroplasty (TKA) in patients aged ≥ 90 years by assessing complication and mortality rates. Additionally, we compared these outcomes with those of younger patient groups and identified comorbidities associated with increased morbidity and mortality.

METHODS: Data were obtained from a national arthroplasty registry. The study included 392,929 patients aged ≥ 60 years who underwent primary TKA for osteoarthritis. Of these, 1,284 patients were aged ≥ 90 years. Patients were divided into four age groups: (1) 60 to 69 years, (2) 70 to 79 years, (3) 80 to 89 years, and (4) ≥ 90 years. Minor and major complications and mortality during follow-up were recorded and compared across age groups and comorbidities; mortality was also compared with the general population using Federal Statistical Office data.

RESULTS: Minor complications occurred in 58% (745 of 1,284) of cases in group 4, compared to 21% (29,867 of 144,978) in group 1 ($P < 0.001$), with postoperative anemia being the most common (38%). Major complications were significantly more frequent in nonagenarians (18%, 233 of 1,284) than in sexagenarians (5.2%, 7,524 of 144,978; $P < 0.001$). Overall mortality increased with age, occurring in 4.0, 9.6, 17, and 36% of patients in Groups 1 to 4, respectively. Kaplan-Meier estimates showed significantly higher mor […]

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Is It Safe to Give Meloxicam or Celecoxib to Chronically Anticoagulated Patients Following Primary Total Knee Arthroplasty?

Lee GC, Zhong H, Nocon A, Poeran J, Sculco PK, Liu J

BACKGROUND: The nonsteroidal anti-inflammatory drugs (NSAIDs) are foundational components of multimodal pain protocols and have been shown to improve pain and range of motion following total knee arthroplasty (TKA). Patients receiving chronic anticoagulation therapy undergoing TKA are potentially disadvantaged because there are concerns with concurrent NSAID use due to bleeding concerns. Evidence suggests that cyclo-oxygenase 2 (Cox-2) inhibitors may not be associated with increased bleeding risk. The purpose of this study was to evaluate the safety of concurrent administration of NSAIDs in chronically anticoagulated patients undergoing primary TKA.

METHODS: Using an administrative claims database, we reviewed the records of 232,542 patients undergoing TKA between 2017 and 2021. Within this cohort, there were 2,299 patients who were chronically anticoagulated and 508 patients who were prescribed either celecoxib or meloxicam following TKA. The outcomes of interest included 30- and 90-day transfusion and readmission rates, rates of manipulation, and reoperations up to 180 days, and 30-, 90-, and 180-day opioid consumption measured as oral morphine equivalents between the two groups.

RESULTS: Concurrent administration of NSAIDs was not associated with increased risks for blood transfusions and readmissions at 30, 90, and 180 days. The administration of NSAIDs was associated wit […]

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Enhancing Diagnostic Accuracy for Uncertain Periprosthetic Joint Infection: Combining Cultures With Intraoperative Multisite Sonication.

Hua L, Guo W, Mu W, Ji B, Li Y, Parvizi J, Xu B, Cao L

BACKGROUND: Accurate diagnosis of periprosthetic joint infection (PJI) remains challenging, particularly in uncertain cases under the International Consensus Meeting (ICM) 2018 criteria. We explored whether combining cultures with intraoperative multisite sonication could improve diagnostic performance, especially in ambiguous cases. We addressed the following questions: (1) Does multisite sonication combined with standard culture reduce diagnostic uncertainty in PJI? (2) Does multisite sonication improve intraoperative culture positivity? and (3) Do preoperative and intraoperative postsonication bacterial profiles differ?

METHODS: A total of 375 patients who were undergoing revision surgery were enrolled, including 242 who had confirmed PJI and 133 who had aseptic loosening. There were four diagnostic models established and compared: Model 1 applied the ICM 2018 criteria without histological data. Model 2 followed the standard ICM 2018 diagnostic criteria, which incorporate histological results and preoperative cultures. Model 3 employed the ICM 2018 criteria with nonsonicated, multisite intraoperative cultures. Model 4 also adhered to the ICM 2018 criteria, but utilized intraoperative multisite cultures with sonication. We compared uncertain PJI incidence in specimens with or without intraoperative sonication, analyzed bacterial spectra, performed receiver operating characte […]

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Association of Admission Nutritional Status Evaluated by the Controlling Nutritional Status Score With Periprosthetic Joint Infection After Primary Total Joint Arthroplasty: An Exploratory Analysis.

Guo H, Yang Z, Wu D, Li C, Ma M, Wang T, Zhu R, Wang M et al.

BACKGROUND: The relationship between admission malnutrition and periprosthetic joint infection (PJI) after total joint arthroplasty (TJA) remains unclear due to limited evidence. This study aimed to assess the association between admission nutritional status and PJI risk within one year after TJA.

METHODS: This exploratory analysis of the prospective collection of data from TJA patients was derived from a database of a tertiary orthopedic referral center from October 1, 2014, to December 31, 2020, in China. Admission malnutrition is assessed by the controlling nutritional status (CONUT) score. The main outcome was PJI, defined according to the 2014 Musculoskeletal Infection Society (MSIS) consensus criteria, determined by review of index hospitalization medical records, microbiology laboratory reports, and readmission records for PJI treatment within one year after discharge. A restricted cubic spline (RCS) was performed to visualize the dose-response relationship between the admission CONUT score and PJI risk. The Youden index was used to identify the optimal cutoff value. Propensity score matching and univariate conditional logistic regressions were used to evaluate the association of preoperative malnutrition with PJI risk. There were 10,775 patients included, with a PJI incidence of 1.01%.

RESULTS: The RCS curve illustrated a significant positive correlation between the a […]

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History of Venous Thromboembolism Is Not a Contraindication to Aspirin Prophylaxis in Patients Undergoing Total Joint Arthroplasty.

Verhey JT, Tarabichi S, Brinkman JC, Hall RP, Deckey DG, Christopher ZK, Clarke HD, Spangehl MJ et al.

BACKGROUND: Although aspirin has become the prophylactic agent of choice for venous thromboembolism (VTE) prevention in low VTE risk patients undergoing primary total joint arthroplasty (TJA), there remains a paucity of data in the literature on the efficacy of aspirin as a mode of chemoprophylaxis in patients who have previously experienced a VTE episode. The purpose of this study was to determine the efficacy of aspirin versus other anticoagulant medications in primary TJA patients who have a documented history of a prior thromboembolic event.

METHODS: This study used a national database to identify patients who underwent primary TJA between 2010 and 2023. Only patients who had a documented history of a VTE event were included. Using demographic data, medical comorbidities, and preoperative anticoagulation use, primary TJA patients who had a history of VTE who received nonaspirin medications for thromboprophylaxis were propensity score-matched on a 3:1 basis to those who received aspirin. Patients who had either an active or previous history of cancer were excluded. A total of 3,132 matched patients undergoing primary total knee arthroplasty and 2,867 matched patients undergoing primary total hip arthroplasty were included in the analysis.

RESULTS: Among total knee arthroplasty patients, aspirin use was associated with comparable odds of deep vein thrombosis (DVT) (odds rat […]

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Are Patients Who Have Sarcoidosis at Increased Risk of Adverse Postoperative Outcomes Following Total Knee Arthroplasty?

Astavans A, Agarwal S, DePalo J, Garcia D, Suresh S, Thakkar S

BACKGROUND: Sarcoidosis is a multisystem inflammatory disease most observed in middle-aged adults, a population increasingly electing for total joint reconstruction. Musculoskeletal manifestations of sarcoidosis, including arthritis, occur in up to 40% of cases, prompting the need for orthopaedic involvement. Our study investigated whether sarcoidosis patients undergoing total knee arthroplasty (TKA) would be at increased risk for adverse postoperative outcomes.

METHODS: Using a national database, patients who underwent primary TKA over the past 20 years were identified, of which 1,829 had a history of sarcoidosis while 355,773 had no such history. The two cohorts were subsequently 1:1 propensity matched based on demographic factors and comorbidities, including diabetes, kidney disease, obesity, and nicotine dependence. Sarcoidosis patients had a significantly increased prevalence of all comorbidities measured. Upon matching, 1,825 patients were included in each cohort. The primary outcomes were the risks of 2-years revision and mechanical complications. The secondary outcomes were risks of 90-day postoperative stroke, thromboembolic events, pneumonia, pulmonary hypertension, acute kidney injury, and all-cause emergency department visits. Tests of significance (alpha = 0.05) were performed, and risk ratios (RRs) were calculated.

RESULTS: There were no significant differences […]

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Short versus Long Venous Thromboembolism Prophylaxis Following Elective Total Hip Arthroplasty: A Bayesian Network Meta-Analysis of Efficacy and Safety.

Kin Nam RH, Selim A, Gaddoura Z, Choudhary Z, Farhan-Alanie MM, Mohammad HR, Griffin XL, Thomas G

BACKGROUND: Pharmacological venous thromboembolism (VTE) prophylaxis following total hip arthroplasty (THA) lacks consensus regarding optimal duration. We conducted a network meta-analysis (NMA) to evaluate the efficacy and safety of commonly used prophylactic strategies.

METHODS: This meta-analysis included studies up to April 2025 comparing short duration (10 to 14 days) and extended-duration (28 to 35 days) regimens of low molecular weight heparin, direct oral anticoagulants (DOACs), and aspirin after THA. Outcomes included symptomatic deep vein thrombosis (DVT), pulmonary embolism (PE), major bleeding, and 90-day mortality. The search yielded 1,733 records, of which 25 studies (n = 28,772; mean age 62 years; 54.3% women) were included.

RESULTS: Short-duration prophylaxis was associated with higher odds of symptomatic DVT (odds ratio (OR): 1.87 [1.19 to 2.94]; P = 0.006) and PE (1.80 [1.02 to 3.18]; P = 0.043) compared with extended-duration regimens, with no difference in bleeding or 90-day mortality. A sensitivity analysis excluding pre-2000 studies showed that the reduction in symptomatic DVT with extended-duration prophylaxis was less pronounced (OR: 1.79 [1.12 to 2.86]; P = 0.014), and the difference in PE was no longer significant. In the five-arm NMA, DOAC-long had the lowest DVT risk (OR 0.69 [0.42 to 1.14]), while low molecular weight heparin-short (OR: 1.97 [1.18 […]

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Characterizing the Cost Trends of Readmissions After Aseptic Revision Total Hip Arthroplasty.

Hrudka BT, Gordon S, Rodoni B, Fuqua A, Wilson JM, Premkumar A

BACKGROUND: As the prevalence of primary total hip arthroplasty rises in the United States, there is a corresponding increase in revision total hip arthroplasty (rTHA), particularly among younger patients. This study explored the etiologies, financial burdens, and risk factors for 90-day readmission following aseptic rTHA, with an emphasis on distinguishing these factors from septic revision cases.

METHODS: This retrospective analysis used data from a large national database from January 1, 2009, to June 30, 2022. Patients undergoing aseptic rTHA were identified via Current Procedural Terminology codes. Readmissions within 90 days postoperatively were analyzed. Baseline characteristics, comorbidities, and readmission costs were compared, and significant risk factors for readmission were identified. Of the 36,212 patients who underwent aseptic rTHA, 3,822 (10.6%) experienced at least one readmission within 90 days.

RESULTS: The median cost of readmission was $41,148, with costs significantly higher in cases requiring reoperation. The primary surgical complications leading to readmission were periprosthetic joint infection, "other mechanical complications," and dislocation. Notable risk factors for readmission included different indications for index rTHA and comorbidities of morbid obesity, active smoking, and rheumatic disease.

CONCLUSIONS: Readmission following aseptic rTHA […]

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The Impact of Bone Marrow Transplant on Total Joint Arthroplasty Outcomes in Patients Diagnosed With Multiple Myeloma.

Rainwater RR, Siebenmorgen JP, Mears SC, Stronach BM, Barnes CL, Stambough JB

BACKGROUND: As screening and treatment for multiple myeloma (MM) improve, 5-year survival rates continue to rise. Bone marrow transplant (BMT) remains a challenging, but effective therapy with lasting benefit. Although postoperative outcomes after total joint arthroplasty (TJA) are well-studied in solid organ transplant recipients, data for BMT recipients are limited. This study aimed to define the impact of BMT on 90-day TJA outcomes.

METHODS: We used a large national insurance database to compare postsurgical complications in MM patients who underwent TJA, with or without prior BMT. A retrospective database review assessed differences in demographics, perioperative complications, readmissions, and postoperative risks. Perioperative outcomes were compared using multivariate analysis.

RESULTS: The BMT recipients undergo TJA at a younger age (total hip arthroplasty (THA): 61 versus 68 years, total knee arthroplasty (TKA): 66 versus 69 years). The BMT recipients do not have a higher risk of periprosthetic joint infection (THA: $P = 0.32$; TKA: $P = 0.3$), periprosthetic fracture (THA: $P = 0.9$; TKA: $P = 0.98$), or prosthetic instability (THA: $P = 1$; TKA: $P = 1$) within 90 days of TJA. However, BMT recipients have higher rates of pathological fractures (THA: odds ratio (OR): 0.92; TKA: OR: 1.73). All-cause 90-day readmission rates are similar, but BMT recipients have increased odds of […]

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Impact of Obesity on Daily Activity Following Total Knee Arthroplasty.

Wu KA, Kugelman DN, Shah SN, Ryan SP, Bolognesi MP, Seyler TM, Wellman SS

BACKGROUND: Obesity is a major risk factor for knee osteoarthritis, frequently requiring total knee arthroplasty (TKA). Although TKA improves mobility and reduces pain, obesity's influence on postoperative activity and gait metrics remains unclear. This study investigated the impact of body mass index (BMI) on daily activity following TKA.

METHODS: A prospective cohort study included 152 patients undergoing unilateral TKA, stratified into normal-weight (BMI: 18.5 to 25), overweight (BMI: 25.0 to 30), obese (30.0 to 40), and morbidly obese (BMI > 40) groups. Objective metrics-step count, standing hours, gait speed, steadiness, and 6-min walk test (6MWT)-were recorded preoperatively and at six weeks, six months, and 12 months postoperatively using watches. Multivariable regression analyses controlled for age and sex.

RESULTS: Morbidly obese patients had significantly lower preoperative step counts (2,104 ± 1,622 versus 4,893 ± 2,977; $P = 0.017$). Postoperatively, all groups showed improvements in step count, gait speed, and steadiness, but morbidly obese patients maintained lower activity metrics. At 12 months, morbidly obese patients' step counts (2,255 ± 1,639) were significantly lower than normal-weight patients (7,262 ± 3,141; $P = 0.011$). Improvements in step count, gait speed, standing hours, and patient-reported outcomes (Knee Injury and Osteoarthritis Outcome Score, Joint […]

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PhysiotherApeutic Treat-To-Target Intervention After Total Hip and Knee Arthroplasty Does Not Improve Outcomes Compared to Usual Physical Therapy: The PATIO Study.

Groot L, Gademan MGJ, van Loon D, Peter WF, Vliet Vlieland TPM, Reijman M, PATIO study group

BACKGROUND: This nationwide superiority cluster randomized controlled trial evaluated the effectiveness of a personalized, treat-to-target postoperative physical therapy (PPT) strategy compared to usual care after total hip arthroplasty (THA) and total knee arthroplasty (TKA).

METHODS: There were 18 hospitals randomized to treat-to-target or usual PPT, with 624 patients undergoing primary THA or TKA being eligible for participation. Assessments were conducted preoperatively and at six, 12, 24, and 52 weeks after surgery. The primary outcome was change in physical functioning three months after surgery, assessed with the Hip/Knee injury Disability and Osteoarthritis Outcome Score-Physical Function Short Form (HOOS-PS/KOOS-PS). Comparisons between groups were done using marginal models, including hospital as a fixed effect, with adjustment for sociodemographic and clinical characteristics. There were 312 THAs (53% women, aged 67 years (range, 40 to 88)) and 312 TKAs (55% women, aged 67 years (range, 41 to 88)) included. The THA patients received, on average, around 20 PPT sessions, and TKA patients around 30 sessions.

RESULTS: All groups showed significant and clinically relevant improvements in function. There were no statistically significant between-group differences in change scores found for either HOOS-PS (19.9 (95% confidence interval (CI): 17.8 to 22.1) in the intervent […]

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Nationwide Comparison of Cemented Versus Uncemented Hemiarthroplasty for Femoral Neck Fractures in the Elderly: A Propensity Score-Matched Analysis Using Japan's Diagnosis Procedure Combination Database.

Tanaka H, Tarasawa K, Mori Y, Sugaya T, Fukuchi H, Fushimi K, Aizawa T, Fujimori K

BACKGROUND: Femoral neck fractures in the elderly are a major public health issue. Hemiarthroplasty is widely used for displaced fractures; however, the optimal fixation method, cemented versus uncemented, remains controversial. In Japan, despite a high prevalence of osteoporosis, cemented hemiarthroplasty is underutilized. To date, no large-scale Japanese study has directly compared early outcomes between cemented and uncemented fixation.

METHODS: We conducted a retrospective cohort study using the Japanese Diagnosis Procedure Combination database from December 2011 to March 2023. Patients aged more than 65 years who underwent hemiarthroplasty for femoral neck fractures were identified. A one-to-three propensity score matching was applied using age, sex, body mass index, and Charlson Comorbidity Index. After matching, 26,591 cemented and 79,773 uncemented cases were analyzed. Multivariate logistic regressions were used to assess early postoperative complications and in-hospital mortality.

RESULTS: Cement use significantly reduced the risk of infection (odds ratio [OR] 0.747, 95% confidence interval [CI] 0.637 to 0.877, $P < 0.01$), periprosthetic fracture (OR 0.668, 95% CI 0.539 to 0.829, $P < 0.01$), and deep vein thrombosis (OR 0.683, 95% CI 0.628 to 0.743, $P < 0.01$). Conversely, it was associated with increased risk of transfusion (OR 1.463, 95% CI 1.418 to 1.510, $P < 0.01$) a […]

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Effect of Vancomycin Powder in Reducing Infection After Primary and Revision Hip and Knee Arthroplasty and Its Complications: An Umbrella Review and Meta-Analysis.

Salimi M, Keshtkar A, Abdelnaser MK, Mosalamiaghili SA, Saka N, Parvizi J

BACKGROUND: Periprosthetic joint infection (PJI) after hip or knee arthroplasty greatly increases morbidity and costs. Local antibiotic powders may help curb infections while minimizing systemic toxicity. We assessed their efficacy and safety in preventing PJI.

METHODS: We systematically searched PubMed, Scopus, Cochrane Central Registry of Clinical Trials (CENTRAL), Web of Science, Embase, and Cumulative Index to Nursing and Allied Health Literature for studies published from January 1990 to January 2024. Eligible trials compared intraoperative antibiotic powder with no powder in primary or revision arthroplasty. Risk ratios (RRs) with 95% confidence intervals (CIs) were pooled.

Methodologic quality was assessed using A MeaSurement Tool to Assess systematic Reviews (AMSTAR 2), ROB 2, and ROBINS-I; evidence certainty was rated with Grading of Recommendations Assessment, Development, and Evaluation (GRADE). Stratified analyses examined dose, surgical type, and administration technique. There were 10 secondary studies (comprising 22 primary studies with 137,651 patients)-all evaluating vancomycin powder (VP)-that met the inclusion criteria.

RESULTS: Pooled data showed no significant reduction with VP in total infections (RR 0.86, 95% CI, 0.62 to 1.18), superficial infections, or PJI specifically. High-quality, randomized, and prospective studies have likewise demonstrated no pr […]

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Rotating Hinge Implants for Primary Total Knee Arthroplasty: A Systematic Review and Meta-Analysis of 6,554 Knees.

Abdel Khalik H, Cruickshank M, Nadeem SM, Ade-Conde MA, Chalmers BP

BACKGROUND: Implant design improvements and expanded indications for total knee arthroplasty (TKA) have led to the increasing use of rotating hinge implants in primary TKA. Revisions following primary rotating hinge total knee arthroplasty (RH-TKA) are complex and often cause major patient morbidity. This review amalgamated survivorship data on primary RH-TKAs, causes for revision, and patient-reported outcome measures (PROMs).

METHODS: MEDLINE, EMBASE, and CENTRAL were searched for studies assessing implant survivorship following primary RH-TKAs. Secondary outcomes included causes for revision surgery, type of revision surgery, and PROMs. Survivorship at less than five years, five to less than 10 years, and 10 to 15 years was assessed using meta-analysis. There were 31 studies (6,554 knees) included for final analysis (median age, 70 years; median body mass index, 28.0), with 13 unique implants utilized.

RESULTS: Less than 5-year, five to less than 10-year, and 10- to 15-year survivorships were 93.8% (95% confidence interval [CI], 90.5 to 96.0), 91.5% (95% CI, 88.6 to 93.6), and 86.3% (95% CI: 82.2 to 89.5), respectively. Survivorship in the 10 to 15-year group was significantly lower than the less than 5-year ($P = 0.002$) and five to less than 10-year ($P = 0.023$) groups. The overall incidence of complications requiring revision was 7.4% (range, 0 to 36.4), with the most comm […]

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Compatibility of Retrograde Intramedullary Nails With Knee Prostheses: A Comprehensive Analysis.

Richards AE, Deckey DG, Stein MK, Ryan SP, Schwartz AM, Seyler TM

BACKGROUND: Periprosthetic distal femoral fractures pose major challenges in orthopaedic surgery, particularly among aging populations with total knee arthroplasties. Retrograde intramedullary nailing has emerged as a viable fixation option due to its biomechanical stability, less invasive nature, and facilitation of early weight-bearing. However, the compatibility between retrograde nails and various knee prostheses remains a critical determinant of surgical success. This review consolidated existing knowledge and evaluated current nail-prosthesis compatibility and options for lateral femoral periprosthetic plating and provided a compatibility matrix for fixation options for these complex fractures.

METHODS: An evidence-based review was conducted to evaluate the compatibility of retrograde intramedullary nails with total knee arthroplasty prostheses. Additionally, currently available lateral distal femoral plates were evaluated. Data were synthesized from key studies, technical specifications from major manufacturers, and recent advancements in nail and prosthesis design. Technical parameters such as intercondylar dimensions, posterior notch geometry, nail diameter, and proximal sagittal nail bend angles were analyzed. The findings were summarized in comprehensive tables, cross-referencing nail systems with prosthesis models to identify optimal combinations and highlight pote […]

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A Complete Infrapatellar Fat Pad Resection Is Associated With New Postoperative Patella Baja and Persistent Pain, but a Lower Risk of Manipulation Under Anesthesia After Primary Total Knee Arthroplasty.

Young B, McGovern AG, Tantikosol P, Sodhi N, Koltsov JCB, Wong HJ, Maloney WJ, Amanatullah DF

BACKGROUND: Patella baja after total knee arthroplasty (TKA) is associated with postoperative stiffness, and there is conflicting evidence regarding the patellar fat as a source of pain. We hypothesized that a complete infrapatellar fat pad resection during primary TKA does not increase the incidence of new postoperative patella baja while reducing the incidence of persistent pain and improving postoperative function. Specifically, we evaluated the odds of 1) new patella baja, 2) persistent pain, 3) knee society score, 4) range of motion, and 5) manipulation under anesthesia.

METHODS: The primary outcome of this single-institution retrospective study was the incidence of new patella baja comparing preoperative and postoperative lateral knee radiographs using the Insall-Salvati ratio of patients who underwent a primary TKA with either a complete or partial infrapatellar fat pad resection from 2014 to 2025. The final cohort included 592 patients who had either a complete (n = 351) or partial (n = 241) infrapatellar fat pad resection. The secondary outcomes included KSS, persistent pain, and range of motion. Differences in outcomes between complete and partial fat pad resection were assessed via risk-adjusted multivariable binomial and linear regression models.

RESULTS: A complete infrapatellar fat pad resection nearly tripled the odds of new patella baja compared to a partial r […]

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Increased Postoperative Complications in Nontobacco Nicotine Users Following Total Hip Arthroplasty.

Fuller Z, Chopra A, Thomas JJ, Oliver D, Fisher M, Abdo ZE

BACKGROUND: Nicotine use is associated with increased complications after total joint arthroplasty, but data on nontobacco nicotine dependence (NTND), including vaping and smokeless nicotine, are limited. This study evaluated perioperative and postoperative outcomes in NTND patients undergoing total hip arthroplasty (THA).

METHODS: A retrospective cohort study using a large national database identified primary THA patients (2003 to 2023). Patients were one-to-one propensity-matched into NTND (n = 10,503) and non-nicotine user (n = 10,503) groups. Systemic complications (90 days) and prosthesis-related complications (two years) were assessed.

RESULTS: The NTND patients exhibited significantly higher rates of systemic complications within 90 days, including sepsis (1.0 versus 0.8%, P = 0.047), deep vein thrombosis (1.9 versus 1.4%, P = 0.01), stroke (0.9 versus 0.6%, P = 0.007), and pneumonia (1.8 versus 1.1%, P < 0.001). Opioid use was significantly higher in NTND patients (88.9 versus 81.5%, P < 0.001). The NTND patients had increased readmission (38.4 versus 33.3%, P < 0.001) and emergency department visit rates (12.1 versus 8.3%, P < 0.001). Prosthesis-related complications at two years were also significantly more common in NTND patients, including periprosthetic joint infection (3.0 versus 2.0%, P < 0.001), prosthetic joint dislocation (1.5 versus 1.2%, P = 0.031), and re […]

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Fractures of the coronoid process: state of the art.

Marinelli A, Riva M, Coliva F, Minerba M, Carbone G, Guerra E

Coronoid fractures are rarely isolated and are much more frequently associated with other osseous or ligamentous structures injuries. On the basis of the coronoid fracture patterns, described by the O'Driscoll classification, it is possible to recognize three main patterns of injury that differ on traumatic mechanism and on associated lesions: posterolateral rotatory instability, posteromedial rotatory instability, and axial load injuries. The management of coronoid fractures is challenging and varies according to characteristics of the fracture, associated lesions, and amount of elbow instability. In general, operative treatment is indicated in every case the fracture is at least 50% of the whole coronoid, whether the sublime tubercle is involved, and whether the ulno-humeral joint is not perfectly reduced. In conclusion, the correct management of the coronoid, especially in the setting of complex elbow instability, represents a predictive factor for patient outcomes and functional results. The stability of the elbow, rather than the size of the coronoid fragment, is the main parameter for surgical indication, aimed to fix the coronoid and/or repair the associated lesions.

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Total hip arthroplasty in Italy: an observational, population-based study on surgical volume growth from 2001 to 2023 and forecasts until 2050 with six different statistical models.

Ciminello E, Cuccu A, Romanini E, Venosa M, Cazzato G, Tucci G, Boniforti F, Carpanese L et al.

BACKGROUND: The number of total hip arthroplasty (THA) procedures has been steadily increasing worldwide, driven by aging population, improvements in surgical techniques and implant design. This study aimed to analyze the temporal trends of elective THA in Italy since 2001-2023 and forecast THA volumes up to 2050 to provide insights for healthcare planning.

MATERIALS AND METHODS: International Classification of Diseases, 9th Revision, Clinical Modification (ICD9-CM) coding system was used to extract records of interest (elective THA) from the Italian National Hospital Discharge Record database. Six statistical models were applied to forecast future THA volumes: logistic regression; Poisson regression; logarithmic regression; inverse/power regression; Poisson log-normal regression; and hierarchical Poisson regression with temporal effects (HPTE). Model performances were assessed by using error metrics and internal validation on the basis of a rolling-origin approach. An out-of-sample validation was conducted to ensure a robust assessment of forecasting reliability. THA volume forecasts were provided with 95% prediction intervals.

RESULTS: A total of 1,318,400 records for primary elective THAs performed in Italy since 2001-2023 were analyzed. The number of THAs increased by approximately 80%, rising from 68.270 in 2001 to 122.777 in 2023. Among the tested models, HPTE generally […]

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Clinical relevance of patient-reported outcome measures in the surgical management of focal chondral defects of the knee: a systematic review.

Migliorini F, Maffulli N, Memminger MK, Hofmann UK

INTRODUCTION: To evaluate clinical outcome following surgical management of focal chondral defects in the knee, patient-reported outcome measures (PROMs) are used. To give these measures meaning, parameters such as the minimal clinically important difference (MCID), patient-acceptable symptom state (PASS), minimally detectable change (MDC), clinically important difference (CID) and substantial clinical benefit (SCB) have been introduced. This systematic review investigated the MCID, SCB, CID, PASS and MDC of the most commonly used PROMs for assessing patients following surgical repair of focal chondral defects of the knee.

METHODS: This systematic review was conducted in accordance with the 2020 Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines and the recommendations of the Cochrane

Handbook for Systematic Reviews of Interventions. All clinical studies investigating tools to assess the clinical relevance of PROMs in the surgical repair of focal chondral defects of the knee were reviewed. In April 2025, the following databases were accessed: PubMed, Web of Science and Embase. The PROMs of interest included: the International Knee Documentation Committee (IKDC) questionnaire, the Knee Injury and Osteoarthritis Outcome Score (KOOS) and its related subscales activities of daily living (ADL), pain, quality of life (QoL), sports and recreational […]

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Simultaneous versus staged bilateral total hip arthroplasty: a meta-analysis.

Soler F, Hernández J, Lamo-Espinosa JM, Benlloch M, Mariscal G

BACKGROUND: There is debate regarding the optimal timing of bilateral total hip arthroplasty (THA), with simultaneous or staged approaches considered. Cost-effectiveness is an important factor that influences resource allocation. The main aim of this study was to assess the cost implications of bilateral simultaneous (sim-THA) versus staged (st-THA) total hip arthroplasty. The secondary objective was to evaluate the efficacy and safety of sim-THA compared those with of st-THA.

METHODS: A systematic review and meta-analysis were conducted according to the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) guidelines. Studies comparing simultaneous and staged bilateral THA in terms of cost, complications, length of hospital stay, and patient outcomes were eligible. Odds ratios (ORs), mean differences (MD), and standard mean differences (SMD) with 95% confidence intervals (CIs) were calculated. The risk of bias was assessed using the Methodological Index for Non-randomised Studies (MINORS). Meta-analyses were performed using Review Manager version 5.4 (Cochrane, Oxford, United Kingdom).

RESULTS: A total of 20 observational studies including 13,984 patients were included. Sim-THA was associated with significantly lower total costs (SMD -0.54, 95% CI -0.92 to -0.16; $p = 0.005$) and shorter hospital stay (MD -2.90, 95% CI -4.38 to -1.42; $p = 0.0001$) than tho […]

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Long-term outcomes of the modified Dunn procedure in moderate and severe slipped capital femoral epiphysis: a prospective case series with 7-year follow-up.

Fahmy M, Abdelazeem AH, Shawky MA

INTRODUCTION: Slipped capital femoral epiphysis (SCFE) is the most common hip disorder in adolescents and may lead to femoroacetabular impingement, early osteoarthritis, and long-term functional disability if inadequately treated. While in situ pinning remains the standard treatment for mild slips, it fails to correct the deformity in moderate and severe cases, potentially predisposing to degenerative changes. The modified Dunn procedure (MDP) was developed to restore proximal femoral anatomy through surgical hip dislocation while preserving vascular supply. The aim of the study is to evaluate the long-term radiological and functional outcomes of the MDP in patients with moderate (14 cases) and severe (10 cases) SCFE, and to assess the incidence of avascular necrosis (AVN), osteoarthritis, and other complications.

METHODS: A prospective case series was conducted between August 2015 and January 2019 at a single tertiary institution. A total of 24 hips with moderate-to-severe SCFE and open physis were treated using the MDP via surgical hip dislocation. Mild and acute-only slips were excluded. MDP was used as a primary procedure, performed early in severe slips and in selected moderate slips after clinical assessment. Patients were followed clinically and radiologically for a mean duration of 84 ± 2.6 months (range 80-88 months). Functional outcomes were assessed using the Harris […]

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The presence of a positive blood culture should be regarded a risk factor for DAIR failure in cases of haematogenous total knee arthroplasty infection.

Pedemonte-Parramón G, Reynaga E, Molinos S, de Los Ríos JD, Vivero A, García-Oltra E, López-Pérez V, Paredes R et al.

PURPOSE: Late acute haematogenous total knee arthroplasty (LAH TKA) infections represent approximately 20% of all prosthetic infections. Debridement, antibiotics, and implant retention (DAIR) is a widely accepted treatment; however, it carries a significant risk of failure. The study aims to identify risk factors for DAIR failure in LAH TKA infections, focusing on blood cultures (BC) as a potential contributor.

METHODS: A retrospective cohort study was conducted on 37 cases of LAH TKA infections from 2015 to 2023. Patients were divided into two groups: 20 in the success group (SG) and 17 in the failure group (FG). The study analyzes various factors, including demographics, comorbidities, serum and joint fluid biomarkers, blood and intraoperative cultures, prior infection or antibiotic use, and surgical and post-surgical variables.

RESULTS: Positive BC were more frequent in the FG compared with the SG (P = 0.03), and were also more common in patients with a history of prior infection (P = 0.03). Logistic regression identified positive BC as the only significant predictor of DAIR failure (odds ratio (OR) 12, 95% confidence interval (CI) 1.1-18, P = 0.04), even after adjusting for other variables. Positive intraoperative cultures were more frequent in the FG, particularly in those with a prior infection history (P = 0.08), even if they were receiving antibiotics (P = 0.05).

CON […]

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Iatrogenic femoral neck fractures or separation of the proximal femoral epiphysis during closed reduction of irreducible femoral head fracture-dislocations in children: a review of 12 cases.

Lu Y, Xue C, Canavese F, De Salvo S, Yao H, Huang D, Lin R, Zheng J et al.

OBJECTIVES: This study aimed to analyze the radiologic characteristics of irreducible pediatric femoral head fracture-dislocations to prevent potential iatrogenic femoral neck fractures (FNF) or separation of the proximal femoral epiphysis (SPFE).

METHODS: This is a retrospective review of patients who were skeletally immature and diagnosed with traumatic hip dislocations combined with femoral head fractures. The collected data included patient demographics, fracture classification, fragment ratio, combined injuries, urgent reductions, treatment strategies, complications, and final outcomes.

RESULTS: We treated 12 patients with femoral head fractures and dislocations; 11 out of 12 patients underwent urgent closed reduction (91.7%). Six of the patients failed closed reduction and experienced FNF (n = 2, 33.3%) or SPFE (n = 4, 66.7%). Five patients presented with avascular necrosis of the femoral head after open reduction with internal fixation via a surgical hip dislocation approach, and two patients required further surgical treatment. Analysis of radiographs and computed tomography (CT) scans of irreducible femoral head fracture-dislocations revealed that the fractured femoral head was perched on the sharp angle of the posterior wall of the acetabulum, with a fragment ratio of 18-27%. After recognizing the irreducibility, one case with a fragment ratio of 26% underwent immed […]

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Vascularized versus nonvascularized free fibular grafts in reconstruction of post-traumatic critical long bone defects, a comparative study.

Fouaad AA, Massoud AH, Shaban M, Abdel-Naby M, Abulsoud MI

BACKGROUND: Reconstruction of critical-sized long bone defects is a complex orthopedic challenge. Free fibular grafts, vascularized (FVFG) and nonvascularized (NVFG), are established reconstructive options, but comparative clinical outcomes remain uncertain.

PURPOSE: To compare clinical and radiological outcomes of FVFG and NVFG in patients with post-traumatic critical bone defects more than 10 cm.

METHODS: A randomized controlled trial was conducted with 50 patients assigned equally to FVFG or NVFG groups. The primary outcome was time to union, while secondary outcomes included graft hypertrophy, functional scores (DASH and LEFS), complication rates, and donor site morbidity.

RESULTS: The mean time to union was 5.78 months in the FVFG group and 6.17 months in the NVFG group, showing no statistically significant difference (p = 0.447). Rates of graft hypertrophy, functional recovery, and complications were comparable between the groups.

CONCLUSIONS: Both FVFG and NVFG provide effective reconstruction for critical bone defects, with nearly similar healing times and functional outcomes. NVFG may represent a less complex alternative in selected cases.

PubMed · DOI

Do modular tapered long stems differ in stability and function? A prospective comparison of two stems in complex femoral revision THA.

Fahmy M, Shawky MA

BACKGROUND: Revision total hip arthroplasty (rTHA) in the setting of substantial proximal femoral bone loss is technically challenging. Modular tapered fluted stems provide predictable diaphyseal fixation while allowing independent adjustment of version, offset, and limb length. Among these, two commonly used systems-modular tapered fluted stem Type A (Revitan™) and Type B (LIMA Modulus™)-have limited direct comparative evidence. This study aimed to prospectively compare radiographic stem subsidence (primary outcome), as well as functional outcomes, complications, survivorship, and secondary outcomes, between Type A and Type B modular long stems in femoral rTHA.

METHODS: In this single-center randomized prospective study, 110 patients undergoing femoral revision rTHA were randomly assigned to receive either Type A (n = 55) or Type B (n = 55) stems. All procedures were performed by experienced revision surgeons under standardized perioperative and rehabilitation protocols. Radiographs were analyzed for stem subsidence, osseointegration, and limb-length restoration. Functional outcomes were assessed using the Harris Hip Score (HHS), Oxford Hip Score (OHS), and European Quality of Life Visual Analogue Scale (EQ-VAS) at baseline and final follow-up (mean 61.4 months). Complications and stem survivorship were recorded prospectively. Statistical analysis included paired and unpaired […]

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A study on the association between tibial plateau fractures and intra-articular soft-tissue injuries under valgus injury mechanisms.

Duan S, Wang S, Zhu T, Li Q, Zhang M

BACKGROUND: While recent investigations have focused on injury mechanism classifications of tibial plateau fractures (TPFs), the association between valgus TPFs and concomitant soft-tissue damage involving menisci and ligaments remains insufficiently elucidated. This study aimed to characterize intra-articular soft-tissue injuries associated with various valgus TPFs and assess the predictive value of lateral plateau depression (LPD) and widening (LPW). Additionally, the analysis extended to other injury mechanisms.

MATERIALS AND METHODS: This study included adult patients with acute tibial plateau fractures who had complete imaging data, excluding patients with open fractures and multiple fractures throughout the body. Imaging examinations were used to assess the fracture mechanism and intra-articular soft-tissue damage.

RESULTS: The study retrospectively analyzed the clinical data of 138 patients with valgus injury TPFs in our hospital. The study compared the incidence of TPFs with intra-articular soft-tissue damage under different injury mechanisms. The result demonstrated that the incidence of valgus hyperextension TPFs combined with medial collateral ligament injuries was relatively high (61.5%) compared with TPFs with other injury mechanism. Multivariable logistic regression and smooth curve fitting revealed significant dose-response relationships of LPD (odds ratio [OR] […]

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The coronoid as the key fragment of trans-ulnar fracture-dislocations of the elbow: Insights from a retrospective cohort comparison using the coronoid-centric Mayo classification system.

Henssler L, Kerschbaum M, Zanklmaier M, Wieczorek LA, Alt V, Klute L

BACKGROUND: Trans-ulnar fracture-dislocations of the elbow are rare injuries with complex fracture patterns and variable outcomes. Traditional classification systems offer limited prognostic value. A recently introduced coronoid-centric Mayo classification distinguishes injury subtypes based on coronoid attachment and identifies trans-ulnar basal coronoid (TUBC) fractures as a particularly challenging entity. This study evaluated outcomes across Mayo fracture types and explored factors associated with inferior results in TUBC injuries.

MATERIALS AND METHODS: In this retrospective cohort study, surgically treated trans-ulnar elbow fracture-dislocations managed at a level I trauma center between 2010 and 2022 were identified and classified according to the Mayo system. Demographic data, injury characteristics, surgical management, radiographic outcomes, and complications were recorded. Functional outcomes were assessed after a minimum follow-up of 12 months using the Mayo Elbow Performance Score (MEPS); Oxford Elbow Score (OES); Quick Disabilities of Arm, Shoulder and Hand Questionnaire (QuickDASH); European Quality of Life Five-Dimension, Five-Level Version (EQ-5D-5L); and range-of-motion measurements. Radiographs were analyzed for union, instability, heterotopic ossification, and post-traumatic osteoarthritis (OA).

RESULTS: A total of 52 patients were included (14 trans-olecr […]

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Feasibility study of a robot-assisted endoscopic technique for plate osteosyntheses of the acetabulum and anterior pelvic ring.

Hinz N, Althoff G, Thomaschewski M, Bonke LA, Eberenz A, Münch M, Behrends L, Männel G et al.

BACKGROUND: Minimally invasive stabilization of acetabular and pelvic ring fractures using endoscopic techniques has become increasingly important. The logical advancement of conventional endoscopic techniques is a robot-assisted approach, which benefits from the advantages of robot-assistance systems (e.g., more degrees of freedom for instruments and improved visualization). The aim was therefore to investigate the feasibility of a robot-assisted endoscopic technique for plate osteosyntheses of the acetabulum and anterior pelvic ring.

MATERIALS AND METHODS: In this two-part feasibility study, four different plate osteosyntheses were performed endoscopically using the Hugo™ robot-assisted surgery system, first on ten synthetic pelvic models and then on ten human cadavers. During robot-assisted dissection for the preperitoneal approach, identification of ten relevant anatomical landmarks was also assessed. In both parts, the success, number of drilling errors, and time for each plate were described as learning curves and analyzed using linear regression.

RESULTS: The infrapectineal plate could be successfully performed in 100% of synthetic models, the posterior column plate in 100%, the suprapectineal plate in 90%, and the superior pubic ramus plate in 80%. Learning curves could be observed for the number of drilling errors per plate (e.g., 0.67 to 0, $p = 0.009$) and the time r […]

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Favorable subjective clinical outcomes after revision hip arthroscopy for femoroacetabular impingement syndrome despite a high rate of capsular defects and increased capsular thickness.

Cao Z, Gao G, Lin W, Zhu Y, Zhou X, Wang J, Xu Y

BACKGROUND: The capsular healing status and capsular thickness changes in patients with femoroacetabular impingement syndrome (FAIS) following revision hip arthroscopy are poorly documented, and their relationship with subjective clinical outcomes remains unclear. The purpose of this work is to evaluate the incidence of capsular defects and changes in capsular thickness using magnetic resonance imaging (MRI) following hip arthroscopy in patients with FAIS, and to compare the subjective clinical outcomes between patients with and without capsular healing after revision hip arthroscopy.

PATIENTS AND METHODS: Consecutive patients with FAIS who underwent revision hip arthroscopy between 2013 and 2023 were included. Patients were categorized into two groups on the basis of capsular healing status after revision hip arthroscopy. Patient-reported outcomes (PROs) were collected preoperatively and at minimum 2-year follow-up, including the modified Harris Hip Score (mHHS), International Hip Outcome Tool-12 (iHOT-12), Hip Outcome Score-Activity of Daily Living Scale (HOS-ADL), Hip Outcome Score-Sport Specific Subscale (HOS-SSS), and visual analog scale (VAS). Achievements in minimal clinically important difference (MCID) and patient acceptable symptom state (PASS), patient satisfaction, and re-revision rates were evaluated and compared between groups. Capsular thickness was assessed by […]

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From bench to bedside: a novel suture-augmented prosthesis for greater trochanteric fixation in osteoporotic hip arthroplasty.

Yang M, Yang J, Wang X, Wu Z, Bian S, Sheng R, Liao F, Qin K et al.

BACKGROUND: Stable fixation of the greater trochanter during hip arthroplasty for unstable osteoporotic intertrochanteric fractures remains a significant challenge. Conventional techniques depend on securing the bone-implant interface, which is particularly compromised in osteoporotic bone. Here, we propose a novel conceptual approach that establishes a direct mechanical bridge from the abductor tendon to the prosthesis, thereby reducing reliance on the fragile bone-implant interface.

METHODS: In this two-stage translational study, we progressed from biomechanical validation to clinical application. First, using a decalcified caprine femoral model simulating osteoporosis, three fixation constructs were compared: locking plate (LP), suture-augmented locking plate (LPSA), and Kirschner-wire tension band (KWTB). Ultimate load and construct stability were evaluated. Biomechanical testing confirmed the principle of suture-mediated load sharing and highlighted the intrinsic weakness of screw fixation in osteoporotic bone. Guided by these results, we designed a novel femoral prosthesis that eliminates screw fixation, employs sutures as the primary load-bearing element, and incorporates integrated suture anchor tunnels. This prosthesis was then assessed in a retrospective series of 15 consecutive elderly patients with osteoporotic intertrochanteric fractures. Clinical outcomes were ev […]

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Updating radiographic parameters for the healthy growing hip: are current parameters still valid?

Hütter K, Smolle MA, Butter S, Schroedter R, Tschauner S, Kraus T

BACKGROUND: This study aimed to evaluate whether the acetabular angle (AA) by Tönnis and lateral center-edge angle (LCEA) by Wiberg-key radiographic parameters in pediatric hip assessment-have changed in a contemporary pediatric population compared with historical reference values, considering trends in earlier skeletal maturation and body composition.

MATERIALS AND METHODS: This retrospective diagnostic accuracy study assessed changes in AA and center-edge angle (CE) angles in a contemporary cohort. A total of 1774 anteroposterior pelvic radiographs (3548 hips) from patients aged 0-18 years without hip pathology were analyzed. Radiographs were obtained between 2006 and 2018 and measured using the Supervisely digital platform. Standardized anatomical landmarks were applied. Data were stratified by age and sex and compared with historical values using two-sample t-tests ($p < 0.05$). To minimize measurement bias, two independent observers conducted all assessments using standardized digital protocols.

RESULTS: A total of 1774 patients aged 0-18 years (mean age, 8.30 ± 5.20 years; 666 females, 1108 males) were evaluated. AA values did not show statistically significant differences in 11 of 16 age groups compared with historical data ($p > 0.05$). In contrast, LCEAs were significantly higher in the contemporary cohort, especially in the 5-8, 9-12, and 12-16 age groups (all $p < 0.01$) […]

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In-hospital deep vein thrombosis after tibial plateau fractures: incidence, laterality/anatomy, and risk factors in a multicenter retrospective cohort of 3366 patients.

Yang S, Li G, Li Y, Long Y, Zhang J, Liu L, Zhao Z, Sun C et al.

BACKGROUND: We aimed to determine the cumulative in-hospital incidence, anatomic distribution, and predictors of duplex ultrasonography-detected lower-extremity deep vein thrombosis (DVT) in patients hospitalized with tibial plateau fractures under routine thromboprophylaxis and protocolized surveillance.

METHODS: We retrospectively analyzed consecutive adults (≥ 18 years) admitted to two level-I trauma centers (November 2014-January 2024). Bilateral duplex ultrasonography was performed on admission/preoperatively (as soon as feasible after arrival and prior to surgery) and repeated serially during hospitalization (approximately every 5-7 days and when clinically suspected). The primary outcome was cumulative in-hospital DVT from admission to discharge; isolated intermuscular calf-muscle vein thrombosis (soleal/gastrocnemius) was recorded descriptively but excluded from the primary endpoint. Multiple injuries (polytrauma) were defined as ≥ 1 additional acute traumatic lesion beyond the index tibial plateau fracture documented on admission. Pulmonary embolism (PE) was assessed only when clinically suspected and confirmed by computed tomography pulmonary angiography (CTPA). Multivariable logistic regression identified independent predictors; receiver operating characteristic (ROC) analysis evaluated discrimination for key continuous predictors.

RESULTS: Among 3366 patients, 675 […]

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Retained articulating spacers (1.5 stage) in chronic knee periprosthetic joint infections: an analysis of associate factors and outcomes in a single center cohort of patients.

Balato G, Ascione T, Festa E, De Mauro D, Di Gennaro D, Rosa D, Mariconda M

BACKGROUND: Two-stage revision remains the gold standard for chronic periprosthetic joint infection (PJI), but it can significantly impact patients' quality of life and functional recovery. The 1.5-stage revision, involving indefinite spacer retention, has emerged as a potential alternative in selected high-risk patients. Our questions were: (1) What are the early functional outcomes, including their infection-free survival? (2) Which pre- and perioperative factors are associated with spacer retention? and (3) What is the mechanical survivorship of retained spacers?

MATERIALS AND METHODS: This was a retrospective observational study including 108 consecutive patients with chronic PJI between 2016 and 2020 who were followed prospectively for clinical outcomes. Functional status and quality of life were assessed using the EuroQol-5-Dimension-5-Level (EQ-5D-5L), Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC), and Knee Society Score (KSS) before spacer placement and prior to scheduled reimplantation. Infection-free status was defined as the absence of clinical, radiographic, and laboratory signs of infection recurrence over a 24-month follow-up. Univariate and multivariate logistic regression were used to identify factors associated with spacer retention.

RESULTS: All patients initially underwent articulating spacer implantation as part of a planned two-st […]

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Moderate leg length discrepancy: a long-term risk factor for hip osteoarthritis?

Aprato A, Donis A, Via RG, Scandurra A, Tuè F, Audisio A, Fusini F, Massè A

BACKGROUND: Leg length discrepancy (LLD) has been implicated as a biomechanical factor contributing to hip osteoarthritis (OA), yet the extent of its influence remains unclear. This study examines the correlation between moderate LLD (≥ 10 mm) and hip OA progression, focusing on asymmetrical OA distribution in patients over 65 years old.

MATERIALS AND METHODS: A retrospective analysis was conducted from a database of 1672 full-length standing X-rays. Patients under 65 years, with deformities, unilateral limb issues, or prosthetics, were excluded; therefore, the study group was composed of 220 patients. Tibial and femoral lengths were measured bilaterally, and hip OA severity was assessed using the Tönnis classification. Statistical analyses included Pearson's Chi-squared test and linear regression to explore correlations between LLD and OA distribution.

RESULTS: Among the sample, 18% showed an LLD ≥ 10 mm. A significant correlation was found between LLD and the asymmetrical distribution of hip OA ($p = 0.002$), with higher OA severity observed in the hypometric limb. Linear regression analysis suggested that each millimeter of LLD corresponded to a 0.74-unit change in OA severity difference between hips.

CONCLUSIONS: This study highlights a significant association between moderate LLD and contralateral hip OA in the elderly, emphasizing the biomechanical impact of asymmetrical […]

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Authorship, titles and open access as drivers of citation performance in orthopaedics: a scientometric analysis.

Migliorini F, Vaishya R, Rivera F, Eschweiler J, Kobbe P, Betsch M, Oliva F, Maffulli N

BACKGROUND: Bibliometric analyses are increasingly used to explore how scientific knowledge is created, disseminated, and perceived. In orthopaedics, research output has expanded rapidly over the past decade, yet the factors determining whether an article achieves wide visibility and scholarly impact remain poorly understood.

Beyond the inherent quality of a study, elements such as authorship patterns, title construction, and open access (OA) availability may play an essential role in shaping citation performance. However, evidence in this field is still limited and sometimes contradictory, highlighting the need for large-scale, field-specific analyses.

METHODS: Orthopaedic publications from 2010 to 2020 were identified in Scopus using the keyword 'orthopaedic'. After duplicate removal, 97,806 unique articles were included with complete data on authorship, titles, citation counts, study design, and OA status. Citation rates were normalised per year since publication. Associations between bibliographic features and citation performance were assessed using multiple linear regression, while differences across title styles and study designs were evaluated with comparative statistical testing. Exploratory modelling was performed to identify combinations of authorship and title characteristics linked to the highest predicted citation rates.

RESULTS: Larger author teams were associa […]

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Interpreting nonsignificant findings: Reassessing the safety of self-locking standalone cages in octogenarian patients undergoing ACDF.

Abudayeh A, Fishchenko I

PubMed · DOI

A novel, simple, and affordable technique for arthroscopic soft-tissue tenodesis of the long head of the biceps tendon.

Fossati C, Ioppoli A, Ravaglia R, Vaja F, Menon A, Randelli PS

BACKGROUND: The purpose of this study was to describe a novel, simple, and implant-free arthroscopic soft-tissue tenodesis technique of the long head of the biceps tendon (LHBT) to the rotator cuff using an absorbable monofilament suture and to evaluate its clinical and functional outcomes at a minimum 1-year follow-up.

METHODS: A retrospective case series of 23 patients (mean age 58.0 ± 7.8 years) who underwent arthroscopic rotator cuff repair with concomitant LHBT soft-tissue tenodesis between June 2021 and June 2023 was analyzed. Functional outcomes were assessed using Constant-Murley Score (CMS), American Shoulder and Elbow Surgeons (ASES) score, Single Assessment Numeric Evaluation (SANE), visual analog scale (VAS) for pain, and the Long Head of Biceps (LHB) score. Supination strength was measured with a handheld dynamometer and compared with the contralateral side. Incidence of Popeye deformity and tenderness over the bicipital groove were recorded.

RESULTS: Objective Popeye deformity was observed in 13% of patients, with subjective concern reported by only 4.3%. Supination strength and LHB scores were similar to the contralateral side (means of 104.65 versus 104.64 N; LHB 93.7 versus 94.6 points). The mean CMS and ASES scores were 90.3 ± 12.4 and 89.6 ± 15.9 points, respectively. The SANE score averaged 87.4 ± 20.9, and the VAS for pain was low (1.65 ± 2.59 cm).

CONCL […]

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ORIF of three- and four-part proximal humeral fractures: a retrospective study of long-term clinical and complication outcomes of two-plate fixation methods.

Conteduca J, Gasbarra E, Rausa I, Casto A, Valentino L, Solarino G, Meccariello L, Rollo G

BACKGROUND: The aim of this study was to compare clinical outcomes and complication rates during long-term follow-up (≥ 6 years) after treatment for three- and four-part proximal humeral fractures using two different types of locking plates.

MATERIALS AND METHODS: A total of 113 patients with three- and four-part proximal humeral fractures who underwent surgery between September 2012 and January 2019 were enrolled retrospectively. Data for 49 patients [9 males, 40 females; mean age [standard deviation] (SD) 68.9 (5.8) years} treated with a PGR (intrauma) plate (group A) and 51 patients [10 males, 41 females; mean age (SD): 66.0 (7.5) years] treated with a PHILOS humeral plate (PHP, group B) were available at the last follow-up. The mean follow-up periods in groups A and B were 10.8

and 10.2 years, respectively, with a minimum follow-up of 6 years. At the final follow-up evaluation, functional outcomes were assessed using the Oxford Shoulder Scale (OSS), Simple Shoulder Test (SST), and Disabilities of the Arm, Shoulder and Hand (DASH) questionnaires. X-ray evaluation was also performed, and complications were recorded.

RESULTS: The mean OSS score in the PGR and PHP groups at follow-up was 42.9 and 39.1, respectively ($p > 0.05$). The SST score (PGR: 7.5 ± 2.0; PHP: 7.3 ± 4.0) and DASH score (PGR: 22.1 ± 5.5; PHP: 20.5 ± 4.0) were similar in both groups ($p > 0.05$). In our serie […]

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Incidence rates and treatment of the transcervical fracture of the neck of femur in Italy: is total hip arthroplasty an increasingly preferred approach? A population study on trends between 2001 and 2023 based on 1,120,770 hospital discharge records.

Ciminello E, Romanini E, Venosa M, Cazzato G, Tucci G, Boniforti F, Carpanese L, Cuccu A et al.

INTRODUCTION: Transcervical femoral neck fractures (TFNFs) are among the most devastating fragility fractures in the elderly. TFNF are associated with excess 1-year mortality rates ranging from 15% to 30%. Treatments include conservative methods, internal fixation, and arthroplasty (partial or total hip arthroplasty). This study aims to analyze the changes in incidences of TFNF in the Italian population between 2001 and 2023 and the evolution of the choices of treatment.

MATERIALS AND METHODS: Using hospital discharge record (HDR) data from 2001 to 2023, records with ICD9-CM codes for femoral neck fractures (820.0 and 820.1) among diagnoses were selected and categorized into four treatment groups: total arthroplasty, partial arthroplasty, fixation, and conservative. Time series were analyzed with stratification by sex and age.

RESULTS: The extracted data included 1,120,724 records of TFNFs, with 871,161 cases treated surgically (total or partial arthroplasty or internal fixation) and 249,563 treated conservatively; the average patient age was 79.1 years, with a higher proportion of women (72.8%). Partial hip arthroplasty was the preferred treatment overall. For younger patients, in the age classes < 45 and 45-54 years, fixation was the most chosen treatment. Over time, the use of the conservative treatment decreased from 27.5% in 2001 to 14.6% of cases in 2023. The use of par […]

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Does latissimus dorsi tendon transfer provide a normal scapular rhythm and external rotation strength in posterosuperior massive irreparable rotator cuff tears? A kinematic analysis in a retrospective cohort.

Calvo C, Fiumana G, Brigo A, Ruggiero ED, Bonfatti R, Donà A, Micheloni GM, Giorgini A et al.

BACKGROUND: Posterosuperior massive irreparable rotator cuff tears (PMIRT) are rare and disabling conditions. When conservative treatment fails, latissimus dorsi tendon transfer (LDTT) is a viable surgical option for symptom relief in carefully selected patients. However, its effectiveness in restoring glenohumeral function and its influence on scapulothoracic rhythm remain subjects of ongoing debate.

PURPOSE: The purpose of this study was to evaluate the clinical outcomes of LDTT and assess its impact on scapulothoracic rhythm using kinematic and electromyographic analysis.

MATERIAL AND METHODS: A total of 18 patients with PMIRT underwent LDTT. Functional scores consisted of Constant-Murley score (CMS), America Shoulder and Elbow Surgeons (ASES) score, and Quick Disabilities of the Arm, Shoulder, and Hand (QuickDASH) score. Electromyography (EMG) activity of the latissimus dorsi muscle was evaluated in conjunction with three-dimensional (3D) kinematic tracking system. Differences in external rotation strength were compared with and without active adduction.

RESULTS: The mean age was 55.9 years (range 40-69 years), with a mean follow-up of 20.4 months (range 6-39 months). At final follow-up, the mean CMS was 68.2 (95% CI 64.9-71.5), ASES was 76.9 (95% CI 66.1-87.6), and QuickDASH was 17.6 (95% CI 9.5-25.6). A significant difference in external rotation strength was observed […]

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Is there a need for exploration in pulseless supracondylar fractures of the humerus? A systematic review and individual patient data meta-analysis.

Haoyang C, Chen JG, Lai H, Lim AKS, Tan SHS, Hui JHP

BACKGROUND: Different approaches have been proposed to treat patients with pulseless supracondylar humeral fractures (SHF). We aim to analyze the current outcomes of patients who have undergone different treatment options for pulseless SHF, namely watchful waiting versus urgent surgical exploration.

METHODS: Electronic databases including PubMed, Embase, and Cochrane Library were explored. Our study included patients from all age groups but only included English-language articles. Single case reports and case studies with adequate description of patient population, injury, and outcomes were included. An individual patient data meta-analysis was done to evaluate key outcomes of surgical exploration versus no surgical exploration in pulseless patients with SHF.

RESULTS: Overall, the data for 1070 individual patients from a total of 48 studies were included. Patients with pulseless SHF with open fractures have a higher probability of requiring vascular intervention ($p < 0.001$) as they are more likely to have disrupted arteries ($p = 0.050$) and more likely to require vascular repairs ($p < 0.001$) than patients with closed fractures. Similarly, patients with pulseless SHF with pucker sign ($p = 0.003$) and ecchymosis ($p = 0.002$) were more likely to undergo surgical exploration. However, the neurological status had no relation to them undergoing surgical exploration ($p = 0.382$), and al […]

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Increasing complexity and plateauing outcomes in AO type C distal humerus fractures: a 50-year institutional case series.

Hönck K, Wenzelburger P, Eibinger N, Sadoghi P, Puchwein P

BACKGROUND: Distal humerus fractures in adults are rare but complex injuries, particularly AO type C3 fractures, often involving comminution, instability, and osteoporotic bone. Demographic trends indicate rising incidence, especially among elderly patients, with increasing functional demands challenging modern osteosynthesis. Open reduction and internal fixation (ORIF) with bicolumnar plating remains the preferred reconstructive approach, although complications remain substantial. This study aimed to evaluate trends in fracture complexity, patient demographics, and functional outcomes over five decades, focusing on the most recent cohort (2009-2018, series E) and comparing with historical cohorts (series A-D, 1969-2008).

METHODS: This retrospective analysis included five consecutive 10-year institutional cohorts (series A: $n = 43$; B: $n = 29$; C: $n = 47$; D: $n = 58$) of 233 patients with 235 supradiacondylar distal humerus fractures (AO type C1-3) treated with ORIF. Functional outcome parameters (Jupiter and Cassebaum Scores) and demographic data were available for all cohorts; additional parameters (range of motion (ROM), QuickDASH Score) were available for series D and E, Mayo Elbow Performance Score (MEPS), Short Form (SF)-36, and strength were assessed specifically in series E ($n = 56$; follow-up $n = 21$). Cross-cohort comparisons of categorical variables used chi-square tests; […]

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Magnetic resonance imaging-based fat infiltration grading improves reparability prediction in massive rotator cuff tears.

Hu Q, Zhang G, Wang D, Tung TH, Ding J, Zhou X

PURPOSE: This study aims to investigate the extent of fat infiltration (FI) in the rotator cuff (RC) muscles in more medial slices from the Y-shaped view (Y-view) on shoulder magnetic resonance imaging (MRI) and assess whether these slices provide a more accurate prediction of the repairability of massive RC tears (massive RCTs).

METHODS: The retrospective study included 57 patients with massive RCTs who successfully underwent arthroscopic surgery between 1 January 2023 and 31 December 2023. Patients were categorized into two groups: the irreparable group and the reparable group. All patients underwent shoulder MRI covering the region from the acromion to the medial border of the scapula in oblique coronal and oblique sagittal orientations. The FI stage of the RC was assessed across three different views to determine which view most effectively predicts the repairability of patients with massive RCTs.

RESULTS: The FI stage of the infraspinatus (IS) muscle across three distinct views demonstrated a significant correlation with the repairability of massive RCTs. The FI stage was more severe in the irreparable group. Receiver operating characteristic (ROC) analysis revealed that the area under the curve for the 1/2 scapula view (0.737) and the most severe view (0.745) exceeded that of the traditional Y-view (0.644). Paired-sample ROC curve analysis revealed a significant differe […]

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Mid-term clinical and functional outcomes after reverse shoulder arthroplasty with latissimus dorsi transfer.

Colombini AG, Rab P, Macken AA, Soares MN, Kimmeyer M, Shirinskiy IJ, Popescu IA, Lafosse L et al.

BACKGROUND: Although reverse total shoulder arthroplasty (rTSA) with concomitant latissimus dorsi transfer (LDT) has been shown to effectively treat external rotation (ER) deficits, there are limited data regarding its outcomes with modern implants and its impact on activities of daily living (ADLs) requiring ER. The purpose of this study was to assess the mid-term clinical and radiographic outcomes of rTSA with concomitant isolated LDT in patients with an ER lag sign and posterior rotator cuff deficiency.

METHODS: This retrospective cohort study with prospective follow-up included consecutive patients who underwent rTSA with concomitant isolated LDT between 2010 and 2022 with a minimum follow-up of 2 years. Primary outcomes included the resolution of ER lag sign, active ER, and the Activities of Daily Living and External Rotation (ADLER) score. Secondary outcomes included the Auto-Constant Score (CS), Subjective Shoulder Value (SSV), activities of daily living requiring internal rotation (ADLIR) score, visual analog scale (VAS) for pain, and radiographic analysis of standardized radiographs.

RESULTS: In total, 32 procedures in 32 patients were identified. Of these, 22 procedures in 22 patients (68% female, 72.9 ± 8.4 years at surgery) were available for follow-up at 4.8 ± 2.2 years postoperatively (response rate 73%). The ER lag sign resolved in 95.5% of patients, the active […]

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Partial versus full revision in cementless single-stage exchange for hip periprosthetic joint infection: a multicenter comparative study with 10-year survivorship analysis.

Mu W, Jarusriwanna A, Xu B, Brown SA, Zhang X, Dawes A, Parvizi J, Cao L

BACKGROUND: Periprosthetic joint infection (PJI) is one of the most challenging complications following total hip arthroplasty (THA). Traditional management typically involves full revision to ensure comprehensive infection eradication. However, for patients with well-fixed implants, partial revision in a single-stage cementless approach may provide a viable alternative, potentially preserving bone stock and reducing operative time. The relative efficacy of this approach compared with full revision remains to be fully explored. This multicenter study aims to determine whether partial revision in cementless single-stage exchange surgery offers comparable infection control outcomes to full revision for select patients with well-fixed implants.

METHODS: We conducted a retrospective, multicenter cohort study involving 226 patients who underwent cementless single-stage exchange hip arthroplasty for PJI between 1 October 2013 and 31 July 2022. Patients were divided into partial revision ($n = 61$) and full revision ($n = 165$) groups. The primary outcome was treatment success, defined as the absence of clinical symptoms and signs of infection at a minimum follow-up of 2 years.

RESULTS: The success rates were 77.0% for the partial revision group and 80.0% for the full revision group, with no significant difference ($p = 0.629$). Both groups showed comparable 10-year survival rates for ove […]

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Biomimetic Hematoma Promotes Superior Bone Regeneration With Ultra-low-Dose rhBMP-2 in a Goat Large-Defect Model.

Glatt V, Aguilar L, Agarwal A, Woloszyk A

OBJECTIVE: The aim of this study was to evaluate whether an ultra-low dose of recombinant human bone morphogenetic protein-2 (rhBMP-2) delivered within a biomimetic hematoma (BH) scaffold could regenerate a large bone defect in goats more efficiently than a high dose of rhBMP-2 delivered on an absorbable collagen sponge (ACS). The BH is an autologous scaffold, created from whole blood and defined concentrations of coagulants, and is designed to closely mimic the structural and biological properties of a naturally healing fracture hematoma. It initiates the body's intrinsic healing cascade, thereby recapitulating the sequential phases of bone repair.

METHODS: A 2.5-cm defect was created in goat tibias and treated with either a BH scaffold containing ultra-low doses of rhBMP-2 (210 and 42 µg) or ACS with 2.1 mg of rhBMP-2. Empty defects and contralateral tibias served as controls. Bone regeneration was evaluated biweekly using radiographs and further assessed with micro-CT and histology at 8 weeks.

RESULTS: A total of 12 specimens, three per group, were used for radiographic, micro-CT, and histological evaluations. By 8 weeks, radiographic scores showed complete regeneration in the BH+210 µg (5.0 ± 0.0) and BH+42 µg (4.7 ± 0.2) groups. The ACS+2.1 mg group scored 4.3 ± 0.4 with some unmineralized regions while the ED group showed limited healing (2.6 ± 0.2; P < 0.001 vs. all ot […]

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Breaking Down the Evidence: A Multicenter Analysis of Venous Thromboembolism Among Trauma Patients with Lower Extremity Fractures.

Knowlton LM, Guorgui JG, Wang S, Arnow K, Knudson MM, CLOTT Study Group

OBJECTIVES: To determine whether chemoprophylaxis initiation within 24 hours reduces venous thromboembolism (VTE) risk among trauma patients with lower extremity long bone fractures.

DESIGN: This was a retrospective cohort study.

SETTING: Seventeen Level I trauma centers as a part of the Consortium of Leaders in the Study of Traumatic Thromboembolism study group.

PATIENT SELECTION CRITERIA: Patients aged 18-40 years with a diagnosis of lower extremity long bone fracture between January 1, 2018, and December 31, 2020, were included.

OUTCOME MEASURES AND COMPARISONS: Primary outcome was VTE during admission. Patients were compared based on VTE chemoprophylaxis initiation within 24 hours (early prophylaxis, E-PROPH) versus not (late or no prophylaxis, L-PROPH) using inverse probability weighted Cox survival analysis.

RESULTS: In total, 120 (5.3%) among 2264 patients with lower extremity fractures developed VTE. 57.5% received E-PROPH and 42.5% received L-PROPH. The E-PROPH group had fewer patients with an injury severity score ≥16 (30.3% vs. 52.4%, P < 0.001) and a lower proportion of associated head injury (8.1% vs. 26.7%, P < 0.001). VTE incidence was significantly higher in the L-PROPH group than in the E-PROPH group (8.6% vs. 2.8%, P < 0.001). In the adjusted model, E-PROPH was independently associated with nearly a half reduction in VTE incidence (hazard ratio: 0.54, 95% […]

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Response to "Competing-Risks Re-estimation of Five-Year Second Hip Fracture Risk: Accounting for Death as a Competing Event".

Konda SR

PubMed · DOI

Competing-Risks Re-estimation of Five-Year Second Hip Fracture Risk: Accounting for Death as a Competing Event.

Shaikh FA, Curran T, Nemeth ZH

PubMed · DOI

Letter to the Editor Regarding "Proton Pump Inhibitors Are Associated With Increased Risk of Site-Specific Nonunion After Open Reduction Internal Fixation".

Ayas O, Acar A

PubMed · DOI

Response to the Letter to the Editor Regarding "Proton Pump Inhibitors Are Associated With Increased Risk of Site-specific Nonunion After Open Reduction Internal Fixation".

Romoff M, Kim MS, Scolaro JA

PubMed · DOI

The Effect of Preoperative Tranexamic Acid on Blood Transfusions in Geriatric Hip Fracture Surgery: A Randomized Controlled Trial.

Sullivan M, Perea LL, Bradburn E, Guglielmo K, Bresz K, Horst M, Martin B, Heinle C et al.

OBJECTIVES: To evaluate whether preoperative intravenous tranexamic acid (TXA) reduces postoperative blood transfusion rates in geriatric patients undergoing operative treatment for hip fracture.

DESIGN: Prospective, double-blinded, randomized controlled trial terminated early after interim futility analysis.

SETTING: Single-center Level I trauma center.

PATIENT SELECTION CRITERIA: Patients aged ≥65 years who underwent operative treatment for femoral neck (AO/OTA 31-B), intertrochanteric region (AO/OTA 31-A), or subtrochanteric (AO/OTA 32-A/B/C) fractures between June 2019 and June 2022 were included. Procedures included arthroplasty (hemiarthroplasty or total hip arthroplasty) and internal fixation (ORIF) with intramedullary nailing or sliding hip screw fixation. Exclusion criteria were recent thromboembolic events, cancer, hypercoagulable disorders, or TXA allergy.

OUTCOME MEASURES AND COMPARISONS: The primary outcome was postoperative transfusion. Secondary outcomes included hospital length of stay, 30-day readmission, and 90-day complications and mortality. Statistical comparisons were performed using odds ratios, chi-squared, and t-tests, with significance thresholds adjusted by O'Brien-Fleming criteria.

RESULTS: A total of 283 patients were analyzed {TXA: 146 [mean age 84 (range 65-100), 71.9% female; placebo: 137 [mean age 83.1 (range 65-100), 77.2% female]}. Baseli […]

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Perioperative Use of Tranexamic Acid Is Associated With Reduced Transfusion Rates in Femoral Shaft Fractures: A Propensity-Matched Cohort Study.

Kim D, Shen V, Norris Z, Ranson R, Sterling R, Villa J

OBJECTIVES: To evaluate the association between perioperative tranexamic acid (TXA) use and transfusion rates, perioperative blood loss, and 90-day postoperative complications in patients undergoing intramedullary nailing of femoral shaft fractures.

DESIGN: Retrospective cohort study.

SETTING: Multicenter academic and community hospitals participating in the TriNetX research network.

PATIENT SELECTION CRITERIA: Adult patients (≥18 years) who underwent open intramedullary nailing of OTA/AO 32 (femoral diaphyseal) fractures between 2010 and 2025 were included. Patients with a history of malignant neoplasm, deep vein thrombosis (DVT), and pulmonary embolism (PE) were excluded. Patients were required to have ≥90 days of postoperative follow-up. The TXA group included patients who received perioperative TXA, while the comparison group did not.

OUTCOME MEASURES AND COMPARISONS: Primary outcomes included the incidence of nonautologous red blood cell transfusion and 90-day postoperative complications. Minor and major complications analyzed were mortality, DVT, PE, myocardial infarction, stroke, and renal failure. Propensity-score matching was performed to balance demographics and comorbidities. Chi-squared tests were used to compare outcomes.

RESULTS: A total of 7644 patients were included in the study. The mean age was 57.0 ± 23.2 in the TXA group (n = 3,822, 54.1% women) and 57. […]

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Long Distally Unlocked Hip Nails Versus Short Distally Locked Nails for Peritrochanteric Fractures: Operative Characteristics and Peri-implant Fracture Risks.

Ayoub M, Tallapaneni J, Robles E, Vue Y, Lindvall E

OBJECTIVES: To compare the primary outcome of surgical time for long, distally unlocked, intramedullary nails (LUIMNs) versus short, distally locked, intramedullary nails (SLIMNs) and to compare secondary outcomes of blood loss, peri-implant fractures, reoperation events, and mortality between these 2 groups.

DESIGN: Retrospective cohort study.

SETTING: One Level I Trauma Center and one community hospital.

PATIENT SELECTION CRITERIA: Patients older than 18 years treated with SLIMNs or LUIMNs for peritrochanteric hip fractures, OTA/AO 31A1, 31A2, 31A3, were included.

OUTCOME MEASURES AND COMPARISONS: The primary outcome was surgical time for LUIMNs versus SLIMNs. Secondary outcomes were estimated blood loss, peri-implant fracture rates, reoperation for nonunion or shortening, all-cause reoperation, and mortality within 2 weeks and 3 months of surgery for LUIMNs versus SLIMNs.

RESULTS: There were 602 patients in the LUIMN group (68.6% female, 31.4% male), with an average age of 77.3 (range, 29-108) years. There were 142 patients in the SLIMN group (63.4% female, 36.6% male) with an average age of 75.3 (range 33-101) years. Average OR time for LUIMNs was 46.9 minutes compared with 53.7 minutes for SLIMNs ($P < 0.001$). Peri-implant fracture rates were 0.6% for LUIMNs and 4.1% for SLIMNs ($P = 0.012$). There was no difference in estimated blood loss between LUIMNs (103.8 ± 55.1) […]

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Comparing the Efficacy Between ChatGPT 5, Grok 3, and Claude 4.5 Sonnet in Analyzing Orthopedic Trauma-Related Imaging.

Holmstrom JA, Braithwaite CL, Alhankawi AR, Moore ML, Patel KA, Miller BH

OBJECTIVES: To evaluate and compare the ability of three popular open-source artificial intelligence platforms to diagnose common trauma-related fractures using radiologic imaging.

DESIGN: Retrospective diagnostic performance comparison study.

SETTING: Publicly accessible online radiologic imaging databases.

PATIENT SELECTION CRITERIA: Five common orthopedic trauma fractures were assessed: ankle, tibial plateau, intertrochanteric, femoral neck, and humerus. Radiographs and computed tomography (CT) images were collected. Images were randomly selected from confirmed diagnoses on Radiopaedia.org .

OUTCOME MEASURES AND COMPARISONS: ChatGPT 5, Grok 3, and Claude 4.5 Sonnet were queried with each image. Diagnostic accuracy, sensitivity, specificity, positive and negative predictive values, and performance by modality (X-ray vs. CT) were assessed. The reference standard was the expert-verified diagnosis provided by Radiopaedia.org , limited to cases labeled with a "diagnosis certain" tag.

RESULTS: Each model was provided with 30 radiographs and 20 CT images whenever possible. ChatGPT 5, Grok 3, and Claude 4.5 Sonnet accurately diagnosed diseased images in 26.8%, 18.8%, and 22.4% of cases, respectively. By fracture type, ChatGPT 5 demonstrated the highest correct classification rates for ankle (10%), femoral neck (38%), humerus (40%), and tibial plateau (44%) fractures. Grok 3 dem […]

PubMed · DOI

Preoperative Quadriceps Muscle Quality Predicts Early Ambulation after Hip Fracture Surgery.

Achebe C, Cureton R, Rothberg D, Higgins T, Marchand L, Haller J

OBJECTIVES: To investigate whether preoperative muscle characteristics measured on CT scans could predict early ambulation capacity after hip fracture surgery.

DESIGN: Retrospective cohort study.

SETTING: Single level I academic trauma center.

PATIENT SELECTION CRITERIA: Patients aged ≥55 years who underwent surgical fixation for hip fractures (OTA/AO types 31A, 31B) between January 2015 and December 2024 were included. Inclusion required preoperative CT imaging and documented ambulation assessments at postoperative day 1 (POD1) and discharge. Patients with neuromuscular disorders affecting muscle mass were excluded.

OUTCOME MEASURES AND COMPARISONS: The primary outcome was distance walked (feet) during physical therapy at POD1 and discharge. Preoperative CT scans were opportunistically analyzed for muscle quality (Hounsfield units) and size (cross-sectional area) in psoas, gluteal, and quadriceps muscles. Associations were analyzed using linear mixed-effects models adjusted for body mass index, Charlson Comorbidity Index, preinjury wheelchair use, time to surgery, and fixation method.

RESULTS: A total of 140 patients (mean age 75 ± 8 years, 59% female) contributed 210 ambulation observations. Ambulation improved from 0 (IQR 0-20) feet at POD1 to 20 (IQR 0-86) feet at discharge ($P < 0.001$). In the fully adjusted mixed-effects model, quadriceps quality was the only muscle […]

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Reliability of Hip Fracture Classification and Treatment Planning on Smartphones Versus Picture Archiving and Communications Systems.

Schultz MJ, Kohring A, Bridges TN, DiCiurcio WT, Smith EB, Pulido S, Deirmengian GK

OBJECTIVES: To compare the reliability of classifying and recommending treatment for femoral neck and intertrochanteric fractures when reviewing radiographs on smartphones versus Picture Archiving and Communications Systems (PACS).

DESIGN: Retrospective cross-sectional radiographic review.

SETTING: Private academic medical center.

PATIENT SELECTION CRITERIA: Consecutive patients ≥18 years with isolated unilateral femoral neck or intertrochanteric fractures (AO/OTA 31A or 31B) and complete radiographs (anteroposterior pelvis, anteroposterior and lateral hip, and anteroposterior and lateral femur) who presented from January 1 to December 31, 2021, were included.

OUTCOME MEASURES AND COMPARISONS: Three orthopaedic surgeons (1 orthopaedic trauma attending and 2 adult reconstruction attendings) independently evaluated radiographs on 2 separate occasions, first on their smartphone and then on PACS using their personal laptop. After reviewing patient images, a survey assessing fracture characteristics and treatment plans was completed. Smartphone images were accessed through an embedded URL link within the survey, providing zooming functionality comparable with that of standard text message. The primary outcome was intraobserver agreement (smartphone vs. PACS) for fracture classification and treatment recommendations (general and specific) using Cohen kappa (k). Additional outcome […]

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Similar Outcomes in Proximal Humerus Fractures Treated With Conventus Intramedullary Cage Device and Open Reduction Internal Fixation.

Kurkowski SC, Taleghani ER, McMillan P, Solomon EG, Keller JE, Gerak SK, Wyrick JD, Grawe BM

OBJECTIVES: To compare clinical outcomes between open reduction and internal fixation (ORIF) with or without Conventus intramedullary cage (CC) for proximal humerus fractures.

DESIGN: Retrospective cohort study.

SETTING: Level 1 trauma center.

PATIENT SELECTION CRITERIA: Included were patients who sustained OTA/AO 11B and 11C fractures treated with ORIF with or without CC with minimum 1 year follow-up.

OUTCOME MEASURES AND COMPARISONS: Primary outcomes included complication and reoperation rates. Secondary outcomes were American Shoulder and Elbow Surgeons and Short Form 12 (SF-12) scores. Outcomes were compared between fixation strategies.

RESULTS: Fifty-six patients were included, 33 ORIF and 23 CC, with mean follow-up of 5.0 (range 1.1-11.4) and 5.3 (range 2.3-7.6) years, respectively. The ORIF and CC groups were 27.3% and 21.7% men, respectively (P = 0.759). The ORIF group had a higher mean age (68.2 ± 5.3 vs. 64.2 ± 8.7 years, P = 0.037). There was a higher proportion of OTA/AO 11C fractures in the CC cohort (65.2% vs. 33%, P = 0.029). CC group had a higher unplanned reoperation rate (18.2% vs. 34.8% for CC, P = 0.158) and complication rate (24.2% vs. 39.1% for CC, P = 0.233) that did not reach statistical significance within this available study population. Unplanned reoperations in the CC cohort included revision fixation (3) and conversion to hemiarthroplasty (2) […]

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Short-Term Outcomes of Supination External Rotation IV Bimalleolar and Deltoid Variant Fractures With No Deltoid Repair.

Malempati M, Danford NC, Chien BY, Greisberg JK

OBJECTIVES: To evaluate whether radiographic and clinical outcomes differ between supination-external rotation (SER) type IV bimalleolar (BM) fractures and deltoid ligament injuries treated without deltoid repair. A secondary aim was to compare outcomes between universal tubular (TB) and fibula-specific (FS) plates for fibular fixation.

DESIGN: Retrospective cohort study.

SETTING: Academic medical center.

PATIENT SELECTION CRITERIA: Skeletally mature patients with SER-IV (AO/OTA 44-B2) ankle fractures (either BM or deltoid ligament injury without medial malleolar fracture) treated between November 2011 and July 2024. Exclusions: pathologic/pilon fractures, congenital deformity, nonunion repair, posterior malleolar fixation, or <3 months of follow-up. Secondary comparison included rotational ankle fractures (SER, pronation external rotation) stabilized with TB or FS plates.

OUTCOME MEASURES AND COMPARISONS: Radiographic outcomes included medial clear space (MCS), superior clear space (SCS), tibiofibular clear space (TFCS), talocrural angle, and Takakura score. Clinical outcomes included complication rates, secondary surgeries, infections, and conversions to arthroplasty. Comparisons were made between intraoperative and final postoperative radiographic measurements.

RESULTS: A total of 132 patients with SER-IV fracture were included in the primary study arm, with 89 in the B […]

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"Throwing the Flag": Patient Behavior Reporting Affects Outcomes After Orthopaedic Trauma Surgery.

Mercer NP, Egol AJ, Jacobi S, Padon B, Lashgari A, Egol KA

OBJECTIVES: To evaluate the association between electronic health record-based behavioral flag designation and postoperative outcomes in orthopaedic trauma patients undergoing surgical fixation for acute fractures.

DESIGN: Retrospective cohort study with 1:1 propensity score matching.

SETTING: Level I trauma center.

PATIENTS SELECTION CRITERIA: Adult orthopaedic trauma patients who underwent operative fixation for an acute fracture and received a long-term behavioral flag issued for documented disruptive, threatening, or violent behavior toward health care staff after institutional review either before surgery or within 1 year postoperatively were included. Those with pathologic fractures and those with inadequate follow-up were excluded. Each flagged patient was matched to an unflagged control based on age, sex, body mass index, smoking status, comorbidity burden, and fracture type.

OUTCOME MEASURES AND COMPARISONS: Primary outcomes included 1-year rates of major postoperative complications (eg, fracture-related infection, nonunion, painful hardware) and reoperation. Subgroup analyses examined outcomes by timing of flag assignment.

RESULTS: A total of 116 patients were included (58 flagged patients, 58 unflagged). Flagged patients had a mean age of 53.4 ± 18.1 years (range, 19-74) and were 55.2% male; unflagged controls had a mean age of 49.0 ± 13.6 years (range, 21-71) a […]

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Novel UHMWPE sleeve for stabilizing loose suture anchors in rotator cuff repair.

Lin CK, Jaw FS, Sam SS, Siow TF, Yii SC, Chang YJ, Young TH

Suture anchors are frequently used for repairing torn rotator cuffs. However, post-operative anchor loosening is a common complication often caused by low bone density. The purpose of this study was to evaluate if a novel UHMWPE cortex locking sleeve could improve the anchor pullout strength after revision in synthetic bone. The initial failed anchor fixation was simulated by inserting a 4.5 mm anchor and pulling it out of the bone, which created an enlarged insertion hole for the revision anchor. After revision, two loosening conditions were simulated; (i) intra-operative anchor loosening where the revision anchor was directly pulled out from the block after insertion, and (ii) post-operative anchor loosening where the revision anchor was first subjected to cyclic loading and then pulled out. For both loosening conditions, the pullout strength and energy absorbed was recorded for the following configurations: (i) Group A, same-diameter anchor (SDA; 4.5 mm × 10 mm); (ii) Group B, larger-diameter anchor (LDA; 5.5 mm × 10 mm;); and (iii) Group C, same-diameter anchor combined with the novel sleeve (SDA + NS). For intra-operative anchor loosening, the highest pullout strength was recorded in the SDA + NS group, with a mean value of 370.077 ± 32.699 N, which was significantly greater than both SDA and LDA groups ($p < 0.05$). Similarly with post-operative anchor loosening, the SDAP […]

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Impact of rheumatoid arthritis on complications and hospital outcomes following reverse shoulder arthroplasty: evidence from 389,135 hospitalizations in the national inpatient sample.

Mahamid A, Elgabsi M, Khatib M, Murad H, Qawasmi F, Lavon E, Zahalka S, Yassin A et al.

OBJECTIVE: To evaluate the impact of rheumatoid arthritis (RA) on perioperative complications and hospital outcomes following reverse total shoulder arthroplasty (RTSA) using the National Inpatient Sample (2016-2021).

METHODS: Adult patients undergoing RTSA were identified using ICD-10 procedure codes. Outcomes included prolonged length of stay (LOS), high hospital charges, inpatient complications, and mortality. Multivariable logistic regression adjusted for demographic, clinical, and hospital characteristics. Survey weights were applied to generate nationally representative estimates.

RESULTS: A total of 389,135 patients underwent RTSA, including 18,140 (4.66%) with RA. In unadjusted analyses, RA patients had higher rates of prolonged LOS (17.8% vs. 14.8%) and acute blood loss anemia (10.7% vs. 8.6%). After multivariable adjustment, RA remained an independent predictor of prolonged LOS (adjusted OR 1.18, 95% CI 1.14-1.23, $p < 0.001$) and acute blood loss anemia (adjusted OR 1.25, 95% CI 1.18-1.33, $p < 0.001$), but not urinary tract infection. In-hospital mortality was rare and could not be modeled due to zero events in the RA group. RA patients also had slightly longer hospitalizations and higher hospital charges.

CONCLUSION: RA is independently associated with prolonged hospitalization and increased risk of perioperative blood loss anemia following RTSA. These findings high […]

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Long-term results of thumb carpometacarpal joint arthrodesis.

Vogler P, Benedikt S, Bode S, Keiler A, Stock K, Arora R

INTRODUCTION: Thumb carpometacarpal (CMC 1) arthrodesis remains an established treatment option for patients with high functional demands and strength requirements. Long-term data on mobility, complication rates and clinical outcomes remain limited. This study aimed to analyze long-term clinical and radiological results after CMC 1 arthrodesis.

MATERIALS AND METHODS: A single-center, retrospective study included patients who underwent CMC 1 arthrodesis between 2004 and 2024. Following medical record analysis, eligible patients were invited to a standardized clinical and radiological follow-up examination. Outcomes included DASH, PRWHE and MHQ scores, pain intensity, mobility, strength and patient satisfaction. Radiological assessment covered fusion signs, implant position and adjacent joint arthrosis. The median follow-up was 9 years.

RESULTS: 34 patients were included, 16 participated in follow-up. The median DASH score was 14 points, PRWHE 4 points and MHQ 80 (operated) vs. 84 (contralateral). Grip strength reached 85% of the contralateral side, with preserved pinch strength. Compensatory adaptation of adjacent joints was observed regarding range of motion. Patients reported no relevant resting pain and only minimal load-related discomfort. 15 of 16 patients were satisfied with the outcome. Non-union occurred in 4 of 34 cases (11.8%), associated with absent cancellous bone […]

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Does the tendon retract in flexor hallucis longus muscle laceration? A cadaveric study.

Ayd■n M, ■ahin Y, Ya■ar H, Çiçek F, Cerano■lu FG, Mert A, Ç■naro■lu S

INTRODUCTION: Flexor hallucis longus (FHL) tendon injuries are common, particularly in open injuries caused by sharp objects. However, data regarding FHL tendon retraction following hallux-level FHL tendon lacerations remain limited. This study investigated the retraction and shortening of the FHL tendon using cadaveric models.

MATERIALS AND METHODS: Six fresh-frozen below-knee amputated cadaveric feet (mean age: 65 years, range: 62-88) were utilized. FHL tendons were dissected, and lacerations were simulated at the hallux level. Tendons were subjected to various tensile forces (20-120 N) using a mechanical device. Force was measured in Newtons via a loadcell, and retraction distance was recorded in millimeters using an encoder. Data were summarized as mean ± standard deviation (SD), median, and quartiles.

RESULTS: The FHL tendon retracted gradually as applied force increased. An average retraction of 7 mm was measured at 20 N, reaching 21 mm at 60 N. Beyond 60 N (> 21 mm retraction), the other four toes also flexed. This occurs due to the anatomical connection between the FHL and flexor digitorum longus (FDL) tendons.

CONCLUSIONS: Only slight retraction was observed in hallux-level FHL tendon lacerations. At most, the findings suggest that distal FHL lacerations may exhibit limited proximal retraction under the tested conditions, which may have implications for surgical pla […]

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Early-onset rice body synovitis of the knee in children under three years: a case series and review of the literature.

Zhu L, Liu C, Liu Y, Mao B, Huang Y, Wen J, Liu Y, Qian T

OBJECTIVE: Rice body synovitis reports have been rarely reported in children under three years of age. We aimed to explore the clinical characteristics and treatment experience of rice body-like synovitis in children's knees.

METHODS: We retrospectively analyzed the clinical and imaging data of 4 cases (5 knees) of rice body-like synovitis in children treated at our hospital from January 2014 to December 2021. Among the patients, 1 was male and 3 were female, with an average age of 22.5 months. The average length of symptoms before admission was 1.86 months. Clinical presentations included swelling and limping, with no fever or rash.

RESULTS: Three patients with unilateral onset and one patient with bilateral onset underwent unilateral arthroscopic excision of synovial lesions. All 4 patients had pathological findings consistent with chronic inflammation, leading to a postoperative diagnosis of juvenile idiopathic arthritis (JIA). Postoperatively, all patients received standardized pharmacological treatment and were followed for 24 to 48 months in collaboration with the rheumatology department. None of the 4 knees undergoing surgery experienced recurrence, and the symptoms of the knee treated conservatively resolved spontaneously. In the last follow-up session, MRI and ultrasound showed only mild synovial thickening, with effusion and loose bodies disappearing.

CONCLUSION: R […]

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Extravertebral temperature exposure during spinal radiofrequency ablation: an experimental surrogate assessment of neural injury risk.

Weigert A, Hoffmann RT, Büttner A, Zaccaria R, Wegener V, Holzapfel BM, Wegener B

BACKGROUND: Radiofrequency ablation (RFA) has become an important minimally invasive option for the treatment of painful spinal tumors and benign bone lesions. While its effectiveness within the vertebral body is well established, there is ongoing concern about unintended heat exposure of nearby neural structures. Experimental data describing how heat propagates beyond the vertebra during spinal RFA are still limited. This experimental in vitro study aims to assess the extravertebral temperature exposure during spinal radiofrequency ablation as a surrogate parameter for potential neural injury risk under controlled conditions.

METHODS: This experimental in vitro study was performed on four fresh-frozen human lumbar spines. Radiofrequency ablation was carried out using a commercially available system and standard clinical protocols. Two anatomical configurations were examined: ablation within the vertebral body and ablation within the pedicle. Temperatures were measured at predefined locations outside the vertebra, including the spinal canal and cortical boundaries, using multiple thermocouples. To aid interpretation of radiofrequency-related measurement artifacts, an additional control setup using an electrothermal heating system without radiofrequency energy was included. Temperature data were evaluated descriptively.

RESULTS: In the control experiments, temperature profiles […]

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Giant cell tumor of bone: exploring HtrA1 as a potential predictor of recurrence.

Mirioglu A, Dalkir KA, Gokmen MY, Bagir M, Ates KE, Deveci MA, Gönlü■en G, Ozbarlas HS

INTRODUCTION: Giant cell tumor of bone (GCTB) is a locally aggressive tumor with a considerable risk of recurrence, but reliable predictive markers are limited. High-temperature requirement A serine peptidase 1 (HtrA1) has been implicated in the progression of several cancers and may also play a role in GCTB recurrence.

MATERIALS AND METHODS: This retrospective study included 34 patients with GCTB treated with curettage between 2002 and 2016. HtrA1 expression was evaluated separately in giant cells (GC) and mononuclear cells (MNC) using immunohistochemistry. Based on high or low expression in each cell type, patients were categorized into four expression patterns: High GC / High MNC, High GC / Low MNC, Low GC / High MNC, and Low GC / Low MNC. Clinical data, including recurrence status and time to recurrence, were analyzed.

RESULTS: The High GC / Low MNC group showed the highest recurrence rate at 71.4%, compared with 28.6% in the High GC / High MNC group and 31.6% in the Low GC / Low MNC group. No recurrence was observed in the Low GC / High MNC group, which included only one patient. The High GC / Low MNC group showed a non-significant trend toward higher recurrence compared with all other groups combined, with a similar trend toward shorter recurrence-free survival.

CONCLUSION: HtrA1 expression patterns in GCTB may provide preliminary insight into recurrence risk. Although […]

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Coronal hamate body fractures with dorsal fourth and fifth carpometacarpal joint instability: fracture patterns, injury mechanisms, surgical strategies, and outcomes.

Lee SH, Chung HJ, Cha SM

INTRODUCTION: Coronal hamate body fractures are rare injuries, and studies regarding their morphology and management are limited. This study analyzed fracture patterns, injury mechanisms, associated injuries, surgical strategy, and clinical outcomes in patients with coronal hamate body fractures involving more than one-third of the articular surface and concomitant carpometacarpal (CMC) joint instability.

METHODS: Fifty-eight male patients were retrospectively reviewed. Fractures were classified as dorsal oblique or coronal splitting types based on computed tomography findings. Surgical treatment involved open reduction and multiple screw fixation of the hamate using a K-wire-guided technique, with supplementary Kirschner wire stabilization for associated metacarpal base fractures or CMC joint instability. Radiographic and functional outcomes were evaluated after a minimum follow-up of 6 months.

RESULTS: Dorsal oblique fractures (63.8%) occurred more frequently than coronal splitting fractures (36.2%) and were most commonly associated with punching (OR, 4.9; $p = 0.01$), whereas falls from height were more common in coronal splitting types. Concomitant fractures occurred in 82.8% of cases, most frequently involving the fourth metacarpal base. Among 35 patients with a minimum 6-month follow-up, 34 achieved successful healing. In the 33 patients with functional outcome data, the […]

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Effect of spinal fusion length on hip and knee osteoarthritis: a comparative cohort study of no fusion, short-segment fusion, and long-segment fusion.

Sülek Y, Demirkale ■, Erto■rul R, Özdemir HM

PURPOSE: Lumbar fusion alters spinal-pelvic biomechanics and may influence degenerative changes in the lower extremities. This study investigated whether fusion length is associated with radiographic progression of hip and knee osteoarthritis.

METHODS: In this retrospective, single-center cohort study, patients aged 40-65 years who underwent lumbar surgery between 2010 and 2021 were grouped as no fusion (N), short fusion (S; ≤3 levels), or long fusion (L; ≥4 levels). Hip joint space width was assessed using minimum joint space (MJS) and superointermediate joint space (SJS), including femoral head-standardized measures (sMJS and sSJS). Knee medial joint space width (KMJS) was measured when available. Spinopelvic parameters (LL, PI, PT, SS, PI-LL) and functional outcomes (ODI, HHS, WOMAC) were recorded. Multivariable linear and logistic regression analyses were performed to identify independent predictors of joint space narrowing and osteoarthritis risk.

RESULTS: Hip joint space narrowing differed significantly among groups, with the highest annual narrowing rates in the long-fusion group, followed by the short-fusion group, and the lowest in the no-fusion group ($p < 0.001$). In contrast, knee KL grade change and annual KMJS narrowing did not differ significantly between groups ($p > 0.05$). In logistic regression, each additional fused level increased the odds of postoperative MJ […]

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Biomechanical comparison of inverted triangle and L-shaped screw configurations with medial buttress and anteromedial support plate in Pauwels type III femoral neck fractures.

Erem M, Demirta■ U, Y■ld■r■m S, Selçuk E, Çopuro■lu C

INTRODUCTION: Surgical stabilization of Pauwels Type III femoral neck fractures remains a significant challenge due to high vertical shear forces. While the medial buttress plate is a recognized solution, it requires extensive deep dissection. This study aims to compare the biomechanical performance of various screw configurations combined with either a medial buttress or anteromedial support plate.

MATERIALS AND METHODS: Twenty-five third-generation synthetic femurs were used to create a standardized 70-degree (Pauwels III) fracture model. Specimens were divided into five groups ($n = 5$): (A) Inverted triangle (IT) screws with a Pauwels screw, (B) Inverted triangle (IT) with medial buttress plate (MBP), (C) Inverted triangle with anteromedial support plate (ASP), (D) L-configuration with medial buttress plate (MBP), and (E) L-configuration with anteromedial support plate (ASP). Axial loading was applied at 2 mm/min until construct failure, defined objectively by the real-time force-distance curve.

RESULTS: Although no statistically significant difference was found between groups ($p = 0.102$), a large effect size was observed ($\eta^2 = 0.309$). Group C (IT + ASP) demonstrated the highest mean failure load (1695 ± 494.6 N). Conversely, Group E (L-configuration + ASP) exhibited the lowest stability (977.2 ± 195.4 N) with a remarkably narrow standard deviation. The majority of failures […]

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Reverse total shoulder arthroplasty is a surgical option for patients over 60 years of age with locked shoulder dislocation.

Sebastia-Forcada E, Miralles-Muñoz FA, Martinez-Mendez D, Albero-Catala L, Climent-Peris V, Lizaur-Utrilla A

BACKGROUND: in locked dislocation of the shoulder, instead of reducing back to the glenoid, the humeral head remains incarcerated on the glenoid in a locked fashion. This clinical situation is fairly uncommon. It is essential to conduct an individual evaluation of each patient to determine the appropriate treatment.

OBJECTIVE: the aim of this study was to evaluate the functional outcomes of reverse total shoulder arthroplasty (rTSA) in the treatment of locked shoulder dislocation.

METHODS: patients with locked shoulder dislocation who underwent reverse shoulder prosthesis surgery and were admitted to our center between 2007 and 2023 were reviewed. The primary outcome was the Constant score. Secondary outcomes included the adjusted Constant, UCLA and DASH scores. Additionally, any signs of radiologic loosening were also documented.

RESULTS: the series consisted of 10 patients, six men and four women, with a mean age of 68.0 years. The average time from the traumatic injury to surgery was 7.5 months. All patients showed improved Constant, Adapted Constant, UCLA, and DASH scores compared to their preoperative values. When comparing the outcomes of chronic posterior and anterior dislocations, no differences in functional outcomes or shoulder motion were observed after rTSA implantation. There were no complications during or after surgery.

CONCLUSION: The results of the present […]

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Does the order matter? Comparing the order of stem placement and fracture reduction in revision total hip arthroplasty for Vancouver B2 and B3 periprosthetic femur fractures.

Antonioli SS, Khury F, Kennedy MF, Haider MA, Duke A, Schwarzkopf R, Aggarwal VK

BACKGROUND: Vancouver B2 and B3 periprosthetic femur fractures (PPFF) have posed significant treatment challenges due to stem instability and lack of adequate femoral bone stock. This study investigated subsidence, survivorship, and outcomes of Vancouver B2 and B3 fractures, based on the order in which revision stem placement and fracture reduction occurred during revision total hip arthroplasty (rTHA).

METHODS: This retrospective, cohort study included 46 rTHAs between June 2011 and April 2023. Included patients underwent rTHA for Vancouver B2 or B3 PPFF with minimum one-year radiographic and two-year clinical follow-up. All patients were treated with diaphyseal-engaging tapered fluted titanium stems and stem modularity decisions were based on surgeon preference. Cohorts were separated based on if stem placement (SF, n = 19), or fracture reduction (RF, n = 27) occurred first. Patient demographics, intraoperative information, and clinical and radiographic outcomes were collected.

RESULTS: The SF and RF cohort showed no statistically significant differences in rate of subsidence ≥5 mm [26.3%[SF], 22.2%[RF], P = 0.749), rate of subsidence ≥ 10 mm (15.8%[SF], 14.8%[RF], P = 0.928), nor average subsidence (4.1 mm[SF], 4.4 mm[RF], P = 0.861). We found no statistically significant differences in surgery-related clinical outcomes or all-cause revision rates within a two-year follow- […]

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Offset reconstruction in stemless and stemmed shoulder hemiarthroplasty: influence on long-term outcomes.

Trefzer R, Gurda A, Deisenhofer J, Weishorn J, Hariri M, Koch KA, Wolf M, Bülhoff M

BACKGROUND: Shoulder hemiarthroplasty (HA) is a well-established treatment for various humeral-side pathologies. Overstuffing should be avoided because it is associated with glenoid wear; however, specific radiological reconstruction targets remain unclear. This study assessed the association between radiological joint restoration in stemmed and stemless HA and long-term functional outcomes.

METHODS: Patients who underwent stemmed or stemless HA between 2001 and 2011 were included in a long-term follow-up. Cases with incomplete or non-calibratable digital radiographs or concomitant glenoid procedures were excluded. Clinical outcomes were evaluated using the Constant-Murley Score (CMS) and satisfaction questionnaires. Radiographic parameters-lateral glenohumeral offset (LGHO), lateral humeral offset (LHO), and center of rotation (COR)-were measured on standardized anteroposterior radiographs pre- and postoperatively by two independent observers. Differences from baseline to final follow-up (ΔLHO, ΔLGHO, ΔCOR) were calculated. Interobserver reliability was determined using the intraclass correlation coefficient (ICC). Associations between radiological measures and clinical outcomes were analyzed using linear regression; intergroup comparisons used unpaired t-tests ($p < 0.05$).

RESULTS: Forty-eight patients (12 stemmed, 36 stemless HA) were examined after a mean follow-up of 16.6 […]

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Emergency access to the subclavian vessels by non-thoracic surgeons: a cadaver-based learning model for orthopedic trauma surgery.

Grechenig P, Gänsslen A, Dauwe J, Wittig U, Sagmeister M, Koutp A, Puchwein P, Sadoghi P et al.

PURPOSE: The aim of this study was to evaluate the procedural time and safety of an infraclavicular approach to the subclavian vessels for damage control of major upper extremity vascular injuries. The study specifically focused on the performance of non-thoracic surgeons using a cadaveric model.

METHODS: A sample of 100 orthopedic trauma surgeons (residents, specialists, and attendings) was recruited from an AO cadaveric dissection course. Emergency infraclavicular access was performed on 50 cadavers over two consecutive days. The procedural time, successful vessel clamping, participant self-assessment, and the resulting learning curve were analyzed.

RESULTS: On day one, 27% (9/33) of participants who successfully achieved correct clamping of both subclavian vessels reported feeling confident. By day two, this proportion increased to 71% (55/77). Comparing day 1 to day 2 we found an improvement of 35% (subclavian artery) and 37% (subclavian vein) in the correct identification of subclavian vessels in our sample. The evaluations of this study show that there is no correlation between surgical experience and successful emergency access.

CONCLUSION: Anatomic dissection is of paramount importance for teaching rare and demanding surgical techniques. This study demonstrates that anatomical workshops significantly improve procedural safety and self-assessment when accessing subcla […]

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Operative timing and surgical complexity in hand trauma: a multicenter analysis of replantation and non-replantation centers in Germany.

Nicklas A, Will Marks P, Dragu A, Schmidt H, Rotärmel A, HandTraumaRegister DGH

BACKGROUND: Hand injuries are common traumatic conditions that often require specialized surgical care. Differences in hospital structures and levels of specialization may influence the timeliness and complexity of treatment. This study evaluates quality of care across different hospital types, focusing on time-to-skin-incision and surgical management of finger and hand injuries.

MATERIALS AND METHODS: This retrospective multicenter study is based on data from the HandTraumaRegister of the German Society for Hand Surgery (HTR DGH). A total of 16,726 surgically treated cases with documented finger or hand injuries recorded between 2018 and 2023 were analyzed. Hospitals were categorized as non-replantation centers, replantation centers, or FESSH-accredited replantation centers. Demographic characteristics, injury-patterns, treatment modalities, and time-related parameters were assessed. Statistical analysis included descriptive statistics, group comparisons, and multivariate regression models to evaluate factors associated with time-to-skin-incision.

RESULTS: The study population consisted predominantly of male patients (75.79%), and 91.6% were right-handed. Soft tissue injuries without fractures or amputations accounted for 46.57% of cases. Non-replantation centers demonstrated the longest time-to-skin-incision (median 13 h), whereas replantation centers (median 3.75 h) and FE […]

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Neglected acromioclavicular joint injury in coracoid process fractures: a retrospective cohort study.

Lee MS, Lee CH, Jo YH, Cho WT, Seo YW, Choi WS

INTRODUCTION: Coracoid process fractures are frequently associated with injuries to the superior suspensory shoulder complex, including the acromioclavicular (AC) joint. However, concomitant AC joint injuries may be overlooked in the acute phase, potentially limiting treatment options. This study aimed to investigate the association between coracoid process fractures and SSSC injuries and to identify imaging-based predictors of concomitant AC joint injury.

METHODS: A retrospective single-center cohort study including 60 patients with coracoid process fractures was conducted between February 2016 and April 2023. Coracoid fractures were classified based on anatomical location, and associated SSSC injuries were categorized according to the involved structures. These injuries were evaluated using 3D shoulder CT. Morphological deformity of the SSSC was assessed by measuring the glenoclavicular distance (GCD)-a longitudinal parameter-on final follow-up clavicle AP radiographs. In base fractures, the medial displacement gap was measured on sagittal CT. Statistical analyses included logistic regression and ROC curve analysis to identify predictors of AC joint injury.

RESULTS: Among the 60 patients, fractures were located anteriorly in 27 cases, in the middle region in 5 cases, and at the base in 28 cases. AC joint injury was identified in 15 patients, all of whom had base-type fractu […]

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Subclinical median nerve alterations after distal radius fractures: a comparative analysis of volar plate fixation and cast immobilization using dynamic ultrasonography and electrophysiology.

Karapinar SE, Arslan E, Oguzlar FC, Hatip AY, Akgun VO, Dincer R, Asci S, Kizilkaya V

BACKGROUND: Distal radius fractures may affect the median nerve through trauma-related and treatment-related mechanisms. While overt neurological deficits are uncommon, subclinical median nerve alterations may occur following both conservative and surgical treatment.

PURPOSE: To determine whether treatment modality influences subclinical median nerve behavior using combined dynamic ultrasonographic and electrophysiological assessment.

METHODS: This retrospective observational study included 42 patients with unilateral distal radius fractures. 18 patients were treated conservatively with circular casting, and 24 patients underwent volar plate fixation. Ultrasonographic assessment of the median nerve was performed at the distal radius level in neutral wrist position, as well as during wrist flexion and extension, using a dynamic ultrasonographic approach, and was combined with electrophysiological evaluation including sensory and motor nerve conduction studies. In the surgically treated group, only patients with Soong type 1 or type 2 plate positioning were included.

RESULTS: Clinical functional outcome scores were comparable between groups, despite a higher frequency of mild clinical signs such as positive Tinel test and night pain in the plate group. However, ultrasonographic evaluation demonstrated significantly greater position-dependent median nerve enlargement and flatte […]

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Impulsivity and inattention in hand tendon injuries: a case-control study revealing distinct profiles for work-related accidents.

A■ta■ Ertan E, Demir Karak■l■ç G, Ertan MB

INTRODUCTION: The impact of attention deficit, impulsivity, and anger on various types of injuries is a subject of ongoing research. Hand tendon injuries are frequently encountered clinical conditions; however, their potential relationships with the aforementioned factors have not been previously investigated. This study aimed to examine the mechanisms of hand tendon injuries from certain psychiatric perspectives.

MATERIALS AND METHODS: 32 patients presenting to the physical therapy and rehabilitation outpatient clinic with hand tendon injuries and 32 healthy controls were evaluated. The assessment included sociodemographic data, causes of injury, Quick Disabilities of the Arm, Shoulder, and Hand (Q-DASH) and Visual Analog Scale (VAS) scores, as well as the physical and functional consequences of the injury. Patients and controls were assessed for attention deficit, impulsivity, and anger using the Adult ADHD Self-Report Scale (ASRS), Barratt Impulsivity Scale-11 (BIS-11), and State-Trait Anger Expression Inventory self-report scales. Statistical analyses of the results were performed.

RESULTS: Patients with tendon injuries were found to have significantly higher scores in ASRS attention and BIS-11 Motor impulsivity compared to the control group ($p = 0.048$, $p = 0.040$). Individuals with hand tendon injuries resulting from work accidents demonstrated significantly lower ASRS to […]

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Comparative analysis of the improvement in shoulder rotation in complex shoulder fractures treated by reverse shoulder arthroplasty compared to open reduction and internal fixation: a systematic review and meta-analysis.

Albadran A, Alsugair R, AlHarthi A, Alshangiti H, Alqahtani H, Alsugair G, Ababtain R, Alhajress D et al.

PURPOSE: To compare the clinical and functional outcomes of open reduction and internal fixation (ORIF) and reverse shoulder arthroplasty (RSA) in patients aged ≥ 65 years with complex proximal humeral fractures (PHFs).

METHODS: A systematic review and meta-analysis was conducted according to the PRISMA guidelines. PubMed, Web of Science, ScienceDirect, EBSCO, and the Cochrane Library were searched for randomized controlled trials (RCTs) and cohort studies published in English without date restrictions. Eligible studies compared RSA and ORIF in elderly patients with PHFs and reported functional, radiographic, or complication outcomes. Pooled data were analyzed using a random-effects model.

RESULTS: Twenty-two studies involving > 34,000 patients were included. RSA was associated with greater forward flexion and a trend toward improved abduction, whereas internal rotation favored ORIF without reaching significance. No significant differences were observed in external rotation. The functional scores (Constant-Murley, Oxford Shoulder Score) were similar, and the complication and reoperation rates did not differ significantly between the groups.

CONCLUSION: RSA offers modest advantages in forward flexion compared with ORIF but does not consistently improve overall functional scores or reduce complications. RSA may be preferred in elderly patients with poor bone stock or rotator c […]

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Changes in kinematic behavior and collateral ligament strain after medially stabilized TKA using a novel intraoperative navigation platform: a cadaveric study.

Van de Vyver A, Taylan O, Peersman G, Demeulenaere M, Bialy M, Lauwagie T, Scheys L

OBJECTIVES: This study aimed to analyze to what extent a novel intraoperative navigation platform (Next-AR, Medacta) for total knee arthroplasty (TKA) allows to restore the native knee joint kinematics and strains in the medial collateral ligament (MCL) and lateral collateral ligament (LCL) throughout a squatting motion.

MATERIALS AND METHODS: Computed tomography (CT) scans of 6 native cadaver legs were used to design patient-specific guides. Bony landmarks and virtual single-line collateral ligaments were identified to acquire real-time intraoperative feedback on bone resection, implant alignment, tibiofemoral kinematics, and collateral ligament elongations using the Next-AR system. The specimens were subjected to squatting (35°-100°) motion using a physiological ex vivo knee simulator while maintaining a constant vertical ankle load of 110 N through active quadriceps and bilateral hamstring controls. Subsequently, each knee underwent a medially-stabilized TKA (Medacta) with mechanical alignment technique and was retested under the same conditions as in their native state. The tibiofemoral and patellofemoral kinematics, along with collateral ligament strains, were computed from 3D marker trajectories using a six-camera optical system (Vicon). MCL and LCL insertions-ant, mid, and post bundles-were identified in relation to bone-pin markers using a wand.

RESULTS: Both native a […]

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Regional differences in the management of patients with mild traumatic brain injury and antithrombotic therapy-an Austrian survey.

Sebek B, Dammerer D, Putzer D, Moser L

BACKGROUND/PURPOSE: The number of patients with mild traumatic brain injury (TBI) receiving oral anticoagulation is increasing. Up-to-date diagnostic and treatment algorithms are missing in Austria. The aim of this survey is to collect and analyse Austria-wide data on the care of this patient group.

METHODS: The survey was carried out using "Jotform" between 27 November 2023 and 31 January 2024, asking questions relating to the diagnosis and treatment of patients with mild TBI receiving antithrombotic therapy. Patient groups comprising individuals with alcohol addiction and hemophilia were incorporated as well with a view to the paucity of available literature on these groups. The survey was sent to all orthopedic-traumatological and neurosurgical departments in Austria.

RESULTS: 39 of 67 (58.2%) orthopedics & traumatology departments and 4 of 10 (40%) neurosurgery departments participated in the survey. The share of departments reporting routine submission of different patient groups to cranial computed tomography (CCT) was 38 (95%) for patients taking antiplatelet agents, 39 (97.5%) departments for patients taking vitamin K antagonists (VKAs), 39 (97.5%) for patients taking direct oral anticoagulants (DOACs), 11 (27.5%) for patients with alcohol addiction, and 28 (70%) for patients with known hemophilia. Considering the separate groups of patients not living in a nursing fa […]

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Extracorporeal shockwave therapy in bone fractures: a systematic review.

Santos TIR, Souza E Silva LC, Bonifacio M, Campos Melo CM, Sena BG, Simões MC, Risso Parisi J, da Silva Fernandes AC et al.

INTRODUCTION: The increasing incidence of fractures and cases of delayed union or nonunion has encouraged the search for non-invasive therapies. Extracorporeal shockwave therapy (ESWT) has been proposed as a strategy to stimulate bone regeneration and improve fracture healing.

MATERIALS AND METHODS: A systematic review was conducted following PRISMA guidelines, with searches performed in PubMed, Scopus, Web of Science, and Embase (2005-2025). Clinical studies evaluating ESWT for fracture treatment with radiographic assessment of bone healing and the presence of a control group were included.

RESULTS: A total of 834 studies were initially identified, of which 7 were included in the final analysis. Most studies demonstrated positive effects of ESWT, including improved bone healing, pain reduction, and functional recovery. However, there was considerable heterogeneity in treatment protocols and relatively small sample sizes.

CONCLUSION: ESWT shows potential as a non-invasive therapy to stimulate fracture healing, particularly in cases of delayed union or nonunion. However, further controlled clinical trials and standardized protocols are needed to confirm its clinical effectiveness.

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Clinical outcomes and complication rates of minimally invasive plate osteosynthesis (MIPO) versus open reduction and internal fixation (ORIF) in midshaft clavicular fractures: a comparative study.

Lin G, Zheng S, Zhang Y, Liu Q, Liu H

BACKGROUND: Conventional open reduction and internal fixation (ORIF) remains the standard approach for midshaft clavicle fractures, yet its traumatic nature has driven the need for minimally invasive techniques. Locking minimally invasive plate osteosynthesis (MIPO) demonstrates clinical potential through biomechanical superiority and tissue-preserving characteristics.

OBJECTIVE: To compare the clinical efficacy and safety of MIPO versus ORIF for midshaft clavicular fractures.

METHODS: A retrospective cohort study included 203 patients (MIPO group:101 cases; ORIF group:102 cases) treated between 2022 and 2024. Baseline characteristics, functional outcomes (DASH and Constant-Murley scores), and complications were analyzed using Mann-Whitney U and chi-square tests.

RESULTS: No baseline differences were observed ($P > 0.05$). Functional outcomes were equivalent between groups (DASH score: $P = 0.906$; Constant-murley score: $P = 0.646$). The overall complication rate was significantly lower in the MIPO group than in the ORIF group (7.92% vs. 36.27%, $P < 0.001$), particularly in nonunion (0% vs.

5.88%, P = 0.039) and sensory disturbance (6.93% vs. 27.45%, P < 0.001). This advantage is also evident in the subgroup of patients with AO type 2 C fractures, where MIPO similarly demonstrates a significantly lower incidence of these two complications.

CONCLUSION: MIPO achieves comparable fun […]

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Isolated scaphoidectomy for type II SLAC and SNAC wrists: retrospective case-series at long-term follow-up.

Altissimi M, Pataia E, Saracco M, Braghiroli L

BACKGROUNDS: Stage II scapholunate advanced collapse (SLAC) and Scaphoid non-union advanced collapse (SNAC) wrists are currently treated either by scaphoidectomy and Four Corner Fusion or by Proximal Row Carpectomy. In our retrospective study, isolated scaphoidectomy demonstrated clinical non-inferiority when compared to standard procedures and could be supplemented by a Four Corner Fusion only at a later time if needed.

METHODS: Between 2006 and 2011, 13 patients / 14 wrists (mostly males, manual workers) underwent the surgical procedure. 8 patients were affected by SNAC and 5 by SLAC. The average age was 56,5. All patients were manual workers. The mean follow-up was 15 years. These patients were reviewed a first time in 2012 and once again in 2024. Subjective outcome, DASH score, return to work, ROM and grip strength were evaluated. Radiographic assessment was done in all 14 wrists at the first follow-up and in 6 at the last long-time review. Statistical analysis was conducted using the SPSS software.

RESULTS: 12 patients had resumed the same work at a mean of 3 months after surgery. Only one patient required a Four Corner Fusion after 2 years. 11 of the 13 enrolled patients declared their satisfaction for the procedure at the longest follow-ups. Clinical results remained stable over time. Radiological deterioration was seen in 6/13 wrists at the first follow-up and in all […]

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Loop versus anchor tenodesis of the long head of the biceps tendon: a prospective randomized controlled pilot study.

Bischofreiter M, Kölblinger C, Gattringer M, Azesberger A, Gruber MS, Rittenschober F, Kindermann H, Ortmaier R

BACKGROUND: The long head of the biceps tendon is a frequent source of anterior shoulder pain, often treated with tenodesis. While anchor tenodesis is well established, it carries risks of implant-related complications. This study evaluates an implant-free "loop tenodesis" technique as a potentially safer and cost-effective alternative.

METHODS: In this prospective, double-blinded, randomized exploratory study, 48 patients undergoing arthroscopic biceps tenodesis were randomized to either loop (n = 24) or anchor (n = 24) tenodesis. Clinical outcomes were assessed using ASES, Constant, and DASH scores, as well as range of motion, supination strength, satisfaction, aesthetic appearance, and return-to-sport. Statistical analysis was performed using the Mann-Whitney U and Wilcoxon signed-rank tests to compare non-parametric data.

RESULTS: Both groups showed significant improvements with similar gains in functional scores. Supination strength recovery was greater in the loop group (103% vs. 93%). There were no significant differences in range of motion or complications. Notably, only the anchor group showed cases of Popeye deformity (2 patients). Return to sport was achieved in 95.3% of patients, typically within 3 months. No revision or implant-related issues occurred.

CONCLUSION: Loop tenodesis offers equivalent clinical outcomes to anchor-based techniques, with fewer aesthetic […]

PubMed · DOI

Split hamate fractures as part of complex ulnar carpometacarpal injuries: radiographic and functional outcomes.

Glaser J, Schmidt E, Aman M, Bickert B, Harhaus-Wähner L, Cordts T

BACKGROUND: Split hamate fractures, which most often are a complicating part of fracture dislocations of the 5th or 5th and 4th carpometacarpal joints, are rare and challenging injuries that can severely affect hand function. The injury usually requires surgical treatment. However, both short- and long-term data on the effectiveness of these interventions are limited. The objective of this study is to advance understanding of this rare hand surgical condition by investigating its epidemiology, etiopathogenesis, subsequent complications, and functional outcomes.

METHODS: Between 2010 and 2024, a total of 29 patients with a split hamate fracture were treated at a level-1 trauma center, of which 12 patients are included in this study. Using patient surveys (DASH, SF-36, PRWE), along with clinical and radiological examinations, the hand function of a total of 12 patients with split hamate fractures is evaluated.

RESULTS: Twelve male patients with a split hamate fracture were included, with a mean age of 28.1 years at the time of surgery. In 10 patients (83%) the dominant right hand was affected, most involving, the injuries resulted from direct trauma or a fall; surgery was performed on average of 12 days after injury. Functional outcomes were favourable, with a mean DASH score of 14.1 and PRWE score of 14.8, indicating minimal residual reduction of function in most cases. Radiol […]

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Outcomes and short-term survivorship of fixed-bearing all-poly unicompartmental knee arthroplasty. A retrospective single-center study of 42 prostheses with a main follow-up of 32.3 months.

Gaggiotti S, Gaggiotti G, Lustig S, Batailler C, Monllau JC, Gaggiotti S, Combalia A

PURPOSE: All Poly (AP) tibial unicompartmental knee arthroplasty (UKA) seems to have advantages in comparison with metal-based (MB) UKA, such as greater polyethylene thickness, wear resistance, bone stock preservation and lower implant costs. The objective of this retrospective study was to evaluate the short-term clinical-functional and radiographic results, complications and survivorship of AP UKA performed consecutively with a minimum follow-up of 2 years.

METHODS: Retrospective cohort analysis all consecutive knees operated with medial and lateral AP UKAs, including bilateral cases, with a minimum follow-up of 2 years. Demographic data were analyzed at the patient level and clinical-functional results were assessed per knee using the KSS, KOOS and VAS scales. Radiographically, mechanical axis (HKA) correction and the presence of radiolucency > 2mm were evaluated. Complications incidence and prosthetic survivorship was determined. Pre-postoperative comparisons were performed using linear mixed-effects models with random intercept per patient to account for the non-independence of bilateral cases. Subgroup analyses involving fewer than 10 observations were considered exploratory and no formal statistical inference is drawn. A value of $p < 0.05$ was considered statistically significant with a confidence interval (CI) 95%.

RESULTS: 42 AP UKAs were included, 5 lateral (11.9%) a […]

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Mix and match in reverse total shoulder arthroplasty: overview of current prostheses systems and potential combinations.

Seiß D, Schlichter A, Graf JM, Stange R, Ohlmeier M

INTRODUCTION: Mix and Match is considered a viable option for minimizing morbidity in total hip arthroplasty (THA). However in reverse total shoulder arthroplasty (rTSA) this strategy remains comparatively new. This study provides an overview of currently available implant systems and the potential combinations of glenoid and humeral components.

MATERIALS AND METHODS: A review of commercially available shoulder arthroplasty systems was performed. Nineteen manufacturers were contacted and asked to provide detailed information on glenoid and humeral component sizes for their rTSA systems. The reported component sizes were compiled and analyzed for possible combinations.

RESULTS: Thirteen manufacturers (68%) consented to the publication of their component size data. Two manufacturers (11%) do not produce rTSA systems, and four (21%) did not respond to repeated inquiries. A total of 21 implant systems were reported, comprising 10 distinct implant sizes. Glenosphere and humeral inlay diameters ranged from 32 mm to 48 mm, with 36 mm being the most commonly available size (16 of 21 systems, 76%).

CONCLUSION: Mix-and-match implantation in rTSA is likely employed more frequently than documented, yet remains poorly investigated. This study demonstrates that numerous systems can theoretically be combined during revision surgery, although the biomechanical implications are not yet under […]

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Radiolucent lines after robotic-assisted surface-matching kinematically aligned UKA: short-term incidence and lack of association with tibial alignment.

Itou J, Kuwasawa A, Nihei K, Okazaki K

PURPOSE: This study investigated the clinical outcomes of robotic-assisted unicompartmental knee arthroplasty (UKA) performed using a surface-matching kinematic technique that restores native joint morphology without alignment constraints. The primary aim was to determine the incidence and distribution of radiolucent lines (RLLs), and the secondary aim was to evaluate short-term revision rates and clinical outcomes.

METHODS: This retrospective cohort study reviewed 242 medial UKAs performed between November 2021 and August 2023 by the same high■volume surgeon using a surface■matching technique. Radiographic parameters and patient■reported outcome measures (PROMs) obtained before and 1 year after surgery were assessed, with radiographic evaluation performed independently of the clinical outcome assessment. A cemented JOURNEY II Uni implant was used in all cases, with intraoperative surface mapping used to guide kinematic placement of components without alignment constraints.

RESULTS: RLLs were seen in 21.9% of knees, exclusively on the tibial side and mainly in the most medial (zone 1: 14.8%, 95% confidence interval [CI] 10.9-19.9) and posterior regions (zone 9: 15.3%, 95% CI 11.3-20.3). A larger tibial component was the only independent predictor of RLLs (odds ratio 1.54, 95% CI 1.27-1.88; p = 0.04). All PROMs improved, with no differences between knees with and without RLLs. […]

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Radiographic parameters and mechanical complications of articulating vs. non-articulating hip spacers.

Lenz J, Baumann F, Baertl S, Straub J, Rupp M, Alt V, Freigang V

BACKGROUND: Periprosthetic joint infection (PJI) is a serious complication of total hip arthroplasty. Standard treatment for PJI is a two-stage revision surgery with antibiotic-loaded cement spacers to control infection and provide temporary joint stability. This study compares the radiological outcomes and complication rates between articulating and non-articulating hip spacers in patients undergoing treatment for PJI and native joint infections.

METHODS: We retrospectively reviewed 71 hip spacers (34 articulating, 37 non-articulating) in 38 patients treated between 2004 and 2022. Data on leg length discrepancy, femoral offset, infection control, and mechanical complications were obtained. For infection analysis, only patients treated exclusively with one spacer type were included. After excluding eight patients with mixed spacer types (29 spacers), 30 patients with 42 spacers were included in this subgroup.

RESULTS: Articulating spacers were significantly better at preserving leg length, with a mean discrepancy of -3.7 mm compared to -16.9 mm for non-articulating spacers (p = 0.025). However, non-articulating spacers maintained femoral offset (1.1 vs. 0.7, p < 0.001) closer to physiological offset. The rate of mechanical complications was higher in the articulating spacer group, with spacer dislocations occurring in 45% of cases compared to 10% in the non-articulating group […]

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The effect of Poller screw or monocortical plate augmentation on the stability of intramedullary nailing in proximal extra-articular tibial fractures: a biomechanical study.

■ahin MA, Durgut F, Yi■it ■, Akar MS, Özkul E, Atiç R, Çiftçi S

BACKGROUND: Proximal extra-articular tibial fractures present considerable biomechanical and technical challenges due to the short proximal fragment, the wide metaphyseal canal, and the deforming forces acting around the knee. These factors predispose isolated intramedullary (IM) nailing to malalignment, prompting the use of adjunct reduction aids such as Poller screws or supplemental plates. This study aimed to compare the mechanical behavior of IM nailing alone with that augmented by Poller screws or a monocortical plate in a standardized proximal tibia fracture model.

METHODS: Fifteen synthetic tibiae with an AO/OTA 41-A2.3 extra-articular proximal fracture model were randomized into three groups (n = 5 each): IM nailing alone, IM nailing plus two Poller screws, and IM nailing plus a four-hole medial monocortical plate. All specimens were instrumented with a 10 × 340-mm titanium nail and two proximal and two distal locking screws. Mechanical testing included axial compression to 800 N, axial loading to failure, and torsion at a rate of 25°/min. Displacement, stiffness, maximum load, and torque parameters were compared across groups.

RESULTS: Under 800 N axial load, the plate-augmented group showed the lowest displacement and highest stiffness (2.40 ± 0.22 mm; 332.4 ± 21.7 N/mm), followed by the Poller screw group (2.48 ± 0.25 mm; 321.6 ± 19.4 N/mm) and the IM-only group (3 […]

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Clinical cutoff value of lumbar bending range for predicting residual pelvic obliquity after total hip arthroplasty in dysplastic hip osteoarthritis.

Yokoi H, Osawa Y, Ozawa Y, Funahashi H, Takegami Y, Imagama S

INTRODUCTION: Residual pelvic obliquity (PO) after total hip arthroplasty (THA) for dysplastic hip osteoarthritis (DHOA) may adversely affect coronal balance and postoperative outcomes. Although lumbar bending range (LBR) is considered critical for PO improvement, the clinical cutoff value for predicting residual PO remains unclear. This study aimed to determine the cutoff value of LBR associated with residual PO after THA.

MATERIALS AND METHODS: Among 382 patients who underwent primary THA for unilateral DHOA between July 2019 and June 2024, 98 patients with preoperative downward PO of ≥ 2° on the affected side were included. Patients were classified into a residual group (postoperative PO ≥ 2° at 1 year) and an improvement group (postoperative PO < 2°). Demographic data and radiographic parameters of the hip, lower limbs, and spine were compared. Multivariate logistic regression analysis was performed to identify factors associated with residual PO, and receiver operating characteristic (ROC) analysis was used to determine the optimal LBR cutoff value.

RESULTS: Residual PO was observed in 28 patients (29%). Compared with the improvement group, the residual group demonstrated significantly greater Crowe index values, preoperative radiographic leg length discrepancy, and preoperative PO. Affected-side LBR was significantly smaller in the residual group (4.3° ± 3.0° vs. 7.7° ± […]

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Hidden blood loss after proximal femur fractures fixation: analysis of the first three postoperative days and influence of anticoagulants.

Mayor J, Winkelmann M, Schwinn LS, Özbek H, Aktas G, Omar-Pacha T, Oberthür S, Krammig RH et al.

PURPOSE: Hidden blood loss (HBL) is an underrecognized contributor to total perioperative blood loss after hip fracture surgery. This study aimed to quantify HBL during the first three postoperative days following proximal femur fracture fixation and to assess the impact of preoperative anticoagulant therapy on blood loss and transfusion requirements.

METHODS: A retrospective cohort study was conducted including patients aged ≥ 65 years who underwent surgical fixation of proximal femur fractures at a level I trauma center between January 2020 and December 2022. Total blood loss was estimated from perioperative hemoglobin changes and calculated blood volume. Patients were stratified according to anticoagulant use, and logistic regression analysis was performed to identify independent predictors of increased blood loss.

RESULTS: Among 260 patients analyzed, median total blood loss was 1184 ml, and 25% experienced losses exceeding 1675 ml. High blood loss correlated significantly with anticoagulant use, higher BMI, and longer operative time ($p < 0.001$). In multivariate analysis, anticoagulant therapy (particularly direct oral anticoagulants) remained an independent predictor of high blood loss (OR 3.06; 95% CI 1.47-6.37; $p = 0.003$). Patients in the highest quartile required significantly more transfusions, though short-term mortality did not differ.

CONCLUSION: Hidden blood los […]

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Visual processing and interference performance influences on knee angular impulse in ACLR individuals: a cognitive-biomechanical analysis of drop-jumps.

Santamaria-Guzman K, Holmes H, Kosek J, Peoples B, Harrison K, Campos-Vargas S, Weimar W, Neely K et al.

INTRODUCTION: Visual processing speed and cognitive interference control are critical for athletic performance and Anterior Cruciate Ligament (ACL) injury risk. However, how these cognitive functions relate to biomechanical performance after Anterior Cruciate Ligament Reconstruction (ACLR) remains unclear. This gap is clinically relevant given persistent re-injury rates of 20-25% in young athletes returning to sport, suggesting that rehabilitation may insufficiently address cognitive demands. This study examined cognitive differences between ACLR individuals and controls during drop-jump tasks, and their relationship with knee angular impulse.

MATERIALS AND METHODS: Thirty-two females (16 ACLR, 16 controls; 20 ± 1 years) completed cognitive assessments including the Stroop Color and Word Test, Trail Making Test, Digit Span Memory Test, and visual/auditory reaction time tests. Participants performed drop-jumps under four conditions: standard, choice, visual-cued, and audio-cued. Knee angular impulse was calculated for eccentric, concentric, and net phases. Binomial logistic regression identified cognitive predictors distinguishing groups, followed by factorial ANOVA to assess biomechanical differences. Spearman correlations examined relationships between cognition and knee angular impulse.

RESULTS: Three cognitive variables distinguished groups: cognitive interference score, v […]

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Perceived leg length discrepancy after total hip arthroplasty is associated with global spinopelvic coronal flexibility.

Minoji S, Ishikura H, Kudo M, Hatano M, Nishiwaki T, Tanaka S

BACKGROUND: Perceived leg length discrepancy (pLLD) is a common source of dissatisfaction after total hip arthroplasty (THA), even in the absence of significant radiographic discrepancies. While spinopelvic factors have been increasingly recognized, most previous studies have focused on sagittal alignment or lumbar mobility, and the impact of preoperative global spinopelvic coronal flexibility on pLLD has remained unclear.

METHODS: We retrospectively reviewed 114 patients who underwent primary unilateral THA for osteoarthritis between January and December 2023. pLLD was assessed using a four-point scale at 6 months postoperatively. Patients who reported no perception of discrepancy were classified as the non-pLLD group, while those reporting mild, clear, or strong perception were grouped as the pLLD cohort. Preoperative spinal flexibility was measured on coronal radiographs during maximal lateral bending, calculating changes in spinopelvic angle (ΔSPA), reflecting thoracic-to-pelvic coronal flexibility, and lumbosacral angle (ΔLSA), reflecting lumbar-to-pelvic coronal flexibility. Secondary parameters included radiographic leg length discrepancy, leg lengthening, and sagittal spinopelvic alignment. Multivariable logistic regression analysis was used to identify independent predictors of pLLD.

RESULTS: pLLD was reported in 47 patients (41.2%). The pLLD group exhibited signific […]

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Sex-related differences in final knee balance during robotic-assisted unicompartmental knee arthroplasty.

Stimolo D, Leggieri F, Secci G, Massenzi M, Civinini R, Innocenti M

INTRODUCTION: In robotic medial unicompartmental knee arthroplasty (mUKA), a preoperative alignment strategy can be defined and subsequently adjusted through stress testing to achieve optimal soft-tissue balance. However, the application of uniform planning principles does not necessarily yield equivalent balance in all patients. The aim of this study was to examine differences in intraoperative balance according to patient sex and the degree of preoperative varus deformity.

MATERIALS AND METHODS: A single-center retrospective cohort study including 254 MAKO® robotic-assisted medial UKAs (2018-2025) was performed using a uniform intraoperative planning protocol. Gap widths were measured in millimeters at full extension and 90° of flexion (F90) and classified as tight (< 0.5 mm and negative values), balanced (0.5-2 mm), or loose (> 2 mm). Patients were stratified by sex and preoperative varus severity, and gaps were analyzed quantitatively and categorically. The independent effects of sex, varus deformity, and operating surgeon on extension and flexion gaps has been assessed.

RESULTS: Mean preoperative alignment was 174.2 ± 2.8°, with no sex-related difference in baseline varus ($p = 0.058$). Flexion gaps were significantly tighter in female patients ($p = 0.0045$). A tight F90 gap was observed in 17.2% more female than male patients ($p = 0.023$), with an absolute mean difference o […]

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A novel pinless augmented reality-based navigation system for total hip arthroplasty in the supine position: a comparative study of acetabular cup angle accuracy.

Kurishima H, Ogawa H, Yamada N, Noro A, Tomata Y, Tsukada S

INTRODUCTION: We developed a novel pinless augmented reality (AR)-based portable navigation system for total hip arthroplasty (THA) in the supine position to eliminate pin-site complications and intraoperative navigation abandonment, and to reduce registration time. This study compared the measurement accuracy of this new pinless portable navigation system with that of a conventional AR-based portable navigation system requiring pin insertion.

MATERIALS AND METHODS: This retrospective cohort study compared primary unilateral THAs performed using the novel pinless AR-based navigation system and a conventional AR-based navigation system. The primary outcome was the absolute difference between the acetabular cup angles displayed on the navigation screen and those measured on postoperative computed tomography. Secondary outcomes included target achievement rates within 5° and 10°. Pin-site complications, intraoperative navigation abandonment, and dislocations were also recorded.

RESULTS: A total of 228 hips were analyzed (Pinless: $n = 115$; Conventional: $n = 113$). While there was no significant difference in the median absolute difference for inclination (2.5° vs 2.4°; $p = 0.366$), the median absolute difference for anteversion was significantly larger in the pinless group than in the conventional group (2.9° vs 1.9°; $p = 0.011$). There were no significant differences in the percent […]

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Impact of modified Henry approach with preservation of Flexor Carpi Radialis Tendon Sheath on wrist function in the treatment of distal radius fractures.

Ma D, Ji J, Zhu Y, Fang X, Wang Y, Fan J

OBJECTIVE: To compare the clinical efficacy of a modified Henry approach (preserving the FCR tendon sheath) versus the traditional approach (incising the sheath) for distal radius fractures (DRFs) treated with volar locking plates, and to evaluate the outcomes within 12 months postoperatively.

METHODS: A retrospective cohort study was conducted on 165 patients with unstable DRFs who underwent volar locking plate fixation from January 2023 to March 2025, who were retrospectively divided into the Preservation Group ($n = 79$, FCR tendon sheath preserved via the modified approach) and the Control Group ($n = 86$, FCR tendon sheath incised via the traditional approach). Perioperative indicators, radiological parameters, Visual Analog Scale (VAS) scores, Disabilities of the Arm, Shoulder and Hand (DASH) scores, grip strength, wrist range of motion (ROM) and complications were compared at the follow-up time points of 1 week, 1, 2, 3, 6 and 12 months postoperatively.

RESULTS: There were no significant differences in baseline data and postoperative radiological reduction quality between the two groups ($P > 0.05$). The Preservation Group had significantly lower VAS scores at all follow-up time points ($P < 0.001$), and lower DASH scores from 1 week to 6 months ($P < 0.001$), with no statistical difference at 12 months ($P = 0.550$). Grip strength of the Preservation Group was significantly super […]

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Clinical Outcomes, Return to Sport and Psychological Readiness After Arthroscopic Bankart Repair Using Knotless All-Suture Anchors at a Minimum 2Year Follow-up.

Rab P, Rupp MC, Pfarrmaier A, Vieider RP, Shirinskiy IJ, Scheiderer B, Siebenlist S, Lacheta L

PURPOSE: To report the clinical outcomes, return to sport (RTS) and psychological readiness of patients who underwent arthroscopic Bankart repair with knotless all-suture anchors with a minimum follow-up of 2 years.

METHODS: In this retrospective case series, consecutive patients who underwent primary arthroscopic Bankart repair using knotless all-suture anchors between 08/2019 and 07/2022 were included. Patient-reported outcomes were assessed using the Western Ontario Shoulder Instability Index (WOSI), American Shoulder and Elbow Surgeons (ASES) score, Disabilities of the Arm, Shoulder and Hand questionnaire (DASH), Shoulder Instability-Return to Sport after Injury (SI-RSI) scale, subjective shoulder value (SSV), and the visual analogue scale (VAS) for pain. Patient satisfaction, RTS, return to preinjury level of sport, instability recurrence and revisions were recorded. Receiver operating characteristic (ROC) curve was calculated to assess the discriminative performance of the SI-RSI scale, and the Youden's index was employed to determine the optimal cutoff for prediction of return to preoperative level of sports.

RESULTS: Of 57 patients eligible for inclusion, 46 patients (11.1% female, 28.7 ± 6.8 years at surgery) were available at a follow-up of 2.9 [2.3-3.4] years. Three patients (6.5%) reported a redislocation, one patient (2.2%) underwent a revision and was excluded f […]

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Impact of frailty on 12-month patient-reported outcomes after posterior lumbar fusion surgery.

Haffer H, Muellner M, Chiapparelli E, Dodo Y, Camino-Willhuber G, Zhu J, Pumberger M, Shue J et al.

INTRODUCTION: Frailty might increase the risk for postoperative complications and death. It is not well understood how frailty influences patient-reported outcomes (PROs) in spine surgery. The objective is to investigate on the impact of frailty on PROs 1-year after lumbar fusion.

MATERIALS AND METHODS: This was a prospective observational study of patients undergoing open posterior lumbar fusion for degenerative conditions with 12-month follow-up (12 M-FU). The modified 5-item frailty index (mFI-5) score was determined preoperatively. PROs included the Oswestry-Disability-Index (ODI), 12-item Short-Form-Healthy-Survey with Physical (PCS-12) and Mental-Component-Score (MCS-12) and Numerical-Rating-Scale (back/leg pain) before surgery and 6 weeks, 3, 6 and 12 months (6 W, 3 M, 6 M, 12 M) postoperatively. PROs were compared between groups stratified as frail (mFI-5 ≥ 2), pre-frail (mFI-5 = 1) and non-frail (mFI-5 = 0). PRO changes at 12 M-FU were evaluated with substantial clinical benefit (SCB) metrics. Regression models were used to determine associations between mFI-5 and PROs.

RESULTS: 135 patients (52.6% female, 62.1 years, BMI 29.1 kg/m2) were analyzed. 31.3% and 51.8% of the frail and non-frail patients reached ODI-SCB, respectively. Frailty patients reached a significantly smaller ODI change at 12 M-FU than pre- and non-frail patients (15.2% vs. 19.3% and 19.0%, p < 0.0 […]

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Rates of union and risk factors for continued nonunion following exchange nailing of tibial nonunion.

Mastracci JC, Averkamp B, Braswell M, Yu Z, Chen AT, Natoli RM, Farooq H, Mir H et al.

INTRODUCTION: The objective of this study was to evaluate the rate of nonunion repair success in tibia nonunions treated with exchange nailing. This retrospective cohort study was conducted across five academic Level 1 trauma centers and included 63 patients with tibia nonunions.

MATERIALS AND METHODS: Patients who sustained a tibia fracture (AO/OTA 42) treated with intramedullary fixation that developed a nonunion and were subsequently treated with exchange nailing were retrospectively reviewed. The primary outcome measure was nonunion repair success based on osseous union. Additional analyses included union rate by AO/OTA classification, nonunion type, implant(s) used, graft used, time from initial procedure, and infection status.

RESULTS: All patients sustained an AO/OTA type 42 fracture. Out of 63 patients, 47 tibias (75%) achieved osseous union after the index exchange nail procedure. The rates of nonunion revision success were similar across nonunion types and time from initial procedure until exchange nailing. There was no significant difference in union rate when infection was present. Complications included re-operation, readmission, infection, and implant failure.

CONCLUSIONS: This large, multicenter study with contemporary implants, instruments, and techniques for exchange nailing tibia nonunions demonstrates a nonunion repair success rate of 75%, consistent with […]

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Robotic-assisted unicompartmental knee arthroplasty is associated with lower odds of prolonged hospitalization and no higher odds of high-charge admission during the index hospitalization.

Sun Q, Xu Y, Hong X, Dai M, Wang J, Xie H, Fu Z

BACKGROUND: Robotic-assisted unicompartmental knee arthroplasty (RA-UKA) may improve implant positioning, but its impact on index-hospitalization length of stay (LOS), billed charges, and early inpatient outcomes in routine practice remains unclear.

METHODS: Using the National Inpatient Sample (2016-2022), we identified primary medial UKA for osteoarthritis. RA-UKA was defined by ICD-10-PCS robotic-assisted lower-extremity procedure codes. Primary outcomes were prolonged hospitalization (LOS ≥ the 75th percentile, ≥ 2 days), high-charge admission (total hospital charges [TOTCHG] ≥ the 75th percentile, ≥ $72,321), and in-hospital mortality. We performed univariable comparisons and multivariable logistic regression adjusting for demographics, payer, admission type, hospital teaching status, comorbidities, and calendar year fixed effects (YEAR 2016-2022) using the unweighted NIS discharge sample.

RESULTS: Among 7,154 UKAs, 1,297 (18.1%) were RA-UKA and 5,857 (81.9%) were C-UKA. Median LOS was 1 day (IQR 1-2) in both groups; however, prolonged hospitalization (LOS ≥ 2 days) occurred less frequently in RA-UKA (412/1,297 [30.7%] vs. 2,162/5,857 [37.2%]; $P < 0.001$). In adjusted analyses including YEAR fixed effects, RA-UKA was associated with lower odds of prolonged hospitalization (aOR 0.750, 95% CI 0.647-0.870; $P < 0.001$) and was not associated with high-charge admission (TOTCHG ≥ […]

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Is biceps augmentation effective for scope-assisted lower trapezius tendon transfer in posterosuperior irreparable rotator cuff tears? A retrospective short-term clinical comparison.

Baek CH, Elhassan BT, Lim C, Kim JG, Kim BT, Kim SJ

INTRODUCTION: Although favorable clinical outcomes of scope-assisted lower trapezius tendon transfer (SALTT) have been reported in posterosuperior irreparable rotator cuff tears (PSIRCTs) patients, the limited restoring of static stability can be supported by biceps augmentation as a static stabilizer. This study aimed to compare the outcomes of scope-assisted lower trapezius tendon transfer (SALTT) with and without biceps augmentation in patients with posterosuperior irreparable rotator cuff tears (PSIRCTs).

MATERIAL AND METHODS: This retrospective clinical comparative study was performed with the inclusion criteria: patients who underwent SALTT for PSIRCT from January 2022 to April 2023; a follow-up period of more than 2 years; and availability for clinical assessment and MRI evaluation preoperatively and at 2 years postoperatively. The patients were grouped according to the management of biceps; Non-augmentation group ($n = 26$) and Augmentation group ($n = 27$). The clinical outcomes were evaluated using shoulder pain, patient-reported outcome measures (PROMs), and active range of motion (aROM). Radiological outcomes were performed to evaluate the progression of shoulder joint arthritis, biceps integrity, and graft integrity.

RESULTS: Both groups showed significant improvements in VAS score, PROMs, and aROM. Although postoperative PROMs of Non-augmentation group were signific […]

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Robotic-arm assisted total hip arthroplasty using a short metaphyseal filling collared femoral implant through a direct anterior approach: minimum two-year outcomes.

Marchand R, Kaczynski E, Taylor K, Smitterberg C, Mont MA

INTRODUCTION: Short, collared, metaphyseal-filling femoral stems preserve proximal bone and achieve stable fixation while reducing stress shielding compared to diaphyseal-engaging stems. Early six-month data for one such stem demonstrated improved patient-reported outcomes with no implant-related complications. This study extends follow-up of the same cohort to a minimum of two years to evaluate (a) functional outcomes, (b) implant-related complications, and (c) survivorship.

MATERIALS AND METHODS: The prospective cohort of 120 patients underwent robotic-arm assisted total hip arthroplasty (THA) through a direct anterior approach (DAA) using a short, collared, metaphyseal-filling stem. Patients had a mean age of 69.2 years (range 37-90) and a body mass index of 28.2 (range 18.1-49.1), and they were 71% women and were followed for a mean of 2.6 years (range 2.0-3.4). The Hip Disability and Osteoarthritis Outcome Score for Joint Replacement (HOOS-JR) was compared with baseline using Student's t-tests. Complications included all-cause revision, periprosthetic joint infection, periprosthetic fracture, and dislocation at final follow-up.

RESULTS: At mean follow-up, HOOS-JR scores improved from a mean of 54.5 ± 13.9 preoperatively to 94.4 ± 10.3 ($P < 0.001$). The overall survivorship was 98.1% following two septic revisions.

CONCLUSIONS: This short, collared, metaphyseal-filling st […]

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Second contralateral hip fractures reduce survival, mobility and daily activity : a matched pair analysis.

Blattner A, Sabath F, Röttinger T, Lisitano L, Mayr E, Fenwick A

BACKGROUND: Hip fractures in older adults are associated with substantial morbidity, functional decline, and mortality. Patients who experience a first fragility fracture are at high risk of subsequent fractures, with repeated events further exacerbating functional impairment and survival outcomes. Second contralateral hip fractures, while clinically important, remain under-characterized in terms of timing, long-term functional impact, and mortality.

OBJECTIVES: To investigate the incidence, timing, survival, mobility, and daily activity outcomes of second contralateral hip fractures using a matched pair analysis, and to situate these findings within the broader context of repeated fragility fractures.

METHODS: A retrospective cohort study was conducted at a single Level I trauma center, including all patients treated operatively for hip fractures (AO 31A1-A3, 31B) between January 2016 and June 2020. Patients with a second contralateral hip fracture were matched 1:1 with patients with a single fracture based on age, sex, fracture type, and Charlson Comorbidity Index. Demographic data, functional status (Barthel Index, Parker Mobility Score), walking aid use, living situation, and mortality were assessed at admission, discharge, and follow-up (minimum 3 years, maximum 7 years). Statistical analysis included paired tests and survival analysis.

RESULTS: Of 1,933 patients, 148 ([…]

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Short-term clinical outcomes of robotic-assisted total knee arthroplasty at 12-month follow-up: a prospective, multicenter, concomitant comparison to conventional total knee arthroplasty.

Nöth U, Rivkin G, Caldora P, Heller KD, Thienpont E, Hannouche D, Perets I, Cholewa J

INTRODUCTION: There is limited data available on short-term outcomes on a cut-block positioning robotic system. The purpose of this study was to compare 12-month clinical outcomes between robotic-assisted (raTKA) and conventional total knee arthroplasty (cTKA) with multiple outcomes and surgical centers.

METHODS: This was a non-randomized controlled trial of patients who received either raTKA ($n = 120$) or cTKA ($n = 101$) at 6 different surgical centers. Variables of interest included occurrence of soft tissue release, complications and revisions at minimum one-year follow-up. Satisfaction, pain (numeric rating scale [NRS]), 5-dimensional European Quality of Life (EQ-5D-5 L) questionnaire (index and visual analog scale [VAS]), Oxford Knee Score (OKS), and the Forgotten Joint Score (FJS-12) were collected pre-operatively, and at six weeks, three months, and 12 months post-operative.

RESULTS: There were significantly less soft tissue releases with raTKA (28/120, 23.3%) vs. cTKA (51/99, 51.5%), $p < 0.0001$). There were significantly fewer cases of medial/lateral instability in the raTKA group at six-weeks ($p = 0.038$) and three-months ($p = 0.007$) post-operative. At one-year follow-up, 96.3 and 92.5% of raTKA and cTKA patients were satisfied with the overall results of their surgery, respectively. Significantly more raTKA patients were very satisfied (32.1% vs. 14.6%) with their abil […]

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Clinical and radiological outcomes of using locked intramedullary nails in the treatment of severe frontal plane lower limb deformity in adolescents with hypophosphatemic rickets (mid-term results).

Galal S, Al-Abri AM, Al-Habsi G, Al Ghammari M, Elissawi A, Abou Senna WG, Arafa AS

INTRODUCTION: Lower limb deformities in patients with hypophosphatemic rickets are multi-apical and require multiple osteotomies for correction. Intramedullary nails (IMNs) are used to fix multiple osteotomies. Authors of this studyaimed to assess the clinical and radiographic outcomes of using IMNs for correcting lower limb deformity in adolescents with hypophosphatemic rickets.

METHODS: This prospective study included patients with hypophosphatemic rickets who underwent deformity correction between November 2020 and November 2023 using IMNs, with a minimum follow-up of 2 years. Clinical outcomes were assessed using the Lower Limb Deformity-Scoliosis Research Society (LD-SRS) score. Radiographic outcomes measured included the mechanical tibiofemoral angle (mTFA), mechanical axis deviation (MAD), mechanical lateral distal femoral angle (mLDFA), mechanical medial proximal tibial angle (mMPTA), and Stevens' knee joint zoning.

RESULTS: Twenty patients (25 limbs and 35 bones) were included (mean age: 16 years; range: 13-22 years). The LD-SRS score improved from 3.2 ± 0.4 preoperatively to 4.3 ± 0.7 postoperatively. Preoperative mechanical tibiofemoral angle was 27.8° ± 12.1° and 17.33° ± 7.9° in the varus and valgus groups, respectively, improving to 3.4° ± 6.4° and 2° ± 5.2°, respectively. Preoperative mechanical axis deviation was 82.2 ± 35.6 mm and 41.8 ± 22.2 mm in the varus […]

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Risk prediction and surgical decision-making for pathological fractures in phalangeal enchondroma: validation of the 0.51 threshold and outcome comparison of prophylactic intervention in 65 patients.

Zhao J, Mo Z, Lao Y, Chen J, Li Q, Lin J

BACKGROUND: Enchondroma is the most common benign bone tumor of the hand, predominantly affecting the phalanges and metacarpals, and is frequently complicated by pathological fractures. Although previous studies have explored risk factors based on tumor size, a precise quantitative cut-off value remains undefined. Furthermore, quantitative research regarding the impact of “delayed treatment” on long-term functional outcomes is lacking. This study aims to establish a risk threshold for pathological fractures in phalangeal enchondroma and to provide an evidence-based rationale for optimizing surgical decision-making by comparing the clinical outcomes of prophylactic surgery versus post-fracture management.

METHODS: We retrospectively analyzed the clinical data of 65 patients with phalangeal enchondroma treated surgically at our institution between 2017 and 2025. Patients were categorized into a Fracture Group (n = 35) and a Non-fracture Group (n = 30) based on the presence of a complete preoperative fracture. The tumor-to-phalanx length ratio was measured using the Picture Archiving and Communication System (PACS), and the degree of cortical thinning was assessed according to the Takedani classification. Multivariate logistic regression analysis was employed to identify independent risk factors, and Receiver Operating Characteristic (ROC) curve analysis was used to determine the […]

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Assessing PHILOS plate as an alternative fixation method for pediatric femoral neck fractures: a biomechanical comparison with cannulated screws.

Lim BS, Wang YP, Chung ST, Wang CH, Chang SW, Wang TM, Hong CK

BACKGROUND: Pediatric femoral neck fractures require stable fixation to avoid complications. It remains unclear whether fixation with the Proximal Humeral Internal Locking System (PHILOS) can serve as an alternative to cannulated screw fixation. The purpose of this study was to compare the biomechanical properties of PHILOS and cannulated screws for stabilizing unstable pediatric femoral neck fractures using a synthetic bone model.

MATERIALS AND METHODS: Twelve fourth-generation synthetic composite femurs were randomly assigned to screw fixation (Group S) or PHILOS fixation (Group P) (n = 6 each). A standardized vertically oriented Delbet type II osteotomy was created in all specimens. Group S was fixed with three 6.5-mm cannulated screws, whereas

Group P received a PHILOS plate with 3.5-mm locking screws. Each specimen underwent a standardized loading protocol using a universal testing machine. Axial stiffness, cyclic displacement, ultimate failure load, and failure modes were recorded and statistically compared between groups.

RESULTS: No statistically significant difference was found in axial stiffness between Group P (746 ± 300 N/mm) and Group S (753 ± 256 N/mm) ($p = 1.000$). Displacement after cyclic loading was significantly greater in Group P (1.42 ± 0.3 mm) compared with Group S (0.57 ± 0.2 mm) ($p = 0.004$). The ultimate failure load was higher in Group S (2378 ± 513 N) […]

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Rerupture after flexor tendon repair of the hand and wrist: a retrospective risk factor analysis.

Marwieser SF, Vavricka S, Kaiser P, Schmidle G, Lindtner RA, Arora R, Konschake M

INTRODUCTION: Rerupture after primary flexor tendon repair of the hand and wrist is a serious complication that can substantially impair hand function and restrict activities of daily living, occupational performance, and recreation. Identifying factors associated with rerupture is essential to optimize surgical techniques and postoperative rehabilitation strategies and to improve patient outcomes.

METHODS: A retrospective observational cohort study was conducted including patients who underwent surgical repair of flexor tendon injuries of the hand and wrist at a single institution between 2010 and 2022. The study included 292 patients with 429 complete tendon injuries. Collected data comprised demographic characteristics, injury mechanism and type, anatomical location, surgical timing and technique, and postoperative immobilization and rehabilitation protocols. Associations between potential risk factors and rerupture were explored using univariable analyses, followed by a multivariable logistic regression model to identify independent predictors.

RESULTS: The overall rerupture rate was low but clinically relevant, occurring in a small proportion of patients and tendon lesions. Rerupture was more frequently observed in male patients, older individuals, and those with specific injury mechanisms. Surgical factors, particularly the need for pulley reconstruction, were associate […]

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Defining orthoplastic limb salvage centers: a systematic review.

Moussa O, Pacheco FJ, Raasveld FV, Gonzalez MR, Oflazoglu K, Ritt MJPF, Valerio IL, Rakhorst H et al.

INTRODUCTION: Limb salvage centers have increased in number over time, but lack standardized defining criteria. This systematic review aimed to assess organizational features of limb salvage centers and determine whether orthoplastic centers, in comparison to vascular limb salvage centers, represent a distinct care model that may benefit from standardization.

METHODS: We conducted a systematic review of publications related to limb salvage centers by searching MEDLINE, Embase, Web of Science, and Cochrane databases from their inception through 2024. We quantified binary data extraction as a reporting score of 26 organizational features across six structural care domains for limb salvage centers, based on a validated quality measurement framework. Organizational features differentiating distinct center types were identified to establish a quality framework for orthoplastic centers. Statistical comparisons between center types were performed using appropriate tests ($p < 0.05$).

RESULTS: Of 118 included studies, orthoplastic ($n = 43$) and vascular ($n = 48$) centers represented 77% of all studies. Recent increases in orthoplastic publications show substantial variability in organizational features. Orthoplastic center literature more frequently reported plastic surgery consultation criteria ($p < 0.001$), surgical outcomes ($p < 0.001$), and centralized network integration ($p \leq 0.006$), […]

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Male incidence rates outpacing females: results from a nationwide analysis on acetabular fractures in Germany.

Baumann F, Bärtl S, Szymski D, Hierl K, Alt V, Freigang V

INTRODUCTION: Acetabular fractures are uncommon, but serious injuries. Demographic changes may have a significant impact on planning healthcare structures to improve treatment outcomes. Aim of this nationwide, registry-based retrospective controlled study was to identify incidence trends, demographic characteristics, and care structures of patients with acetabular fractures in Germany.

MATERIALS AND METHODS: We analyzed inpatient data from the Institute for the Hospital Remuneration System (InEK). Based on 52 095 patients with primary diagnosis of an acetabular fracture between 2019 and 2024, we calculated incidence rates for different age-groups and put a spotlight on geriatric acetabular fractures (> 65 years of age).

RESULTS: Incidence rates in patients under 65 years remained stable, whereas patients over 65 years showed a significant age-dependent increase with an exponential rise in men aged 80 + with the highest incidence being 122.4/100 000 inhabitants annually. We recorded high levels of co-morbidity and nursing care dependency for elderly patients after acetabular fracture. Although 43% of patients were treated in hospitals > 500 beds, acetabular fractures were managed across all hospital sizes.

CONCLUSIONS: There is a rapidly increasing incidence of geriatric acetabular fractures, predominantly driven by elderly male patients over 80 years. Patients over 65 years […]

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Cemented total hip arthroplasty reduces early complications: a Japanese nationwide propensity-matched study.

Tanaka H, Tarasawa K, Mori Y, Baba K, Kurishima H, Fushimi K, Aizawa T, Fujimori K

INTRODUCTION: The optimal fixation method in total hip arthroplasty (THA) remains under debate. While cemented fixation has been associated with a lower risk of periprosthetic fracture, uncemented fixation predominates in Japan. This study aimed to compare early postoperative complications between cemented and uncemented fixation in elective THA using a nationwide inpatient database.

MATERIALS AND METHODS: We identified 198,102 patients aged ≥ 65 years who underwent primary THA for osteoarthritis, osteonecrosis, or rheumatoid arthritis between December 2011 and March 2023 from the Japanese Diagnosis Procedure Combination (DPC) database. After 1:1 propensity score matching for age, sex, body mass index (BMI), and Charlson Comorbidity Index, 36,859 patients were included in each fixation cohort. Surgical and medical complications, and in-hospital mortality were compared using multivariate logistic regression.

RESULTS: Cemented fixation was associated with a significantly lower risk of periprosthetic fracture (odds ratio [OR], 0.40; 95% confidence interval [CI], 0.30-0.53; $p < 0.001$), blood transfusion (OR, 0.76; 95% CI, 0.74-0.78; $p < 0.001$), and deep vein thrombosis (OR, 0.79; 95% CI, 0.74-0.84; $p < 0.001$). There were no statistically significant differences based on the predefined threshold ($p < 0.001$) in dislocation, infection, pulmonary embolism, cardiac or cerebrovascular […]

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Healthcare utilization following hip fractures based on social vulnerability status in the US: an analysis of 2016-2020 nationwide readmissions data.

Tilve R, Zhou G, Khan ST, Koroukian SM, Deren M, Piuzzi NS

INTRODUCTION: Social vulnerability (SV) influences rehabilitation and postoperative care for patients with hip fracture. However, most previous work relies on area-level measures that overlook interindividual variation. The recent adoption of ICD-10 Z-codes allows clinical identification of patient-level SV and may offer a better understanding of its impact. This study aimed to evaluate healthcare utilization, including readmissions, discharge disposition, and length of stay (LOS) in surgically treated hip fracture patients with and without clinically acknowledged SV.

METHODS: Adults surgically treated for hip fracture between 2016 and 2020 were included from the Nationwide Readmissions Database. SV was defined as having at least one documented relevant ICD-10 Z-code. Primary outcome measures included complications, LOS, discharge disposition, and 30- and 90-day readmissions, stratified by SV and evaluated using chi-square analyses. Multivariable logistic regression assessed long LOS (≥ 5 days) and discharge to home, adjusting for age, insurance/income status, and substance use.

RESULTS: Patients with SV were younger (35.6% with SV vs. 50.1% without SV were 81+), had a lower median household income (38.8% with SV vs. 25.7% without SV were in the lowest quartile), and were more often insured by Medicaid (19.3% vs. 3.8%). Alcohol/drug use disorders were significantly more preva […]

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Antibiotic prophylaxis for open lower extremity fractures: a comparative propensity-scored matched analysis of cefazolin vs. piperacillin-tazobactam.

Mody KS, Ponna AK, Santilli RT, Laychur A, Galloway J, Adams M, Reilly M, Lin S

INTRODUCTION: Optimal antibiotic prophylaxis for open fractures remains controversial, particularly regarding whether broader-spectrum regimens offer clinical advantages over cefazolin monotherapy. Despite guideline recommendations supporting first-generation cephalosporins for most open fractures, many centers routinely administer piperacillin-tazobactam. This study aims to evaluate differences in postoperative outcomes between cefazolin and piperacillin-tazobactam in lower extremity open fractures across all Gustilo-Anderson types.

MATERIALS AND METHODS: This retrospective cohort study was performed using the TriNetX database to identify adult patients with Gustilo-Anderson type I-III lower extremity open fractures who received either cefazolin or piperacillin-tazobactam. 1:1 propensity score matching controlled for age, sex, demographics, and relevant comorbidities. Outcomes were assessed at 90 days and 1 year using risk ratios (RR) with corresponding 95% confidence intervals.

RESULTS: A total of 47,692 patients met inclusion criteria prior to matching. After matching, 1,527 patients remained in each treatment group for the combined type I/II/III cohort and similar matched pairs were obtained for type I/II and type III subgroups. At 90 days, cefazolin was associated with significantly lower rates of surgical site infection (RR 0.569), osteomyelitis (RR 0.292), sepsis (RR 0 […]

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Venous thromboembolism prophylaxis in total shoulder arthroplasty: a matched cohort analysis.

Katakam A, Joshi T, Soussou T, Sirch F, Jones T, Yedikian T, Erickson J

BACKGROUND: Venous thromboembolism (VTE), including deep vein thrombosis (DVT) and pulmonary embolism (PE), is a potentially life-threatening complication following total shoulder arthroplasty (TSA). Despite the increased use of chemical prophylaxis, its effectiveness and safety in TSA populations remain unclear. This study evaluated the incidence of VTE, bleeding events, and related outcomes among TSA patients with and without postoperative chemical prophylaxis.

MATERIALS AND METHODS: A retrospective cohort study was conducted using the TriNetX database, identifying patients who underwent TSA, stratifying into two cohorts: those who received chemical VTE prophylaxis and those who did not. Propensity score matching (1:1) was employed to balance demographics, comorbidities, and other confounding variables. Outcomes, including VTE, bleeding events, prosthetic joint infection (PJI), revision, and mortality, were assessed at 30 days, 90 days, and 6 months postoperatively.

RESULTS: After matching, 9,859 patients were included in each cohort. There was no significant difference in the incidence of VTE, PE, or DVT at any time point between the groups. Patients who received prophylaxis showed a reduced risk of ischemic stroke at 30 days (HR 0.539; 95% CI 0.367–0.793; $p = 0.001$) and myocardial infarction (MI) at both 30 days (HR 0.469; 95% CI 0.306–0.718; $p < 0.001$) and 90 days (HR 0. […]

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Advancing pelvic and acetabular trauma education.

Pfeifer R, Gaensslen A, Lindahl J, Enocson A, Hildebrand F, Pape HC

Pelvic and acetabular injuries are uniquely demanding due to complex three-dimensional anatomy, the proximity of critical neurovascular structures, and the incidence of these fractures in elderly, a well as and polytrauma patients, is increasing. While existing courses focus heavily on acute surgical management and established techniques, they often omit critical subjects such as complication management, revision strategies, and interdisciplinary care. To establish an international benchmark for surgical excellence in pelvic surgery, the proposed program may utilize a competency-based framework divided into three parts: Part 1: Theoretical Foundation: Independent study of biomechanics, pathology, and operative planning, validated by a rigorous multiple-choice examination. Part 2: Surgical Skills Assessment: Evaluation of technical precision and intraoperative decision-making within a controlled wet lab environment. Part 3: Clinical Reasoning & Reflection: Expert-led discussions centered on the participant's own clinical cases, focusing on evidence-based reasoning and the management of complex revisions. By standardizing these advanced competencies, the program aims to ensure surgeons possess both the technical proficiency and the mature clinical judgment required to deliver high-quality care for complex pelvic and acetabular trauma cases.

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Mid- to long-term outcomes and survival of total hip arthroplasty using a Kerboull-type acetabular reinforcement plate: an analysis of associated factors.

Gosho S, Kaku N, Hosoyama T, Shibuta Y, Tanaka K

INTRODUCTION: Revision total hip arthroplasty (THA) for acetabular bone loss is challenging. The modified Kerboull-type (KT) plate has been adopted; nonetheless, mid- to long-term clinical outcomes remain poorly understood. We evaluated survival rates and radiographic outcomes of acetabular reconstruction using the KT plate and investigated risk factors for plate breakage or re-revision.

MATERIALS AND METHODS: We retrospectively included 120 patients (130 hips) who underwent acetabular reconstruction using the KT plate (1997-2024) and evaluated perioperative outcomes, Harris Hip Score (HHS), survival rates, and key radiographic parameters.

RESULTS: Mean age at surgery and follow-up duration were 69.0 ± 10.4 years and 107.4 ± 71.7 months, respectively. Mean blood loss and operative time were 630.2 mL and 284.3 min, respectively. HHS improved from 56.2 to 86.4. Fractures (5.3%) and dislocations (3.0%) were observed. Ten-year survival rates were 95.6% for re-revision and 93.0% for plate breakage. Failure and head migration occurred in 11 (8.5%) and 6 (4.6%) hips, respectively. Multivariate analysis identified younger age, use of morselized bone chips alone, and postoperative head migration as independent predictors of failure. In revision THA cases with ≥ 5-year follow-up and Paprosky classification type 3 A or 3B defects, age, Knight classification, and head migration differed […]

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Scaphoid waist fractures in the presence of carpal coalitions: a consistent biomechanical weak point?

Schmitt S, Winterholer D, Fritsche E

Carpal coalitions may alter intercarpal kinematics and load transfer. Reported coalition-associated scaphoid fractures show an exclusive predilection for the scaphoid waist, raising the possibility of a non-random biomechanical mechanism with focal stress concentration. These injuries have been associated with delayed union and nonunion. This pattern may justify CT-based assessment and consideration of a lower threshold for early surgical stabilization, including percutaneous screw fixation even in selected nondisplaced waist fractures. However, the currently available evidence is limited to case reports and does not allow firm treatment recommendations. Awareness of coalition anatomy and careful follow-up may be reasonable.

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Prophylactic percutaneous intra-aortic balloon occlusion reduces blood loss during anterior acetabular fracture surgery: a propensity score-matched study.

Kanezaki S, Miyazaki M, Hino A, Kawagishi M, Nishine J, Shigemi T, Kaku N

INTRODUCTION: Intra-aortic balloon occlusion (IABO) is used to control pelvic hemorrhage and has recently been adapted to elective surgery to limit intraoperative blood loss. Its role in acetabular fracture fixation, particularly via anterior approaches, remains unclear. This study evaluated whether prophylactic percutaneous IABO is associated with reduced blood loss during anterior open reduction and internal fixation (ORIF) of acetabular fractures, using propensity score matching (PSM) to minimize selection bias.

MATERIALS AND METHODS: We conducted a retrospective cohort study including 80 consecutive adult patients who underwent anterior ORIF for acetabular fractures between 2013 and 2024. Twenty-four patients received prophylactic percutaneous IABO, and 56 served as controls. One-to-one PSM (caliper 0.2 SD of the logit) was performed on demographics, fracture characteristics, and surgical factors, yielding 20 matched pairs. Outcomes included intraoperative blood loss (IBL; primary), total blood loss (TBL; gross formula), operative time, transfusion, reduction quality, perioperative complications, and IABO-related parameters.

RESULTS: IABO was associated with significantly lower blood loss both before and after matching. In matched pairs, median IBL was 525 g versus 1070 g in controls ($p = 0.004$), representing a 51% reduction, while TBL was 601 g versus 921 g ($p = 0.003$), […]

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Comparison between the accuracy of conventional and portable computed tomography-based navigation systems in total hip arthroplasty.

Asai H, Osawa Y, Takegami Y, Funahashi H, Imagama S

BACKGROUND: Although conventional computed tomography (CT)-based navigation provides excellent placement accuracy and clinical outcomes, whether recently introduced portable systems can achieve comparable results remains unclear. This study aimed to evaluate the placement accuracy and surgical outcomes of portable CT-based navigation systems.

METHODS: This study assessed 56 hips of patients that underwent total hip arthroplasty (THA) using portable CT-based navigation. Using propensity score matching based on age, sex, and body mass index, we identified 51 hips treated with portable CT-based navigation (portable CTN group) and 51 hips with conventional CT-based navigation (CTN group). The evaluation parameters included cup orientation accuracy, cup positioning, operative time, blood loss, preoperative and postoperative Japanese Orthopaedic Association scores, and complications.

RESULTS: Regarding accuracy error, the portable CTN (radiographic inclination [RI]: 2.8 ± 2.8°, radiographic anteversion [RA]: 3.8 ± 3.2°) and CTN groups (RI: 2.7 ± 1.9°, RA: 3.0 ± 2.1°) did not significantly differ. For navigation error, the portable CTN group (RI: 3.2 ± 3.2°, RA: 3.5 ± 3.1°) had significantly inferior results to the CTN group (RI: 2.2 ± 1.7°, RA: 2.3 ± 1.7°) regarding anteversion. The portable CTN group demonstrated a significantly lower accuracy, as the proportion of hips with a nav […]

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Analysis of mental health outcomes in major versus minor upper extremity amputations: a retrospective national database study.

Nedder V, Wang J, Peek K

BACKGROUND: Upper extremity amputations are associated with developing psychiatric conditions. The aim of this study is to identify differences in mental health outcomes for patients undergoing major versus minor upper extremity amputations.

METHODS: Data were obtained from the PearlDiver database between 2010 and 2022 using Current Procedural Terminology and International Classification of Diseases codes. Patients aged 10 and above without prior mental health diagnoses or antidepressant prescription records who underwent upper extremity amputations were stratified by major (shoulder disarticulation, arm, forearm, wrist, transmetacarpal) versus minor (single metacarpal, digit, phalanx) amputations. Demographic characteristics and rates of mental health diagnoses, antidepressant prescriptions, and psychotherapy care were assessed for 90 days and one year postoperatively, using Welch's T-tests and Pearson's chi-squared tests.

RESULTS: Patients with major amputations had increased risk of mental health diagnoses at 90 days (OR: 3.29, $p < 0.001$) and at one year (OR: 2.01, $p < 0.001$). Both groups had higher rates of mental health diagnoses than the general population. Patients undergoing major amputations had higher odds of starting antidepressants at 90 days (OR 3.82, $p < 0.001$) and at one year (OR 2.38, $p < 0.001$). Psychotherapy care was significantly increased after major amput […]

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20 years of blood management and VTE prevention in orthopedic surgery in the Netherlands - a nationwide survey.

Kok RY, Gerritsen JTM, Verheyen CCPM, Németh B, Ettema HB

INTRODUCTION: Advances in arthroplasty care have progressively improved patient outcomes, including reduced bleeding and venous thromboembolism rates. Consequently, blood-saving measures applied pre-, peri- and postoperatively as well as thromboprophylaxis regimens have evolved over time. To evaluate these developments, we conducted a 2023 follow-up to the national surveys from 2002, 2007 and 2012, examining current blood-saving strategies and thromboprophylaxis use in The Netherlands.

MATERIALS AND METHODS: In 2023, a questionnaire designed to match the previous surveys was distributed to all orthopedic departments in the Netherlands. The collected data were summarized and compared with previous findings.

RESULTS: Responses were received from 61% of the Dutch orthopedic departments. Blood-saving practices have shifted significantly toward the use of tranexamic acid, accompanied by a decline in other measures. In particular, the use of drains, blood transfusions, and erythropoietin supplementation has decreased substantially. Thromboprophylaxis protocols showed fewer changes, although the average duration of prophylaxis has shortened, with low-molecular-weight heparin remaining the most commonly used anticoagulant.

CONCLUSIONS: Advances in surgical techniques and postoperative care around arthroplasty appear to have reduced concerns about perioperative blood loss to a level […]

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Optimal timing for hemodialysis in relation to hip fracture surgery for patients with end-stage renal disease.

House HE, Shimizu MR, Ma IH, Cohen JB, Summers HD, Levack AE

INTRODUCTION: End-stage renal disease (ESRD) is associated with a significantly increased risk of hip fractures and a higher rate of postoperative complications and mortality, particularly in patients on hemodialysis (HD). The timing of HD relative to surgery may influence outcomes by altering acid–base balance, electrolyte levels, and fluid status. This study aimed to evaluate the effect of perioperative HD timing on postoperative complication rates in ESRD patients undergoing hip fracture surgery.

MATERIALS AND METHODS: This retrospective cohort study included patients with ESRD on HD who underwent operative treatment for hip fractures at a single Level I academic trauma center. Patient demographics, comorbidities, and perioperative variables were collected. Timing of HD was calculated as the interval in hours from the start of HD to the start of surgery. Disruptions in patients' routine HD schedules were also recorded. Univariable logistic regression was used to assess associations between HD timing and 90-day postoperative complications.

RESULTS: Forty-one patients met inclusion criteria; 25 (61.0%) experienced a postoperative complication. The interval between HD and surgery did not significantly predict postoperative outcomes. Disruption of routine HD schedules, observed in 31 patients, was also not associated with increased risk of complications (OR = 0.94, 95% CI = [0 […]

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The use of communication applications in orthopaedics: a scoping review.

Olivelle A, Borra P, Sati R, Gupta MS, Oochit K, Patel A

BACKGROUND: Effective communication between clinicians is vital in orthopaedic practice, particularly in trauma and emergency care where optimal decision making is necessary. Although communication application use in orthopaedics is common within practice, there is little research to back up its ongoing use. This scoping review aims to synthesise the existing evidence on the use of clinician-to-clinician communication applications in orthopaedic practice, focusing on its prevalence, perceived utility, diagnostic reliability and associated challenges.

MATERIALS AND METHODS: A scoping review was conducted in accordance with the PRISMA-ScR guidelines. This involved a literature search of MEDLINE, Embase, and the Cochrane Library from inception to September 2025. Studies involving clinician-to-clinician communication applications used within orthopaedic care were included. Data was extracted and synthesised using a methodological approach.

RESULTS: Seven studies published between 2016 and 2023 were included. The evidence demonstrates widespread adoption of communication applications, particularly WhatsApp, for informal clinical communication, with consistent perceptions that these platforms improve communication efficiency and coordination of care. Four studies evaluating smartphone transmitted radiographic images reported good to excellent intraobserver diagnostic agreement when […]

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Double-row wired anchors combined with "8" cross tension bands: an innovative procedure for the treatment of inferior pole fractures of the patella and its finite element and biomechanical evaluation.

Deng Z, Zhang W, Chen J, Yang Z, Gao W, Liu Y, Zhang X, Zeng W et al.

OBJECTIVE: To propose and verify the biomechanical properties of a new double-row suture anchor combined with figure-of-eight cross tension band technology for the treatment of inferior pole fractures of the patella, in order to overcome the shortcomings of traditional fixation methods in achieving stable fixation and reducing implant-related complications.

METHODS: (1) Based on clinical CT data, a three-dimensional finite element model of patellar inferior pole fracture was reconstructed, and the new fixation technology (deep double-row anchors in the proximal bone fragment combined with figure-of-8 sutures) and Kirschner wire tension band wire fixation (TBW) were simulated and analyzed. Mechanical performance in the treatment of inferior pole fracture of the patella; (2) Divide the model into two groups according to surgical methods: suture anchor (SA) fixation group and TBW fixation group; (3) Carry out finite element analysis to compare the two groups at 500 Displacement and stress distribution of the knee joint at 45°and 135°of flexion under N load.

RESULTS: Finite element analysis and biomechanical test results showed that there was no significant difference in displacement and stress between the two groups at different flexion angles.

CONCLUSION: Double-row anchors combined with figure-of-eight suture technology exhibited excellent biomechanical properties, and is a f […]

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Coordinated Return-to-Work model reduces sickness absences after hip or knee arthroplasty: a registry-based study.

Kangas P, Pamilo K, Soini S, Hirvonen M, Kervinen V, Kinnunen ML

INTRODUCTION: Working-age adults increasingly undergo total hip (THA) and knee (KA) arthroplasties. We evaluated how post-arthroplasty referral to occupational health services (OHS) using the Coordinated Return-to-Work (CRTW) model, affects return to work (RTW).

MATERIALS AND METHODS: The CRTW model was evaluated through a benchmarking controlled trial. We used the electronic records of four hospitals in Finland to identify working-age THA and total/unicondylar KA patients before (control group, N = 668) and after (intervention group, N = 536) the CRTW model was implemented. We combined these data with sickness benefits registry data. The differences between the study groups' RTW were analyzed using a Cox regression model, adjusting for age, sex, body mass index, number of special reimbursement entitlements for medicines, and earnings as covariates. Subgroup analyses included intervention participants whose sick leaves were prescribed by a surgeon according to the CRTW protocol.

RESULTS: After THA, the control group's mean RTW-duration was 87.8 days, compared to 74.9 days for the intervention group, with the mean difference being 12.9 days (95% CI 5.7–20.2). The intervention group was associated with earlier RTW; HR 1.35 (95% CI 1.13–1.61). After KA, the control group's mean RTW-duration was 107.8 days, while the intervention group's was 93.4 days, with the mean difference be […]

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Minimum 10-year outcomes of cementless total hip arthroplasty using a 32-mm femoral head with CT-based navigation: a comparative study with manual technique.

Tone S, Naito Y, Kobayashi G, Sudo A, Hasegawa M

INTRODUCTION: Computed tomography (CT)-based navigation improves the accuracy of component placement in total hip arthroplasty (THA). To date, evidence of its long-term clinical impact is limited. This study aimed to compare minimum 10-year outcomes of cementless THA performed using CT-based navigation and a manual technique with a 32-mm femoral head.

MATERIALS AND METHODS: This study included a minimum follow-up of 10 years. The CT-based group included 58 hips and the manual group included 53 hips. Postoperative cup orientation was assessed using CT. Clinical outcomes were evaluated using the Merle d'Aubigné and Postel score, Harris Hip Score (HHS), and

Forgotten Joint Score-12 (FJS-12). Kaplan–Meier analysis was used to estimate the 10-year implant survival with revision for any reason as the endpoint.

RESULTS: The CT-based group achieved a significantly higher cup placement accuracy (100% within Lewinnek's safe zone versus 49% in the manual group, $p < 0.001$). At the final follow-up, the CT-based group showed higher Merle d'Aubigné and Postel total score (16.8 vs. 16.1, $p = 0.035$) and HHS (90.3 vs. 85.0, $p = 0.015$), although the difference in HHS did not exceed the minimal clinically important difference. The FJS-12 scores did not differ significantly (71.1 vs. 62.2, $p = 0.051$). Ten-year survival rates were 98.2% in the CT-based group and 100% in the manual group (n.s.).

C […]

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Long-term outcomes and revision-free survival following robotic-assisted unicompartmental and total knee arthroplasty.

Fleisher A, Kennedy M, Trudeau M, Piergrossi D, Madrid I, Meftah M

BACKGROUND: While robotic assistance improves precision in unicompartmental (R-UKA) and total knee arthroplasty (R-TKA), long-term comparative data are limited. This study evaluated perioperative outcomes and mid- to long-term survivorship for R-UKA versus R-TKA.

METHODS: A retrospective review was performed of 163 patients who underwent primary, elective, R-UKA (n = 84) or R-TKA (n = 79) between 2011 and 2020 at a single, tertiary academic medical center, with a minimum follow-up of five years. Demographics, perioperative characteristics, and postoperative outcomes were compared between cohorts. Revision-free implant survivorship was evaluated using Kaplan–Meier analysis.

RESULTS: R-TKA procedures had longer operative times than R-UKA (131 vs. 112.5 min; $P < 0.001$) and both were performed primarily for primary osteoarthritis. Cohorts had similar 90-day ED visit rates (3.8 vs. 4.8%; $P = 1$), for both surgical (2.5 vs. 3.6%) and medical causes (1.3 vs. 1.2%), while readmission rates were also similar (2.5 vs. 3.6%, $P = 1$). Median follow-up (9.55 vs. 6.07 years) and time since surgery (12.33 vs. 7.48 years) were significantly longer for R-UKA (both $P < 0.001$). At final follow-up, revision occurred in 3.8% of R-TKA and 4.8% of R-UKA cases ($P = 1$). Time to revision was shorter for R-TKA (1.36 vs. 5.22 years; $P = 0.114$). Kaplan–Meier analysis demonstrated 5-year revision-free survi […]

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Comparison of 20-year results of total hip arthroplasty using first-generation annealed highly cross-linked polyethylene and zirconia heads in patients aged ≤ 50 and > 50 years.

Naito Y, Tone S, Kobayashi G, Hasegawa M

INTRODUCTION: This study investigated 20-year outcomes of total hip arthroplasty (THA) using zirconia heads on annealed highly cross-linked polyethylene (HXLPE) liners, comparing patients aged ≤ 50 and > 50.

METHODS: We reviewed 117 hips (100 patients) who underwent cementless THA between 2001 and 2004. After excluding follow-up < 15 years, 22 hips (≤ 50) and 61 hips (> 50) were analyzed. Merle d'Aubigné and Postel score, University of California, Los Angeles activity score, Kaplan–Meier survivorship, femoral head penetration, and oxidative degradation in retrieved liners were evaluated.

RESULTS: Patients ≤ 50 had higher activity. Three revisions for infection occurred in the > 50 group; none due to wear. Survivorship was 93% overall, 100% in ≤ 50, and 91% in > 50, with no difference. Steady-state wear rate was 0.01 mm/year in both groups. Retrieved liners showed oxidation.

CONCLUSION: Zirconia-on-annealed HXLPE THA showed excellent 20-year wear resistance and survivorship, with comparable outcomes between ≤ 50 and > 50 groups.

PubMed · DOI

Effect of a one time guideline based educational video intervention on osteoporosis related knowledge.

Hauser AL, Noriega Urena JF, Saatashvili L, Esmail F, Lange T, Schulte TL, von Glinski A

INTRODUCTION: Osteoporosis remains underdiagnosed and undertreated, partly due to insufficient patient knowledge. This study aimed to assess osteoporosis-related knowledge and to evaluate the immediate and short-term effects of a one-time educational intervention through a guideline-based video in an adult online population.

MATERIALS AND METHODS: In this prospective, randomized controlled trial, adults aged ≥ 18 years were recruited via an established online panel. Participants were randomized to an intervention group, which viewed a standardized educational video on osteoporosis, or a control group without intervention. Osteoporosis knowledge was assessed using the Facts on Osteoporosis Quiz (FOOQ) and the Osteoporosis Knowledge Assessment Tool (OKAT). Knowledge retention was evaluated in the intervention group after one week. Between-group comparisons were performed using Man-Whitney U tests, χ^2 tests, and odds ratios.

RESULTS: A total of 513 participants were included (intervention group: n = 198; control group: n = 315), with no significant differences in baseline demographic or clinical characteristics. Immediately after the intervention, osteoporosis-related knowledge was significantly higher in the intervention group compared with controls, as measured by both OKAT (11.6 vs. 8.8 points, $p < 0.0001$) and FOOQ (13.2 vs. 11.1 points, $p < 0.0001$). The proportion of partici […]

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Modified rotational wedge distal metatarsal osteotomy versus chevron osteotomy for hallux valgus: long-term radiographic and clinical outcomes.

Aybar A, Gokkus K, Özden E, Çetin MH, Çetin MÜ

BACKGROUND: Distal Chevron osteotomy is commonly used for mild-to-moderate hallux valgus, but long-term loss of correction and radiographic recurrence remain concerns, particularly when distal articular alignment and sesamoid position are not adequately restored. We compared long-term radiographic and clinical outcomes of a modified rotational wedge distal metatarsal osteotomy versus standard distal Chevron osteotomy in adults with mild-to-moderate symptomatic hallux valgus.

METHODS: In this single-center retrospective cohort study, 100 feet (100 patients) treated between 2010 and 2019 were analyzed (Modified, n = 46; Chevron, n = 54) at a mean follow-up of 101.2 ± 11.5 months. Soft-tissue balancing was standardized, with an intra-articular lateral release performed in both groups. Outcomes included radiographic measures (hallux valgus angle [HVA], intermetatarsal angle [IMA], distal metatarsal articular angle [DMAA], and medial sesamoid position), clinical scores (AOFAS, VAS), recurrence, and complications. Radiographic recurrence was defined as final HVA > 15°.

RESULTS: Final AOFAS scores were similar between groups ($p = 0.621$), and the between-group difference in final VAS pain scores did not remain significant after Benjamini-Hochberg false discovery rate (BH-FDR) adjustment ($q = 0.057$). Compared with the Chevron group, the Modified group demonstrated superior final radio […]

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Tendon gel using the film model method promotes ligament healing in rabbits.

Yoshimizu R, Nakase J, Yoshioka K, Kuzumaki T, Torigoe K, Demura S

INTRODUCTION: Tendon gel is a translucent gel-like material secreted from the ends of a severed tendon. When mechanical stress is applied to a 3-day in vivo-preserved tendon gel, it matures into type I collagen–dominant tissue similar to normal tendon. This study aimed to evaluate the effects of transplanting a 3-day in vivo-preserved tendon gel into a knee medial collateral ligament (MCL) injury site in rabbits to promote intrinsic ligament regeneration.

MATERIALS AND METHODS: Tendon gel was prepared from rabbit Achilles tendons using the film model method and harvested after 3 days of in vivo preservation. The 3-day tendon gel was transplanted into the knee MCL injury sites in another set of rabbits (n = 48). Additionally, the healing process was assessed at 1, 2, and 4 weeks postoperatively using mechanical and histological analyses. Ultimate load, peak stress, and elastic modulus were measured. Histological maturity was semi-quantitatively scored, and collagen type I and III expressions were examined by immunofluorescence staining.

RESULTS: At 2 weeks, the tendon gel group demonstrated significantly higher ultimate load than the control group (12.25 ± 4.90 vs. 5.25 ± 2.40 N; p = 0.02). The tendon gel group had greater peak stress than the control group (3.54 ± 1.44 vs. 1.69 ± 0.78 MPa; p = 0.02). Histological scores were higher in the tendon gel group than in the control […]

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Does the preoperative CPAK classification influence clinical outcomes after medial and lateral unicompartmental knee arthroplasty? Short-term outcomes and phenotype distribution in a retrospective series of consecutive cases.

Campi S, Esposito C, Manfreda A, Franceschetti E, Rizzello G, Vadalà G, Longo UG, Papalia R

INTRODUCTION: Recent evidence suggest that mechanically aligned TKAs may yield heterogeneous outcomes across CPAK categories, but whether this applies to unicompartmental knee arthroplasty (UKA) remains unclear. This retrospective study evaluated the influence of preoperative CPAK category on postoperative clinical outcomes after UKA, with additional analyses assessing CPAK distribution, residual varus alignment, and their relationship with patient-reported outcomes.

MATERIALS AND METHODS: A total of 312 consecutive UKAs (280 patients) with ≥ 1-year follow-up were analyzed. Preoperative HKA and JLO were measured on weight bearing long-leg radiographs, and patients were classified according to the CPAK system. Outcomes were compared across the six most represented CPAK groups (I – VI) in 253 patients with complete PROM data.

RESULTS: Mean postoperative OKS was 40.5 ± 5.3, with a mean ∆OKS of 21.3 ± 8.7 and 97% patients' satisfaction. No significant differences were observed among CPAK categories for OKS (p = 0.80), ∆OKS (p = 0.90), or satisfaction (p = 0.70), and alignment-based grouping (varus, neutral, valgus) showed similarly homogeneous outcomes. Residual varus ≥ 5° did not adversely affect results (p > 0.10), and no meaningful correlation was found between aHKA and postoperative OKS (r = – 0.003, p = 0.96). CPAK distribution differed significantly by sex (p < 0.001). Five […]

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Managing Vancouver B2 periprosthetic femoral fractures: is open reduction and internal fixation superior to stem revision? : A comparative analysis with a minimum 5-year follow-up.

Wang T, Xu H, Wu J, Wang X

PURPOSE: The aim of this study was to compare the long-term outcomes of stem revision (SR) and open reduction and internal fixation (ORIF) for the treatment of Vancouver B2 periprosthetic fractures of the femur.

METHODS: From June 2013 to May 2023, 56 consecutive patients were studied at our institution. Four patients were lost to follow-up, four had incomplete data. Thus, 48 cases were included in the analysis. The patients were divided into a stem revision group (SR group with 25 patients) and an open reduction and internal fixation group (ORIF group with 23 patients). The surgical complications, perioperative parameters, and 1-year mortality rates were assessed, the functional outcomes were assessed with the Harris Hip Score (HHS), and the radiographic outcomes were assessed in accordance with the Beals and Tower criteria.

RESULTS: In SR group, the mean follow-up time was 61.2 months, 36% of patients experienced complications, the mean HHS was 75.27, and 92% of the patients had "excellent–good" radiographic outcomes. In ORIF group, the mean follow-up time was 63.7 months, 21.7% of patients experienced complications, the mean HHS was 73.56, and 91.3% of the patients had "excellent–good" radiographic outcomes. The total number of postoperative complications, dislocation rate, blood loss volume, operation time and transfusion rate were lower in ORIF group, and two patients in […]

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Outcomes and associated factors of revision procedures after failed total ankle arthroplasty: a comparative cohort analysis.

Pfahl K, Eder J, Simon D, Beckers G, Holzapfel BM, Walther M

INTRODUCTION: After failed total ankle arthroplasty (TAA), revision arthrodesis (RAA) and revision arthroplasty (RTAA) are treatment options, but comparative outcome data remain limited. The aim of this study was to compare mid-term survival of RAA and RTAA after failed primary TAA and to explore clinical and radiographic factors associated with failure.

MATERIALS AND METHODS: In this retrospective cohort study, 124 patients (RAA = 72, RTAA = 52) were reviewed after failed TAA between 2006 and 2020, with a minimum follow-up of 12 months. Kaplan-Meier analysis and Cox regression were used to assess survival, and decision tree models were used as exploratory tools to identify factors associated with failure.

RESULTS: Mean follow-up was 71.6 ± 42.7 months. Surgical failure occurred in 6.45% of patients, with no significant difference (RAA 6.9%, RTAA 5.8%). Five-year survival was slightly higher for RTAA (97% vs. 93%), although durability decreased beyond 87 months. Female sex, higher BMI, and younger age were associated with failure after RAA, whereas large periprosthetic cysts and elevated BMI were related to failure after RTAA.

CONCLUSIONS: RAA and RTAA demonstrate comparable mid-term survival after failed TAA. Revision strategy selection should be individualized, considering bone stock, patient characteristics, and failure patterns.

LEVELS OF EVIDENCE: III.

PubMed · DOI

Intramedullary headless compression screw fixation for metacarpal fractures: a retrospective clinical and radiological study.

Ku■cu B, K■na■ M

BACKGROUND: Intramedullary headless compression screw fixation has gained increasing popularity for the surgical treatment of metacarpal fractures owing to its minimally invasive nature and potential for early mobilisation without prolonged postoperative immobilisation. However, comprehensive data on clinical and radiological outcomes remain limited.

METHODS: This retrospective study included patients with metacarpal shaft or neck fractures treated with retrograde intramedullary headless screw fixation. No postoperative immobilisation was applied, and early active mobilisation was initiated in all patients on the day of surgery. Demographic data, fracture characteristics, operative details, pre- and postoperative angulation, radiological union time, functional outcomes, QuickDASH score, and complications were recorded. Preoperative and postoperative angulation values were compared using a paired Student’s t-test.

RESULTS: A total of 37 patients were included, with a mean age of 37.5 ± 15.4 years. The majority of fractures involved the fifth metacarpal (70.3%). The most common mechanisms were direct impact/blunt trauma and falls. Mean operative time was 12.8 ± 3.9 min (range, 7–21 min). Mean preoperative angulation of 41.2° ± 12.2 improved significantly to 0.2° ± 0.71 postoperatively ($p < 0.001$). Mean radiological union time was 5.6 ± 0.8 weeks (range, 4–7 weeks). At final fol […]

PubMed · DOI

Persistent extensor strength deficit despite acceptable clinical outcomes after surgical treatment of patella fractures: a mid- to long-term follow-up study.

Dalaslan RE, Yücel MO, Sa■lam S, Ar■can M, Karaduman ZO, Sav ■

INTRODUCTION: To evaluate mid- to long-term clinical outcomes together with objective isokinetic measurements of knee flexor and extensor muscle strength in patients who underwent surgical treatment for AO/OTA type C patella fractures, and to investigate the relationship between fracture severity, clinical scores, and strength deficits. We hypothesized that significant quadriceps strength deficits would persist despite acceptable clinical outcomes and that these deficits would be associated with worse pain and functional scores.

MATERIALS AND METHODS: This retrospective cohort study included 58 patients treated surgically for patella fractures between 2015 and 2024, with a median follow-up duration of 60 months (IQR: 32-94; range: 13.1-132.0 months). The primary outcome was extensor muscle strength deficit measured by isokinetic dynamometry. Secondary outcomes included flexor strength deficit, Lysholm score, visual analog scale (VAS), and knee range of motion. Isokinetic dynamometry at 60°/s was used to measure concentric flexor and extensor peak torque values, and strength deficits were calculated by comparison with the contralateral limb. Outcomes were compared among fracture subtypes (C1, C2, C3), and correlations between strength deficits and clinical scores were analyzed. Non-parametric tests (Kruskal-Wallis) and partial correlation analysis (Pearson, adjusted for age) we […]

PubMed · DOI

Periacetabular osteotomy of the hip: an 8-year follow-up of 96 consecutive cases.

Enocson A, Wallensten R, Lundblad H

INTRODUCTION: A periacetabular osteotomy (PAO) is a joint-preserving surgical option for treatment of acetabular dysplasia. The procedure aims to prevent, or at least delay, the development of osteoarthritis, and subsequent need for total hip arthroplasty (THA). The conversion rate to THA differs widely in the literature, but most of the studies have few patients, and the follow-up time is often short for THA as an endpoint. The aim of this study was to evaluate the long-term outcome after PAO surgery with the rate of conversion to THA as the primary outcome.

MATERIALS AND METHODS: Patients ≥18 years that underwent a PAO operation at the Karolinska University Hospital in Stockholm, Sweden from 2006 to 2022 were included. Radiological signs of hip osteoarthritis, and the lateral center-edge angle (LCEA) was calculated on pre- and postoperative radiographs or CT-scans. The national Swedish Arthroplasty Register was used to find cases who had a secondary operation with THA.

RESULTS: The number of cases included was 96. Median age was 30 (18–46) years, and 84% (n = 81) were females. Median follow-up time was 99 (17–227) months (8 years). A total of 21 (22%) cases had a secondary THA. Cox regression analyses identified that age ≥ 30 years and smoking was associated with THA reoperation in both uni- (HR 2.8, CI 1.1–7.3, HR 3.7, CI 1.0–13) and multivariable (HR 4.0, CI 1.1–15, HR 7. […]

PubMed · DOI

Time-zero stability of uncemented standard versus intermediate revision stems after extended trochanteric osteotomy (ETO): a biomechanical study.

Hax J, Zderic I, Gueorguiev B, Kalbas Y, Leunig M, Pape HC, Rüdiger HA

INTRODUCTION: This time-zero biomechanical study investigates femoral stem stability after extended trochanteric osteotomy (ETO) without fragment refixation, representing a worst-case scenario. ETO is used in complex femoral revisions to improve exposure while preserving soft tissue and neurovascular structures. In the setting of a short ETO, surgeons may choose between standard (STD) and intermediate revision (REV) stems, although their primary stability remains unclear. Clinically, STD are still used after short ETO with low rates of aseptic loosening, aiming for proximal bone loading to reduce stress shielding. We therefore hypothesized that STD would demonstrate non-inferior axial and torsional stability compared with REV in a short open ETO model.

MATERIALS AND METHODS: A biomechanical model using SYNBONE® femurs and uncemented, collared, fully coated triple-tapered STD and REV 135° CCD was established. A standardized 60 mm ETO (vastus ridge to distal end of ETO) was performed and left unrepaired to simulate a worst-case scenario. Four groups (n = 6 each) were tested: (1) STD and (2) REV under axial loading (500 cycles at 20° adduction up to 500 N at 1 Hz, then static loading to failure), and (3) STD and (4) REV under torsional loading (500 cycles up to 10Nm internal rotation at 1 Hz, then failure testing). Axial stiffness (N/mm), torsional stiffness (Nm/deg), and failure […]

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Functional and quality-of-life outcomes after metal-on-polyethylene articulating spacer implantation for periprosthetic knee infection: a retrospective evaluation of a prospectively collected cohort.

Balato G, Festa E, Ascione T, De Mauro D, Di Gennaro D, Mariconda M

INTRODUCTION: This study aims to evaluate functional and quality-of-life (QoL) outcomes in patients with periprosthetic joint infection (PJI) undergoing implantation of a metal-on-polyethylene articulating knee spacer as part of a planned two-stage revision, and to explore preoperative predictors of meaningful postoperative improvement.

METHODS: This was a retrospective observational study including 108 consecutive patients with chronic PJI treated with a metal-on-polyethylene articulating spacer between 2016 and 2020 and followed prospectively for clinical outcomes. QoL and joint-specific function were assessed preoperatively and before planned reimplantation using EQ-5D-5 L, Western Ontario and McMaster University (WOMAC), and Knee Society Score (KSS). Minimum clinically important difference (MCID) thresholds were derived using a distribution-based method. Receiver operating characteristic (ROC) analysis identified baseline predictors of achieving MCID.

RESULTS: All outcome measures improved significantly after spacer implantation ($p < 0.001$), exceeding MCID thresholds for pain and function. The largest EQ-5D-5 L domain gain was for pain/discomfort. Predictive thresholds for achieving MCID were an EQ-5D-5 L index ≤ 0.44 and WOMAC index ≥ 42.5. Female sex was associated with better preoperative and postoperative WOMAC and KSS scores.

CONCLUSION: Metal-on-polyethylene articu […]

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Different sagittal cutting plane orientations do not affect posterior tibial slope in open wedge high tibial osteotomy: a Sawbone model study.

Gurbanov E, Cochard B, Bauer D, Miozzari H, Tscholl P

PURPOSE: Control of the posterior tibial slope (PTS) is crucial in high tibial osteotomy (HTO), since unintended changes can alter the normal knee kinematics and ligament loading. This study investigated the influence of sagittal cutting plane orientations on PTS in bi-planar medial open wedge HTO (MOWHTO) on a Sawbone model.

METHODS: Sixty Sawbone® left-tibia models (A Pacific Research Company, USA) were used to perform biplanar MOWHTO with three sagittal orientations: (1) parallel to the medial tibial plateau (PO); (2) 10° anterior-inclined osteotomy (AIO); (3) 10° posterior-inclined osteotomy (PIO). The biplanar cut was performed distal to the tibial tuberosity in half the specimens and proximal in the remainder. Customized 3D-printed cutting guides ensured reproducibility. PTS and valgus correction were measured pre- and post-osteotomy using a navigation system. Group- and subgroup-specific ∆PTS were analyzed using a Bayesian multilevel model. Minimal clinically important difference (MCID) was set at 2.5°. Bayesian “power” analysis (assurance) was used to assess the sensitivity of ∆PTS and pairwise differences relative to the MCID.

RESULTS: Across the three sagittal orientation and their distal/proximal subgroups, ∆PTS and pairwise ∆PTS did not exceed the predefined MCID of 2.5°. Bayesian assurance of equivalence exceeded 86% in every unit and pairwise comparison, indicat […]

PubMed · DOI

Comparative assessment of the reaming characteristics of two different technologies for intramedullary bone graft harvesting.

Guttau S, Lange M, Finze R, Beimel C, Saifzadeh S, Gospos J, Wille ML, Kobbe P et al.

INTRODUCTION: The study aimed to compare the biomechanical impact and reaming performance of two intramedullary bone graft harvesting techniques: the established Reamer-Irrigator-Aspirator 2 (RIA 2) system and a novel aspirator+reaming-aspiration (ARA) concept.

MATERIALS AND METHODS: In a preclinical in vivo sheep model, sixteen femora were assigned to either the RIA 2 or ARA group (n = 8 each). Biomechanical testing, computed tomography (CT)-based 3D bone geometry analysis, and fracture line assessment were performed postoperatively. Bending stiffness of the reamer shafts, cortical wall thickness pre- and post-reaming, and torsional stability of reamed femora were evaluated. Statistical analysis included TOST, Mann-Whitney U, and Shapiro-Wilk tests, considering values of $p < 0.05$ as statistically significant.

RESULTS: Both reaming systems produced the greatest cortical thinning medially in the midsections of the analyzed region. No statistically significant differences were observed in postoperative cortical wall thickness, bone removal rates, or torsional stability between groups (all $p > 0.05$). Bending stiffness was significantly higher for the RIA 2 system ($p = 0.004$). In both groups, fractures predominantly initiated medially or at sites of previous drill holes. The ARA concept demonstrated equivalent biomechanical and reaming performance compared to the RIA 2 system, wi […]

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External fixation versus reverse shoulder arthroplasty for proximal humerus fractures in the elderly: a retrospective comparative study.

Vadalà A, Benelli C, Suraci F, Carta B, Baldassari G, Maffulli N

BACKGROUND: The optimal management of displaced proximal humerus fractures (PHFs) in the elderly remains a subject of ongoing debate. This study aims to compare the clinical and functional outcomes of biologically driven external fixation (EF) versus functional joint replacement via reverse shoulder arthroplasty (RSA) in patients aged 65-80 years.

METHODS: A retrospective comparative study was conducted on patients with displaced Neer two- or three-part PHFs treated between 2015 and 2022. The final analysis included 67 patients: Group A (EF; n=34) and Group B (RSA; n=33). Clinical and functional outcomes were quantified using the Constant-Murley Score (CMS), Simple Shoulder Test (SST), American Shoulder and Elbow Surgeons (ASES) score, Disabilities of the Arm, Shoulder and Hand (Quick-DASH) score, and Range of Motion (ROM) assessment. Patient satisfaction was evaluated via a 5-point Likert scale.

RESULTS: Both cohorts were homogeneous regarding mean age (72.2 ± 3.5 vs. 73.4 ± 3.8 years). Group A demonstrated significantly shorter mean operative times (48.2 ± 10.5 vs. 92.4 ± 15.2 min; $p < 0.001$) and a lower requirement for blood transfusions (0% vs. 12.1%; $p < 0.05$). At the final follow-up, both cohorts achieved comparable CMS (58.4 ± 8.2 vs. 55.2 ± 10.1; $p = 0.42$), SST (7.6 ± 1.5 vs. 7.0 ± 1.4; $p = 0.38$), and ASES scores (69 ± 7.8 vs. 65 ± 9.4; $p = 0.51$). Group A exhibited si […]

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Biomechanical comparison of palmar plate plus headless compression screw versus radiopalmar double plating in AO/OTA 23-C2.1 distal radius fractures.

Riegner P, Spiegel C, Kohler FC, Kielstein H, Zderic I, Gueorguiev-Rüegg B, Lenz M, Weschenfelder W

INTRODUCTION: This biomechanical study compared the fracture stability of a palmar plate combined with a headless compression screw (HCS) with that of a radiopalmar double-plate construct for AO/OTA 23-C2.1 distal radius fractures with metaphyseal defect zones.

MATERIALS AND METHODS: Eleven matched pairs of cryopreserved human radii were prepared with standardized AO/OTA 23-C2.1 fractures. Left radii (n = 11) were fixed with a palmar plate plus HCS, while right radii (n = 11) received a radiopalmar double-plate construct. Construct stiffness and axial displacement were assessed using a universal testing machine. Interfragmentary range of motion (ROM) and rotation (ROT) were quantified using an optical three-dimensional motion-tracking system. Measurements were obtained before and after 5000 cycles of dynamic axial loading at 150 N. Two specimen pairs were excluded due to early failure or incomplete data acquisition.

RESULTS: Both constructs demonstrated comparable stiffness and axial displacement, with no implant loosening or hardware failure observed. Interfragmentary ROM did not differ significantly between groups. However, the plate–HCS construct showed greater variability in rotational parameters. Initial radial-shaft rotation was significantly greater in the plate–HCS group (1.14° vs. 0.51°, $p = 0.02$). After cyclic loading, ulnar-shaft rotation increased significantly in […]

PubMed · DOI

Differences in pelvic tilt and spinal alignment according to age in patients with hip dysplasia.

Funahashi H, Osawa Y, Takegami Y, Ido H, Asamoto T, Imagama S

INTRODUCTION: We aimed to establish age-related characteristics of posture-dependent pelvic tilt and standing spinopelvic alignment in patients with symptomatic hip dysplasia (HD) from adolescence to middle age.

MATERIALS AND METHODS: A total of 101 patients with HD indicated for eccentric rotational acetabular osteotomy (ERAO) between 2016 and 2024 were included. Pelvic tilt in supine and standing positions was quantified using the anterior pelvic plane (APP) angle estimated from computed tomography-based 3D pelvic models matched to supine and standing anteroposterior hip radiographs. The change from supine to standing was calculated as standing minus supine; negative values indicated a posterior change. Three-dimensional acetabular coverage during postural transition was compared across age groups. Standing radiographic hip and spinopelvic parameters were also examined using multivariable regression analysis to identify factors associated with age.

RESULTS: Age was correlated with the supine-to-standing change in the APP angle ($r = 0.47$, $p < 0.001$). The change shifted closer to zero with increasing age, with younger patients showing a larger posterior change compared with older patients. Patients in their teens showed a more pronounced decrease in anterosuperior acetabular coverage at the 2 o'clock position compared with those in their 40 s ($p = 0.017$) and 50 s ($p = 0.040$). […]

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Are new implants for patellar fractures reasonable? A biomechanical comparison of intramedullary locking nails versus tension-band osteosynthesis.

Gercek N, Arand C, Glockner C, Nienhaus M, Gercek E, Hopf J, Rommens PM, Gruszka D

BACKGROUND: Patellar fractures are predominantly treated surgically by open reduction and internal fixation using classic tension band osteosynthesis. Biomechanical testing was conducted to examine transverse patellar fracture stabilization using locked intramedullary nail systems in comparison to classic tension band osteosynthesis.

METHODS: Twenty-four Sawbones® with transverse patellar fractures type AO 34-C1.1 were divided equally into three test groups. Three principles for fracture stabilization were used: classical tension band osteosynthesis and newly developed locked intramedullary double nail- and single nail- prototypes. Knee motion (0°/0°/90°) was simulated in a dynamic test set-up using a “single muscle model” by applying tensile forces up to 300 N. The widening and the symmetry of the fracture gap were investigated over 1000 motion cycles using a video-optical system. The significance level was set at $\alpha = 0.05$ for all statistical evaluations using the Bonferroni–Holm corrected unpaired unilateral t-test and mixed ANOVA with post-hoc Tukey test.

RESULTS: The fixation principles showed different maximum fragment displacements of $M = 2.04 \pm 0.67$ mm using classical tension band osteosynthesis, $M = 1.34 \pm 0.74$ mm using intramedullary single nail system and $M = 0.55 \pm 0.31$ mm using intramedullary double nail system. There was a statistically significant change in widt […]

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Post-traumatic Candida infection leading to bilateral wrist rice body tenosynovitis: a case report and literature review.

Wang S, Ro H, Xu H, Fang L, Li X

BACKGROUND: Rice body tenosynovitis is a rare condition of uncertain etiology characterized by the formation of multiple fibrinous particles within the synovium or synovial fluid. It is frequently associated with chronic inflammation due to infection or rheumatoid arthritis and most commonly affects large joints such as the shoulder and knee. Simultaneous bilateral wrist involvement following trauma is exceptionally uncommon.

CASE PRESENTATION: A 59-year-old man presented with persistent bilateral wrist swelling for 10 months after a fall. MRI demonstrated extensive fluid accumulation and synovial thickening within the flexor tendon sheaths of both wrists. Surgical exploration revealed numerous yellow rice-like loose bodies and inflammatory proliferative tissue in the tendon sheaths. Pathology confirmed inflammatory granulation tissue with fibrinous exudate, and fungal culture identified Candida species infection. The left wrist recovered well after the initial surgery, but the right wrist recurred after 6 months and required a second thorough debridement, after which it resolved completely.

CONCLUSIONS: This report describes a rare case of bilateral wrist rice body tenosynovitis secondary to post-traumatic Candida infection with unilateral postoperative recurrence. It emphasizes that incomplete removal of inflammatory tenosynovium remains the primary cause of recurrence, eve […]

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In-hospital mortality among orthopedic trauma patients during and outside the COVID-19 pandemic: a nationwide population-based analysis of 2.67 million hospitalizations in Poland.

Jojczuk M, Naylor K, Filipek K, Dolliver I, Toborek K, Goniewicz K

INTRODUCTION: Orthopedic trauma is a leading contributor to morbidity and mortality worldwide. The COVID-19 pandemic created unprecedented disruptions to hospital services, but its indirect effects on in-hospital mortality among orthopedic patients remain poorly quantified.

METHODS: We conducted a nationwide retrospective analysis of 2 674 263 orthopedic hospitalizations in Poland between 2019 and 2023. Hospital admissions were classified as occurring during COVID-19 (19 months across five national pandemic waves) or non-COVID periods (41 months). We examined overall, age-stratified, and injury site–specific mortality using descriptive statistics, Z-tests for proportions, and effect size estimates including risk difference, relative risk, odds ratios, and number needed to harm.

RESULTS: Overall in-hospital mortality was 1.06% (28 441 deaths). Mortality was significantly higher during the COVID-19 period than in non-COVID months (1.35% vs. 1.01%; RR 1.34, 95% CI 1.30–1.38; NNH 290). Age strongly modified outcomes: excess risk was minimal in young adults (18–40 years; NNH ~ 3 700), moderate in middle-aged patients (41–65 years; NNH ~ 621), and greatest among older adults (65 + years; NNH ~ 111). Mortality rose across all injury sites, with the largest absolute excess in head trauma (RD 0.68% points; NNH 148) and the steepest relative increase in chest injuries (RR 1.48, 95% CI […]

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Extra-long femoral heads as a surrogate marker for revision risk in primary total hip arthroplasty.

Beckers G, Simon D, Grimberg A, Wu Y, Steinbrück A, Holzapfel BM

AIMS: The effect of femoral head length on implant survival in total hip arthroplasty (THA) has been little studied so far. Longer heads may increase taper corrosion and reflect intraoperative complexity. This study evaluated factors associated with the use of extra-long heads (≥ XL) and their impact on implant survival.

METHODS: We analyzed 562,001 primary THA from the German Arthroplasty Registry. Subgroup analyses were performed by hospital annual primary THA volume (≤ 250, 251–500, ≥ 501), surgical indication (primary osteoarthritis [OA] vs. femoral neck fracture [FNF]), and fixation method (cemented vs. cementless). Logistic regression identified factors associated with ≥ XL head use, and implant survival was compared between head lengths using Kaplan–Meier analysis in both subgroups and the overall cohort.

RESULTS: The use of ≥ XL femoral heads decreased with increasing hospital volume (5.4% low, 4.5% medium, 3.0% high; $p < 0.001$). Rates were higher in FNF than OA across all volumes (8.1% vs. 4.7% in low-volume hospitals; 5.0% vs. 2.7% in high-volume hospitals). Cemented fixation was independently associated with higher odds of ≥ XL head use (OR 1.14, 95% CI 1.09–1.18, $p < 0.001$), with additional predictors including male sex (OR 2.13, 95% CI 2.06–2.19), BMI ≥ 40 (OR 1.94, 95% CI 1.77–2.12), higher Elixhauser comorbidity score (OR 1.09, 95% CI 1.04–1.15), and surgery fo […]

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Lateral unicompartmental knee arthroplasty through a medial parapatellar approach: surgical technique and mid-term results in 108 patients.

Leggieri F, Braconi L, Chirico M, Salari P, Baldini A

INTRODUCTION: Lateral unicompartmental knee arthroplasty (L-UKA) is typically performed through a lateral parapatellar approach. Few studies of L-UKA performed through a medial approach are published. We aimed to present surgical tips, and mid-term clinical results and survivorships of L-UKA performed through a medial approach.

METHODS: We retrospectively reviewed single-centre single-surgeon data from L-UKA using medial parapatellar approach between 2016 and 2023. Patients with follow-up < 12 months were excluded. Primary endpoint was implant survivorship (time from index surgery to revision or last follow-up). Secondary endpoints included the rates of patients achieving Patient Acceptable Symptom State (PASS) thresholds: ≥ 67.5 for Knee Society Score-Knee (KSS-K) and ≥ 70.5 for Knee Society Score-Function (KSS-F). Survival analysis was used for implant survivorship. Logistic regression was performed for factors potentially associated with failure to achieve the PASS thresholds. Significance was set at $P < 0.05$.

RESULTS: Among 110 patients initially identified, two died before 12-month follow-up and were excluded from clinical outcome analysis, leaving 108 patients for functional assessment at a mean 47.6 months ± 24.5 of follow-up (range 12–96.7 months). No complications were reported during hospitalization. There were four failures (3.7%) overall: two cases (1.8%) of OA pr […]

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Machine learning-based identification and ranking of risk factors for lumbar paraspinal muscle atrophy.

Schönnagel L, Folkerts T, Guven A, Chiapparelli E, Zhu J, Camino-Willhuber G, Caffard T, Arzani A et al.

BACKGROUND: Recent studies highlight the crucial role of the paraspinal musculature (PM), particularly the multifidus (MF), in spinal health and patient outcomes. However, factors associated with PM atrophy and their relative importance remain unclear. To address this gap, we analyzed factors linked to PM atrophy in patients undergoing lumbar fusion using machine learning, aiming to clarify the multifactorial mechanisms underlying this condition.

METHODS: Fatty infiltration (FI) of the lumbar MF was measured as a proxy for muscular atrophy in patients undergoing lumbar spinal fusion. Two machine learning models, logistic regression and extreme gradient boosting (XGBoost), were trained to predict severe FI (> 50%) of the MF. Model performance was evaluated on unseen test data using receiver operating characteristic (ROC) analysis and Brier score, and predictor importance was assessed via SHAP (SHapley Additive exPlanations).

RESULTS: The study included 316 patients, primarily treated due to lumbar spinal stenosis. Both machine learning models effectively predicted severe MF atrophy, with an area under the curve (AUC) of 0.83 (95% CI 0.74–0.83) for the logistic regression and 0.88 (95% CI 0.81–0.88) for the XGBoost. In the logistic regression model, only sex, age, and facet joint degeneration were significant predictors. The XGBoost model identified the same top three variables […]

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Artificial intelligence in orthopedic trauma surgery: a scoping review of current applications and research gaps.

Wurm LM, Ertel W, Laue D

BACKGROUND: Artificial intelligence (AI) is rapidly transforming clinical decision-making, yet its role in orthopedic trauma surgery remains fragmented and unevenly validated in clinical practise.

METHODS: We conducted a PRISMA-SCR–compliant scoping review using a systematic search of the Semantic Scholar, OpenAlex and PubMed corpus via Elicit (497 records). Studies were eligible if they applied AI or machine-learning methods to traumatic orthopedic conditions, included ≥ 10 human subjects, reported quantitative performance metrics, and represented original research. After title/abstract and full-text screening, 146 studies were included. Data on study characteristics, AI methodology, clinical application, validation strategy, performance metrics, explainability and translational maturity were synthesized descriptively.

RESULTS: Research output increased sharply after 2017, with 52% of all studies published since 2022. Most studies were retrospective (≈ 99%). Deep learning dominated the field (61%), particularly for fracture detection and classification, while classical machine-learning models were mainly used for outcome prediction. Internal validation was reported in 85% of studies, whereas only 15% clearly performed external or multicenter validation; true prospective clinical testing was rare (1.4%), and only a small subset of models had been implemented in practice (3.4% […]

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Comparative evaluation of additively manufactured PLA models as cost-effective, non-inferior alternatives to composite bones for pull-out testing.

Demirta■ U, ■yibilgin O, Bayam L, Genç E

INTRODUCTION: The purpose of this study was to perform biomechanical pull-out comparisons between composite bone blocks and additively manufactured bone models using locked plate and screw systems. The hypothesis posited that the mean pull-out force of the 3D-printed models would demonstrate non-inferior pull-out strength compared to standard composite bones within a clinically acceptable 15% margin. A standardized pull-out mechanism was developed to evaluate its biomechanical reliability and its potential as a customizable alternative.

MATERIALS AND METHODS: The study compared three groups: standard composite (Sawbones) blocks and two types of additively manufactured polylactic acid (PLA) blocks (grid and gyroid infill pattern), each measuring the same dimensions with 30% infill and mimicking a Sawbones block with 2 mm cortical thickness. A 6-hole LC-DCP titanium plate was fixed onto bone block models with dimensions of 29 × 170 × 42 mm using four screws, each measuring 36 mm in length, which were inserted perpendicularly and tightened to the same torque levels. Steel cables placed under the plate were connected to a mechanical testing setup, and pull-out tests were conducted to evaluate fixation strength. Seven replicate specimens were fabricated to ensure repeatability and reliability of the results.

RESULTS: Mean pull-out forces were 1784.4 ± 79.9 N for Sawbones, 1698.5 ± […]

PubMed · DOI

Can EOS and lateral pelvic radiographs reliably identify patient with reduced pelvic roll back at risk for dislocation ?

Spilo K, Miller TT, Sterneder CM, Myers A, Boettner F

INTRODUCTION: Hip instability and dislocation post-THA is a prevalent complication leading to revision surgeries. Patients are considered "high-risk" if the sit-to-stand sacral slope change is less than 10 degrees. The current study analyses if lateral radiographs and EOS imaging result in similar measurements and identify the same at-risk patients.

MATERIALS AND METHODS: In a retrospective study consisting of 59 hips (58 patients), spinopelvic measurements were obtained from lateral sitting and standing pelvic radiographs and EOS sitting and standing (whole body) films taken during the same preoperative assessment. Measurements on each images set were performed to assess variation in measurements between the two modalities.

RESULTS: The average measured sacral slope difference between the two imaging methods was 4.5° (standing) and 6.6° (sitting), with a sit-to-stand difference of 7.2°. Lateral radiographs identified 16 patients at high-risk for dislocation and EOS identified 15 high risk patients; however, only 9 patients overlapped between the two modalities. There was a significant postural difference between the two modalities: average spinal angle difference was 10.4°(p= 0.0026).and femoral angle off the horizonal was 7.4°(p= 0.16).

CONCLUSION: There are discrepancies between EOS and conventional lateral radiographs in measuring sacral slope and change in sacral slope […]

PubMed · DOI

Prevalence and risk factors for stem subsidence following reverse shoulder arthroplasty using on-lay press-fit short stems.

Nishiura R, Nakazawa K, Manaka T, Hirakawa Y, Ito Y, Terai H

INTRODUCTION: In reverse shoulder arthroplasty (RSA), press-fit short stems offer benefits such as reduced operative time and bone preservation. However, stem subsidence has been reported, although the risk factors remain unclear. This study aimed to determine the prevalence and risk factors for stem subsidence in RSA with on-lay press-fit short stems, hypothesizing that a lower filling ratio (FR) would correlate with subsidence.

MATERIALS AND METHODS: This retrospective analysis included patients who underwent RSA using on-lay press-fit short stems between 2017 and 2023, with a minimum follow-up of 2 years. Subsidence was defined as a ≥ 5 mm reduction in the distance from the tip of the greater tuberosity to the distal end of the stem on anteroposterior radiographs between the immediate postoperative and final follow-up evaluations. FR was assessed at the metaphyseal (FRmet) and diaphyseal (FRdia) levels. Neutral alignment was defined as a stem-shaft angle within 5°. Clinical outcomes included range of motion, Constant score, and American Shoulder and Elbow Surgeons score.

RESULTS: This study enrolled 151 patients: 65 with cuff tear arthropathy, 54 with massive rotator cuff tears, 29 with glenohumeral osteoarthritis, and three with shoulder dislocations. Osteoporosis was identified in 37 patients. Subsidence occurred in 20 patients (13.2%). The mean FRmet was significantly l […]

PubMed · DOI

Exploratory analysis of depressive symptom trajectories before and after hip or knee arthroplasty in geriatric patients.

Schiegl J, Bammert P, Maderbacher G, Reinhard J, Pagano S, Michalk K, Grifka J, Kappenschneider T

BACKGROUND: Depressive symptoms are prevalent among patients with osteoarthritis (OA), particularly in geriatric patients. Pain and mood are closely interconnected, with chronic joint pain contributing significantly to psychological distress. Total hip arthroplasty (THA) and total knee arthroplasty (TKA) are established procedures that improve pain and physical function and may also influence depressive symptoms. The objective of this study was to evaluate changes in depressive symptoms following THA and TKA in geriatric patients.

METHODS: In this prospective pilot study, we analysed data from 143 participants enrolled in the ongoing Special Orthopaedic Geriatrics (SOG) trial, funded by the German Federal Joint Committee (G-BA). Depressive symptoms were assessed using the Geriatric Depression Scale (GDS) preoperatively and at 3 days, 7 days, 4 weeks, and 3 months after surgery. Depressive symptoms were analysed in two predefined groups: the total sample (GDS 1–15) and patients with elevated baseline symptoms (GDS 6–15). Statistical analysis included the Friedman test for repeated measures, followed by post-hoc testing.

RESULTS: In the overall cohort, median GDS scores decreased from 3 at baseline to 2 at the 3-month follow-up ($p < 0.001$). In the subgroup with elevated baseline symptoms (GDS 6–15), median scores declined from 8.5 at baseline to 4 at 3 months ($p < 0.001$). Impro […]

PubMed · DOI

Concurrent validation of a novel intraoperative navigation platform for total knee arthroplasty.

Demeulenaere M, Taylan O, Bia■y M, Van de Vyver A, Louwagie T, Peersman G, Scheys L

INRODUCTION: This cadaveric study aimed to assess the accuracy of the Next-AR navigation platform in terms of implant alignment, kinematics, and collateral ligament elongations in comparison to gold standard measurement tools.

MATERIALS AND METHODS: Computed tomography of 7 fresh-frozen legs were acquired for patient-specific guide design, identification of bony landmarks and collateral ligaments to acquire real-time feedback for bone resection, alignment and collateral ligament elongations using Next-AR system. The specimens underwent medially-stabilized TKA (Medacta) based on the mechanical alignment technique. The difference between target and achieved implant alignment parameters was assessed. The relative change in tibiofemoral kinematics, medial collateral ligament- (MCL), and lateral collateral ligament (LCL) elongations acquired with the Next-AR system were compared to reference values acquired with bone-pin mounted motion trackers (Vicon, UK) while performing quasistatic squatting in an ex vivo knee simulator.

RESULTS: The mean differences between achieved and target hip-knee-ankle angle, femoral varus, femoral flexion, femoral rotation, tibial varus, and tibial slope measured -1.5°, 0.5°, 0.07°, -0.14°, 0.79° and -0.36°, respectively, with no statistically significant differences found ($p \geq 0.12$). Knee flexion ($p=0.32$) and valgus orientation ($p=0.63$), obtained from […]

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Combined iPACK and adductor canal block versus two-level erector spinae plane block in elderly patients undergoing total knee arthroplasty: a randomized, triple-blinded clinical trial.

Pietraszek P, Reysner T, Gola W, Kluzik A, Perek A, Marsza■ek-Buko J, Reysner M

BACKGROUND: Effective postoperative pain management after total knee arthroplasty (TKA) in elderly patients is challenging because of their increased susceptibility to opioid-related adverse events. Motor-sparing regional anesthesia techniques such as the adductor canal block (ACB), the interspace between the popliteal artery and the posterior capsule of the knee (iPACK) block, and the erector spinae plane block (ESPB) may improve analgesia while preserving quadriceps strength and modulating the systemic stress response.

METHODS: In this single-center, prospective, randomized, triple-blinded trial, 60 patients aged 65–100 years undergoing primary TKA under spinal anesthesia were allocated to receive either combined iPACK + ACB (20 mL of 0.2% ropivacaine for each block) or a two-level lumbar–sacral ESPB at L1 and S1 (20 mL of 0.2% ropivacaine at each level). The primary outcome was total opioid consumption within 48 h, expressed in morphine milligram equivalents (MME). Secondary outcomes included time to first rescue opioid, Numerical Rating Scale (NRS) pain scores, neutrophil-to-lymphocyte ratio (NLR), platelet-to-lymphocyte ratio (PLR), and quadriceps strength (Medical

Research Council scale).

RESULTS: Compared with two-level ESPB, iPACK + ACB significantly reduced total opioid consumption (9.7 ± 5.4 vs 12.9 ± 6.9 mg MME; p = 0.0348) and prolonged the time to first rescue op […]

PubMed · DOI

Minimally invasive versus open talonavicular arthrodesis: comparable functional outcomes in a retrospective comparative cohort.

Werneburg F, Arbab D, Zeh A, Delank KS, Gutteck N

BACKGROUND: Talonavicular (TN) arthrodesis is an established procedure for symptomatic TN joint pathology and medial column dysfunction. Open techniques provide direct visualization but may be associated with approach-related soft-tissue morbidity. Minimally invasive surgery (MIS) has been proposed to reduce tissue disruption; however, comparative clinical evidence focusing on functional outcomes remains limited. This study compared functional outcomes between MIS and open TN arthrodesis.

METHODS: This retrospective cohort study included 56 feet (55 patients) treated with TN arthrodesis between January 2021 and January 2025. Thirty-two feet underwent MIS TN arthrodesis and 24 feet an open approach. The primary endpoint was the American Orthopaedic Foot & Ankle Society (AOFAS) ankle–hindfoot score assessed preoperatively and at final follow-up. Secondary endpoints were operative time and complications. Radiographic follow-up was performed as part of routine care to assess osseous fusion and to guide postoperative rehabilitation.

RESULTS: Mean follow-up was 361 ± 69 days. Functional outcomes improved substantially in both groups. Mean AOFAS increased from 53.2 ± 17.6 to 89.4 ± 9.5 in the MIS group (mean gain 36.2 ± 13.3) and from 56.9 ± 14.5 to 87.7 ± 12.7 in the open group (mean gain 30.9 ± 9.1). Between-group differences were not significant for postoperative AOFAS (p = 0.853 […]

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Pelvic insufficiency fractures after radiation therapy for pelvic cancer in female patients: an updated meta-analysis of 11,272 patients.

Gao M, Qiu Z, Xu Z, Zhao C, Zhong F, Gao T

BACKGROUND: Pelvic insufficiency fractures (PIFs) are a debilitating complication of pelvic radiotherapy (RT) for pelvic cancer in female patients, which could lead to a sharp decline in patients' quality of life. A 2019 meta-analysis by Lucas Gomes Sapienza reported a PIF rate of 14%, but advancements in RT techniques (IMRT, VMAT, C-ion RT) and new published studies necessitate an updated analysis. This study aimed to re-evaluate PIF incidence, fracture location, and associated factors using recent high-quality evidence.

METHODS: A systematic search of PubMed, Web of Science and Cochrane Library (1980–April 2025) with cohort studies for gynecologic/anal/rectal cancers, pelvic insufficiency fractures. Data on PIF incidence, location, and RT methods were extracted. Random-effects models were used to address expected high heterogeneity. Subgroup analyses were explored in detail.

RESULTS: The pooled PIF incidence was 17% (95% CI: 15–20%), higher than the prior 14% estimate. Sacral bone/sacroiliac joint involvement dominated (71%), followed by pubic bone (11.6%). Subgroup analyses revealed higher PIF rates with definitive RT (25%) vs. adjuvant RT (6%). Modern RT techniques (IMRT: 8%, 3D-CRT: 11%) reduced PIF risk vs. historical methods (AP/PA: 22%, 4-field: 18%). However, VMAT and C-ion RT showed elevated rates (30%), likely due to higher doses (≥ 55 Gy). Regional difference indi […]

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The ulna osteotomy locking plate II in patients with ulnocarpal impaction syndrome: a retrospective evaluation.

Benedikt S, Seeher U, Stricker M, Bode S, Stock K, Arora R

INTRODUCTION: Ulnocarpal impaction syndrome is one general cause of ulnar-sided wrist pain. If conservative therapy fails, various surgical procedures and different instruments and implants are available. The aim of this study was to evaluate the results of the ulna osteotomy locking (UOL) plate II from the manufacturer I.T.S.

MATERIALS AND METHODS: Thirty-five cases (34 patients) with primary or secondary ulnar impaction syndrome treated with the UOL plate II were evaluated retrospectively. Demographic, radiologic and clinical data were collected focusing on the postoperative ulnar variance and the complication rate, especially non-unions and signs of implant failure.

RESULTS: Median age was 48 years, with 20 female and 15 male cases. The measured median postoperative ulnar variance was – 0.8 mm (Q1: – 2; Q3: 0). There was no case of a non-union or a postoperative infection. Postoperative complications included one case of secondary dislocation, one case of a neuroma of the dorsal branch of the ulnar nerve and one mild CRPS. A total of six cases wished for an implant removal in the follow-up, of which five complained about local irritation caused by the plate.

CONCLUSIONS: The UOL plate II has proven to be a reliable option for ulnar shortening osteotomy with a low implant associated complication rate, especially no case of non-union. The occasional need for implant removal […]

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Tranexamic acid reduces transfusion requirements after pelvic osteotomy: a nationwide propensity score-matched analysis.

Tanaka H, Tarasawa K, Mori Y, Baba K, Kurishima H, Kanabuchi R, Kawamata H, Fukuchi H et al.

INTRODUCTION: Pelvic osteotomy is frequently associated with substantial perioperative blood loss due to the highly vascularized pelvic anatomy. Tranexamic acid (TXA) is widely used to reduce blood loss in total hip arthroplasty; however, evidence supporting its use in pelvic osteotomy, particularly in Asian populations, remains limited. This study evaluated the association between perioperative TXA use, transfusion requirements, and postoperative complications in patients undergoing pelvic osteotomy in Japan.

MATERIALS AND METHODS: Using the Japanese Diagnosis Procedure Combination nationwide administrative database, we retrospectively identified patients who underwent pelvic osteotomy between December 2011 and March 2023. Patients receiving perioperative TXA were compared with those who did not, using one-to-one propensity score matching. Outcomes included transfusion rates on postoperative days 0–2, cumulative transfusion volume, and in-hospital complications, including infection, deep vein thrombosis, pulmonary embolism (PE), and reoperation.

RESULTS: After matching, 3,542 patients were included in each group. TXA use was independently associated with a significantly lower likelihood of blood transfusion on postoperative days 0 (95% CI, 0.552–0.693; $p < 0.05$) and 1 (95% CI, 0.472–0.579; $p < 0.05$). The cumulative transfusion volume over the first three postoperative days w […]

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Revision rate of large head diameter metal-on-metal total hip arthroplasty: long-term results.

Wakabayashi H, Tone S, Naito Y, Sudo A, Hasegawa M

BACKGROUND: To assess the long-term outcomes (minimum 10 years) of total hip arthroplasty (THA) using a large head diameter metal-on-metal (MoM) acetabular prosthesis.

METHODS: In total, 108 primary THAs (98 patients) were performed using a Cormet MoM acetabular prosthesis. Clinical hip function, radiographic evaluation and revision rate were evaluated.

RESULTS: Of the 98 patients, 15 (16 hips) followed up for < 10 years were excluded. Among the remaining 92 hips (83 patients), 35 (38.0%) had pseudotumors (PTs) detected by MRI at an average of 5.0 years after the initial THA. On radiographic evaluation, osteolysis (OL) was noted in 10 hips (10.8%) on the cup side and six hips (6.5%) on the stem side. Loosening was observed in six hips (6.5%) on the cup side, whereas no loosening was noted on the stem side. Twenty-three hips (22 patients) were switched to metal-on-polyethylene articulation between 1.2 and 14.7 years (mean: 6.3 years) postoperatively due to pain, swelling, infection, and/or implant failure. In this analysis, cup failure (OL, loosening, or both) was the highest in patients with revision surgery compared to those with PTs but not revision surgery and those without PTs.

CONCLUSIONS: The prevalence of PTs was 38.0%, and that of revision surgery was 25%. Some cases required revisions even after 10 years following surgery. We advise against its continued use and rec […]

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Influence of different preservation methods on the mechanical in vitro stability of bones: a comparative study using porcine metatarsals.

Bäumlein M, Kühlein-Lehr M, König AM, Ziller V, Ruchholtz S, Paletta JRJ, Dos Santos V

INTRODUCTION: Large quantities of bone specimens are required for the mechanical testing of orthopedic and trauma implants. Currently, fresh-frozen or chemically preserved bones—such as formalin-fixed—are commonly used. This study aimed to evaluate the effects of different conservation methods on mechanical bone stability under standardized conditions.

MATERIALS AND METHODS: Porcine metatarsals were divided into seven groups based on the preservation technique: fresh, fresh-frozen (– 18 °C), frozen with repeated freeze-thaw cycles, vacuum-sealed and refrigerated (4 °C), and chemically fixed using formalin, Thiel-fixation, or glycerol-based methods. All specimens underwent three-point bending tests to assess mechanical stability. Following induced fractures, bones were stabilized with a plate osteosynthesis (DCP® small fragment set, Synthes®, Germany) and retested. Bone mineral density (BMD) was also measured via dual-energy X-ray absorptiometry (DXA).

RESULTS: Results revealed no significant differences in BMD, maximum load, or breaking strength between groups. However, chemically fixed bones—especially formalin and glycerol—showed significantly reduced deformation, indicating increased stiffness. Additionally, chemically fixed specimens exhibited more consistent fracture patterns compared to fresh or physically preserved bones. After osteosynthesis, bones preserved with glyc […]

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Medial open-wedge high tibial osteotomy alters sagittal tibial tubercle-trochlear groove distance.

Schmidt S, Leite CBG, Franco D, Krabb N, Gravius S, Jacobs CA, Lattermann C

PURPOSE: Medial open-wedge high tibial osteotomy (MOWHTO) is widely used to treat varus knee osteoarthritis, but its impact on patellofemoral biomechanics remains incompletely understood. In particular, the sagittal tibial tubercle–trochlear groove (sTT-TG) distance, a novel parameter linked to patellofemoral contact pressure, has not been evaluated in this context. This study aimed to assess changes in sTT-TG following MOWHTO and identify anatomical predictors of its postoperative magnitude.

METHODS: In this retrospective study, 34 knees from 33 patients (mean age 36.6 ± 9.5 years, mean BMI 26.2 ± 4.3 kg/m²) undergoing ascending biplanar MOWHTO with pre- and postoperative MRI and radiographs were analyzed. The sTT-TG, Caton–Deschamps Index (CDI), posterior tibial slope (PTS), and tibiofemoral rotation angle (TFRA) were measured. Correlation and multivariable regression analyses were performed to identify predictors of postoperative sTT-TG.

RESULTS: MOWHTO significantly decreased the sTT-TG distance from 6.25 ± 5.34 mm to 3.74 ± 6.81 mm ($p = .009$), indicating anteriorization of the tibial tubercle. Patellar height (CDI) decreased from 1.14 ± 0.20 to 0.99 ± 0.15 ($p < .001$), and TFRA was reduced from 4.74 ± 5.54° to 2.62 ± 5.50° ($p = .017$). Multivariable regression identified preoperative sTT-TG, postoperative medial PTS, and CDI as independent predictors of postoperative sTT-T […]

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Favorable midterm clinical results of medial unicompartmental knee arthroplasty guided by coronal limb alignment using an image-free navigation system.

Shimozaki K, Shimozaki S, Shimozaki E

INTRODUCTION: Medial unicompartmental knee arthroplasty results in good postoperative motion and high patient satisfaction. However, inaccurate bone resection during manual procedures can cause tibial fractures, implant subsidence, or lateral compartment osteoarthritis. To improve accuracy, we performed medial unicompartmental knee arthroplasty using an image-free navigation system, with a target postoperative alignment of 2°–5° varus. This study evaluated midterm clinical outcomes and safety.

MATERIALS AND METHODS: This single-center retrospective case series included 144 consecutive knees (25 in men and 119 in women; mean age, 72.2 years) that underwent medial unicompartmental knee arthroplasty for varus osteoarthritis between 2011 and 2021, with at least three years of follow-up (mean, 63.1 months). A fixed-bearing implant (Triathlon PKR; Stryker) was used in all cases. The navigation system guided tibial and femoral resections to achieve a hip–knee–ankle angle of 2°–5° varus, while minimizing tibial resection. Pre- and postoperative outcomes were compared for range of motion, Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC), ability to sit in the seiza position, implant survival, and complications.

RESULTS: Mean knee extension improved from – 5.7° to – 0.6°, flexion from 141.6° to 145.9°, and the mean WOMAC score improved from 41.5 ± 9.9 to 4.9 ± 3.5 […]

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Pre-operative activity level and a sport or recreation injury mechanism are associated with 2-year clinical outcome after proximal hamstring tendon repair.

Ebert J, Edwards P, Klinken S, Ricciardo B, Annear P, D'Alessandro P

INTRODUCTION: Surgical repair of proximal hamstring tendon ruptures has demonstrated encouraging outcomes and satisfaction rates, superior to non-operative management. This study sought to investigate injury, surgical and post-operative factors associated with clinical outcome and satisfaction 2-years after proximal hamstring tendon repair of acute tendon injuries.

METHODS: This study included 59 patients undergoing proximal hamstring repair for acute tendon ruptures. Clinical assessment pre-operatively and 2-years post-operatively included the Perth Hamstring Assessment Tool (PHAT), Lower Extremity Functional Scale (LEFS) and Tegner Activity Scale (TAS). Regression analysis assessed the contribution of pre-operative patient (age, sex, body mass index, TAS), injury/surgery (time from injury to surgery, injury mechanism, semimembranosus and conjoint tendon retraction) and post-operative (peak isokinetic knee flexor strength) variables, to the 2-year PHAT and reporting complete satisfaction.

RESULTS: From baseline to 2-years, the PHAT improved by 47.1 points (95% CI, 41.9 to 52.3; $p < 0.001$), with 97.1% of patients meeting the PHAT minimal important change of 8.6 points by 2 years post-surgery. Furthermore, the LEFS improved by 44.5 points (95% CI, 39.9 to 49.1; $p < 0.001$) and TAS by 2.6 points (95% CI, 2.1 to 3.2; $p < 0.001$). Forty-six (78.0%) patients were ‘very satisfied’ wi […]

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Serotonergic antidepressants are associated with increased bleeding events within 30-days after total shoulder arthroplasty: a propensity-matched analysis of 54,291 patients.

Strony JT, Moyal AJ, Adelstein JM, Burkhart RJ, Imbrogno AM, Abid R, Gillespie RJ, Chen RE

INTRODUCTION: Selective serotonin reuptake inhibitors (SSRIs) and serotonin-norepinephrine reuptake inhibitors (SNRIs) are associated with bleeding events following orthopaedic surgery. However, their effect on total shoulder arthroplasty (TSA) outcomes is unknown.

METHODS: Patients undergoing primary anatomic or reverse TSA in the United States were identified in the TriNetX database using Current Procedural Terminology (CPT) and International Classification of Disease (ICD-10) codes. TSA for fracture was excluded. Two cohorts were created based on preoperative serotonergic antidepressant use. Cohorts were matched to 14 demographic, comorbidity, laboratory and medication parameters. Outcomes were analyzed at one week, one month and three months postoperatively. Significance was set at $p < 0.05$.

RESULTS: 7,374 matched patients were included per cohort. The SSRI/SNRI cohort had significantly higher odds for post-hemorrhagic anemia and transfusion on the day of surgery (Anemia - OR 1.50, 95%CI 1.28–1.76; Transfusion - OR 2.55, 95%CI 1.27–5.13), at seven-days post-op (Anemia - OR 2.78, 95%CI 1.68–4.58; Transfusion - OR 2.58, 95%CI 1.39–4.79) and at one month post-op (Anemia - OR 2.73, 95%CI 1.37–5.46; Transfusion - OR 2.31, 95%CI 1.09–4.88). From one through three-months post-operatively, the two cohorts did not differ in rates of postoperative anemia, hematoma/hemorrhage, or re […]

PubMed · DOI

Surgical therapy of conservatively exhausted rhizarthrosis - total joint replacement or resection arthroplasty? A systematic review.

Boever J, Unglaub F, Spies CK, Cavalcanti Kußmaul A, Wulf J, Böcker W, Ayache A

UNLABELLED: Rhizarthrosis is a common degenerative disorder of the carpometacarpal joint of the thumb, significantly impairing hand function due to pain and loss of mobility. While conservative and joint-preserving measures are the first-line treatment in early stages, advanced disease often requires surgical intervention. The aim of this study was to evaluate and compare the clinical outcomes, functional results, and complication profiles of resection arthroplasty versus latest generations of thumb carpometacarpal joint prostheses. Resection arthroplasty has proven to be a reliable therapy, providing effective pain relief and high patient satisfaction. Latest generations of thumb carpometacarpal joint prostheses demonstrate low rates of loosening and dislocation, along with advantages such as faster rehabilitation, preservation of thumb length, and good functional outcomes. The choice of procedure should be individualized, considering age, activity level, and functional requirements.

SUPPLEMENTARY INFORMATION: The online version contains supplementary material available at 10.1007/s00402-025-06180-5.

PubMed · DOI

Examining the underreporting of social determinants of health in randomized controlled trials published in spine journals.

Megafu MN, Ayers JT, Evans LC, Newell SA, Uwefoh M, Guerrero OD, Hecht AC, Lin J et al.

INTRODUCTION: Social determinants of health (SDOH) are nonmedical factors that influence patient outcomes. This study aimed to determine the prevalence of randomized controlled trials (RCTs) reporting SDOH factors in their patient cohorts within several of the highest-impact spinal surgery journals. We hypothesize that SDOH factors will be infrequently reported and often miss variables such as socioeconomic status, employment status, insurance, and education level.

METHODS: From 2020 to 2024, the PubMed database was queried to analyze spine RCTs reporting SDOHs from four high-impact journals. For each study, we determined the age, sex, and/or gender, body mass index (BMI), the year the article was published, the country of origin of the senior author, and self-reported SDOH factors.

RESULTS: Of the 147 included RCTs, age (95.9%), sex/gender (94.6%), and BMI (72.8%) were reported by the majority of studies. There were a total of 33 countries represented by the RCTs reporting spine outcomes, where 25 (17.0%) RCTs were conducted in the United States, 22 (15.0%) RCTs conducted in China, 11 (7.5%) RCTs in the Netherlands, 9 (6.1%) RCTs in Japan and 9 (6.1%) RCTs in Turkey.

CONCLUSION: This paper provides valuable insights into the landscape of spine research and emphasizes the necessity of consistently reporting all SDOH factors. Future research can better inform clinical practic […]

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Obesity and long bone fractures in children. Systematic review.

Aly A, Aly T

BACKGROUND: Childhood obesity is a pressing global health issue with potential implications for musculoskeletal injury risk and recovery. Although the influence of obesity on bone metabolism is acknowledged, its specific connection to the incidence, patterns, and treatment results of long bone fractures in the pediatric demographic requires further clarification.

PURPOSE: This systematic review aims to evaluate the existing literature on the relationship between obesity and long bone fractures in children and adolescents, with a focus on fracture risk, anatomical distribution, management approaches, and clinical outcomes.

METHODS: We conducted a systematic literature search of PubMed, MEDLINE, Cochrane Library, and Google Scholar for studies published from January 2000 to March 2025. Search terms included “childhood obesity”, “bone health”, and “long bone fractures”. We included English-language original research that analyzed the relationship between pediatric obesity (BMI ≥ 95th percentile) and long bone fractures. Data on study characteristics, fracture patterns, treatments, and outcomes were extracted.

RESULTS: Out of 2,152 articles screened, 14 met the inclusion criteria. Children who were overweight or obese had higher odds of lower extremity fractures, particularly of the tibia and femur (odds ratios 1.5–3.3). Obesity was linked to more complex fracture patterns, incl […]

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TGCT of the shoulder - a case series and literature review.

Stadelmeier J, Bijeljac F, Klein A, Winden F, Reidler P, Dürr HR

INTRODUCTION: Tenosynovial giant cell tumor (TGCT), also known as pigmented villonodular synovitis (PVNS), involving the shoulder is extremely rare and can present with a challenging clinical course. Due to the complex anatomy of the shoulder, both diagnosis and treatment are demanding. This study aims to evaluate the diagnostic and therapeutic management of shoulder TGCT based on a case series and a review of the literature.

MATERIALS AND METHODS: Between 2005 and 2021, four patients (2 females, 2 males) with shoulder TGCT (1 localized, 3 diffuse) underwent surgical treatment at our institution. The minimum follow-up was 39 months (range: 39–233 months). Functional outcomes were assessed using the Disabilities of the Arm, Shoulder and Hand Questionnaire (DASH), the Oxford Shoulder Score (OSS), and the Short Form 36 Health Survey (SF-36). Additionally, a literature review of 65 studies comprising 108 patients was performed.

RESULTS: Treatments consisted of arthroscopic and open resections, tendon repair, adjuvant radiosynoviorthesis, and radiotherapy, as appropriate. All patients were recurrence-free at the last follow-up, except for one with stable residual disease after radiotherapy. The mean interval from symptom onset to diagnosis was 23.6 ± 28.1 months. In the literature cohort, the mean patient age was 50.3 ± 19.7 years, with a nearly equal gender distribution. Diffuse […]

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Comparing outcomes of total hip arthroplasty between cirrhotic and non-cirrhotic patients through a propensity-matched analysis.

Haider M, Khury F, Katzman J, Connolly P, Sarfraz A, Schwarzkopf R, Lajam CM

BACKGROUND: The impact of liver cirrhosis on surgical outcomes is well-known. This study aimed to compare postoperative outcomes of total hip arthroplasty (THA) in patients with versus without cirrhosis.

METHODS: A retrospective review was conducted of all patients who received a THA between 2012 and 2021 with a minimum of two years of clinical follow-up at a single, urban tertiary health center with lab results available to calculate Model for End-stage Liver Disease (MELD) scores. Using demographic variables, patients with and without cirrhosis underwent a 10:1 propensity score match. Short-term clinical outcomes were compared between cohorts. Cirrhotic patients were stratified based on their MELD score as mild (MELD < 10, n = 39) or moderate-to-severe (MELD ≥ 10, n = 10).

RESULTS: Of the 539 patients included in this study, 49 patients were in the cirrhotic group and 490 patients were in the non-cirrhotic group. Compared to non-cirrhotic and mild cirrhotic, moderate-to-severe cirrhotic THA patients had significantly higher incidence of 30-day (2.9% vs. 2.6% vs. 30.0%, p = 0.011) and 90-day readmissions (5.9% vs. 2.6% vs. 30.0%, p = 0.038) due to periprosthetic joint infection (PJI), and higher incidence of 90-day (3.1% vs. 2.6% vs. 20.0%, p = 0.024) and all-time revisions (1.4% vs. 5.1% vs. 20.0%, p = 0.016) due to PJI. There were no differences in overall 90-day reoperati […]

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Correction: Immunological purification of rat precartilaginous stem cells and construction of the immortalized cell strain.

Zhang S, Chen A, Hu W, Li M, Liao H, Zhu W, Song D, Guo F

PubMed · DOI

The accumulation of technical errors exponentially increases the risk of screw cut-out in femoral intramedullary nailing.

Brochard D, Thepaut M, Daoulas T, Poiri A, Letissier H, Di Francia R

PURPOSE: Proximal femur fractures primarily affect the elderly, with significant morbidity, mortality, and socioeconomic impact. The main complication of short trochanteric intramedullary nailing is the cut-out of the cervicocapital screw through the femoral head. The objective of this study was to analyze the influence of technical errors in short trochanteric intramedullary nailing for the treatment of trochanteric femur fractures on the mechanical failure of osteosynthesis.

METHODS: A total of 540 patients who underwent surgery for a trochanteric femur fracture using short trochanteric intramedullary nailing were included in a single-center, retrospective study conducted between February 2012 and July 2018.

RESULTS: Thirty patients (5.6%) experienced mechanical failure of the osteosynthesis at the 3-month follow-up. An anterior position of the cervicocapital screw, accumulation of technical errors, a tip-apex distance > 25 mm, and an intra-focal entry point were significantly associated with cervicocapital screw cut-out.

CONCLUSION: The mechanical failure rate is 5.6%. Short trochanteric intramedullary nailing requires precise execution to reduce the risk of cervicocapital screw cut-out, which is a source of osteosynthesis failure.

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Recommendations of an international Delphi study group for total knee arthroplasty in obese patients.

Rajgopal A, Palanisami DR, Annapareddy A, Khan M, Easwaran R, Shyam A, Sanghavi S, Sancheti P

INTRODUCTION: Total knee arthroplasty (TKA) in obese patients poses multiple challenges and there is a lack of consensus on various aspects of TKA in this patient population. This study is a modified Delphi consensus study of international experts to provide recommendations on TKA in obese patients.

MATERIALS AND METHODS: The consensus statements were generated using an anonymized two-round modified Delphi questionnaire, sent to an international panel of 53 knee surgeons, with an 80% agreement being set as the limit for consensus. The responses were analysed using descriptive statistics, with median as the measure of central tendency. Anonymized feedback was provided to all panellists based on responses from previous rounds to help generate the consensus.

RESULTS: 9 statements reached a consensus: WHO classification (87%), BMI cut-off for single-stage bilateral TKA (92%), use of tibial stem (84%), medial parapatellar approach (92%), same surgical approach as non-obese TKA (92%), inherent malnutrition (82%), higher risk of aseptic loosening in BMI class 3 (81%), need for time interval between bilateral TKA (88%), and consensus against leaving the knee in slight flexion (82%). The panel was unable to reach a consensus on 17 statements.

CONCLUSION: This indicates lack of consensus on a majority of issues among experts and further research is required in this field to address ev […]

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Impact of distal femoral morphology on short-term outcomes following high tibial osteotomy (HTO) in varus knees.

Zhu R, Wang M, Wang J

INTRODUCTION: Despite the availability of various osteotomies for varus knee osteoarthritis, including proximal tibial, distal femoral, and fibular osteotomies, HTO is the most prevalent in clinical practice. The primary focus of this study is to investigate how distal femoral morphology influences the outcomes of HTO.

METHODS: Demographic characteristics, radiographic parameters, and functional scores were collected preoperatively and at 2-year follow-up from patients undergoing HTO at our institution between January 2022 and July 2022. Patients were categorized into two cohorts based on the lateral distal femoral angle (LDFA): Group I (LDFA ≤ 89°) and Group II (LDFA > 89°). Each cohort was further stratified into three subgroups according to postoperative hip-knee-ankle angle (post-HKA): Subgroup A (HKA = 180° ± 0.5°), Subgroup B (HKA = 181° ± 0.5°), and Subgroup C (HKA = 182° ± 0.5°). Radiographic and functional outcomes were compared.

RESULTS: The study included 191 patients distributed across six subgroups (IA: 33, IB:29, IC:29, IIA:37, IIB:30 and IIC:33). Significant inter-subgroup differences were observed in the change in joint line convergence angle (ΔJLCA) for both Group I (F = 11.876, P = 0.048) and Group II (F = 9.826, P = 0.042). ΔJLCA exhibited a positive correlation with the change in hip-knee-ankle angle (ΔHKA) in both groups (Group I: r = 0.323, P = 0.011; Gr […]

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Septic arthritis following intra-articular corticosteroid injections: a retrospective analysis.

Mueller MR, Cleland T, Hileman CO, Olsen A, Wissner R, Monden KR

INTRODUCTION: Septic arthritis is a rare but devastating complication of intra-articular corticosteroid injection (CSI), associated with significant medical morbidity and poor clinical outcomes. Although previous studies have examined risks associated with CSI, few have tracked patients long-term. This study evaluates the incidence, timing, and patient characteristics related to iatrogenic septic arthritis within 6 months of receiving a large joint CSI, offering new insights to inform clinical practice.

MATERIALS AND METHODS: A retrospective, descriptive cohort study was conducted using SlicerDicer, a software stratification system within Epic, to identify patients diagnosed with septic arthritis within six months of receiving an intra-articular CSI of the hip, knee, or shoulder. Data were collected from a single institution over a 10-year period (July 1, 2010 to July 1, 2020). Individual chart review was used to obtain patient demographics, clinical characteristics, and procedural details for identified cases.

RESULTS: Of 15,021 intra-articular corticosteroid injections performed, 14 cases of septic arthritis were identified within 6 months of the procedure, resulting in an incidence rate of 0.093%. Of the affected patients, 21% had underlying inflammatory arthritis, and 21% had underlying comorbidities resulting in immunosuppression. The median time to diagnosis was 3.5 (ra […]

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Properly designed femoral stem impactors help to avoid overstuffing and make a second trial reduction unnecessary.

Brodt S, Rohe S, Knospe P, Sanz-Ruiz P, Wassilew G, Matziolis G

INTRODUCTION: For cementless implantation of hip stems, it is very important that the original stem fits exactly into the femoral prosthesis bed previously shaped using a stem rasp. If the geometry of the stem rasp does not match the prosthesis geometry, this can result in either overstuffing or post-sintering of the original stem in relation to trial reduction with the stem rasp. Overstuffing results in leg lengthening, while subsidence of the stem results in shortening with the risk of dislocation and impingement. Trial reduction with the original stem and trial head can prevent this, but is associated with additional soft tissue trauma and a longer operating time.

MATERIALS AND METHODS: Three groups were prospectively randomized. Group 0 was treated with the Fitmore B stem and the conventional rasp system, Group 1 with the Optimys stem and Group 2 with the Fitmore B stem with a new, optimized rasp system via a minimally invasive posterolateral approach. Intraoperatively, the differences in leg length and offset between trial reduction with the stem rasp and the original stem were recorded using a hip navigation system.

RESULTS: The conventional rasp system led to significant overstuffing (1.2 ± 0.5 mm, p = 0.024) with the Fitmore B stem, compared with the instruments of the Optimys stem. In contrast, the optimized rasp system with the Fitmore B stem resulted in an equally […]

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I Injury

Volume 57, Issue 7 · July 2026 · 35 new articles

Letter to the editor: Lattakia earthquake 2023: Orthopedic injuries and analysis of pelvic fractures (two years follow up).

Gök M

PubMed · DOI

Elbow hemiarthroplasty is not the "half" of elbow arthroplasty.

Wang Z, Lu Y

Distal humeral arthroplasty (DHA), also termed elbow hemiarthroplasty, replaces the distal humeral articular surface while preserving the native ulnar and radial articular surfaces. Compared with total elbow arthroplasty (TEA), DHA has distinct characteristics, resulting in different surgical indications and clinical outcomes. The prefix "hemi-" may be misleading, as DHA should not be regarded as half--the humeral part of TEA.

PubMed · DOI

Beyond survival: What drives senescence in trauma, shock and sepsis?

Mösenlechner M, Ignatius A, Haffner-Luntzer M, Halbgebauer R, Huber-Lang MS

Cellular senescence is a conserved stress response characterized by stable growth arrest, resistance to apoptosis, and a pro-inflammatory secretome that shapes tissue repair, tumor suppression, and aging. In this review, senescence is framed in the context of trauma, hemorrhagic shock, and sepsis. Severe injuries to brain, lung, abdomen, and bone, as well as ischemia-reperfusion and hemorrhagic shock, rapidly induce p21- and p16-governed senescence-like programs that modulate neurodegeneration, fibrosis, fracture healing, and organ dysfunction. In sepsis, senescence extends beyond immunosenescence to endothelial, epithelial, and stromal compartments, intersecting with ferroptosis, mitochondrial failure, and microbiome dysbiosis, and contributing to post-sepsis syndrome. Integrating mechanistic data with high-dimensional profiling and early interventional studies, this review proposes a context- and time-dependent model in which acute "pseudosenescence" may be organ-protective, whereas persistent senescent reservoirs fuel inflammaging and chronic pathology. Future translational studies are required to define the clinical rationale for phase-specific senotherapeutic interventions in trauma, shock, and sepsis, particularly with respect to long-term outcomes beyond survival.

PubMed · DOI

Potential temporal discrepancies in an analysis of post-operative ketorolac and humeral shaft repair outcomes.

Chow PC, Tsai KW, Wang J

PubMed · DOI

Multidisciplinary analysis of teamwork in orthopaedic trauma: How do expectations differ from reality in the operating room environment?

Menezes M, Kalun P, Whyne C

BACKGROUND: Surgical teamwork is widely recognized as a cornerstone of patient safety and care quality, yet the realities of operating room (OR) collaboration often differ from the expectations codified in policies, protocols, and training models. This study aimed to explore how surgical teams navigate the realities of intraoperative care during hip fracture fixation, using the human factors work-as-imagined versus work-as-done framework.

METHODS: This study was conducted at a teaching hospital and Level 1 trauma centre in Canada and focused on individuals who work in a high-volume orthopaedic trauma OR. An ethnographic approach facilitated a deep exploration of the culture of the OR and a better understanding of surgical team members' lived experiences. Semi-structured interviews were conducted with surgeons, anesthesiologists, nurses, and radiation technologists (RTs). The interview data were analyzed using thematic analysis with primarily inductive coding; this allowed the insights from the experts to be identified organically, using their own words. A work-as-done versus work-as-imagined framework was used to compare and contrast tensions within the data.

RESULTS: Our themes highlighted key mismatches in expectations related to teaching, communication, team functioning, patient care, and the role of RTs in the OR. While participants emphasized psychological safety, clear […]

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Retrograde intramedullary nailing versus locked plate fixation for distal femur fractures: A systematic review and meta-analysis of randomised controlled trials only.

Sain B, Sethi A, Suryawanshi S, Mishra P, Patel V, Rambani R

BACKGROUND: The optimal fixation method for distal femur fractures remains contentious. Retrograde intramedullary nailing (RIN) and locking plate fixation (PLATE) are the two principal surgical options, yet their comparative effectiveness has not been definitively established. This systematic review and meta-analysis aimed to comprehensively compare the operative, radiological, and functional outcomes of RIN versus PLATE for distal femur fractures exclusively from randomised controlled trials (RCTs).

METHODS: A systematic search of MEDLINE, Embase, Scopus and Web of Science was performed from inception to December 2025. Only RCTs directly comparing RIN with PLATE for distal femur fractures in adult patients were included. Risk of bias was assessed using the Cochrane Risk of Bias 2.0 tool. Random-effects meta-analyses were performed for 11 outcomes. Heterogeneity was quantified using I^2 statistics and prediction intervals. Subgroup analyses were conducted by fracture complexity, open fracture status, and follow-up duration.

RESULTS: Eighteen RCTs comprising 1105 patients (555 RIN, 550 PLATE) were included. RIN demonstrated significantly shorter operative duration (mean difference (MD) -13.75 min; 95% confidence interval (CI) -22.89 to -4.62; p = 0.003), faster fracture union time (MD -3.01 weeks; 95% CI -4.33 to -1.69; p < 0.001), and lower infection rates (risk ratio (RR) 0.5 […]

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Postoperative impacts of reamer-irrigator-aspirator (RIA) on femoral strength: A finite element analysis.

Nishida R, Kumabe Y, Sawauchi K, Matsumiya Y, Fukui T, Kondo H, Niikura T, Kuroda R et al.

INTRODUCTION: The reamer-irrigator-aspirator (RIA) system is used to harvest a large amount of autologous bone graft, most commonly from the femur. One of its major complications is postoperative fracture resulting from decreased bone strength. Although the number of reported cases is limited, it is a serious issue because it involves fracture in originally healthy bone. While several cadaveric studies have examined post-RIA bone strength, no studies have investigated it using finite element analysis (FEA). The aim of this study was to elucidate the effects of RIA on postoperative femoral strength using FEA.

METHODS: Quantitative computed tomography (CT)-based FEA was performed using pre- and postoperative CT images from three patients who underwent RIA of the entire femur for autologous bone grafting during nonunion surgery at our institution. Patient 1: a 60-year-old man who underwent reaming with a reamer 4 mm larger than his isthmus diameter. Patient 2: a 71-year-old woman with a wide endosteal diameter, typical of osteoporotic bone. Patient 3: a 64-year-old woman with cortical penetration on the posterior proximal diaphysis and the anterior supracondylar region. Three loading conditions were analyzed for each patient: (a) axial loading, (b) direct force, and (c) rotation. Fracture load was defined as the minimum load at which at least one shell element failed at the compl […]

PubMed · DOI

A prospective review of transfusion practices in paediatric trauma patients at level 1 trauma centre.

Krishna A, Sinha TP, Meher R, Prasanth V, Chopra S, Kumar A, Chaudhary N, Soni KD et al.

BACKGROUND AND OBJECTIVES: Transfusion practices in paediatric trauma patients are frequently based on adult protocols, despite notable differences in physiology and injury mechanisms between children and adults. This study was aimed to evaluate the current transfusion practices, among the paediatric trauma population, which will help in guiding management and planning along with research opportunities among paediatric trauma patients MATERIALS AND METHODS: We conducted a prospective observational study over one year, including trauma patients under 18 years who received transfusions. Data collected encompassed demographics, clinical and injury details, transfusion specifics, and outcomes, with descriptive analyses and assessment of associations with clinical and laboratory parameters.

RESULTS: Of the paediatric trauma patients, 394 (38.7%) received transfusions. Among these, 388 (98.5%) received PRBC, 159 (40.4%) RDP, 170 (43.2%) FFP, and 3 (0.8%) cryoprecipitate. Median requirements were 21.8 ml/kg PRBC, 5.8 ml/kg RDP, 17.5 ml/kg FFP, and 2.6 ml/kg cryoprecipitate. Early PRBC transfusions were associated with delayed admission, tachycardia, and low baseline haemoglobin. RDP and FFP correlated with higher injury scores, shock index, FAST/eFAST positivity, tachycardia, hypotension, Low Hb, and abnormal PT.

Massive transfusion (≥40 ml/kg in 24 h) was required in 102 cases (25.9 […]

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Percutaneous screw fixation for traumatic Tile C pelvic fractures - Does the number of screws matter? A review of clinical and biomechanical outcomes.

Domashevich T, Filley A, Zakusylo A, Higinbotham SE, Schlauch AM, Mauffrey C

BACKGROUND: This review addresses whether one-screw or two-screw percutaneous constructs offer superior outcomes for Tile C pelvic fractures. It compares biomechanical stability and clinical outcomes, weighing potential gains in stability against increased surgical complexity and complication risk.

METHODS: A systematic search of PubMed, Embase, and Web of Science identified clinical and biomechanical studies on posterior fixation in Tile C fractures. Inclusion criteria required adult traumatic injuries, percutaneous posterior screw fixation, and reported primary outcomes. Two reviewers independently screened and extracted data.

RESULTS: Thirty biomechanical, six clinical, and one mixed-design study met inclusion criteria. One third of biomechanical studies favored the use of both a S1 and S2 screw for greater rigidity, while others showed benefit. Clinical studies showed mixed results, with some benefit for radiographic reduction and fixation failure in high-risk patterns but no consistent functional advantage. Significant heterogeneity in methodologies limited direct comparisons.

CONCLUSION: Cumulative biomechanical evidence supports two-screw fixation for increased stability in Tile C fractures, especially unstable patterns. Clinical benefits remain uncertain, emphasizing individualized treatment decisions and the need for standardized future research.

PubMed · DOI

Reconsidering risk pathways: PTSD predictors after major trauma in New Zealand and beyond.

Siregar NI, Nurbaiti N, Putri RD, Ramadhani E

PubMed · DOI

Minimally invasive anatomical fixation of distal humeral fractures: Direct reduction through the anterior MIPO distal window.

Segura-Cobos J, Tejero S, López-Hernández C, Giráldez-Sánchez MÁ

Conventional surgical treatment for distal humeral shaft fractures consists of osteosynthesis through an open posterior approach. This technique requires a wide incision, extensive muscular dissection, and direct handling of the radial nerve, resulting in considerable surgical morbidity. Although the anterior MIPO technique represents a less invasive alternative and avoids these drawbacks, it has traditionally been associated with indirect reduction and relative stability alone. In distal humeral shaft fractures, the distal window of the anterior MIPO approach is located directly above the fracture site, which allows direct reduction manoeuvres. Fragment control through the distal window facilitates reduction and permits the application of absolute stability while preserving a minimally invasive strategy. This technical note describes the integration of these technical innovations together with the use of an inverted PHILOS plate, whose design characteristics enable enhanced distal fixation with a biomechanical profile superior to that of conventional plates. Preliminary results are presented from a case series of Holstein-Lewis-type fractures sustained through a homogeneous torsional mechanism related to arm-wrestling practice.

PubMed · DOI

Does increased preoperative education for patients undergoing external fixation improve overall satisfaction?

Smith M, Alford T, Clark C, Strubel JJ, Hunter N, Achor TS, Munz JW, Choo AM et al.

OBJECTIVES: To determine if increased preoperative education on how to manage activities of daily living with a lower extremity external fixator would improve overall patient satisfaction.

METHODS: Design: Prospective cohort, randomized, double-blinded study.

SETTING: Level I academic trauma center. Patient Selection Criteria: Patients were prospectively identified for enrollment before undergoing external fixation. Adult patients with any lower extremity fracture requiring placement of a knee or ankle external fixator were included. Exclusion criteria consisted of skeletally immature patients, those with a history of external fixation, altered mental status, or placement of multiple external fixators. Patients were randomized via a stratified block method and were blinded as to whether they had received the preoperative educational intervention. Outcome Measures and Comparisons: Patient satisfaction was determined using a validated modified Leeds Satisfaction Questionnaire distributed at two time points: (1) on postoperative day one or two (Survey A) and (2) after external fixation at a follow-up appointment (Survey B). Patient questionnaires were distributed and collected by a research investigator blinded to the participant's intervention status.

RESULTS: A total of 89 patients were included in the analysis. Only one survey question demonstrated a statistically significan […]

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Posterolateral vs. anterolateral approach for posterolateral tibial plateau fractures: A multicenter cohort with four-year outcomes.

Elmi A, Kooshki AM, Hushmand H, Zamani M, Sadighi A, Nikpay A, Atashin H, Sharifzadeh Y

BACKGROUND: Posterolateral tibial plateau fractures (PLTPFs) are challenging due to limited surgical access. The posterolateral approach (PLA) offers direct visualization, while the anterolateral approach (ALA) is more familiar but may compromise reduction quality. This multicenter cohort study compares PLA versus ALA for PLTPFs with four-year follow-up.

METHODS: A multicenter concurrent cohort study (January 2020 - December 2024) included 110 patients with PLTPFs confirmed on CT. Fifty-five underwent ORIF via PLA, and 55 via ALA. Primary outcome was articular step-off on postoperative CT (two blinded radiologists, ICC=0.92). Secondary outcomes included range of motion, IKDC score, Tegner-Lysholm score, complications, revision surgery. Multivariable logistic regression and Kaplan-Meier survival analysis were performed.

RESULTS: Mean articular step-off was 0.9±0.5 mm in the PLA group versus 1.8±0.8 mm in the ALA group (mean difference 0.9 mm, 95% CI: 0.6-1.2, $p < 0.001$). Anatomic reduction (<2 mm) was achieved in 89.1% (49/55) of PLA patients versus 70.9% (39/55) of ALA patients ($p = 0.02$; adjusted OR=3.8, 95% CI: 1.4-10.2, $p = 0.008$). At four years ($n = 98$, 89.1% retention), mean IKDC score was 86.7±6.2 versus 79.3±7.5 ($p < 0.001$), and 78% of PLA patients achieved MCID versus 52% of ALA patients ($p = 0.01$). Transient peroneal neurapraxia occurred in 7.3% (4/55) of PLA versus […]

PubMed · DOI

Should we ORIF all nondisplaced medial malleolar fractures?

Wajda B, Topham J, Buckley R

PubMed · DOI

Selective neurectomy as surgical treatment for infrapatellar pain syndrome: A systematic review and meta-analysis.

Liechti R, Muhammad I, Sinisi M, Panagiotidou A, Fox M, Simpson AI

INTRODUCTION: Injury to the cutaneous nerves around the knee can result in neuropathic pain and functional impairment, commonly referred to as infrapatellar pain syndrome (IPS). This meta-analysis evaluates clinical outcomes of selective neurectomy for the treatment of IPS.

MATERIALS AND METHODS: MEDLINE, Embase, CENTRAL, and Web of Science were systematically searched for studies reporting clinical outcomes following selective neurectomy for IPS. A single-arm meta-analysis was performed using random-effects models. Pooled estimates are presented as mean differences (MDs) for continuous outcomes and event rates for dichotomous variables, each with corresponding 95% confidence intervals (CIs).

RESULTS: After screening 1018 studies, eight retrospective observational case series comprising 261 patients were included. The weighted mean age was 56.0 years (SD 12.9). Women were more commonly affected than men (69.9% vs. 30.1%). Primary and revision total knee arthroplasty were the most frequent precursors to IPS (54.0% of cases). The infrapatellar branch of the saphenous nerve was most affected (96.6% of patients), followed by the medial (44.4%) and lateral (25.7%) retinacular nerves. Weighted mean follow-up was 2.9 years (SD 1.3). All patients underwent local anaesthetic blockade prior to surgery. Pooled analysis of pre- versus postoperative Visual Analog Scale (VAS) scores demons […]

PubMed · DOI

The soft tissue healing diamond (ST-Diamond) concept: A translational framework for tendon and ligament regeneration.

Maffulli N, Giannoudis P

BACKGROUND: The diamond concept, originally articulated for bone fracture healing, defines five essential and interdependent elements for successful repair: growth factors, an osteoconductive scaffold, mesenchymal progenitor cells, an optimal mechanical environment, and adequate vascularisation. Tendons and ligaments are dense, hypovascular collagenous tissues with limited intrinsic regenerative capacity. When injured, they heal by biomechanically inferior fibrotic scarring rather than by regeneration. No unifying biological framework currently exists to guide the development of biological augmentation strategies for soft tissue repair.

FRAMEWORK: The diamond concept can be translated to tendon and ligament healing as a Soft Tissue Diamond (ST-Diamond), with five tissue-specific vertices. At the growth factor vertex, GDF-5 (BMP-14), TGF-beta, and bFGF drive tenogenesis in tendons, whilst FGF-2, TGF-beta1, and BMP-12 govern ligament repair. At the progenitor cell vertex, tendon stem/progenitor cells (TSPCs) and ligament fibroblasts replace bone marrow mesenchymal stem cells, supplemented by exogenous cell delivery where the native population is insufficient. At the scaffold vertex, bioresorbable, mechanically compliant materials, notably three-dimensional bioprinted methacrylated collagen (ColMA), replace osteoconductive constructs. At the mechanical vertex, controlled progress […]

PubMed · DOI

With an operative distal radial fracture, when should the brachioradialis tendon be released, if at all?

Kuljic N, Torresrangel E, Buckley R

PubMed · DOI

Reassessing hand dominance in pediatric lateral condyle fractures: A cohort study of surgical predictors.

Smadi Z, Balaji S, Turner L, McCarver M, Mubarak F, McKay S

INTRODUCTION: Lateral condyle fractures are the second most common elbow fracture in pediatric patients with high rates of articular involvement and displacement leading to surgical management. This study aims to assess whether dominant side fracture predicts conservative treatment change, undergoing surgical management or clinical outcomes in pediatric lateral condyle fractures and to identify independent predictors of surgery, loss of reduction, and complications.

METHODS: Pediatric patients with lateral condyle fractures treated between 2019-2021 with documented hand dominance at injury were included. Outcomes included surgical management, change of conservative treatment, loss of reduction, and complications. Demographic and fracture characteristics were analyzed using descriptive statistics. Continuous variables were compared using Welch's t-test, and categorical variables using chi-square or Fisher's exact tests. Multivariate logistic regression identified independent predictors.

RESULTS: A total of 200 patients were included (mean age 5.98 ± 2.52 years; 65% male; mean BMI 18.58 ± 4.00 kg/m^2; mean follow-up 88.9 ± 153.4 days). Mechanisms of injury included fall (50%), direct hit (35%), and FOOSH (15%). Weiss type 1 fractures predominated (61.5%), and fracture displacement occurred in 47.5%. Dominant-hand fracture was not an independent predictor of initial surgery, over […]

PubMed · DOI

Bouldering-related trauma: Injury patterns and operative burden over 10 years at a UK major trauma centre.

Jemmett H, James C, Davidson E

INTRODUCTION: Bouldering is a rapidly growing discipline within climbing, characterised by short ascents performed without ropes, typically at heights of up to several metres above protective crash mats. While overall injury burden among climbers has been described in the literature, detailed analyses of severe orthopaedic trauma presenting to major trauma centres remain limited. As participation continues to increase, particularly within indoor climbing facilities, a clearer understanding of injury patterns associated with bouldering is needed. This study aimed to characterise the anatomical distribution, injury patterns, and operative burden of bouldering-related trauma presenting to a UK Level I trauma centre.

METHODS: A retrospective cohort study was conducted of adult patients (≥16 years) presenting with climbing-related orthopaedic injuries between January 2015 and May 2025. Cases were identified through trauma registry and electronic record review. Demographic data, injury mechanism, anatomical injury location, and management strategy were collected. The primary outcome was requirement for operative intervention. Secondary outcomes included injury distribution and specific fracture or dislocation patterns.

RESULTS: A total of 176 patients were identified (mean age 29 years; 51% male). The average reported fall height was 2.9 m. Lower limb injuries predominated, account […]

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Letter to the editor: Toward precision use of antibiotic bead pouches in severe open lower extremity fractures.

Zhang Y, Du J

PubMed · DOI

Wound complications following calcaneal fracture surgery using sinus tarsi approach.

Larsen MS, Viberg B, Singh UM, Lind M, Penny JØ, Walsøe L, Gundtoft PH

INTRODUCTION: The sinus tarsi approach (STA) has gained popularity as a minimally invasive alternative to the extensile lateral approach for displaced intra-articular calcaneal fractures (DIACFs), yet wound complications remain a concern. This study aimed to determine the incidence and predictors of wound complications following STA fixation.

METHODS: A retrospective multicenter cohort study was conducted across four tertiary referral hospitals, including all surgically treated DIACFs managed with STA between 2018 and 2022. Demographic, injury-related, and operative variables were extracted from medical records. The primary outcome was wound complications within eight postoperative weeks, defined as inflammation, necrosis, or persistent wound leakage >2 weeks. Secondary outcomes included reoperation and identification of risk factors.

RESULTS: 148 fractures in 143 patients were included. Wound complications occurred in 13 cases (9%), with only one requiring surgical revision; eight were treated with antibiotics. Reoperation within one year occurred in 10% of cases, with a higher, though not statistically significant, rate among patients with wound complications (21% vs. 7.5%). Significant risk factors for wound complications were psychiatric comorbidity ($p = 0.012$), open fractures ($p = 0.032$), and concomitant fractures ($p = 0.021$). Trends toward higher complication rates were […]

PubMed · DOI

Comment on "Predictors of traumatic brain injury amongst secondary school students in England: A retrospective cohort study using electronic health records".

K V VK, Kshirsagar P, Mishra M, Panwar S

PubMed · DOI

From hazard identification to mechanism elucidation: The next step in preventing traumatic ear injuries.

Wu J, Lin J, Huo Y, Zhang J

PubMed · DOI

One-year mortality after first-time long bone fractures in older adults: A multicenter retrospective cohort study.

Ghafarian AM, Wajahath M, McBee K, Nasser E, Ghali A

BACKGROUND: Fragility fractures represent a major public health concern in the aging population, contributing substantially to morbidity, functional decline, and mortality. Although hip fractures are widely recognized for their high early mortality risk, less is known about the comparative one-year mortality associated with first-time long bone fractures at other anatomical sites. This study evaluates one-year all-cause mortality across multiple fracture types in older adults using a large, multicenter research network.

METHODS: We performed a retrospective cohort study using the TriNetX US Collaborative Network, identifying patients ≥ 65 years with a first-time long bone fracture between 2004 and 2024. Fracture types included hip, pertrochanteric, subtrochanteric, distal femur, proximal tibia, distal tibia, proximal humerus, and distal humerus fractures. Patients with pathologic, metastatic, or multiple fractures were excluded. A non-fracture cohort of age-matched older adults served as the comparison group. The primary outcome was one-year all-cause mortality. Propensity score matching (1:1) was used to balance demographics. Relative risks (RR) with 95% confidence intervals (CIs) and Kaplan-Meier survival analyses were performed.

RESULTS: A total of 165,017 fracture patients and 4.55 million non-fracture controls were identified; 162,469 patients remained after matching per […]

PubMed · DOI

Clinical and functional outcomes of primary bipolar hip arthroplasty: A prospective observational study.

Azar MS, Kargar-Soleimanabad S, Kalteh M

PURPOSE: Primary bipolar hip arthroplasty is commonly used to manage displaced femoral neck fractures in elderly patients, offering early mobilization and reduced surgical burden compared with total hip arthroplasty. However, long-term functional outcomes and predictors of recovery remain variable. This study aimed to prospectively evaluate the clinical and functional outcomes of primary bipolar hip arthroplasty.

METHODS: This prospective observational study included 196 patients who underwent primary bipolar hip arthroplasty between January 2016 and September 2024 in three university-affiliated teaching hospitals. Functional outcomes were assessed using the Harris Hip Score (HHS), while postoperative complications were recorded for one year. Patients were stratified by age (<60 vs. ≥60 years) to examine the impact of demographic factors on outcomes. Statistical analysis included correlation and subgroup comparisons with significance set at $p < 0.05$.

RESULTS: The mean age of participants was 68.7 ± 11.9 years, and the mean follow-up was 46.97 ± 25.21 months. No major complications or revision surgeries occurred; 4.0% of patients developed superficial wound infections. Overall, the mean HHS was 50.51 ± 18.59, with 79.6% of patients achieving poor scores. Younger patients (<60 years) demonstrated better functional outcomes, particularly in mobility-related domains ($p = 0.06$), w […]

PubMed · DOI

Establishing synergistic role of acacia honey and vitamin C: A promising strategy for faster wound tissue regeneration.

Majie A, Roy K, Jha B, Lim WM, Gorain B

Being established as an important natural component for wound healing, acacia honey (AH) in combination with ascorbic acid (AA) was evaluated for its tissue regenerative properties to establish a synergistic effect. A thermosensitive hydrogel formulation was developed and characterized for viscosity, mucoadhesion, moisture retention, pH, and morphology, where the parameters were found to be favourable for topical application and longer retention. The porosity and swelling of the hydrogel matrix allow for prolonged and sustained release of AH and AA. The release patterns of AH and AA from the gel matrix followed the Korsmeyer-Peppas and Michaelis-Menten patterns, respectively, indicating sustained release. The optimized formulation revealed antimicrobial properties, as evidenced by the significant increase ($p < 0.05$) in the diameter of the zone of inhibition. Scratch wound assay using HaCaT cell line showed wound healing potential of 90.6±4.5% for the co-loaded formulation. Furthermore, the wound healing potential of the co-loaded hydrogel in the excision wound model in experimental rats could be linked to increased vascularization due to the daily application of the formulation at the wound site. The antioxidant capacity was demonstrated by decreased IC50 values of 53.75±5.9 and 72.71±6.63 µg/mL against ABTS and DPPH, respectively, which additionally supported the wound healing […]

PubMed · DOI

Minimally invasive total thoracoscopic fixation versus open fixation for multiple Rib fractures: Systematic review and meta-analysis of clinical outcomes.

Altamimi A, Musawa I, Alshahrani MA, Alghamdi MA, Almugharrid SM, Alsayed TO, Alansari F, ALHarthi AE et al.

BACKGROUND: Multiple rib fractures cause substantial morbidity through severe pain, impaired ventilation, and pulmonary complications. While open rib fixation is well established, thoracoscopic fixation may reduce soft-tissue trauma and enhance recovery, but comparative evidence remains unclear. This systematic review and meta-analysis aimed to compare thoracoscopic versus open fixation for multiple rib fractures in terms of effectiveness and safety outcomes.

METHODS: We conducted a systematic literature search across PubMed, Scopus and Web of Science to retrieve comparative studies comparing thoracoscopic fixation versus traditional open fixation for multiple rib fractures regarding pain, perioperative outcomes and safety outcomes. Risk of bias of included studies was assessed using the ROBINS-I tool. A meta-analysis was conducted using a random-effects model in R (version 4.5.0).

RESULTS: Nine comparative studies were identified (total participants = 751). Meta-analysis revealed that thoracoscopic fixation was associated with improved postoperative pain compared with open fixation. Pain was significantly lower with thoracoscopy on postoperative day (POD) 1 (SMD= -1.12, 95% CI -1.64 to -0.61) and POD7 (SMD= -1.90, 95% CI -3.08 to -0.73), while POD3 was not significant (SMD= -1.49, 95% CI -3.52-0.54). Thoracoscopy reduced incision length (MD= -4.19 cm) and blood loss (MD= -18 […]

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Frailty as defined by the comprehensive geriatric assessment frailty index (CGA-FI) is associated with in-hospital complications and mortality in geriatric hip fracture patients, A retrospective cohort study.

Rossum du Chattel AMV, Laane DWPM, Aydin FB, DiStefano MT, Hu R, Gonzalez MR, Wignakumar T, Javedan H et al.

BACKGROUND: Increased frailty is associated with higher rates of adverse events after hip fracture surgery. Unfortunately, considerable uncertainty remains about which tool or index best quantifies frailty and therefore the associated risk of complications following hip fracture surgery. This study tried to evaluate whether a Comprehensive Geriatric Assessment-Frailty Index could predict in-hospital complications following hip fracture surgery.

METHODS: A retrospective cohort study was conducted among 1469 patients aged 70 years and older with an operatively managed hip fracture. Patients grouped according to their Comprehensive Geriatric Assessment-based Frailty Index: pre-frail to mildly frail and moderately to severely frail. The primary outcome was the occurrence of one or more complications. Secondary outcomes included specific complications, intensive care unit admission, length of stay and in-hospital mortality. Multivariable regression was used to adjust for confounders and presented as adjusted Odds Ratios (aOR).

RESULTS: Moderately to severely frail was independently associated with an increased risk of having one or more complications (aOR 1.70, 95% CI=1.28-2.26, $p < 0.001$), urinary tract infection (aOR 2.12, 95% CI=1.01-4.47, $p < 0.05$), delirium (aOR 2.05, 95% CI=1.43-2.93, $p < 0.001$), in-hospital death (aOR 3.35, 95% CI=1.00-11.26, $p = 0.05$) and 1-year mortality […]

PubMed · DOI

The fracture orthopedic risk of non-home discharge (FORD) score: A novel bedside predictive tool for non-home discharge in orthopedic trauma patients.

Salman SG, Phadke R, Carlin T, Rana A, Dawson JR, Fitzgerald CA, Seger CP, Zielinski MD et al.

INTRODUCTION: Non-home discharge after orthopedic trauma is associated with worse outcomes, increased costs, and greater resource utilization. Existing prediction tools often rely on hospital course variables unavailable at presentation or are limited to specific fracture populations. This study aimed to develop and validate the Fracture Orthopedic Risk of Non-Home Discharge (FORD) Score, a bedside tool using only emergency department-available variables to predict non-home discharge in adult fracture patients.

METHODS: A retrospective cohort study was conducted of adult fracture patients treated at an ACS-verified Level I trauma center from 2015 to 2023. Patients were randomly split into derivation (67%) and validation (33%) cohorts. Candidate predictors available immediately upon patient arrival were evaluated using univariate logistic regression, followed by multivariate logistic regression after collinearity assessment. Independent predictors were converted into an integer-based point system to construct the FORD Score. Model discrimination, calibration, and classification performance were assessed in the validation cohort and compared with established trauma severity measures.

RESULTS: The final cohort included 8422 patients, of whom 8.1% had non-home discharge. Fifteen independent predictors comprised the FORD Score, including age, physiologic abnormalities, fracture ch […]

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Editorial letter regarding the letter by McLean AL concerning "Biomechanical analysis of odontoid and transverse atlantal ligament in humans with ponticulus posticus variation under different loading conditions: A finite element study".

Kanat A

PubMed · DOI

Comment on "Suture button versus syndesmotic screw fixation in acute ankle fractures with syndesmotic injury: An umbrella review of functional outcomes and clinical relevance based on the minimal clinically important difference".

Juneja M, Ramteke H, Khan R

PubMed · DOI

Comment on "Multiple iliosacral screws in a single osseous fixation pathway: Utility and safety".

Terzi MM, Alemdaro■lu KB

PubMed · DOI

Letter to the Authors of 'Lessons learned: The thematic analysis of eight countries with mature trauma systems'.

Consunji R, El-Menyar A, Peralta R, Abdelrahman H, Rizoli S, Al-Thani H

PubMed · DOI

Comment on "Orthopedic trauma in pregnancy: A literature review" (Fang et al., Injury).

Pecold J, Szarpak L, Al-Jeabory M, Weglowski R

PubMed · DOI

"Dirty Fat Pad" Sign: A novel computed tomography (CT) indicator of injury to the posterior ligamentous complex in acute fractures of the thoracolumbar spine.

Dunsmuir RA, Price MJ, Rankine JJ

OBJECTIVES: To describe a CT finding which indicates Posterior Ligamentous Complex (PLC) injury in acute thoracolumbar spinal fractures and to determine if this has been described previously.

METHODS: The anomaly was first described by the senior author. We reviewed 1235 trauma CTs looking for this sign. We identified all thoracolumbar fractures and classified these as thoracic or lumbar fractures depending on the spinal level of the injured vertebra. We sought to determine if the 'dirty fat pad' sign was present on the mid-sagittal CT images of the spine. If present we looked to see if there was MRI confirmation of disruption of the posterior ligamentous complex. A literature review was performed of the MEDLINE, Embase Classic and EMBASE databases for descriptions of the appearance of the posterior ligamentous complex on CT following spinal trauma, from their respective inceptions in 1946 and 1974, to 21.11.25.

RESULTS: We found 356 thoracolumbar fractures in the 1235 trauma CTs. Twenty scans showed the 'dirty fat pad' sign (5.6%). MRI in these cases confirmed the disruption of the posterior ligamentous complex. The literature review provided titles and abstracts describing CTs with spinal trauma. These papers were screened for relevance, and thirty-two texts then reviewed in full. No paper retrieved described the radiological appearance of the fat pads on CT or MRI, in heal […]

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EJTE

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Prehospital endotracheal intubation and 30-day survival in severe trauma: a matched cohort study from Navarra, Spain.

Zulet Murillo D, Ferraz Torres M, Fortún Moral M, Belzunegui Otano T, Tamayo Rodríguez I

PURPOSE: Prehospital endotracheal intubation (ETI) is frequently used in major trauma, but its association with 30-day survival remains debated. We evaluated the association between prehospital ETI and 30-day survival and assessed whether sex modifies the age-related increase in mortality risk.

METHODS: We conducted a retrospective cohort study using the Navarra Major Trauma Registry (2010-2019). Patients aged ≥15 years with New Injury Severity Score (NISS) ≥15 were included; patients who died before hospital admission or had incomplete records were excluded. To mitigate confounding by indication, intubated and non-intubated patients were matched 1:1 using prehospital Glasgow Coma Scale (GCS), respiratory distress, and prehospital cardiac arrest as matching variables. Because the proportional hazards assumption for Cox regression was not met, parametric Weibull survival models were fitted, including a sex-by-age interaction term.

RESULTS: Overall, 1,909 patients were included; 212 (11.1%) underwent prehospital ETI. Matching yielded 190 pairs (n=380). In the matched cohort, prehospital ETI was not independently associated with 30-day survival (adjusted HR 0.91, 95% CI 0.65-1.27). Mortality risk was independently associated with lower GCS (per 1-point increase: HR 0.86, 95% CI 0.82-0.91) and respiratory distress (HR 7.63, 95% CI 4.34-13.4). A significant sex-by-age interaction […]

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Mechanical and manual cardiopulmonary resuscitation in traumatic cardiac arrest: a retrospective observational study.

El-Menyar A, Ahmed K, Howland IR, Mollazehi M, Mekkodathil A, Rizoli S, Cameron P, de Oliveira Manoel AL et al.

BACKGROUND: Prehospital Traumatic cardiac arrest has a high fatality rate despite advances in trauma systems and resuscitation strategies. Mechanical chest compression devices have been increasingly adopted to optimize cardiopulmonary resuscitation (CPR) during Emergency Medical Services (EMS) transportation. We aimed to evaluate the impact of CPR (mechanical during transportation versus manual CPR only at the scene) on survival among trauma patients.

METHODS: A retrospective analysis was conducted for patients who received CPR at the scene and during transportation to the hospital between 2016 and 2024.

RESULTS: A total of 610 patients sustaining blunt traumatic cardiac arrest were included. The mean age was 34 ± 12.6 years; 94.6% were male, and 5.4% were female patients. Two hundred twenty (36.1%) received mechanical CPR during transportation to the hospital, and 390 (63.9%) received manual CPR only at the scene. Compared with the manual CPR group, the mechanical CPR group had higher rates of primary chest injury (30.5% vs. 20.5%; p = 0.006), bystander CPR (15.9% vs. 6.7%; p = 0.001), adrenaline administrations (95% vs. 26.4%; p = 0.001),

and initial non-shockable rhythm (80% vs. 27.2%; p = 0.001). The two groups were comparable in median Injury Severity Score (ISS), head Abbreviated Injury Score (AIS), and total transport time. Mechanical CPR was associated with a markedly […]

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Cervical spine clearance in pediatric trauma patients - Consensus algorithm of the Pediatric Spinal Trauma Group of the Spine Section of the German Society for Orthopedic and Trauma Surgery (DGOU).

Bolte J, Rüther H, Brecht P, Wiersbicki DW, Youssef Y, Jarvers JS, Disch AC

PURPOSE: Although cervical spine injuries in pediatric patients are rare, clinical and radiological examinations are clinical routine. Currently, no standardized decision algorithm does exist for detection of these injuries in the European region. The proposed algorithm is aiming to safely identify significant injuries through consistent diagnostics while minimizing immobilization and radiation exposure at the same time.

METHODS: The Pediatric Spinal Trauma Group of the Spine Section of the German Society for Orthopedic and Trauma Surgery (DGOU) developed an algorithm based on own scientific investigations and considering primary and secondary literature. The algorithm was developed through a formal consensus process and thereby aligns with international recommendations.

RESULTS: The algorithm uses the initial pediatric Glasgow Coma Scale (pGCS) to group patients to three treatment pathways. Resulting recommendations for imaging, clinical re-assessment, and further management include the child's age, trauma mechanism, and clinical evaluations.

CONCLUSION: Despite their rarity, potential cervical spine injury in children requires accurate diagnosis. Clinical examinations can be challenging, especially in children with compromised consciousness. Increased radiation exposure from X-rays and computer tomography (CT) scans must be considered, particularly as magnetic resonance im […]

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Development of a low-cost external fixation system: an off-label solution for resource-limited settings.

Seiser M, Deininger C, Fierlbeck J, Hollensteiner M, Coles P, Reuter C, Tempfer H, Traweger A et al.

PURPOSE: External fixators are essential for the surgical management of long-bone fractures, particularly in resource-limited settings where access to commercial systems is often restricted by cost and logistics. This study aimed to develop a low-cost external fixation concept using widely available construction materials and evaluate its biomechanical performance under axial loading.

METHODS: An external fixator composed of M10 threaded rods, washers, and nuts was developed. Five configurations were tested: SP (Steinmann pins), KT (threaded K-wires), KS (smooth K-wires), and BKS (bilaterally applied, smooth K-wires). A commercial external fixator served as control (CG). Juvenile bovine ulnae with a standardized fracture gap were subjected to cyclic axial loading. Construct stiffness, elastic deformation, and plastic deformation were assessed.

RESULTS: CG showed the highest stiffness (107.80 ± 14.12 N/mm), followed by SP (87.73 ± 17.92 N/mm), BKS (83.81 ± 10.01 N/mm), KS (49.84 ± 3.80 N/mm), and KT (46.66 ± 2.39 N/mm). Post-hoc analyses showed no significant differences between CG vs. SP and CG vs. BKS, whereas KT and KS were significantly less stiff. Elastic deformation followed a similar pattern. Plastic deformation was greatest in CG, while SP and BKS showed significantly lower values.

CONCLUSIONS: The threaded-rod fixator provides relevant mechanical stability at extreme […]

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Supplementary fixation in multifragmentary midshaft clavicle fractures: do interfragmentary screws or suture cerclage improve outcomes compared with plate-only fixation?

Barthel C, Hinkelmann MA, Schibler D, Frei F, Pape HC, Allemann F

BACKGROUND: Clavicle fractures are common injuries. As displaced or multifragmentary midshaft patterns carry a substantial risk of malunion and nonunion with nonoperative treatment, they have contributed to the increasing use of surgical fixation. Modern operative techniques include plate fixation, intramedullary devices, and hybrid strategies to address comminution and improve fragment stability. Supplementary interfragmentary screws and cerclage constructs have been proposed to enhance fragment control and optimize load transfer in multifragmentary fracture patterns. This study aimed to compare three fixation strategies for operatively treated midshaft clavicle fractures managed with plate osteosynthesis: Group IS (interfragmentary screw), Group CA (cerclage augmentation), and Group POF (plate-only fixation), with respect to operative time, complication rates, and potential biological and mechanical differences.

METHODS: This retrospective study included patients aged ≥ 16 years who underwent plate fixation for multifragmentary clavicle fractures between 2016 and 2024. Patients were categorized into three groups: Group IS (supplementary interfragmentary screw), Group CA (suture cerclage augmentation), and Group POF (plate-only fixation). Demographic, surgical, and radiographic data were collected. Callus formation was assessed by three independent observers. Statistical anal […]

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Cerebral glycerol during haemorrhagic shock in normal and raised intracranial pressure resuscitated with total REBOA: an experimental porcine study.

Bader S, Magnuson A, Brorsson C, Wallin G, Löfgren N, Löfgren F, Blind PJ, Öman M et al.

BACKGROUND: Haemorrhagic shock (HS) is frequently associated with secondary cerebral injury due to compromised cerebral perfusion. Resuscitative endovascular balloon occlusion of the aorta (REBOA) is an effective haemorrhage control strategy; however, concerns persist regarding its cerebral effects, particularly in the presence of elevated intracranial pressure (ICP). Cerebral microdialysis (CMD) derived glycerol (CGly) is a recognised marker of cellular membrane stress and injury.

OBJECTIVE: To investigate CGly dynamics during prolonged total REBOA (tREBOA) in HS and to compare cerebral cellular responses between animals with normal and elevated ICP.

METHODS: In this experimental porcine study, eighteen pigs were subjected to controlled HS and resuscitated with tREBOA for 90 min. Animals were allocated to either a normal ICP group (NICPG, n = 9) or an elevated ICP group (EICPG, n = 9). Continuous monitoring of proximal arterial pressure, ICP, and cerebral perfusion pressure was performed. CGly concentrations were measured using CMD throughout the experimental protocol.

RESULTS: tREBOA effectively restored proximal arterial pressure and cerebral perfusion pressure in both groups. CGly concentrations remained relatively stable during the haemorrhage phase and increased progressively during prolonged aortic occlusion. Importantly, CGly responses did not differ significantly be […]

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Transport and destination hospital for patients with suspected multiple and/or severe injuries - a systematic review and clinical practice guideline update.

Pavlu F, Könsgen N, Bieler D, Goossen K, Gliwitzky B, Häske D, Faul P, Wölfl C et al.

PURPOSE: Our aim was to update the evidence-based and consensus-based recommendations on criteria for transport and destination hospital for patients with suspected multiple and/or severe injuries based on available evidence. This guideline topic is part of the 2025 update of the German Guideline on the Treatment of Patients with Multiple and/or Severe Injuries.

METHODS: MEDLINE and Embase were systematically searched to January/February 2024. Further literature reports were obtained from clinical experts. Randomised controlled trials (RCTs) or observational studies reporting risk-adjusted outcomes were included if they compared different transport strategies or destination hospitals for patients with (suspected) multiple and/or severe injuries. We considered patient-relevant outcomes such as mortality and neurological outcomes. Risk of bias was assessed at outcome level using ROBINS-I for observational studies and RoB 2 for RCTs. We conducted meta-analyses if possible; alternatively, we synthesised the evidence narratively. We used GRADE to rate the certainty of evidence. Expert consensus was used to develop recommendations and determine their strength.

RESULTS: A total of 127 studies were identified. They addressed helicopter vs. ground emergency medical services (EMS) transport, ground EMS with vs. without a physician, on-scene time, and handover on arrival at the hospital […]

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Does the addition of peripheral nerve blocks improve analgesia during distal radius fracture reduction? A prospective, randomized, multicenter clinical trial.

Guerrero-Serrano JA, Santamaría-López A, García-Lamas L, Jiménez-Díaz V, González-Fernández Á, Cecilia D

PURPOSE: This study aimed to evaluate whether adding a median nerve and a superficial branch of the radial nerve block to hematoma block improves analgesia during distal radius fracture reduction and to identify factors associated with increased pain perception.

METHODS: A prospective, randomized, controlled study was conducted in two tertiary hospitals, including 180 adult patients with displaced distal radius fractures requiring closed reduction. Patients were allocated either to an isolated hematoma block group or to a combined block group receiving hematoma block with median nerve and superficial branch of the radial nerve blocks. Within the combined group, half of the median nerve blocks were performed under ultrasound guidance and half using anatomical landmarks. The primary endpoint was digital pain during traction and fracture reduction, assessed using a visual analog scale (VAS, 0-10). Secondary outcomes included wrist pain at different procedural stages and the influence of demographic and clinical variables including age, sex, laterality, participating hospital, body mass index (BMI), fracture type (AO/OTA), and baseline psychiatric or chronic analgesic medication use.

RESULTS: Digital pain was significantly lower in combined group compared with isolated hematoma block during traction (4.54 ± 2.53 vs. 7.39 ± 1.76; $p < 0.001$) and reduction (5.36 ± 2.56 vs. 8.33 ± 1. […]

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Under pressure - a sensor-based analysis of simulated prehospital pressure dressing application for hemorrhage management.

Brauckmann V, Menzel J, Welke B, Decker S, Macke C

BACKGROUND: The concept of "x-ABCDE", with "X" representing exsanguinating hemorrhage, highlights the priority of immediate bleeding control in trauma care. Despite being a fundamental intervention, pressure dressing techniques lack standardization. Studies show high variability in achieved pressures, often below therapeutic targets or exceeding harmful levels, with no correlation between subjective estimates and objective measurements.

OBJECTIVES: To objectively assess current practices of pressure bandage application and identify influencing factors including material type, technique, and provider experience.

METHODS: This prospective, single-center, simulation-based observational study was conducted from August 2024 to January 2025 at a Level I Trauma Center. Emergency medical providers applied pressure dressings according to personal preference using available materials. Pressure distribution was measured via two calibrated capacitive pressure sensors (61 measurement fields). Analysis included correlation tests, group comparisons, and multiple regression.

RESULTS: A total of 124 emergency medical providers (75% male; 36% paramedics, 44% emergency medical technicians, 13% trainees, 7% emergency physicians) completed pressure dressing applications, with 116 datasets included in final analysis. Mean maximum pressure was 169.35 ± 84.33 mmHg (range 50-412 mmHg) with applicati […]

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Clinical impact of early postoperative routine radiology in isolated pelvic injuries: a retrospective study.

Bolierakis E, Ioannou P, Schubert SL, Groven RVM, Bläsius FM, Bode M, Alabdulrahman H, Hildebrand F et al.

PURPOSE: The clinical value of routine immediate postoperative conventional radiography following surgical fixation of pelvic fractures remains debated. This study assessed the impact of early postoperative X-rays on management decisions and revision surgery in patients with isolated pelvic ring or acetabular fractures.

METHODS: We conducted a retrospective cohort study at a Level 1 trauma center, including adults undergoing surgical treatment for isolated pelvic ring or acetabular fractures between 2020 and 2023. All patients received high-quality intraoperative fluoroscopy and postoperative conventional radiographs. The primary outcome was any change in standard postoperative care based on radiographic findings. The secondary outcome was revision surgery prompted by conventional postoperative imaging.

RESULTS: A total of 216 patients were included (median age 63 years), with 156 pelvic ring injuries and 60 acetabular fractures. Postoperative radiographs were obtained at a median of 3 days post-surgery. Changes in postoperative care based on radiographs occurred in 2 patients (3.3%) within the acetabular fracture group. Revision surgery triggered by conventional imaging was required in 1 patient (0.6%) among pelvic ring injuries. In all cases, additional CT imaging preceded definitive management changes.

CONCLUSION: Routine immediate postoperative conventional radiography a […]

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Older age and post-traumatic organ failure a TraumaRegister DGU® analysis of 34,469 elderly major trauma patients.

Bewersdorf TN, Lefering R, Stein S, Streblow J, Findeisen S, Schmidmaier G, Grossner T

PURPOSE: As the number of elderly major trauma patients (EMTP) increases continuously, this study sought to determine the incidence of organ failure (OF) and multiple organ failure (MOF) and their impact on mortality and recovery status in this cohort.

METHODS: Based on the German TraumaRegister DGU® database this study included EMTP aged between 65 and 99 years with an Injury Severity Score (ISS) of ≥ 16 between 2014 and 2023. OF was defined as Sequential Organ Failure Assessment (SOFA) score of ≥ 3 and MOF as failure of ≥ 2 organs.

RESULTS: The database revealed 34,469 EMTP, here 37% suffered from OF (16.0%) or MOF (21.0%). Mortality was significantly higher in OF and MOF patients compared to non-OF patients (non-OF: 7.4%; OF: 45.0%; MOF: 59.0%; $p < 0.001$). 80.9% of all deaths were associated with failure of at least one organ and 51.1% of non-survivors suffered from MOF. Highest odds ratios (OR) for mortality were found for liver (OR: 5.13, $p < 0.001$), central nervous system (OR: 3.63, $p < 0.001$) and renal failure (OR: 3.25, $p < 0.001$), followed by cardiovascular (OR: 1.95, $p < 0.001$) and lung failure (OR: 1.16, $p = 0.010$). Haemostatic failure had no impact on mortality.

CONCLUSION: OR and mortality differ significantly between distinct OFs due to varying degrees of success in treatment options and systemic impact of these OFs. Clinicians should be aware that OF and MOF o […]

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Denis pattern-dependent biomechanical behavior of posterior trans-iliac plate fixation compared with bilateral triangular osteosynthesis in unilateral sacral fractures: a finite element study.

Yang J, Böker KO, Li C, Schilling AF, Ritter Z, Acharya M, Lehmann W

OBJECTIVE: Posterior trans-iliac plate osteosynthesis (TPO) is commonly used for the stabilization of unilateral vertically unstable sacral fractures. However, its biomechanical performance across different Denis fracture patterns remains unclear. This study aimed to investigate the fracture pattern-dependent biomechanical behavior of posterior trans-iliac plate osteosynthesis (TPO) in unilateral vertically unstable sacral fractures, using bilateral triangular osteosynthesis (BTO) as a high-stability reference construct.

METHODS: A validated three-dimensional finite element model of the lumbopelvic complex was developed to simulate unilateral vertically unstable sacral fractures corresponding to Denis type I, II, and III patterns. Two posterior fixation strategies, TPO and BTO, were constructed for each fracture type. Seven physiological loading conditions were applied, including standing, flexion, extension, axial rotation, and lateral bending. Von Mises stress distribution and fracture displacement were evaluated. Fracture micromotion was quantified by calculating relative displacement changes between paired points along the fracture gap.

RESULTS: When interpreted relative to the high-stability BTO reference construct, TPO showed fracture pattern-dependent variation in fracture displacement. The difference was most pronounced in Denis type I fractures and progressively decr […]

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Factors associated with in-hospital mortality in necrotising soft tissue infections. a multicentre retrospective cohort study.

Podda M, Ceresoli M, Virdis F, Rosa F, Pilia T, Murzi V, Dessì A, Tedesco S et al.

PURPOSE: Necrotising soft tissue infections (NSTIs) are rare but life-threatening conditions associated with high mortality rates. This multicentre study aimed to identify admission variables associated with in-hospital mortality.

METHODS: This retrospective study included adult patients with surgically confirmed NSTIs treated at four high-volume academic referral centres in Italy between 2010 and 2024. Demographic, clinical, physiological, and laboratory variables available at hospital admission were analysed. Categorical variables were compared between survivors and non-survivors using the chi-square test or Fisher's exact test, while quantitative variables were compared using the Student's t test or Mann-Whitney U test. Multivariable logistic regression analyses were performed to identify independent predictors of in-hospital mortality. Results were reported as ORs with 95%CI. A prognostic nomogram was developed from the final multivariable model, including variables independently associated with mortality.

RESULTS: A total of 379 patients were included. In-hospital mortality was 16.7%. In subgroup comparisons, mortality was 17.0% in necrotising fasciitis and 22.4% in Fournier's gangrene ($P = 0.275$). In the overall NSTI population, age (aOR 1.061, 95%CI 1.037-1.087) and serum creatinine at admission (aOR 1.301, 95%CI 1.040-1.629) were independently associated with mortalit […]

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Influence of frailty on short-term mortality in older patients with multiple trauma in the emergency department.

Çelik ■, Toker A, Biçer S, Cebecio■lu ■K

PURPOSE: To investigate the association between frailty and 28-day mortality in older patients presenting to the emergency department with multiple trauma and to assess the discriminative performance of PRISMA-7 and the Trauma-Specific Frailty Index (TSFI) in frailty assessment in relation to commonly used trauma severity scores.

METHODS: This prospective observational study included patients aged ≥ 65 years with multiple trauma admitted to the emergency department between July 1 and August 31, 2024. Frailty was assessed using PRISMA-7 and TSFI. Patients were stratified into frail and non-frail groups and further categorized according to age groups (65-74, 75-84, and ≥ 85 years). Clinical characteristics, trauma severity scores, and outcomes, including 28-day mortality, were compared. Receiver operating characteristic (ROC) analysis was used to evaluate discriminative performance, and logistic regression analysis was performed to assess the association between scores and mortality.

RESULTS: A total of 144 patients were included. Frailty was significantly associated with increased 28-day mortality, higher ICU admission rates, and longer hospital stays ($p < 0.001$). Mortality increased with age, reaching 30.0% in patients aged ≥ 85 years ($p = 0.006$). Frailty scores (PRISMA-7 and TSFI) increased significantly with age, whereas ISS did not differ significantly across age groups. R […]

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Mortality prediction in older adults with earthquake-related trauma: a model combining MEWS and laboratory biomarkers.

Ozen A, Acehan S, Satar S, Bulut EA, Gulen M, N■g■z K, Sevd■mbas S, Sonmez GO et al.

PURPOSE: This study aimed to identify laboratory parameters and clinical scores that can predict in-hospital mortality among patients aged 65 years and older who presented to the emergency department (ED) with earthquake-related trauma following the February 6 and 20, 2023 earthquakes.

METHODS: In this retrospective observational study, 362 patients with earthquake-related trauma between February 6-21, 2023, were included. Demographic characteristics, laboratory results, and clinical scores such as the Modified Early Warning Score (MEWS), Injury Severity Score (ISS), and Shock Index (SI) were analyzed.

RESULTS: Of the included patients, 53.9% (n = 195) were female, and the median age was 73 years (IQR: 69-80). The overall mortality rate was 9.4%. Multivariable logistic regression analysis identified MEWS (OR: 3.535; 95% CI: 2.397-5.212; p < 0.001), urea (OR: 1.009; 95% CI: 1.002-1.015; p = 0.008), and hs-Troponin I (OR: 1.001; 95% CI: 1.000-1.002; p = 0.016) as independent predictors of mortality. Receiver operating characteristic (ROC) analysis demonstrated an AUC of 0.936 for the combined model including MEWS, urea, and hs-Troponin I (95% CI: 0.875-0.996; p < 0.001). Among the variables, MEWS showed the highest AUC (0.903; 95% CI: 0.829-0.976). The optimal cutoff value for MEWS was determined as 1.5, with a sensitivity of 70.6% and a specificity of 97.6%.

CONCLUSIONS: In o […]

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The economic burden in terms of cost of illness and generic health-related quality of life of posttraumatic long bone non-unions among the adult population of the Netherlands from a societal perspective.

van der Broeck LCA, Shchurov DG, Gabrio A, Lodewijks AJL, Poeze M, Blokhuis TJ, Evers S

PURPOSE: This study assessed the societal economic burden in terms of cost of illness and health-related quality of life (HRQoL) of posttraumatic long bone non-unions in the Netherlands.

METHODS: An incidence-based bottom-up approach was used, focusing on adult patients with posttraumatic long bone non-unions. The analysis included healthcare costs, patient and family costs, and productivity losses, measured using the iMCQ and iPCQ questionnaires. Cost evaluations followed Dutch costing guidelines, and productivity losses were calculated using the friction cost method. HRQoL was assessed with the EQ-5D-5L. A deterministic one-way sensitivity analysis varied baseline characteristics by ± 10%. Scenario analyses were conducted from healthcare and patient perspectives, as well as for patient subgroups.

RESULTS: Average costs per patient during the three months before the initial visit at a non-union clinic were €8,928 (healthcare, N = 78), €1,360 (patient and family, N = 58), and €4,313 (productivity losses, N = 59), the average of the observed data calculates to 10,831 (N = 44). The mean EQ-5D-5L utility score was 0.390 (± 0.29 SD). Subgroup analysis showed no significant cost increase for patients with infections or open fractures. However, a higher Non-Union Severity Score and a lower HRQoL were significantly associated with higher total costs.

CONCLUSION: Posttraumatic long […]

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Nonoperative treatment of Maisonneuve fractures.

Frühauf J, Bartoní■ek J, Fojtík P, Hor■áková A, Tu■ek M

PURPOSE: The aim of this study is to describe the basic pathoanatomical characteristics of a stable Maisonneuve fracture and mid-term results of its nonoperative treatment.

METHODS: The study included 17 prospectively collected patients with a mean age of 59 years. The postinjury ankle CT had to meet the following criteria: nondisplaced or minimally displaced (up to 1 mm) fracture of medial malleolus, medial clear space less than 3 mm, nondisplaced or minimally displaced (up to 2 mm) fracture of posterior malleolus, anatomical position or minimal malposition of the distal fibula in the fibular notch (widening of the tibiofibular space up to 2 mm or external rotation of the distal fibula up to 10°). The average follow-up period was 34 months, the final follow-up included CT examination and functional evaluation based on AOFAS and FAOS scores.

RESULTS: A medial malleolus fracture was recorded in 12% cases, a posterior malleolus fracture in 29% patients and a Tillaux-Chaput tubercle fracture in 18% cases. All fractures of the proximal fibula, medial and posterior malleolus healed within 3 months. The position of the distal fibula in the fibular notch did not change in 11 cases compared to the post-injury CT scan, improved slightly in 5 cases, and worsened slightly in 1 case. The average final AOFAS hindfoot score was 96.9 points and the average final FAOS score was 98.7%.

CONCL […]

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Age-related differences in prehospital triage and critical interventions: a multicenter TraumaRegister DGU® analysis.

Koch DA, Hagebusch P, Hackenberg L, Pavlu F, Jakobi T, Münzberg M, Lefering R, Schweigkofler U et al.

PURPOSE: Older trauma patients are known to be undertriaged in the prehospital setting. However, it remains unclear whether these disparities translate into differences in the delivery of critical prehospital interventions. This study aimed to evaluate age-related differences in prehospital triage, allocation, and treatment within a physician-staffed emergency medical service system.

METHODS: A retrospective multicenter analysis of the TraumaRegister DGU® was conducted, including trauma patients aged ≥ 20 years between 2020 and 2023 with trauma team activation and subsequent intensive care unit admission. Patients were stratified into two age groups, 20-59 vs. ≥60 years, with additional subgroups up to 90-99 years. Prehospital interventions, transport modality, and trauma center allocation were analyzed. Furthermore, predefined high-risk scenarios (severe motor vehicle trauma, traumatic brain injury with GCS ≤ 8, and pneumothorax) were evaluated.

RESULTS: A total of 67,069 patients were included, with comparable injury severity across age groups (mean ISS 20.2). Older patients were less frequently transported by air and less often allocated to supra-regional trauma centers. While overall intervention rates were similar, differences emerged in high-risk scenarios. Older patients received endotracheal intubation and tranexamic acid less frequently despite comparable injury seve […]

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Prediction of risk of death in severely injured patients: the revised injury severity classification score, version 3 (RISC III).

Lefering R, Imach S, Bieler D

PURPOSE: Trauma Registries with a focus on severely injured patients use survival as their primary outcome. In order to compare hospitals, interventions, and changes over time, a precise risk of death prediction is mandatory. The German TraumaRegister DGU® uses the RISC II model (Revised Injury Severity Classification, version II) since 2013. The aging trauma population and an improved handling of missing data required the present revision.

METHODS: A total of 53,738 seriously injured trauma patients documented in 2022-2023 served as basis for development and validation (3:1 ratio). Missing values should now be considered to be within normal physiological range except in patients with specific findings indicative for an altered physiology. These findings were suggested by clinical experts and validated by registry data. The increasing age of trauma patients was addressed by additional categories and higher point weights. A logistic regression analysis provided new point weights for all predictors. Precision (observed versus predicted mortality) and discrimination (area under the receiver operating characteristic curve, AUROC) were calculated in the development and validation dataset.

RESULTS: Patients in both datasets were well comparable, with a mean age of 55 years, 69% males, and an average Injury Severity Score (ISS) of 18 points. The rate of missing values ranged from 0% […]

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Dislodgement forces and cost effectiveness of prehospital securement of peripheral intravenous catheters on moist skin: a randomized controlled clinical trial in healthy volunteers.

Vogeler J, Schumann S, Heinrich S, Hell J, Schmutz A

OBJECTIVES: Peripheral intravenous catheters (PIVC) are widely used clinical devices for administering fluids or essential drugs to patients. The failure rate of PIVC remains high and Emergency Medicine has been shown to be a risk factor for dislodgement. When the skin is moist from sweat or fluids, standard intra-hospital dressings and securements fail. In emergency situations, a failed catheter can then critically delay intravenous therapies. The most effective dressing to prevent accidental removal of a prehospital PIVC remains unclear. It was the aim of this study to compare the force required to dislodge a PIVC with four different methods of securing PIVCs used in emergency medicine. In addition, the costs were calculated.

METHODS: Artificial sweat was applied to the skin of 180 volunteers. PIVCs were attached onto the forearm using four different securements (elastic gauze, cohesive gauze, clingfilm and a velcro securing device). Continuously increasing traction force was applied until dislodgement of the respective securement. For statistical tests, either Friedman's test or repeated measures ANOVA was used.

RESULTS: Clingfilm showed the greatest resistance to increasing pulling force with cohesive bandages as close second. The velcro securing device was strongest at resisting low level of forces but fell off sharply at higher force. Elastic bandages were the weakest i […]

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Injury to the thorax is a dominant contributor to post-traumatic neutrophil mediated inflammatory response.

van Eerten FJC, Duebel LV, de Fraiture EJ, Hanlo MF, Breuking EA, Brands SR, van den Bosch TD, van Wessem KJP et al.

PURPOSE: Traumatic injury induces a neutrophil associated inflammatory response mediated by the release of damage-associated molecular patterns (DAMPs). This response can be maladaptive, increasing risk of infection. Since blunt thoracic injuries are associated with high infection rates, we hypothesized that these injuries elicit a stronger neutrophil associated inflammatory response than injuries to other body regions.

METHODS: A cohort study was conducted in adult patients presenting at the trauma bay of a level 1 trauma center between March 2020 and November 2022. These patients underwent routine analysis of their neutrophil associated inflammatory response by phenotyping blood neutrophils using automated flow cytometry. Injury per body region was counted when the Abbreviated Injury Severity (AIS) scores were ≥3 to focus on substantial injuries. The primary outcome was an extensive post-traumatic inflammatory neutrophil response.

RESULTS: 861 patients were included. Blunt internal thoracic injuries showed the strongest association with severe neutrophil associated inflammation (OR 5.7), followed by abdominal (OR 2.6), pelvis (OR 2.5) and lower extremities (OR 2.3), even when corrected for ISS, age and hemodynamic instability. In multivariable analysis, only thoracic internal injuries remained a statistically significant contributor to excessive neutrophil associated inflam […]

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Major improvement using peroperative 3D compared to 2D imaging for SI screw fixation in traumatic pelvic fractures.

van Eerten FJC, Benders KEM, van Baal MCPM, Hietbrink F

INTRODUCTION: Pelvic fractures, though accounting for only 3% of skeletal injuries, carry a high mortality rate, primarily due to the associated risk of hemodynamic instability. Sacroiliac (SI) screw fixation is a common surgical intervention, which can be technically challenging due to the complex pelvic anatomy. Imaging techniques have evolved from 2D to 3D imaging for its improved accuracy in screw placement. The impact of this transition for SI screw placement was evaluated.

METHOD: A single-center observational retrospective study was performed between 2013 and 2023. Patients ≥ 18 years who underwent SI screw placement following pelvic injuries were analyzed. 2D imaging was used until 2021, after which 3D imaging was implemented. Outcomes included the need for revision surgery during initial admission and < 18 months after trauma.

RESULTS: 2D imaging was used in 42 patients and 3D imaging in 44 patients. Baseline characteristics and injury severity of both groups were comparable. In the 2D group, 18 patients (42.9%) had suboptimal placement of the SI screw. Of these, 3 (7.1%) required revision surgery due to malpositioned screws. In contrast, no patients (0.0%) in the 3D group had suboptimally positioned screws or underwent revision surgery, ($p < 0.001$, $p = 0.22$).

DISCUSSION: Intraoperative 3D imaging enhanced the accuracy of SI screw placement, leading to an encouragin […]

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Minimally invasive versus open fixation for flail chest in elderly patients: a prospective cohort study on perioperative safety and long-term outcomes.

Tian Y, Zheng Z, Ji Z, Lu W, Song N, Wang J

OBJECTIVE: To systematically evaluate the perioperative safety and short- and long-term efficacy of minimally Invasive fixation using a reversed shape-memory alloy plate versus conventional open fixation for the treatment of flail chest in elderly patients.

METHODS: This prospective cohort study enrolled 130 elderly patients (≥ 60 years) with flail chest between January 2022 and January 2026. After 1:1 propensity score matching, 100 patients were included (minimally invasive group, n = 50; open group, n = 50). Perioperative parameters (operative time, blood loss, C-reactive protein(CRP) at 24 h/48 h/72 h, complications), short-term outcomes (resting/cough-evoked Numerical Rating Scale(NRS) pain scores, non-steroidal analgesic drug duration, rescue analgesic frequency, Patient-Controlled Analgesia Pump(PCA) consumption, time to ambulation, hospital stay), and long-term outcomes (implant dislodgement, 6-month pulmonary function, Douleur Neuropathique 4 Questions Naire(DN4), Medical Outcomes Study 36- Item Short Form Health Survey(SF-36)) were compared.

RESULTS: The minimally invasive group had significantly fewer incision-related complications (P = 0.046), lower resting NRS scores on postoperative days 1 and 3 (P < 0.01; P = 0.041), lower cough-evoked NRS score on day 1 (P = 0.011), shorter non-steroidal analgesic duration (P < 0.01), and lower PCA consumption (P = 0.027). At 6 […]

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Epidemiology and outcomes of traumatic sternal fractures and associated blunt cardiac injury: a nationwide cohort study in the Netherlands.

Klei DS, Benders KEM, Leenen LPH, van Wessem KJP

PURPOSE: Comprehensive data on epidemiology, trauma mechanisms, associated injuries, and outcomes of traumatic sternal fractures are scarce. This study analysed nationwide data to improve diagnosis and management within the Dutch healthcare system.

METHODS: This nationwide retrospective cohort study using the Dutch National Trauma Registry included adult patients admitted with traumatic sternal fractures between 2015 and 2023. Patients with prehospital cardiopulmonary resuscitation or penetrating trauma were excluded. Incidence, patient characteristics, trauma mechanisms, associated injuries, and in-hospital outcomes were analysed. Subgroup analyses evaluated patients with concomitant blunt cardiac injury (BCI).

RESULTS: Of 568,399 adult trauma admissions, 4,765 patients (0.84%) sustained traumatic sternal fractures. Median age was 62 years; 60% were male. Motor vehicle accidents (48%) and falls (28%) were the leading mechanisms. 35% were severely injured (ISS ≥ 16). Associated injuries included rib fractures (51%), spinal fractures (36%), and lung contusions (18%). Critical care unit admission was 40%, with median mechanical ventilation duration of 4 days; median hospital stay was 5 days. In-hospital mortality was 5.7%, and 30-day mortality 6.0%. BCI occurred in 9.5% of patients and was associated with a higher number of injuries and increased injury severity, emergency inte […]

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Trauma patients ≥ 65 years old constitute 40% of severely injured adult trauma patients in Sweden - a 10-year analysis of geriatric trauma prevalence.

Bredberg S, Modalen E, Jöneby C, Nilsson D, Lapidus O

BACKGROUND: The elderly population in Sweden is steadily increasing. Age is an independent predictor of mortality in trauma patients. There is growing evidence that geriatric patients constitute an increasing proportion of the Swedish trauma population. No studies have analyzed temporal trends in the proportion of geriatric trauma patients in the context of the national trauma population in Sweden.

AIM: To determine the proportion of geriatric patients in the Swedish trauma population and analyze temporal trends in geriatric trauma prevalence.

METHODS: The study cohort was 11,969 severely injured trauma patients (NISS > 15) treated in Sweden during 2013-2022, obtained from the Swedish Trauma Registry. The proportion of geriatric patients was compiled annually. Temporal trends were analyzed using weighted linear regression models. Two subgroups were defined based on injury severity (NISS ≥ 25 and NISS ≥ 40).

RESULTS: The proportion of patients ≥ 65 years in the adult trauma population increased from 30% to 40% during 2013-2022 (p = 0.003), while the proportion of patient ≥ 80 years also increased, from 8.6% to 17% (p = 0.009).

Similar increases were also seen in the NISS ≥ 25 and NISS ≥ 40 subgroups.

CONCLUSION: There is likely an increase in geriatric trauma prevalence in Sweden, but the full extent of this temporal trend remains uncertain. Trauma patients ≥ 65 years old co […]

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A totally laparoscopic strategy combining microwave ablation and autologous blood transfusion for patients with initial WSES grade IV traumatic splenic rupture in selected transient responders: a proof-of-concept study.

Feng Z, Xie D, Hong W, Zheng C

PURPOSE: Splenic artery embolization (SAE) is the standard intervention for transient responders with high-grade splenic trauma. When SAE is unavailable, contraindicated, or declined, alternative minimally invasive approaches may be warranted. This study evaluates a totally laparoscopic strategy combining microwave ablation (MWA) for hemostasis with intraoperative autologous blood transfusion (ABT) in selected transient responders with initial WSES Grade IV splenic rupture.

METHODS: This retrospective study included 10 consecutive transient responders (October 2021-June 2024) with initial WSES Grade IV splenic rupture who underwent totally laparoscopic MWA combined with intraoperative ABT. All patients exhibited transient hemodynamic response to limited resuscitation (≤ 1500 mL fluids) and preoperative shock index > 0.9. Intraoperative injury was graded by the AAST system. Outcomes included technical success, transfusion requirements, complications, and follow-up.

RESULTS: All 10 patients underwent successful spleen-preserving laparoscopy without conversion. AAST grades: II (n = 2), III (n = 4), IV (n = 4). Mean operative time was 191.3 ± 101.7 min. Mean blood salvage was 1587.5 ± 663.9 mL, yielding 490.8 ± 80.2 mL autologous red cells. No patient required postoperative allogeneic transfusion. Hemoglobin, platelets, and lactate improved significantly by postoperative day 3 (P […]

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Pediatric pancreatic trauma: a 10-year retrospective analysis from two high-complexity centers in Medellín, Colombia.

Barrientos ML, Calderon SP, Zapata CAL, Lopez CAD

PURPOSE: Pediatric pancreatic trauma is uncommon, but it is associated with significant morbidity and an ongoing debate regarding its management. Hemodynamic stability and pancreatic duct involvement are key factors in treatment decisions.

METHODS: A retrospective study analyzed 25 pediatric patients with pancreatic trauma treated between 2010 and 2020 at two high-complexity hospitals in Medellin. Clinical variables, trauma mechanism, imaging findings, management, and complications were analyzed.

RESULTS: Blunt trauma was the predominant mechanism (64%). Preoperative computed tomography was performed in 12 patients (48%); radiological AAST grading was not retrievable in 13 (52%). Immediate surgery was performed in 13 patients (52%), predominantly driven by associated intra-abdominal injuries rather than the pancreatic injury itself. Non-operative management was initially attempted in 9 patients (36%), with a 56% success rate; failures (44%) were related to clinical deterioration and progression of peripancreatic collections. Clinically relevant postoperative pancreatic fistula (ISGPS Grade B) occurred in 3 of 13 operated patients (23%), with a mean closure time of 55.6 days (range 18-110); no Grade A or Grade C fistulas were recorded. Other complications included post-traumatic pancreatitis (24%), pancreatic pseudocyst (8%), and organ/space surgical site infection (8%); most […]

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Influence of Antithrombotics on Severely Injured Patients Over 50 With Blunt Abdominal Trauma.

Eibinger N, Limberger B, Hörlesberger N, Puchwein P, Lefering R, Bayer TP, Zumsteg JS, Pape HC et al.

INTRODUCTION: Severe abdominal trauma represents a critical subset of injuries, particularly in the aging European population, where the prevalence of preexisting anticoagulant or antiplatelet therapy is increasing. While the impact of such medications on outcomes in traumatic brain injury is well studied, limited data exist regarding their influence in patients with abdominal trauma.

MATERIALS AND METHODS: This retrospective cohort study used data from the TraumaRegister DGU® between January 2015 and December 2023. Inclusion criteria were patients aged ≥ 50 years with severe blunt abdominal trauma (AIS Abdomen ≥ 3) without relevant head injury (AIS Head ≤ 3) from Austria, Germany, or Switzerland. Patients were grouped according to pre-injury antithrombotic medication (no medication [NM] vs. antithrombotic medication [AM], with subgroups). Statistical analysis included descriptive comparisons, standardized mortality ratios (SMR) using RISC III, and multivariate logistic regression to identify independent predictors of hospital mortality.

RESULTS: Of 328,281 patients, 4,069 met inclusion criteria (2,831 NM; 1,238 ATM). Patients in the AM group were older (74.1 vs. 62.3 years) and had higher mortality (25.0% vs. 10.4%), particularly in the DOAC subgroup (30.3%). SMR analysis demonstrated increased mortality in the AM group compared to expected rates. In multivariate logistic re […]

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Predicting limb amputation in trauma-related necrotizing fasciitis of the extremities: insights from a 10-year cohort.

Yang S, Ren Z, Li Y, Zhao Z, Sun C, Liu L, Jin L, Hou Z

BACKGROUND: Necrotizing fasciitis (NF) is a rapidly progressive, life-threatening soft tissue infection that in most cases requires aggressive surgical management. Major limb amputation is a devastating limb-related outcome. We retrospectively analyzed NF cases to quantify the rate of amputation and identify associated risk factors.

METHODS: We retrospectively reviewed the medical records of all patients with trauma-related extremity NF treated at Hebei Medical University Third Hospital from Jan 2014 to Dec 2024. Demographics, comorbidities, admission laboratories and key time intervals were compared between patients who did and did not undergo major limb amputation. Multivariable logistic regression identified independent predictors, and receiver operating characteristic (ROC) analysis assessed the discriminative power of continuous variables.

RESULTS: Of 134 patients, 22 (16.4%) required amputation. Compared with the non-amputation group, amputees had a markedly longer pre-hospital delay (median 242 h vs. 65 h), more multiple injuries (63.6% vs. 13.4%) and a higher prevalence of diabetes (59.1% vs. 21.4%). Admission lactate dehydrogenase (LDH) was substantially higher (median 589 vs. 182 U/L). In multivariable analysis, elevated LDH, prolonged time from traumatic injury to hospital admission, multiple injuries and diabetes were independently associated with amputation. LDH […]

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Is 18 F-FDG PET-CT-guided geographic debridement superior to conventional debridement in appendicular chronic osteomyelitis? a retrospective comparative cohort study.

Elsheikh A, Elseidy H, Saad H

BACKGROUND: Determining the surgical extent of chronic osteomyelitis and fracture-related infection (FRI) remains challenging. Adequate resection margins prevent recurrence; however, no standardised preoperative tool objectively defines resection borders.

OBJECTIVE: To compare PET■CT-guided surgical planning versus the local standard■of■care imaging pathway (predominantly plain radiographs, with selective CT or MRI in a minority of patients) for chronic limb osteomyelitis and FRI outcomes.

METHODS: Retrospective comparative cohort study of 126 patients treated 2019-2024. Group 1: 84 patients underwent 18 F-FDG PET-CT imaging and map-based geographic debridement. Group 2: 42 patients received surgery after local standard■of■care imaging pathway (predominantly plain radiographs, with selective CT or MRI).

PRIMARY OUTCOME: infection recurrence at 12 months. Multivariable logistic regression controlled for confounders.

RESULTS: PET-CT group achieved 3.6% recurrence versus 16.7% in the conventional group ($p = 0.011$), representing 78.5% relative risk reduction and a number-needed-to-treat of 8. Critically, the PET-CT cohort had significantly worse baseline complexity: 3.3-fold longer infection duration (43.1 vs. 13.1 months, $p < 0.001$) and 3.6-fold more prior surgeries (2.48 vs. 0.69, $p < 0.001$). Multivariable analysis confirmed PET-CT-guided surgery as an independent protective […]

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Digital coordination in mass casualty incidents: a retrospective analysis of prehospital distribution times before and after IVENA-MANV implementation.

Brauckmann V, Ringe B, Caviglia M

BACKGROUND: Digital coordination tools have been proposed to improve management during mass casualty incidents (MCIs). In 2021, the IVENA-MANV system was introduced in the Hannover region as an extension of the established IVENA eHealth platform to enable digital patient tracking and hospital capacity management. However, no real-world evaluations have yet quantified their operational impact.

OBJECTIVES: This study aimed to provide the first real-world evaluation of the IVENA-MANV digital coordination system and to determine how its implementation affected prehospital distribution processes and time intervals during mass casualty incidents.

METHODS: A retrospective observational study to assess the impact of the IVENA-MANV digital coordination platform on prehospital times during MCIs in Germany was conducted using dispatch records and IVENA-MANV logs from all MCIs between January 2018 and July 2025 in the Hanover-Region in Germany. Primary outcomes were the triage-to-evacuation (TtE) and total prehospital time (TPHT) per patient. Mann-Whitney U tests compared pre- and post-implementation groups, and multiple linear regression models examined associations between IVENA-MANV use, MCI-level, and time intervals.

RESULTS: Seventy-three MCIs met inclusion criteria, including 188 documented individual casualty characteristics. 41.1% of incidents used IVENA-MANV. Most MCIs were tra […]

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The real-world management of acute mesenteric ischemia in Spain: results from a multicenter national survey.

González-Castillo AM, Calsina Juscafresa L, Gelabert A, Velescu A, Marrano E, Vikatmaa P, Tolonen M

BACKGROUND: Acute mesenteric ischemia (AMI) remains one of the most lethal vascular emergencies, with in-hospital mortality rates frequently exceeding 50%. Although early diagnosis and timely revascularization are critical, clinical practice remains highly variable. This study aimed to evaluate the real-world management of AMI in Spain, identifying gaps in resources, protocols, and interhospital coordination.

METHODS: A national cross-sectional survey was conducted between March and August 2024 using the Survio® platform. The questionnaire was distributed to general surgeons through national surgical societies and included 27 items covering hospital infrastructure, clinical protocols, diagnostic and therapeutic availability, personal experience, and perceived system-level barriers.

RESULTS: A total of 291 surgeons responded. The median age was 40 years (IQR: 17). Most were consultants (76.3%) working in tertiary (42.6%), secondary (33%), or community hospitals (24.4%). While 97.6% reported access to 24/7 multiphasic CT and 90.4% to round-the-clock radiology, only 51.9% had 24/7 interventional radiology and 60.1% vascular surgery. Just 26.8% had institutional AMI protocols. The median distance to a referral center was 25 km (range: 2-250 km), and 68.4% reported difficulty in patient transfers. While 96.9% felt competent managing AMI, only 36.4% were familiar with the term "int […]

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Changes in operative and in-hospital outcome of proximal femoral fractures before and after the implementation of work hour restrictions in Switzerland.

Tewes A, Canal C, Schöb O, Neuhaus V

PURPOSE: Surgeon fatigue is a recognized risk factor for patient safety; however, reduced working hours may limit surgical training exposure. In 2005, Switzerland introduced a maximum 50-hour workweek for resident surgeons. In view of current discussions on further reductions (42 + 4 h), concerns persist regarding training quality and workforce capacity. This study evaluates the impact of the 2005 Swiss labor law on patient outcomes, operative characteristics, and intraoperative teaching utilizing a standardized trauma procedure.

METHODS: A retrospective analysis of anonymized data from a national database was performed. All patients undergoing operative fixation of trochanteric femur fractures (ICD-10 S72.1) were included. Two four-year periods were compared: pre-regulation (2001-2004) and post-regulation (2016-2019). Primary endpoints were in-hospital complications, mortality, and length of stay. Secondary endpoints included patient characteristics, surgeon seniority, teaching status, operative duration, and time to surgery.

RESULTS: Post-regulation patients were older and exhibited higher American Society of Anesthesiologists classifications despite fewer comorbidities. Complications increased from 9.1% to 17.7%, and mortality from 1.6% to 3.5%. Length of stay decreased from 14 to 9 days. Operative duration decreased by 10 min across all surgeon levels. Resident surgeons i […]

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Early discharges in severely injured patients by ISS score: exploring injury patterns and coding practices.

Parouei F, Krüsemann EJZ, Poeze M, de Jongh MAC, Hietbrink F, Landelijke Beraadsgroep Traumachirurgen (LBTC)

PURPOSE: This study examines the prevalence of early discharge (ED) among patients classified as severely injured in the Dutch National Trauma Registry (DNTR) and evaluates whether identifiable patient-, injury-, and system-related factors are associated with the occurrence of ED in this population.

METHODS: This retrospective cohort study analyzed DNTR data from 2015 to 2022, focusing on severely injured patients (Injury Severity Score [ISS] ≥ 16). Patients were grouped as early discharge (ED; discharged within 48 h without in-hospital mortality) or non-ED (NED). Patients' characteristics were compared using descriptive statistics, and a mixed-effect model identified factors associated with ED.

RESULTS: From 2015 to 2022, a total of 37,626 severely injured patients were registered including 1,454 (3.9%) ED patients. This proportion increased from 3.2% in 2015 to 4.8% in 2022. ED was more common in younger patients, males, general practitioner (GP) or self-referred cases, those with severe head injury (Abbreviated Injury Scale [AIS] ≥ 3) and fewer registered injury codes. In the mixed-effects model, younger age (OR up to 2.0), increasing head injury severity (OR: 1.16 per AIS point), less than 3 registered AIS codes (OR: 0.60), referral by a general practitioner (OR 1.68) or self-referral (OR 1.92), and admission year (OR: 1.08 per year) independently predicted early discharg […]

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Trauma team activation for pediatric patients in Denmark: a multicenter study of criteria, organization, and injury severity.

Nordestgaard CH, Gude MF, Leth SV, Raaber N

BACKGROUND: Pediatric trauma team activation (TTA) is intended to ensure timely management of severely injured children, yet both prehospital visitation practices and in-hospital TTA criteria vary across trauma systems. The aim of this study was to describe and compare pediatric TTA criteria, organizational models, and injury severity across all Danish level I trauma centers.

METHODS: We conducted a retrospective multicenter cohort study including all pediatric trauma patients (< 18 years) admitted via TTA at the four Danish level I trauma centers between 2014 and 2024. Center-specific prehospital and in-hospital TTA protocols were obtained through structured inquiries. Injury severity was assessed using Injury Severity Score (ISS) and Abbreviated Injury Scale (AIS). Overtriage was defined as TTA in patients with ISS < 15.

RESULTS: A total of 3,452 pediatric trauma patients were included. Two distinct TTA models (single-criterion and point-based) and four organizational structures were identified. Overall, 14% of patients had ISS ≥ 15, corresponding to high overtriage across all centers. Overtriage rates ranged from 81% to 95% and were highest at centers using point-based TTA models. Mortality was low and did not differ significantly between centers. Among the most severely injured patients (ISS ≥ 25), head and thoracic injuries predominated.

CONCLUSION: Pediatric trauma tri […]

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Does the operative technique (MIPO vs. ORIF) influence surgical outcomes and complications in proximal humerus fractures? A matched-pair analysis.

Barthel C, Rudolph F, Kraus M, Kalbas Y, Russek R, Pape HC, Allemann F

INTRODUCTION: Proximal humerus fractures account for approximately 4-6% of all fractures and are among the most common fractures in the elderly. Surgical treatment is increasingly applied for displaced and unstable fracture patterns. The two most commonly used techniques are open reduction and internal fixation (ORIF) via a deltopectoral approach and minimally invasive plate osteosynthesis (MIPO) using a deltoid-split or anterolateral approach. ORIF allows direct fracture visualization and anatomical reduction but is associated with more extensive surgical exposure. MIPO aims to preserve soft tissues and vascular supply through indirect reduction techniques. This study compares ORIF and MIPO in proximal humerus fractures with regard to surgical parameters, radiographic outcomes, and complications.

METHODS: A retrospective single-center cohort of 392 proximal humerus fractures was analyzed. Seventy-nine patients treated with MIPO were matched to 79 patients treated with ORIF using exact matching for fracture type (Mayo FJD classification) and propensity score matching for age and sex. Primary outcomes included operative time, reduction quality, and complication rates.

RESULTS: The MIPO group showed significantly shorter operative times (97.3 ± 30.8 min vs. 112.2 ± 34.0 min, $p = 0.004$). No significant differences were observed in reduction quality or overall complication rates […]

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Risk factors and predictors of in-hospital mortality in necrotizing fasciitis: a five-year retrospective cohort study from a tertiary care center in Colombia.

Barrientos ML, Cordoba ASR, Valencia MDML, Valencia MMM, Toro DAM, Zapata CAL, Delgado CA

PURPOSE: Necrotizing fasciitis (NF) is a life-threatening soft tissue infection with limited epidemiological data from Latin America. This study aimed to characterize clinical outcomes, identify factors associated with in-hospital mortality, and develop a clinical prediction model in patients with NF at a tertiary care center in Medellín, Colombia.

METHODS: A retrospective cohort study was conducted including all adult patients (≥ 18 years) with surgically or histopathologically confirmed NF between January 2019 and December 2024. Demographic, clinical, microbiological, and therapeutic variables were collected. Bivariable analysis and multivariable logistic regression were performed to identify independent mortality predictors. A clinical risk score was developed and internally validated using bootstrap resampling and 10-fold cross-validation.

RESULTS: Of 112 identified patients, 111 met inclusion criteria. Overall in-hospital mortality was 23.4% (26/111). Deceased patients were significantly older (median 69.5 vs. 52 years; $p < 0.001$). Serum lactate demonstrated strong prognostic value (AUC 0.719; 95% CI 0.596-0.842; 2.9 vs. 1.8 mmol/L; $p = 0.003$). Septic shock was present in 34.2% and doubled mortality risk (34.2% vs. 17.8%; $p = 0.060$). Polymicrobial infections predominated (74.8%), with a predominantly gram-negative profile and high antimicrobial resistance rates (carbapen […]

PubMed · DOI

ESTES recommendations regarding distal radius fractures.

Lameijer CM, Teuben MPJ, van Delft EAK, Tomazevic M, Kastelec M, Nau C

Distal radius fractures (DRFs) account for 15% of all fractures in adults. Optimal treatment of DRFs depend on patient and fracture characteristics. Non- or minimally displaced DRFs can be treated nonoperatively with a cast. Although surgical treatment is becoming more popular, the majority of patients with a DRF are still treated nonoperatively. In addition, these considerations may be different for elderly patients. Conflicting evidence exists on the different aspects of treatment of DRFs. These recommendations regarding distal radius fractures describe considerations regarding diagnostic modalities, associated soft tissue injuries, different treatment options, rehabilitation protocol, considerations for treatment of elderly patients, follow-up decisions and possible complications. The aim is to aid the orthopedic traumasurgeon in decision making and treatment of distal radius fractures.

PubMed · DOI

Small traumatic intracranial hemorrhages identified in routine radiology reports are associated with a low risk of adverse events: a retrospective cohort study.

Gerdås K, Wigermo A, Offenbartl M, Vestlund S, Rezk F, Vedin T

PURPOSE: To determine whether routinely available radiology reports, together with basic clinical data, can identify patients with traumatic intracranial hemorrhage (TICH) who are at low risk of adverse events.

METHODS: This retrospective cohort study of adults with TICH in Region Jönköping, Sweden (2019-2021). Clinical data, findings from radiology reports and outcomes were extracted from medical records. Hemorrhage size was classified as small (≤ 4 mm or described as minimal/very small/discrete) or larger. Adverse events were defined as neurosurgical intervention or death directly attributable to the TICH. Risk difference (RD), relative risk (RR), and Firth's penalized logistic regression were used to assess associations with adverse events.

RESULTS: Among 527 included patients, 195 (37%) had small TICHs. None of these patients experienced adverse events, compared with 13.6% neurosurgical interventions and 13.0% trauma-related deaths in the group with larger TICHs (RD 24.5% points, 95% CI 18.4-29.6; RR 97.1, 95% CI 6.1-1557.1; $p < 0.001$). Small TICH size had the strongest association with absence of adverse events. Normal neurological status and GCS 14-15 were also associated with a low risk of adverse events. Anticoagulant or antiplatelet therapy showed no significant association with adverse events.

CONCLUSION: Routinely available radiology reports, combined with basic c […]

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From fracture to joint injury: association between CT-based fracture characteristics and surgically treated non-bony injuries in tibial plateau fractures.

Neidlein C, Colcuc S, Ullerich F, Schöllmann H, Steinhausen ES, Kröpil P, Brinkmann N, Dudda M et al.

PURPOSE: Tibial plateau fractures (TPF) are often associated with non-bony injuries that may impair knee stability and long-term outcomes. However, it remains unclear in which patients preoperative MRI is indicated to detect these injuries, and clear recommendations for surgical decision making are lacking. This study aimed to develop a CT-based predictor of surgically relevant non-bony injuries.

METHODS: This retrospective single-center study included all intra-articular TPF treated between January 2022 and May 2025 at a Level I trauma center. Fractures were classified according to the 10-segment and three-column classifications. Operative reports were reviewed to identify non-bony injuries requiring surgical treatment. Descriptive statistics and multivariate logistic regression were performed to determine predictors of surgically relevant non-bony injuries.

RESULTS: Among 243 patients (mean age 49,3 ± 14,9 years), 34,6% had at least one surgically treated non-bony injury. Posterior column involvement was significantly associated with a higher injury rate (41,4%), particularly affecting meniscus (20,7%), Anterior cruciate ligament (ACL) (17,2%), and medial collateral ligament (MCL) (12,3%). The posterolateral-central (PLC) segment showed the highest segment-specific injury rate (37,6%) and was an independent predictor of ACL injury. Increasing fracture complexity was associa […]

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Transferrals and clinical pathways of unstable pelvic fractures over the last 10 years - a retrospective analysis of the Trauma Register DGU®.

Weuster M, Pfeifer R, Seekamp A, Lippross S, Lefering R, Dgu T, Menzdorf L

PURPOSE: Pelvic fractures are classified as stable or unstable. They correlate with a severity of trauma and the initial medical treatment is decisive. This study evaluated the transferrals of such fractures and described the initial treatment as well as the clinical course.

METHODS: We analysed retrospective data from a large cohort of the TraumaRegister DGU® (TR-DGU), covering the period from 2014 to 2023, comprising a total of n = 397,910 patients. All patients aged ≥ 16 years were included. Injury patterns were described according to the Abbreviated Injury Scale (AIS), the mechanically unstable fractures were classified with an AIS ≥ 3. We considered all participating hospitals within Germany. The patients were subdivided in three groups: Group 1 = primary admitted patients with outcome, group 2 = pre-treated patients transferred in from other hospitals, and group 3 = primary admitted and early (< 48 h) transferred out.

RESULTS: The majority of the patients was male and about 53 years old. Blunt trauma was the leading trauma mechanism. Concomitant injuries (AIS 2+) affected thorax (56%), spinal cord (41%), lower extremities (38%), head (31%) and abdomen (24%). Among primary admitted cases with pelvic fractures (n = 36,398), 21,091 cases (57.9%) had an unstable pelvic fracture (AIS pelvis 3-5). Level 1 trauma centers not only treated 12,836 primary admitted cases with unst […]

PubMed · DOI

Surgical treatment of coronal shear fractures: short- to mid-term results and risk factor analysis.

Bauer A, Cornel L, Sauter M, Jakobi T, Münzberg M, Klug A

BACKGROUND: Coronal shear fractures of the distal humerus are rare but severe injuries. Reconstruction is often challenging, especially in comminuted cases, which is why many surgeons opt for an elbow arthroplasty in those cases. However, total elbow arthroplasty is associated with a variety of potential problems itself. Therefore, the aim of this study was to present the functional and clinical outcome of coronal shear fractures treated by osteosynthetic reconstruction in a short- to mid-term follow-up, and to identify possible risk factors for an inferior outcome.

METHODS: We performed a retrospective follow-up assessment of 51 consecutive patients (30 women; median age 56 years, (IQR 39-62)) who underwent osteosynthetic reconstruction for coronal shear fractures between 2012 and 2022 after a minimum follow-up period of two years. The Mayo Elbow Performance Score, Oxford Elbow Score, and Disabilities of the Arm, Shoulder and Hand score were evaluated, and all available radiographs were analyzed. All complications and revision procedures were assessed. Univariable and multivariate regression analyses were performed to identify potential risk factors for a poor outcome following osteosynthetic reconstruction.

RESULTS: After a median follow-up period of 43 (IQR 28-78) months, the median Mayo Elbow Performance Score was 100 (IQR 85-100), the median Oxford Elbow Score was 42 (IQ […]

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Publisher Correction: Experience with patient-specific guides, instead of general surgical experience, improves the accuracy of 3D-guided corrective osteotomies.

Buist ML, Meesters AML, Colaris JW, Doornberg JN, Ten Duis K, Tabernée Heijtmeijer SJC, Jutte PC, Pijpker PAJ et al.

PubMed · DOI

Assessment of a new technique for biological augmentation in sternoclavicular joint dislocations: a cadaveric feasibility and biomechanical evaluation.

Somberg O, Förster E, Rosteius T, Bernstorff M, Schildhauer T, Königshausen M

AIMS: Biological augmentation is necessary in chronic sternoclavicular dislocations. However, the placement of drill holes for a graft can be difficult at that region. The aim of this study was to evaluate, as an early proof-of-concept, the feasibility of using a locally available clavicular periosteal flap (CPF) or sternal periosteal flap (SPF) for sternoclavicular joint (SCJ) stabilization and to biomechanically test the periosteal flaps.

METHODS: A cadaveric anatomic feasibility and biomechanical study (level of evidence V) was performed on nine SCJs. The CPF was mobilized from three sides by lifting the flap off the bone, preserving its attachment near the SCJ. Two 1.6 mm drill holes were created in the sternum, and the CPF was folded over the SCJ and fixed with a non-resorbable wire. Additionally, the non-resorbable wire was used in terms of a figure-of-8-bracing through a 1.6 mm clavicular drill hole to secure the joint position. Flap dimensions and sternal overlap were measured. Vice versa SPFs were prepared in selected specimens and fixed in a similar fashion. The CPFs were biomechanically tested using a uniaxial tensile testing machine to obtain values for stress at 50% strain (σ (50%)), maximum force (F_{max}), maximum stress (σ_{max}), strain at maximum force ($\varepsilon(F_{max})$), and strain at failure (ε at failure).

RESULTS: CPF mobilization and fixation were successfully achieve […]

PubMed · DOI

Differentiating injury patterns and outcomes in accidental, suicidal and occupational falls from heights.

Beyersdorf C, Hussmann B, Wergen N, Lefering R, Schiffner E, Maus U, Jaekel C, Dgu T

BACKGROUND: Falls from significant heights are a leading cause of severe injuries and can result from accidents, suicide attempts, or occupational incidents. While previous studies have explored general injury patterns associated with falls, little is known about the specific differences between accidental, suicidal, and occupational falls. This study aims to identify distinct injury patterns and outcomes among these fall types, thereby enhancing primary care and decision-making algorithms.

METHODS: We conducted a retrospective analysis using data from the German TraumaRegister DGU® (2014-2023), focusing on patients who experienced falls from heights. Only patients admitted to trauma centers with trauma team activation and a Maximum Abbreviated Injury Scale (MAIS) severity ≥ 3 were included. All falls were categorized in accidental, suicidal, or occupational and distinguished between high (≥ 3 m) and low (< 3 m) falls. Statistical analyses included chi-squared tests for categorical variables and Mann-Whitney U tests for continuous variables, with significance set at $p < 0.05$.

RESULTS: Among 188,708 patients, 29,991 (15.9%) experienced high falls, while 50,674 (26.9%) fell from low height. High falls were predominantly accidental (82.3%), with suicidal falls accounting for 17.7%. Accidental high falls mainly caused thoracic (58.9%) and head injuries (46.2%), whereas suicidal f […]

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Predictive performance of ISS and NISS for clinical outcomes in severely injured trauma patients: a retrospective registry study.

Säkkinen A, Partio N, Mattila V, Mäenpää H, Riuttanen A

BACKGROUND: The Injury Severity Score (ISS) and New Injury Severity Score (NISS) are widely used in evaluation of injury severity in trauma patients. The aim of this study is to assess the predictive accuracy of these scoring systems on clinical outcomes.

METHODS: We conducted a retrospective registry study of 1,112 severely injured trauma patients (NISS ≥ 16) treated at Tampere University Hospital (TAUH) between 2015 and 2024. Outcomes included in-hospital mortality, prolonged hospital and ICU stay (≥ 75th percentile), in-hospital intubation, and blood transfusions. Predictive performance was assessed using the area under the receiver operating characteristic curve (AUC) and the Hosmer-Lemeshow (H-L) test for calibration. Subgroup analyses were performed for patients with significant (AIS ≥ 3) head, thorax, and extremity injuries.

RESULTS: A total of 848 (76%) patients had a higher NISS than ISS with a median increase of 9 points. Both scoring systems generally demonstrated fair to considerable ($0.7 < AUC < 0.9$) discrimination for mortality and intubation, and poor to fair ($0.6 < AUC < 0.8$) discrimination for blood transfusions and prolonged hospital/ICU stay. ISS had superior discrimination for blood transfusions in the total cohort (AUC: 0.669 vs. 0.630, $p = 0.019$), head (AUC: 0.737 vs. 0.672, $p = 0.004$), and thorax (AUC: 0.656 vs. 0.613, $p = 0.006$) subgroups, and for intu […]

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Surgical and anesthetic activity in a french emergency field hospital responding to devastating floods.

Gaudray E, Brac L, Agopian P, Arnaud I, Ribelles P, Buland S, Couret A, Simonet J et al.

PURPOSE: Floods are the most frequent natural disasters, causing widespread mortality and devastation. In 2023, Storm Daniel triggered catastrophic flooding in Derna, Libya, leading to over 11,300 deaths and leaving more than 10,000 people missing. In response, the French Emergency Medical Team (EMT-FRA) ESCRIM, deployed via the European Union Civil Protection Mechanism, arrived to provide critical medical aid. METHODS: The team set up a field hospital equipped with adult and pediatric care, ICU, radiology, and laboratory services, adhering to strict aseptic conditions. Data collection was performed daily by designated physicians, with records transcribed for analysis. Over the course of the 22-day mission, the EMT-FRA provided care to over 1,600 patients, with 7.6% undergoing surgery. RESULTS: The anesthesia team performed 64 procedures, including 11 critical care assessments and 65 pre-anesthetic consultations. Surgical cases were primarily related to infections (49%), visceral issues (27%), and orthopaedic injuries (25%), including wound debridement, VAC therapy, amputations, and laparotomy. The surgical team carried out 64 surgeries and 126 consultations, with 83% of patients receiving emergency care for the first time. A significant number of procedures (78%) were dirty or infected, reflecting the local health challenges. CONCLUSIONS: The deployment of EMT-FRA demonstrated […]

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Hook plate versus locking plate fixation in lateral clavicle fractures: operative efficiency and cumulative surgical burden in a propensity score-matched cohort study.

Schibler D, Hinkelmann MA, Frei F, Pape HC, Allemann F, Barthel C

INTRODUCTION: Hook plates remain controversial in the management of lateral clavicle fractures despite their biomechanical advantages. Concerns include implant-related complications, subacromial irritation, and the need for routine implant removal. The aim of this study was to compare operative efficiency and surgical burden of hook plate fixation with locking plate fixation in lateral clavicle fractures. METHODS: A retrospective cohort analysis was conducted at a single tertiary trauma center between January 1, 2016, and December 31, 2024. Surgically treated lateral clavicle fractures were identified. Propensity score matching was performed based on age, sex, and AO fracture classification to reduce confounding, and matched pairs were compared between hook plate fixation (Group HP) and locking plate fixation (Group LCP). Outcome measures included operative time, complications, implant removal, radiographic callus formation, and refracture rates. RESULTS: A total of 101 lateral clavicle fractures were identified. After propensity score matching, 44 matched pairs (Group HP vs. Group LCP) were analyzed, with comparable baseline characteristics between groups. Operative time was significantly shorter in Group HP (77.8 ± 28.4 vs. 93.5 ± 30.2 min, $p = 0.018$). Complication rates were higher in Group HP but did not reach statistical significance (18.2% vs. 4.5%, $p = 0.093$). Implant re […]

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Diagnostic value of hypoxia-inducible factor 1 alpha and adrenomedullin in acute mesenteric ischemia.

Bayindir E, Küçükceran K, Özer MR, Ayranci MK, Kilinç ■, Kökbudak N, Bayindir GK

BACKGROUND: This study aims to evaluate the diagnostic significance of adrenomedullin and hypoxia-inducible factor 1 alpha (HIF1-alpha) levels in an animal model of acute mesenteric ischemia.

MATERIALS AND METHODS: In our study, a total of 21 male New Zealand rabbits were used, and the animals were divided into three groups for the study. Blood samples were taken for adrenomedullin and HIF1-alpha levels at 0, 1, 3, and 6 h from the control, sham, ischemia groups.

RESULTS: HIF levels of the ischemia group at the 3rd and 6th hours were statistically significantly higher than those in the sham ($p = 0.002$, $p < 0.001$, respectively) and control groups ($p = 0.002$, $p < 0.001$, respectively). Adrenomedullin levels of the ischemia group at the 1st, 3rd, and 6th hours were statistically significantly higher than those in the sham ($p = 0.004$, $p < 0.001$, $p < 0.001$, respectively) and control groups ($p < 0.001$, $p < 0.001$, $p < 0.001$ respectively).

CONCLUSION: In this experimental model of acute mesenteric ischemia, hypoxia-inducible factor-1 alpha and adrenomedullin levels were significantly higher in the ischemia group compared with the control and sham groups. Adrenomedullin demonstrated an earlier increase, while hypoxia-inducible factor-1 alpha increased significantly at later time points. These biomarkers may provide supportive diagnostic value when interpreted alongside clinical and im […]

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Videolaryngoscope-assisted fibreoptic intubation in anticipated difficult airways of maxillofacial trauma: a prospective randomised comparison with videolaryngoscope-guided nasotracheal intubation.

Jain K, Nair R, Goyal A, Makkar JK, Singla K, Arora S, Bhatia N

BACKGROUND: Maxillofacial trauma poses airway management challenges due to edema, distorted facial anatomy and associated bleeding. Further, its fixation requires unhindered access to facial fractures, often necessitating securing the airway naso-tracheally during the intraoperative period. Traditional technique of nasotracheal intubation is associated with 17%-77% incidence of epistaxis. Fibreoptic bronchoscope helps guide nasotracheal tube under vision, decreasing incidence and severity of epistaxis. Moreover, as patients with maxillofacial trauma have difficult airways, every effort should be made to secure these challenging airways with best possible technique that increases first-attempt success rate. Use of fibreoptic bronchoscope as a part of hybrid videolaryngoscope-assisted fibreoptic intubation(VAFI) technique may provide these benefits. Hence, we conceptualized this study comparing the VAFI technique with traditional videolaryngoscope-assisted technique for securing nasotracheal tube in patients with anticipated difficult airways following maxillofacial trauma. METHODS: Following a written, informed consent, we included adult patients with maxillofacial trauma, scheduled for fracture fixation under general anaesthesia, requiring nasotracheal intubation. All patients were randomised into two groups of 30 patients each, with patients in group-VLS undergoing videolaryng […]

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Malignancy risk and management outcomes in adult intussusception: a single-institution retrospective study.

Chang HC, Kang JC, Pu TW, Chen CY

PURPOSE: Adult intussusception is rare and frequently caused by a pathological lead point, including malignancy. We assessed malignancy rates by CT-based anatomical subtype/location, identified predictors of malignancy, and summarized management outcomes to support risk-stratified decision-making in the emergency setting. METHODS: We conducted a single-institution retrospective study at Tri-Service General Hospital (Taipei, Taiwan), including consecutive adults with CT-diagnosed intussusception from January 2015 to August 2024. We collected demographics, symptoms, hemoglobin, CT-based type/location, management, and histopathology. Malignancy was defined by index-admission pathology (malignant vs. benign). Anemia was defined as hemoglobin < 13 g/dL in men or < 12 g/dL in women. Predictors of malignancy were evaluated using univariable and multivariable logistic regression. Primary analyses included pathology-confirmed cases; sensitivity analyses additionally included presumed benign cases. RESULTS: Forty-seven patients were identified; 38 had pathology-confirmed outcomes. In the primary cohort (n = 38), 18 (47.4%) were malignant. Lower hemoglobin was associated with malignancy (OR per 1 g/dL increase, 0.709; 95% CI, 0.529–0.950; p = 0.021), and anemia strongly predicted malignancy (OR, 11.667; 95% CI, 2.438–55.830; p = 0.002). In multivariable analysis adjusting for age and colo […]

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Operative burden and resuscitation resource utilisation in patients with traumatic shock requiring urgent surgical or endovascular intervention.

Noonan M, O'Loan M, Hendel S, Fitzgerald M, Ban EJ, Huynh F, McKay S, Scorer A et al.

BACKGROUND: Patients with traumatic shock represent a high-risk subgroup within major trauma populations, yet the operative burden and resuscitation resource utilisation associated with urgent surgical and endovascular intervention in trauma systems managing predominantly blunt injury remain incompletely described. AIMS: To describe the epidemiology, resuscitation resource utilisation, and operative burden of patients presenting with traumatic shock who require urgent surgical or endovascular intervention. METHODS: We conducted a retrospective observational study at an Australian level 1 trauma centre using prospectively maintained trauma registries. Adult major trauma patients (≥ 16 years, Injury Severity Score ≥ 13) meeting institutional shocked trauma activation criteria between December 2022 and December 2024 were included. Resource utilisation, blood product transfusion, operative and endovascular interventions, procedural timing, and critical care outcomes were described. Comparisons were performed between shocked patients requiring urgent surgical or endovascular intervention and those managed without urgent procedures. RESULTS: Of 3667 major trauma patients, 324 (8.8%) met shocked trauma criteria, and 138 (42.6% of shocked patients) underwent urgent surgical or endovascular intervention. These patients demonstrated substantial operative complexity; 37.7% required combin […]

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Incidence of ROTEM®-defined coagulopathy at hospital arrival among polytrauma patients receiving fibrinogen concentrate in prehospital setting: a prospective observational study.

Vrbica K, Keller F, Stingl J, Vanickova K, Stepanova R, Dolecek M, Hrdy O, Gal R

PURPOSE: Trauma-induced coagulopathy affects approximately one-quarter of polytrauma patients and is associated with increased mortality. Given the central role of fibrinogen and its early depletion after injury, supplementation is recommended after hospital admission in the presence of hypofibrinogenaemia or suspected major haemorrhage. As coagulopathy develops within minutes of trauma, prehospital fibrinogen administration may help mitigate its progression before hospital arrival. This study evaluated the incidence of coagulopathy at hospital admission, assessed using ROTEM®, among polytrauma patients receiving fibrinogen concentrate in the prehospital setting. METHODS: In this single-centre, prospective observational study, adult patients with polytrauma (Injury Severity Score ≥ 16) were enrolled after prehospital administration of fibrinogen concentrate. The primary outcome was the presence of coagulopathy at hospital arrival, defined according to predefined ROTEM® Delta thresholds (FIBTEM MCF ≤ 8 mm and/or EXTEM CT ≥ 80 s and/or EXTEM MCF ≤ 49 mm). The secondary outcomes were laboratory coagulation status, 28-day all-cause mortality, and thromboembolic complications. RESULTS: Thirty patients were enrolled, of whom 28 were included in the final analysis. Twenty-seven patients received 4 g of fibrinogen concentrate and one received 2 g due to short transport time. ROTEM®-def […]

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Timing of laparoscopic cholecystectomy after percutaneous transhepatic gallbladder drainage and the risk of bailout surgery: a restricted cubic spline analysis.

Yoshitake H, Uchida K, Kataoka N, Mizobata Y

BACKGROUND: The optimal timing of laparoscopic cholecystectomy (LC) after percutaneous transhepatic gallbladder drainage (PTGBD) for acute cholecystitis remains controversial. Previous studies have often categorized surgical timing using arbitrary cutoffs and have not adequately evaluated potential non-linear associations. In particular, the interval from PTGBD to LC has rarely been analyzed as a continuous variable when assessing operative difficulty. This study aimed to assess the relationship between the interval from PTGBD to LC and intraoperative technical difficulty using restricted cubic spline (RCS) analysis. METHODS: This retrospective single-center study included patients with acute cholecystitis who underwent LC after PTGBD between November 2014 and November 2024. The primary outcome was bailout surgery (conversion to open surgery, subtotal cholecystectomy, or fundus-first approach). Secondary outcomes included operative time, intraoperative blood loss, postoperative length of stay, and major postoperative complications (Clavien–Dindo ≥ III). The interval from PTGBD to LC was analyzed as a continuous variable using multivariable regression models incorporating RCS. Models were adjusted for age, C-reactive protein (CRP) at diagnosis, Tokyo Guidelines 2018 severity grade, previous abdominal surgery, anticoagulant therapy, and Charlson Comorbidity Index. RESULTS: A tota […]

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Early orthoplastic coverage reduces fracture-related infections in Gustilo III B/C tibial fractures - a systematic review and meta-analysis.

Ruprecht N, Klingebiel FK, Hübner CT, Hax J, Teuber H, Jensen KO, Hierholzer C, Pfeifer R et al.

BACKGROUND: Gustilo-Anderson type III B/C open tibial fractures are among the most complex and high-risk injuries in trauma care. Their management demands a multidisciplinary approach, and early intervention is critical. While early debridement has been established as essential, the optimal timing for definitive orthoplastic coverage remains less clearly defined. OBJECTIVE: To evaluate whether early orthoplastic coverage reduces the risk of fracture-related infections (FRIs) and complication rates compared to late coverage in type III B/C open tibial fractures. METHODS: We conducted a PRISMA-compliant systematic review and meta-analysis of studies comparing early versus late orthoplastic treatment in patients with Gustilo-Anderson III B/C open tibial fractures. Databases included PubMed and EMBASE. Primary outcome was FRI; secondary outcomes included amputation, non-union, and revision surgery rates. Risk of bias was assessed using the Newcastle-Ottawa Scale. RESULTS: Of 3746 studies screened, five studies with a total of 454 patients met the inclusion criteria. Cutoff for early vs. late treatment was defined variably across studies (either within 72–168 h). Meta-analysis showed that early treatment significantly reduced the odds of infection (OR: 0.34; 95% CI: 0.23–0.52; $p < 0.001$) compared to late treatment. Subgroup analyses confirmed consistent benefits regardless of timing […]

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Long-term renal function after high-grade renal trauma: a nuclear imaging based cohort study.

Saxena M, Dheer Y, Roy S, Singh U, Deswal S, Kumar N, Singh A, Jaiswal V et al.

PURPOSE: Renal trauma constitutes approximately 1% of all trauma cases, yet data on long-term renal function following severe injury remain scarce. This study aimed to evaluate long-term functional and morphological outcomes in patients sustaining Grade 3 or higher renal trauma, and to identify predictors of residual renal function using nuclear imaging. METHODS: This retrospective study included patients with Grade ≥ 3 renal injuries who presented for follow-up ≥ 6 months after trauma at a tertiary care centre between January 2018 and June 2023. Patients with nephrectomy or comorbidities affecting renal function were excluded. Serum renal function tests, DTPA scans (differential GFR), and DMSA scans (cortical morphology and dye uptake) were performed. Multivariable linear regression was used to determine predictors of residual function based on injury grade, gender, and duration of follow-up. RESULTS: Thirty-six patients (mean age 20.50 ± 12.08 years; 75% male) were included with equal representation across Grades 3–5. Mean follow-up was 35.92 ± 18.09 months. The affected kidney demonstrated reduced function (mean GFR 24.96 ± 12.50 mL/min; split function 31.83 ± 12.69%; dye uptake 35.71 ± 13.40%). Grade 5 injuries were significantly associated with lower differential GFR and dye uptake ($p < 0.01$), while longer follow-up duration and female sex predicted better renal recovery ([…]

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Experience with patient-specific guides, insteatientad of general surgical experience, improves the accuracy of 3D-guided corrective osteotomies.

Buist ML, Meesters AML, Colaris JW, Doornberg JN, Ten Duis K, Tabernée Heijtmeijer SJC, Jutte PC, Pijpker PAJ et al.

PURPOSE: This study investigated whether the accuracy of 3D-guided osteotomies is influenced by surgeon experience. Two types of experience were assessed: (1) general surgical experience, defined as self-reported years in practice as an orthopaedic trauma surgeon, and (2) prior use of patient-specific guides (PSGs). Secondary aim was to evaluate whether accuracy varies by anatomical location. METHODS: 24 orthopaedic-trauma surgeons performed 75 corrective osteotomies on cadaveric long-bones (43 single-cut, 32 double-cut; 107 cuts in total). Each osteotomy was preoperatively planned on CT-derived 3D-reconstructions, and PSGs were manufactured to guide the cuts. Postoperative CT-scans were used to compare planned and executed osteotomy planes. Surgeons completed a questionnaire reporting years of practice and annual PSG usage. A Spearman's rank-order correlation was used to test differences in accuracy versus experience, a Kruskal-Wallis test was performed to explore differences across anatomical locations. RESULTS: General surgical experience ranged from 0 to 25 years (median 9), and PSG-usage from 0 to 20 cases annually (median 2). No association was found between general surgical experience and osteotomy accuracy ($p = 0.406$ and $p = 0.548$). In contrast, PSG-experience correlated with improved accuracy for angular and translational deviation of the osteotomy plane ($p = 0.027$ and […]

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Analysis of DRG remuneration of acute fracture procedures versus elective total arthroplasty for the German health care system - Is elective total arthroplasty really more cost-effective?

Hierl K, Kerschbaum M, Baumann F, Alt V

PURPOSE: Elective arthroplasty is considered a cost-effective procedure compared to trauma-related surgical interventions. This study aims to compare remuneration, certain ratios regarding implant costs, operation (OR) time and length of stay (LOS) as well as contribution margins including personnel costs for acute fracture interventions and elective arthroplasty procedures at the hip and shoulder joint in the aG-DRG system. METHODS: Based on billing data of 212 inpatient treatment cases from a University Orthopaedic Trauma Department reimbursements of nail fixation and hemi-arthroplasty (HA) for proximal femoral fractures versus elective total hip arthroplasty (THA) for osteoarthritis were analysed. Locking plate fixation and reverse total shoulder arthroplasty (rTSA) for proximal humeral fractures were compared to elective rTSA for osteoarthritis. Different ratios of the DRG remuneration in relation to LOS, OR time and implant costs but also personnel costs and contribution margins were calculated. RESULTS: DRG reimbursement was highest for fracture arthroplasty of the shoulder (9.050 EUR, rTSA) and the hip (6.930 EUR, HA). Regarding LOS, DRG reimbursement was highest for elective rTSA (1.641 EUR/day). Nail fixation of proximal femoral fractures showed the highest DRG reimbursement regarding OR time (111 EUR/min) and implant costs (15 EUR/EUR). The contribution margin was hig […]

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Percutaneous screw osteosynthesis for the treatment of intra-articular displaced calcaneus fractures.

Wilmsen L, Neubert A, Deichsel A, Schulz D, Hoelscher-Doht S, Windolf J, Icks A, Thelen S

PURPOSE: This systematic review investigates how percutaneous screw osteosynthesis for the treatment of intra-articular displaced calcaneus fractures differs from open reduction and internal fixation (ORIF). METHODS: Randomized controlled trials and controlled clinical trials investigating adults with closed intra-articular displaced calcaneus fracture (Sanders type II - IV) treated with percutaneous screw osteosynthesis or ORIF were included. Severe vascular and neurological diseases or polytrauma patients were excluded. On January 29, 2024, five databases and two trial registries were searched. The Risk of Bias Tool of Cochrane was used. The protocol was registered on PROSPERO (CRD42021244695). RESULTS: Five studies with 654 participants (682 calcaneus fractures) were included. The risk of bias was moderate to high. None of the single studies found a significant difference regarding severe complications. However, meta-analysis revealed a lower risk for the percutaneous screw osteosynthesis group compared to ORIF (relative risk (RR) 0.40 (95% Confidence Interval [0.17; 0.92]; I^2 = 0%; p = 0.03). One study showed less pain in the percutaneous screw osteosynthesis group up to 4 weeks postoperatively. No study assessed health-related quality of life. CONCLUSION: Percutaneous screw osteosynthesis in intra-articular displaced calcaneus fractures might be beneficial to ORIF in terms […]

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Conservative versus operative treatment of LC-1 pelvic ring fractures: radiographic and long-term functional outcomes.

Metzger F, Buschbeck S, Swartman B, Schweigkofler U

PURPOSE: To compare radiographic and long-term functional outcomes of conservatively and operatively treated traumatic lateral compression type 1 (LC-1) pelvic ring fractures, considering fracture morphology and injury severity. METHODS: This single-center observational cohort study included adult patients with traumatic LC-1 pelvic ring injuries treated between 2012 and 2020. Radiographic outcomes were assessed using the Matta–Tornetta system, and functional outcomes were evaluated at long-term follow-up using the normalized Majeed Pelvic Score and SF-12. Baseline characteristics and complications were analyzed descriptively. RESULTS: Seventy-seven patients were included (38 conservative, 39 operative). Operatively treated patients had higher injury severity and more complex fracture morphology. Despite worse initial alignment, radiographic outcomes at follow-up were similar between groups, with most patients achieving excellent or good results. Radiographic consolidation was observed in nearly all patients. Long-term functional outcomes did not differ significantly between groups. CONCLUSION: In this observational cohort with substantial baseline differences, no significant differences in long-term functional outcomes were observed between treatment strategies. These findings support an individualized, morphology-based treatment approach. However, results should be interprete […]

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Correction: Clinically relevant bleeding risk in low-energy fragility fractures of the pelvis in elderly patients.

de Herdt CL, Loggers SAI, van Embden D, Bijlsma T, Joosse P, Ponsen KJ

PubMed · DOI

A novel concept for splint fixation of midshaft clavicle fractures: a biomechanical feasibility study.

Ziegenhain F, Zderic I, Richards RG, Gueorguiev B, Groven RVM, Varga P, Hildebrand F, Pape HC et al.

BACKGROUND: Minimally invasive treatment of clavicle shaft fractures has become increasingly popular due to reduced soft tissue damage, smaller incisions, lower rates of infections, and minimal scarring. Most commonly described options are titanium elastic nailing and plate osteosynthesis. Clinical studies report promising results with good healing tendencies, particularly in children and adolescents. However, there are still some downsides to consider—titanium elastic nails have been associated with soft-tissue irritation, implant associated discomfort, and notable complications rates including lateral perforation of the cortex and medial protrusion of the implant. Implant removal after a few months is standardly carried out, leading to a second surgery with corresponding risks and costs. Furthermore, biomechanical studies report a low rotational stability for these implants. Therefore, the aim of this biomechanical feasibility study was to explore a new concept for minimally invasive splint fixation of clavicle shaft fractures in comparison to the titanium elastic nailing. METHODS: Twelve anatomical composite clavicles with standardised oblique midshaft fractures AO/OTA 15.2 A were randomized to two groups for fixation with either a 2.5 mm titanium elastic nail (Group 1) or a 5.0 mm rib splint (Group 2). Each specimen underwent (1) non-destructive quasi-static pure-bending te […]

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Development and external validation of a novel prediction model for the TraumaTriage App.

Gulickx M, Lokerman RD, van der Sluijs R, Tuinema RM, van Vliet R, Hietbrink F, Groenwold RHH, van Heijl M

PURPOSE: Accurate pre-hospital trauma triage is essential for optimizing survival and functional outcomes within inclusive trauma systems. A prediction model incorporated into the TraumaTriage App (TTApp) has been shown to improve patient allocation in the Netherlands, but was developed based on a relatively small cohort. This study aims to redevelop and improve the performance and applicability of the TTApp's prediction model before nationwide implementation. METHODS: In this prospective multicenter cohort study, data from all trauma patients transported within two emergency medical service (EMS) regions (i.e., Brabant and Utrecht), between February 2015 and October 2019, were used to develop and externally validate a prediction model to identify severely injured adults (Injury Severity Score [ISS] ≥ 16) using routinely available pre-hospital data. A Gradient Boosting Decision Tree (GBDT) algorithm was applied, and model performance was evaluated in terms of discrimination and calibration. RESULTS: The development cohort included 51,001 patients (median age 63.8 years, median ISS 9), and the external validation cohort included 29,737 patients (median age 62.1 years, median ISS 9). In external validation, the GBDT model showed excellent discrimination (c-statistic 0.850; 95% CI, 0.837–0.863) and good calibration (calibration-in-the-large 0.009; slope 0.952). Sensitivity was 91. […]

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Association between high ratio of platelet concentrate to red blood cell and survival among massively transfused patients.

Aoki M, Okada Y, Fujiwara G, Katsura M, Matsumoto S, Kiyozumi T, Tomura S, Matsushima K

PURPOSE: Although balanced transfusion protocols (platelet concentrate (PC): red blood cell (RBC) = 1:1) are standard for severe trauma, the impact of a higher PC: RBC ratio (> 1:1) on survival remains unclear. This study aimed to investigate the effectiveness of a high PC to RBC among severely injured patients requiring massive transfusion. METHODS: We performed a retrospective cohort study using the Japan Trauma Data Bank (2019–2023). Adult patients receiving massive transfusion (> = 10 units of RBC) within the first 24 h of injury were included. Patients were classified into high PC (> 1) and low PC groups (< = 1). The primary outcome was 24-hour mortality, analyzed using inverse probability of treatment weighting and modified Poisson regression. Additionally, we performed spline curve analysis to investigate the nonlinear relationship between the PC: RBC ratio and outcome. RESULTS: Among 3,067 patients (mean age 57.1 years, 67.2% male), 934 were in the high PC group and 2,133 in the low PC group. The high PC group had lower 24-hour mortality (8.9% vs. 15.7%). Adjusted analysis showed a significantly lower risk of mortality in the high PC group (risk ratio 0.54; 95% CI 0.43–0.69). A nonlinear analysis suggested that increasing PC: RBC ratio was associated with lower mortality. CONCLUSION: A high PC: RBC ratio was associated with lower 24-hour mortality in severely injured pa […]

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Treatment and outcomes of periprosthetic femoral fractures in patients aged 90 years or older - a retrospective bicentre cohort study with 107 patients.

Müller F, Wulbrand C, Probst A, Langenhan R

PURPOSE: The surgical treatment of periprosthetic femoral fractures (PFFs) is highly demanding, especially in very old patients. For patients aged 90 years or older, data regarding internal fixation (IF) and revision arthroplasty (RA) are lacking. METHODS: This retrospective case series was conducted in two different regional centres in one country. We enrolled patients with an age of 90 years or older, and who had PFFs with cemented or cementless standard stems. The treatment period was from 01/2010 to 12/2024. The exclusion criteria were interprosthetic femoral fractures or periprosthetic acetabular fractures. Treatment was performed with IF (locking plates) or RA (noncemented revision stems). The primary end point was any revision within the follow-up period. The secondary end points were death, serious adverse events, and function with Parker score. The minimum follow-up for living patients was one year after surgery. Living patients received a final contact by phone. RESULTS: A total of 107 patients were included in this study. The mean age of the cohort was 92.0 years (range 90 to 100), and most patients were female (n = 85). Following the Vancouver classification and its recommendations, 17 out of 22 patients with type B1 fractures were treated with IF, 44 out of 48 patients with type B2/3 fractures were treated with RA, and 34 out of 37 patients with type C fractures we […]

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The graft-reposition-on-flap technique for fingertip amputations: a patient-reported outcomes study of 111 patients.

Thushara KR, Sankaran A, Kolleri MN

PURPOSE: Graft-reposition-on-flap (GRF) is a reconstruction method described for non-replantable fingertip amputations. We hypothesised that this technique provides acceptable outcomes for such an injury, as measured by a patient-reported survey.

METHODS: This retrospective study evaluated 111 patients [118 digits], operated between January 2019 and March 2023 for Allen 3 or 4 fingertip amputations. All patients had crush avulsion amputations not fit for replantation and had undergone flap cover with grafting of the amputated nail bed and bone. 80 digits had undergone a flap with bone and nail bed graft, while 38 digits received only a nail bed graft with the flap. The flaps used were based on the defect, including local and pedicle flaps. Outcomes were assessed using an original patient-reported questionnaire administered telephonically at least 1 year after injury. The survey contained six items rated on a 1-10 Likert scale, with higher scores indicating better outcomes.

RESULTS: Nail growth was present in 105 of the 118 digits (89%). 107 patients of 111 (96%) were willing to undergo a similar procedure again if indicated. The median rated score for the Length of the finger was 9 [Interquartile range (IQR) 8-10], for Fingertip shape 8 [IQR 7-9], for Donor appearance 9 [IQR 8-10], for Nail appearance 8 [IQR 5-10], and for Sensibility 9 [IQR 7.5-10]. The median Overall Satisf […]

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Mechanical comparison of different plate-and-screw configurations for repairing forearm diaphysis fractures: a systematic review.

Zdero R, Brzozowski P, Schemitsch EH

PURPOSE: This is the first systematic review of all studies that adjusted plate and/or screw variables to determine the most mechanically stable implant configuration for forearm fracture repair. METHODS: PubMed and Scopus archives were searched using the phrase “biomechanics” AND “fracture” AND “plate” AND (“forearm” OR “radius” OR “ulna”). Eligibility criteria were applied: (i) studies that mechanically analyzed implant performance by modulating plate and/or screw variables, such as plate geometry, plate material, screw number, etc.; (ii) forearm diaphysis “shaft” fractures of the radius and/or ulna; (iii) studies published at any time; (iv) English language studies. RESULTS: There were 34 eligible papers. There was greater mechanical stability for these configurations: (i) plates used alone or double-stacked; (ii) plates that were longer, wider, thicker, straight or curved at the bottom, non-ribbed at the bottom, and/or non-grooved at the sides; (iii) plates made of material with higher elastic modulus; (iv) plates that were locking; (v) plates not elevated above bone; (vi) plates affixed to the anterior side of bone; (vii) screws that were longer, thicker, and/or numerous. There was a range of numerical outcomes: (i) interfragmentary motion; (ii) peak stress in cortical bone or plates; (iii) cortical bone stress under the plate to assess bone “stress shielding”; (iv) fatigu […]

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Postoperative complications of xenogenic dural substitutes in hemicraniectomy.

Tanrikulu L, Alhasan AA, Dömer P, Woitzik J

PURPOSE: Decompressive hemicraniectomy (DHC) is a treatment option for refractory intracranial pressure (ICP) elevation in cases of cerebral infarction, hemorrhage, and traumatic brain injury. Several surgical techniques are used for musculocutaneous, bony, and dural exposure. The method of dural closure, specifically whether to use expanding duraplasty with or without xenogenic dural substitutes, remains a topic of debate. This study focuses on the occurrence of postoperative surgical side effects associated with these approaches. METHODS: A total of 258 patients (mean age 52.8 ± 16.7 years; 140 male, 118 female, 79 infarctions, 57 traumatic brain injuries, 55 subarachnoid hemorrhages, 32 intracranial hemorrhages, 30 subdural hematomas, 5 other indications) who underwent DHC between 2015 and 2024 at our institution were reviewed. Complete clinical and radiological data were available for all patients. The patient population was divided into two groups: • Group A: 201 patients who received the xenogenic dural substitute. • Group B: 57 patients who did not receive a dural substitute. RESULTS: Baseline demographics (age, gender, underlying pathology) were comparable between both groups. No significant differences were found in the following parameters: • Duration of DHC surgery: 102.0 min (Group A) vs. 108.0 min (Group B); p = 0.24. • Postoperative CSF leakage: 3.5% (7/201) in Gr […]

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Rethinking routine chest radiography after central venous catheterisation: a rapid review of current evidence and alternatives.

Orso D, Guglielmo N, Pangallo R, Montanar V, Favaro S, Bove T

BACKGROUND: Despite growing evidence supporting ultrasound (US)- and ECG-guided protocols, chest radiography (CXR) remains the most widely used method for post-procedural confirmation of central venous catheter (CVC) position in clinical practice. METHODS: We conducted a rapid review of observational studies, randomised controlled trials, systematic reviews, and guidelines comparing CXR with US- and ECG-based confirmation modalities in adult patients across ICU, emergency, and perioperative settings. Systematic searches were performed in PubMed, Scopus, and Web of Science through December 2025. Outcomes of interest included diagnostic accuracy for tip position and complications, procedural time, and clinical and economic impact. FINDINGS: Ninety-four studies were included, comprising more than 20,000 CVC placements. Evidence from meta-analyses, multicentre cohorts, and comparative studies suggests that US-based protocols – particularly those incorporating agitated saline (“bubble test”) and right-atrial visualisation – and intracavitary ECG (IC-ECG) can achieve high diagnostic performance for tip confirmation in appropriately selected patients, with several reports describing sensitivity and specificity of approximately 95% or higher. Across studies, bedside strategies enabled faster confirmation than CXR (often 15–40 min), avoided radiation exposure, and facilitated earlier de […]

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Comparison of CATCH, PECARN and CHALICE rules in children with minor head trauma.

Ferhatlar ME, Cumhur BA, Koca Y, Bozan O, Kalkan A

PURPOSE: The aim of this study is to prospectively compare the diagnostic performance of the PECARN, CATCH, and CHALICE clinical decision rules in pediatric patients presenting with minor head trauma, within a rural secondary-level emergency department characterized by limited resources, high patient volume, and restricted access to specialist physicians.

METHODS: The study was designed as a prospective observational study. The PECARN, CATCH, and CHALICE clinical decision rules were applied to the patients included in the study, and the data were recorded. The data included in the study were statistically analyzed using the SPSS (Statistical Package for Social Sciences; SPSS Inc., Chicago, IL) 27 program. A value of $p < 0.05$ was considered statistically significant.

RESULTS: A total of 1,032 patients were included in the study. The PECARN did not miss any of the 15 patients with pathological brain CT and had 100% sensitivity and 52.1% specificity. CATCH failed to detect pathological brain CT in 1 patient and had 93.3% sensitivity and 53.3% specificity. CHALICE failed to detect pathological brain CT in 2 patients, with 86.7% sensitivity and 87.8% specificity. Overall, PECARN was more successful in detecting pathological brain CT, but it also had a high false-positive brain CT rate.

CONCLUSION: PECARN criteria are more sensitive than other criteria in predicting brain CT patho […]

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Effect of head injury on blood transfusion requirements in severe trauma: a post-hoc analysis of the RESTRIC trial.

Nagasawa H, Omori K, Tanaka N, Maekawa C, Hamada M, Ota S, Ohsaka H, Yanagawa Y et al.

PURPOSE: To evaluate whether head-region injury, as defined by Abbreviated Injury Scale (AIS) coding, is associated with transfusion requirements in patients with severe trauma. METHODS: This study is a post-hoc analysis of the RESTRIC trial, a multicentre, cluster-randomised, crossover, non-inferiority trial comparing restrictive (7–9 g/dL) and liberal (10–12 g/dL) blood transfusion strategies for patients with severe haemorrhagic trauma. Adult patients with severe trauma anticipated to require transfusion were included. Head injury was defined as an AIS score ≥ 2 in the head region. The primary outcome was cumulative red blood cell (RBC) transfusion volume within 28 days. Key secondary outcomes included RBC transfusion volume within 24 and 48 h. Propensity score matching (PSM) was used to balance baseline injury severity using Injury Severity Score (ISS) and, separately, AIS scores for body regions excluding the head. Sensitivity analyses were conducted using generalised linear mixed models (GLMM) with random intercepts for study centre. RESULTS: Among the 411 patients, 157 (38.2%) had head AIS ≥ 2. In unadjusted analyses, 28-day RBC transfusion volumes were similar between patients with and without head injury (median 10 vs. 10 units). After adjustment using PSM and GLMM, no evidence of an independent association between head-region injury and RBC transfusion volume was obse […]

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Emergency laparotomies in trauma management: a six-year case series.

Ahrens G, Hoffmann M, Proksch N

INTRODUCTION: Emergency laparotomies, defined as procedures performed within 24 hours of hospital admission following trauma, are associated with considerable mortality. Notably, blunt abdominal injuries carry higher mortality rates compared to penetrating injuries. The primary aim of this study was to evaluate whether emergency trauma laparotomies performed at our Level I trauma center—serving a population with a comparatively higher proportion of penetrating injuries—achieve outcomes comparable to contemporary benchmark data. METHODS: We conducted a retrospective analysis of patients treated between 2019 and 2024 in the surgical trauma bay at Asklepios Klinik St. Georg, Hamburg, who underwent laparotomy within 24 hours of hospital admission. Mortality was defined as survival until hospital discharge. Functional outcomes were assessed using the Glasgow Outcome Scale (GOS). RESULTS: A total of 64 patients were included. Of these, 39.0% sustained penetrating trauma and 61.0% underwent laparotomy following blunt trauma. Among penetrating injuries, stab wounds predominated (92.0%), while 8.0% resulted from gunshot wounds. The most frequent mechanisms of blunt trauma were traffic accidents (30.8%) and falls from height (25.6%). Overall survival was 71.9% (mortality 20.0% for penetrating trauma; 33.3% for blunt trauma). Prehospital cardiopulmonary resuscitation (CPR) had been perfor […]

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Correction: Comparison of total intravenous anesthesia vs. inhalational anesthesia on brain relaxation, intracranial pressure, and hemodynamics in patients with acute subdural hematoma undergoing emergency craniotomy: a randomized control trial.

Preethi J, Bidkar PU, Cherian A, Dey A, Srinivasan S, Adinarayanan S, Ramesh AS

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Non-scaphoid carpal fractures: lessons learned from a series of 86 cases.

Herrero Alonso S, Fernández González-Cuevas J

PURPOSE: To describe the epidemiology, injury mechanisms, diagnostic approach, management, and clinical outcomes of non-scaphoid carpal fractures, which are frequently overlooked or diagnosed late. METHODS: A retrospective review was conducted of patients with non-scaphoid carpal fractures treated at our institution between 2013 and 2020. Demographic data, fracture characteristics, time to diagnosis, treatment strategy, rehabilitation requirements, and clinical outcomes, including complications, were analyzed. RESULTS: 76 patients with 86 fractures were included; 54 (71%) were male. The most commonly affected bones were the triquetrum (37%) and hamate (33%). Isolated fractures accounted for 89% of cases. Advanced imaging was frequently required, with CT performed in 53% and MRI in 7% of patients. The median diagnostic delay was 6 days (range: 1–40). Conservative treatment was used in 84% of cases. Surgical management was significantly more frequent in lunate (75%) and hamate fractures (25%) ($p = 0.008$). Surgically treated patients required rehabilitation more often than those managed conservatively (58% vs. 31%, $p = 0.012$). Functional range of motion (> 40° flexion/extension) was achieved in 97% of patients. However, 16% developed chronic wrist pain, 8% reported prolonged soreness, and one patient developed reflex sympathetic dystrophy. CONCLUSIONS: Although functional outcomes […]

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A machine-learning-based tool for stroke risk prediction in blunt cerebrovascular injury: development and preliminary evaluation.

Wagner V, Preston JD, De Leon Castro A, Mueller WF, Nguyen J, Garcia-Toca M, Benjamin ER, Todd SR et al.

PURPOSE: Stroke risk correlates with the Biffl grading system in blunt cerebrovascular injury (BCVI). Although anti-thrombotic therapy is the mainstay of stroke prevention, no point-of-care clinical decision-support tool exists to guide timing for therapy. We sought to develop an interactive online calculator that incorporates patient-specific demographic and injury characteristics to estimate stroke risk and risk reduction with anti-thrombotic (AT) administration.

METHODS: Data from BCVI patients (n = 1,197) at a Level I Trauma Center were retrospectively collected. Six machine learning methods were employed to predict stroke risk with and without AT therapy. Class imbalance was addressed using downsampling and/or class weighting. Model performance was assessed using 10-fold cross-validation. The model was implemented as an R-based Shiny online application.

RESULTS: Stroke rate among the population was 4%, and the strongest predictors for stroke were the greatest Biffl grade of carotid (aOR [95%CI] = 2.02 [1.62-2.53]) and vertebral injuries (1.44 [1.18-1.77]). The least absolute shrinkage and selection operator (LASSO) model outperformed all others, achieving 66% [33%-100%] sensitivity and 74% [62%-82%] specificity for stroke prediction, with an area under the receiver operating characteristic curve of 0.79 [0.57-0.95]. This model was integrated into an interactive online to […]

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Association between screw angulation and union after inverted-triangle cannulated screw fixation of femoral neck fractures.

Çatal T, Bayrak A, T■ng■r M, Sapmaz ■E, K■l■çkaya ME, Mercan N, Kural C, Duramaz A

PURPOSE: The aim of this study was to evaluate whether screw angulation relative to the inter-ischial(horizontal) axis and femoral anatomical axis was associated with union after inverted-triangle cannulated screw fixation of femoral neck fractures and to explore angle intervals associated with higher union rates. METHODS: Between January 2017 and December 2022, 83 patients operated on with closed reduction-percutaneous cannulated screws by the same surgical team were retrospectively analyzed. The angles of the calcar, anterosuperior, and posterosuperior screws were measured according to both the femoral anatomical axis and the horizontal axis. Multivariable logistic regression analyses adjusted for lateral angulation and Garden or Pauwels classification were additionally performed for the horizontal-axis screw-angle parameters. Exploratory angle intervals were assessed using quartile and sliding-window analyses. RESULTS: A total of 64 patients achieved union, with a mean union time of 8.69 ± 3.03 weeks. The nonunion group had higher lateral angulation and greater postoperative neck length(p = 0.006, p = 0.042, respectively). In angle values relative to the horizontal axis, calcar and anterosuperior screws exhibited higher measurements in the union group(p = 0.027, p = 0.013, respectively). In exploratory sliding-window analyses, high union intervals were identified as follows: […]

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Response to the letter to the editor to discuss "Factors to consider when selecting an injured control group in case-control studies on skiing injuries".

Rutsch N, Gewiess J, Albers CE, Tinner C

PubMed · DOI

A study on prolonged use of resuscitative endovascular balloon of the aorta (REBOA) with shunt flow in a porcine hemorrhagic shock model.

Kiriu N, Saitoh D, Yamamura K, Sasa R, Kiyozumi T, Tomura S

PURPOSE: To evaluate the future efficacy and prolonged availability of double-balloon resuscitative endovascular balloon occlusion of the aorta (DB-REBOA) to determine whether bypassing upstream blood at the occlusion site improves survival in a porcine hemorrhagic shock model. METHODS: Twenty-five pigs (approximately 40 kg, estimated circulating blood volume 25%) were subjected to hemorrhagic shock by controlled phlebotomy. Animals were treated with standard REBOA (control, n = 12), internal shunt DB-REBOA (internal shunt, n = 4; an intraluminal conduit was set parallel to the DBREBOA in the aorta), or external shunt DB-REBOA (external shunt, n = 9; upstream blood from the occlusion site was shunted by connecting the internal carotid and femoral arteries with a pressure-resistant tube). Zone 1 aortic occlusion was maintained for 90 min, followed by balloon deflation. Survival, systolic blood pressure, and serum lactate levels were analyzed. RESULTS: Ten of 12 control animals developed cardiac arrest shortly after balloon deflation. All animals in the internal shunt group survived, whereas 5 of 9 animals in the external shunt group experienced cardiac arrest. Internal shunt DB-REBOA maintained stable systolic BP during prolonged occlusion, resulting in significantly lower serum lactate levels at 90 min compared to controls. External shunt DB-REBOA resulted in unstable hemodynam […]

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Checklist- and algorithm-based simulation training produce comparable improvements in teamwork-related shared mental models: a cluster-randomized study in trauma resuscitation teams.

Beaugé Y, Kusch S, Thorsson L, Zentgraf M, Schauwienold J, Muenzberg M, Verboket R, Lefering R et al.

PURPOSE: Shared mental models (SMMs) reflect the extent to which team members hold a similar understanding of roles and team processes. In trauma resuscitation, higher SMM alignment is considered a prerequisite for coordinated interprofessional teamwork. This study compared the effects of checklist-based versus algorithm-based in situ simulation training on teamwork-related SMMs in trauma room teams. METHODS: In this multicenter, cluster-randomized pre–post study, 29 interprofessional trauma teams (186 participants) from six hospitals received either checklist-based training (n = 15 teams) incorporating the Trauma Room Manual checklists or algorithm-based training (n = 14 teams), both grounded in ATLS principles. Teams completed an SMM questionnaire adapted to the trauma setting, based on a previously published instrument, assessing similarity in (1) task responsibility and (2) team communication. Team-level similarity scores were calculated pre- and post-training. Training effects were analyzed using mixed repeated-measures ANOVA. RESULTS: Across both training formats, teamwork-related shared mental models increased significantly from pre- to post-training in total similarity, task responsibility, and communication. No significant between-group differences or time × group interactions were observed. The interaction for task responsibility approached significance (p =.06; $\eta^2 p$ = […]

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The accuracy of the injury severity score in patients below the age of 18 years, a trauma registry data study.

Edwards BHJ, van Heumen JJP, Nelen SD, Edwards MJR, Jaarsma RL, de Blaauw I, Hermans E

PURPOSE: The Injury Severity Score (ISS) is an anatomical score that estimates both severity of trauma and risk of mortality in adults. Although widely used, its applicability in pediatric trauma remains uncertain. This retrospective cohort study evaluated the predictive value of ISS for pediatric mortality and clinically relevant outcomes. METHODS: Patients (< 18 years) treated at a level-1 trauma centre with traumatic injury between 2015 and 2023 were included in the Dutch National Trauma Registry. Receiver operating characteristic (ROC) analyses were used to assess the discriminative performance of ISS for mortality, prehospital advanced care, emergency interventions, imaging, length of stay, surgical burden, and functional outcome. RESULTS: 1733 patients were included. ISS showed excellent discrimination for mortality (AUC 0.97) and good performance for emergency interventions, prehospital advanced care, functional outcome, and admission to higher levels of care, but limited value for predicting number of surgeries. Optimal thresholds for ISS were 16 or higher, except for CT imaging (ISS ≥ 10). CONCLUSION: ISS reliably predicts mortality and several clinically relevant outcomes in children. While ISS ≥ 16 remains appropriate for identifying severe injury, optimal thresholds vary by outcome. These findings emphasize the need for outcome-specific interpretation of ISS, for be […]

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Epidemiology of acute ice-hockey-related orthopedic fractures in high school- and college-aged players in the United States from 2006-2023.

Reiad T, Dinh P, Khan H, Bruni D, Chang E, Owens B, Marcaccio S

BACKGROUND: This investigation quantified the incidence and injury patterns of acute orthopedic fractures attributed to ice hockey among high-school and college-aged athletes in the United States. METHODS: We performed a retrospective analysis of the National Electronic Injury Surveillance System (NEISS) database focusing on ice hockey fractures in patients aged 14–23 years from 2006 to 2023. The cohort was stratified into high-school-aged (14–18) and college-aged (19–23) groups. Weighted national estimates were calculated to evaluate anatomical sites, fracture subtypes, and hospitalization rates. RESULTS: An estimated 24,350 ice hockey-related fractures occurred in the study population. Males accounted for 94.8% of cases. Upper extremity and trunk fractures comprised 81.6% of all fractures, with shoulder (27.0%), wrist (19.3%), and lower arm (11.3%) being the most common sites. In comparison to high-school-aged players, college-aged athletes were significantly more likely to sustain hand fractures (OR 2.82; p = 0.001) and lower leg fractures (OR 3.67; p = 0.042), while less likely to sustain lower arm fractures (OR 0.31; p = 0.039). Hospitalization was required for 4.2% of fractures overall, with the neck (82.6%) and upper leg (76.3%) having the highest admission rates, and several areas like the elbow, hand, finger, and knee having admission rates < 1%. Temporal analysis demo […]

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Incidence, mortality, and factors associated with fat embolism syndrome in patients with long bone fractures at a trauma referral center in Bogotá, Colombia: 2016-2019 and 2022-2023.

Bernal OYG, Lozano NRC, Cordero JFB, Tenjo EAA, Lazaro JS, Acosta DCB, Aguilera CSQ, Valero JDG et al.

PURPOSE: To estimate the incidence, mortality, and factors associated with the development of fat embolism syndrome (FES) in patients with long bone fractures treated at a trauma referral center in Bogotá, D.C., Colombia. METHODS: Retrospective cohort study including patients >18 years with diaphyseal fractures of humerus, femur, tibia, or fibula, treated between 2016-2019 and 2022-2023. COVID-19 pandemic years (2020-2021) were excluded. Statistical analyses included univariate, bivariate, and multivariate logistic regression. RESULTS: Among 3,475 patients with long bone fractures, FES incidence was 4.3% (n=148) with 5.4% in-hospital mortality. Multivariable analysis identified femoral fractures as the strongest risk factor (aOR=4.63; 95%CI:2.99-7.22; p<0.001), while early surgical fixation (<24h) demonstrated significant protection (aOR=0.42; 95%CI:0.25-0.71; p<0.001). Obesity independently increased FES risk (aOR=2.63; 95%CI:1.33-4.93; p=0.004). Most cases (97%) resulted from high-energy trauma, predominantly motorcycle accidents. Respiratory compromise was severe, with 93.9% of FES patients requiring advanced ventilatory support versus 16.8% in controls (p<0.001). CONCLUSION: This study demonstrates that early surgical fixation significantly reduces FES risk and identifies humeral fractures as an underrecognized risk factor. These findings support immediate implementation of […]

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Association between fracture location and one-year mortality in patients aged 65 years and older after lower extremity trauma: a retrospective cohort study.

Helen A, Michelle Z, Christoph H, Thomas L

PURPOSE: Fractures in older patients are associated with increased mortality. Our study evaluated 30-day and one-year mortality among older patients based on fracture location. METHODS: A descriptive retrospective cohort analysis was conducted including all patients aged 65 years or older hospitalized at a level I trauma center with a pelvic or lower extremity fracture between January 1, 2016 and December 31, 2019. Mortality rates were compared with age- and gender-matched data from the Swiss general population. In addition, an exploratory multivariable logistic regression analysis was performed to assess the association between fracture location and one-year mortality, adjusting for age and sex. RESULTS: A total of 1,739 patients were included. The overall 30-day mortality rate was 12%, and the one-year mortality rate was 34%. In comparison, the age- and gender-matched one-year mortality rate of the general Swiss population was 8% (p < 0.001). Unadjusted one-year mortality was highest after femur fractures (41%), followed by pelvic fractures (33%) and lower leg or foot fractures (11%). Among specific fracture types, pertrochanteric femur fractures showed the highest one-year mortality (47%), followed by Tile type C pelvic fractures (43%), distal femur fractures (40%), and femoral neck fractures (39%). In contrast, patients with fractures of the lower leg or foot had a one-year […]

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Proximal femoral replacement for oncologic and non-oncologic indications: a retrospective study over a 13-year period.

Ramón R, Gökmen A, Mayor J, Özbek H, Niemann A, Marín L, Omar Pacha T, Omar M et al.

BACKGROUND: Proximal femoral replacement (PFR) is a critical limb-salvage strategy for managing massive proximal femoral bone loss due to oncologic and non-oncologic conditions. Although oncologic outcomes are well described, evidence surrounding PFR for infection, periprosthetic fracture, and mechanical failure remains limited.

This study evaluates short- to mid-term outcomes following PFR for multiple indications and compares primary with revision procedures. METHODS: We retrospectively reviewed all patients undergoing PFR at a Level I trauma center from 2009 to 2023. Cases were categorized by procedure type (primary vs. revision) and indication (tumor, infection, other causes). Outcomes included operative time, complications according to the Henderson classification, reoperation rates, and functional outcomes. RESULTS: A total of 81 patients (84 procedures) were included with a median follow-up of 7 months (IQR 0–31); 57.1% were lost to follow-up. The overall complication rate was 42.9%, significantly higher in revision procedures than in primary PFRs. Infection (Henderson Type IV) was the most frequent complication and occurred predominantly in revision surgeries. No cases of aseptic loosening (Type II) were identified. Reoperation was required in 29% of all procedures, with the highest rates observed in infection cases. CONCLUSIONS: PFR remains a reliable limb-salvage opti […]

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Interleukin-6 to identify mildly injured patients in the trauma resuscitation room - a clinical feasibility study.

Bewersdorf TN, Wimmer M, Popp E, Streblow J, Baumann L, Leowardi C, Stein S, Schmidmaier G et al.

PURPOSE: Overtriage in the trauma resuscitation room (TRR) is often a consequence of admission based solely on trauma mechanism criteria. As interleukin-6 (IL-6) can assess injury severity in critically injured patients, the hypothesis of this study was that IL-6 can differentiate upon admission between mild (Injury Severity Score (ISS) 0-8) and moderate (ISS 9-15) trauma and that the parameter correlates with injury severity in lower injury severity groups (ISS < 16) during the first 24 h. Subsequently, we evaluated the ability to define IL-6 cut-off values with an adequate sensitivity, specificity, and negative predictive value to distinguish between moderate and mild injuries.

METHODS: Fifty patients admitted to the TRR solely based on trauma mechanism criteria and with an ISS < 16 were included in the study. IL-6 was measured at admission, 1, 6, 12 and 24 h later.

RESULTS: Thirty-one patients were mildly injured (ISS 0-8), while nineteen patients sustained moderate injuries (ISS 9-15). IL-6 correlated significantly with ISS at all time points. Moderately injured patients showed significantly higher IL-6 levels than mildly injured patients during the first 6 h. IL-6 levels at admission predicted moderate injuries for the IL-6 cut-off value of 2.9pg/ml with a sensitivity of 0.933, specificity of 0.370 and a negative predictive value of 0.970 resulting in undertriage rates o […]

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Attenuating pulmonary injury and inflammation with C5/CD14 inhibition therapy: results from a porcine polytrauma model with blunt chest trauma.

Mert Ü, Groven RVM, Qin K, Greven J, Balmayor ER, Mollnes TE, Huber-Lang M, van Griensven M et al.

BACKGROUND: Thoracic injury is prevalent among polytrauma patients, affecting up to 45% of these patients and often leading to complications such as pulmonary dysfunction up to acute respiratory distress syndrome (ARDS). This study investigates the impact of surgical invasiveness and C5/CD14 inhibition therapy on pulmonary tissue damage and the local immune response of the lungs in a porcine polytrauma model. METHODS: A post hoc analysis focusing on the pulmonary consequences of blunt thoracic trauma and multiple trauma was performed in a previously published porcine trauma model, in presence or absence of immunomodulation. A control group (n = 6; CG) was used that received identical instrumentation, anesthesia, nutrition, and fluid intake, but no trauma. For the trauma model, male pigs underwent a standardized polytrauma, followed by allocation into three treatment groups: early total care (n = 8; ETC), damage control orthopedics (n = 8; DCO), and ETC with C5/CD14 inhibition (n = 4). Pulmonary histopathological changes were assessed at 72 h after trauma through lung injury scores and wet/dry ratios, while ex vivo cytokine expressions from isolated alveolar macrophages (AMs) were analyzed using enzyme-linked immunosorbent assays. RESULTS: Histological results revealed significant lung damage in both the ETC and DCO groups compared to the control as well as the C5/CD14 inhibitio […]

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Hemodynamic rescue after blast injury: intravenous administration outperforms intramuscular vasopressin in a swine model.

Renberg M, Gustavsson J, Rehn M, Günther M

BACKGROUND: Hemorrhage remains the leading preventable cause of trauma mortality. Blast trauma combines primary blast effects with complex secondary and tertiary injuries, precipitating profound hemodynamic instability and microcirculatory disruption. While intramuscular (IM) arginine vasopressin (AVP) stabilizes arterial pressure in isolated hemorrhagic shock, vasopressin's efficacy in blast-associated hemorrhagic shock is unknown. METHODS: In a randomized, blinded study, 22 swine underwent femoral blast injury and controlled class II hemorrhage. Animals received IM AVP 40 U (two doses 60 min apart due to short half-life and transient effects; n = 5), IM terlipressin 2 mg (n = 5), intravenous (IV) terlipressin 2 mg (n = 5), or saline control (n = 7). All subjects received a 500 mL autologous whole blood transfusion and were observed for 120 min. The primary outcome was systolic arterial pressure (SAP). Secondary outcomes included systemic vascular resistance index (SVRI), cardiopulmonary and metabolic variables, and serum AVP to assess IM uptake. RESULTS: IV terlipressin rapidly stabilized hemodynamics, increasing SAP (mean difference 24 mmHg, p = 0.01) and SVRI (mean difference 44593 dynes·s·cm■■·kg, p = 0.004) versus controls, and increased mixed venous oxygen saturation and urine output. Respiratory and metabolic variables were similar across groups. Neither IM AVP nor IM t […]

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Prevalence and outcomes of frailty in emergency laparotomy: a single-centre cohort study.

Rival PM, Bellgrove C, Ejaz MU, Van Weel JM, Stokes MAR, Pilgrim CHC

PURPOSE: Frailty is common among patients undergoing emergency laparotomy and is associated with adverse postoperative outcomes, yet routine frailty assessment remains inconsistently implemented despite international guideline recommendations. This study evaluates the prevalence of frailty using three rapid assessment tools and examines their associations with postoperative outcomes following emergency laparotomy. METHODS: We conducted a single-centre retrospective cohort study of adults undergoing open emergency laparotomy over a 12-month period. Frailty was assessed retrospectively using the Clinical Frailty Scale (CFS ≥ 5), Emergency Surgery Frailty Index (EmSFI ≥ 7), and five-item Modified Frailty Index (mFI-5 ≥ 2). Primary outcomes were 30- and 90-day mortality. Secondary outcomes included postoperative complications, ICU admission, hospital length of stay, and discharge destination. Unadjusted analyses were descriptive. Multivariable logistic regression models were constructed with frailty measures specified as the primary exposures and adjusted for age, sex, operative indication, and anaesthetist-assigned ASA grade. RESULTS: Among 102 patients (median age 67 years, IQR 54–79; 53% female), frailty prevalence was 26% by CFS, 24% by EmSFI, and 27% by mFI-5, with 37% meeting at least one frailty threshold. In unadjusted analyses, patients living with frailty experienced high […]

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Timing of surgery and predictors of canal reduction after short-segment fixation for thoracolumbar burst fractures.

Aono H, Takenaka S, Okuda A, Kikuchi T, Takeshita H, Nagata K, Ito Y

PURPOSE: In patients with thoracolumbar burst fractures, surgical delay is often unavoidable due to concomitant injuries. Concerns persist that delayed surgery may compromise indirect reduction of retropulsed fragments via ligamentotaxis. This study aimed to determine whether the timing of surgery affects the quality of intracanal fragment reduction and identify independent predictors of successful reduction following short-segment posterior fixation. METHODS: This multicenter study included 252 patients who sustained a single traumatic thoracolumbar burst fracture and underwent short-segment fixation. The canal compromise ratio (CCR) was measured on pre- and postoperative CT scans, and the reduction rate was calculated using this formula: Reduction rate = (Preoperative CCR – Postoperative CCR) / Preoperative CCR × 100. Multiple linear regression analysis identified predictors of reduction quality including age, sex, affected level, surgical timing, AO classification, load-sharing score (LSS), preoperative CCR, and concomitant vertebroplasty, RESULTS: The study included 141 males and 111 females. The mean time from injury to surgery was 3.8 days. The mean CCR improved from 45% to 24%, yielding a mean reduction rate of 44%. Multiple linear regression analysis revealed that T11 to L1 and AO type B fractures were associated with a significantly higher reduction rate. Notably, the […]

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Outcomes following implementation of a structured diagnostic and treatment approach for combat abdominal trauma in a hybrid war setting: a retrospective cohort study.

Oleh H, Cerkauskaite D, Kashtalian M, Yaroslav H, Kvasnevskyi I, Okolets A, Dulskas A

OBJECTIVE: This study evaluated whether implementation of an optimized diagnostic and treatment protocol improved early and intermediate outcomes among combat casualties.

BACKGROUND: High-energy ballistic and blast mechanisms during the war in Ukraine caused a surge of complex abdominal injuries requiring intensive resource utilization and rapid coordination of surgical, critical care, and evacuation capabilities.

METHODS: A retrospective cohort study was conducted including 496 male soldiers with combat-related abdominal injuries treated at military hospitals from 2014 to 2017. Patients were stratified into pre-implementation (n = 161) and post-implementation (n = 335) groups. The structured protocol included perfusion index-guided triage, expanded use of FAST, diagnostic laparoscopy, and standardized damage control surgery principles. Primary outcomes were postoperative complications, in-hospital mortality, and return-to-duty rates; secondary outcomes included missed intra-abdominal injuries and hospital length of stay.

RESULTS: After protocol implementation, FAST utilization increased (68.1% vs. 19.3%) and diagnostic laparoscopy increased (30.7% vs. 14.9%), while the rate of missed intra-abdominal injuries decreased from 7.5% to 3.6% ($p = 0.04$). Median decision-to-surgery time shortened by 26 min ($p = 0.001$). Postoperative complications declined (39.1% vs. 27.8%, $p < 0.05$ […]

PubMed · DOI

Association between preoperative NT-proBNP levels and mortality among patients undergoing acute high-risk abdominal surgery: a prospective cohort study.

Fattori S, Kanstrup CTB, Serup CM, Kleif J, Lundstrøm LH, Bertelsen CA

PubMed · DOI

Systemic injury extension and mechanism of trauma determine ICU admission in CT-confirmed maxillofacial fractures.

Usta T, Bayramlar OF, Yüce S, Demirtakan T, Sert E, Fettaho■lu S, ■ncealt■n O

PURPOSE: To evaluate whether intensive care unit (ICU) admission in CT-confirmed maxillofacial trauma is primarily associated with facial fracture burden or with radiologically defined systemic injury extension. METHODS: This retrospective cohort study included 1,511 patients with CT-confirmed maxillofacial fractures treated at a tertiary trauma center. Injury mechanism, fracture multiplicity (mFISS-face), and systemic extension (cranial and thoracic involvement) were assessed. Multivariable Firth penalized logistic regression was performed to identify independent predictors of ICU admission. Model discrimination, calibration, and internal validation were evaluated using AUC, calibration metrics, Brier score, and bootstrap resampling. RESULTS: ICU admission occurred in 37 patients (2.45%). In multivariable analysis, age (aOR 1.04 per year, $p < 0.001$), traffic-related mechanism (aOR 2.74, $p = 0.038$), thoracic injury (aOR 25.08, $p < 0.001$), and cranial extension (aOR 98.24, $p < 0.001$) were independently associated with ICU admission. Fracture multiplicity demonstrated borderline significance (aOR 1.32, $p = 0.071$). The model showed good discrimination (optimism-corrected AUC 0.874), acceptable calibration (calibration slope 0.98), and low overall prediction error (Brier score 0.016). CONCLUSION: In CT-confirmed maxillofacial trauma, ICU admission is predominantly associated with s […]

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Better & systematic trauma care (BEST) in Norway: from a local quality improvement project to a national collaborative.

Brattebø G, Wisborg T

PURPOSE: Healthcare professionals in Norwegian local hospitals have focused on the need for improved trauma care since the late 1990s. Because of lack of national guidelines the hospital management of patients with serious trauma varied. The authors and colleague health care providers felt a need for improvement. Therefore, a project called Better & Systematic Trauma care (BEST) was established. The goal was to improve trauma care through training and preparation for infrequently occurring situations, with a focus on the team. METHODS: The project began as a bottom-up initiative in four local hospitals. Through cross-professional team training and local simulation, staff built competence in trauma care and practised non-technical skills such as communication, cooperation, leadership, and decision-making in their local setting. Creating a national meeting place for all hospitals was essential for fostering collaboration across institutions. RESULTS: A 1-day team course was developed and adopted nationwide. Network meetings have enabled the collaborative development of tools and training materials without national strategic leadership or external influence. Over time, the focus shifted from individual professions to the trauma team. A damage-control surgery course was created for all surgical teams. All materials remained freely available, and the project operated independently o […]

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Conservative management of 900 pediatric distal radius fractures using the schede cast: a multicenter study.

Scherer S, Schultz J, Rausch T, Schoof B, Altmeier L, Hertel J, Sahin M, Esser M et al.

BACKGROUND: Distal forearm fractures are among the most common pediatric injuries and are typically managed with cast immobilization. Volar-flexion ulnar-deviation splints (Schede casts) have been proposed to reduce displacement and prevent secondary dislocation. However, data on efficacy and safety in children remain limited. This study aimed to evaluate the clinical outcomes of Schede cast immobilization across four pediatric trauma centres over a 16-year period. METHODS: We conducted a retrospective analysis of patients under 18 years with distal forearm fractures treated with Schede casts. Demographic data, fracture characteristics, initial and post-cast displacement, incidence of secondary dislocation, and complications were recorded and analyzed. RESULTS: A total of 900 patients (mean age 9.47 years) were included. The most common mechanism of injury was sports-related trauma, and transverse metaphyseal fractures predominated. Mean initial displacement was 17.7° (± 9.4°), reduced to 5.9° (± 5.5°) after cast application and 6.9° (± 5.4°) at consolidation after a mean of 35 (± 16) days. Secondary dislocations were effectively prevented at flexion angles > 50° ($p < 0.001$). Complications were rare: tingling paresthesia occurred in 22 patients (2.4%), and prolonged movement restriction in 8 patients (0.9%). All adverse events resolved within days to weeks without long-term seq […]

PubMed · DOI

Publisher Correction: Clinical outcomes and quality of life of patients after surgical treatment of a tibia plateau fracture.

Höller LF, Höller S, Klinger E, Henkies D, Leha A, Lehmann W, Dobroniak CC, Hoffmann DB

PubMed · DOI

Does the Hertel criteria correlate with postoperative avascular necrosis and failure in proximal humerus fractures? A retrospective cohort study.

Ismailov S, Bacaksiz T, Maden M, Akan I, Onder Y, Kazimoglu C

PURPOSE: Avascular necrosis and failure are significant complications following surgical fixation of proximal humerus fractures. The Hertel criteria are a widely used morphological classification system intended to predict humeral head ischaemia. This study aimed to evaluate the correlation of Hertel criteria with the development of postoperative avascular necrosis and failure and to assess the prognostic value of other radiological parameters. MATERIALS AND METHODS: In this retrospective cohort study, 94 patients with proximal humerus fractures treated with locking plate fixation and a minimum 2-year follow-up were included. Preoperative radiographs and computed tomography scans were used to assess Hertel criteria, Neer classification, and proximal humeral cortical thickness. Postoperative radiographs were evaluated for avascular necrosis and failure, defined as a composite of nonunion, severe avascular necrosis, screw penetration, severe deformity, or significant arthrosis. Clinical outcomes were assessed using the Visual Analog Scale and Disability of Arm, Shoulder and Hand scores. RESULTS: The presence of an anatomical neck fracture and tubercle displacement greater than 10 mm were significantly associated with both avascular necrosis and failure ($p < 0.05$). In contrast, other factors in the classical Hertel criteria, such as metaphyseal head extension less than 8 mm, media […]

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Learning gain of an ATLS®-based interprofessional and multidisciplinary in-situ simulation training of trauma resuscitation.

Hammer S, Lock JF, König S, Happel O, Paul MM

PURPOSE: Team performance in polytrauma management determines patient outcome and is crucially shaped by Crew Resource Management (CRM). This study explored patterns of self-perceived learning following an interdisciplinary, interprofessional, in-situ, simulation- and ATLS®-based trauma team training with a focus on CRM principles. METHODS: We conducted a retrospective analysis of routine educational data collected immediately after 36 training sessions (03/2022–11/2023) including 238 participants at a single Level I trauma center. Participants completed a retrospective pre-test/post-test self-assessment questionnaire. Exploratory analysis, including EFA, were used to descriptively structure self-perceived learning patterns and subgroup differences. RESULTS: Participants came from anesthesiology, surgery and radiology in equal proportions and differed in working experience, professional role, and exposure to polytrauma management. Exploratory factor analysis identified three descriptive dimensions of self-assessed CRM-related concepts: (i) personal operational competence, (ii) team communication, and (iii) decision making. Patterns of self-perceived change varied by experience level, with larger reported gains in personal operational competence among less experienced participants. CONCLUSION: This exploratory analysis describes patterns of self-perceived learning following in-s […]

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Multifaceted intestinal defense following experimental blunt abdominal trauma.

Meisen S, Rayatdoost F, Oßwald K, Palmer A, Breunig M, Huber-Lang M, Halbgebauer R

PURPOSE: Blunt abdominal trauma (AT) can cause significant intestinal injury and act as a trigger for remote posttraumatic organ dysfunction, including the development of multi-organ dysfunction syndrome (MODS). A deeper understanding of its impact on intestinal barrier integrity and immune regulation is essential for developing therapeutic strategies that support posttraumatic recovery. METHODS: We assessed jejunal tissue 24 hours following a standardized murine blast-induced direct blunt AT. Histological, immunohistochemical, proteome profiling, Western Blot, and molecular analyses were applied to assess epithelial integrity, barrier-associated proteins, intestinal mucus composition, and immune responses. RESULTS: The jejunum showed no trauma-specific histomorphological changes. Consistently, the expression of key epithelial barrier proteins, such as E-cadherin and tight-junction protein 1, remained stable or were even upregulated. Local inflammatory responses were evident; AT induced activation of defense-related pathways, accompanied by a marked increase in S100A8 expression and enhanced infiltration of CD3+ T cells. Mucus-covered areas of the intestinal surface were reduced, although transcriptional levels of MUC2 and C1GALT1 were elevated. In addition, the antimicrobial peptide CRAMP was upregulated in jejunal tissue following injury. CONCLUSIONS: These findings suggest t […]

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From surgery to rehabilitation: a cross-sectional survey on the feasibility of live digital ward rounds between acute care and rehabilitation clinics in trauma patients.

Schreiber L, Halvachizadeh S, Schäfer FP, Tschui F, Sturzenegger C, Johannes S, Pape HC, Heining SM

INTRODUCTION: Telemedicine is increasingly embraced by both patients and healthcare professionals. By bridging the gap between acute care hospitals and rehabilitation clinics, it offers a promising approach to enhance communication and continuity of care. This study aimed to assess the feasibility of telemedicine in the rehabilitation of patients with musculoskeletal injuries or polytrauma. METHODS: A cross-sectional survey was conducted involving healthcare professionals engaged in the telemedical management of patients with musculoskeletal injuries or polytrauma during acute care and rehabilitation clinics. We established digital ward rounds between an acute care hospital and rehabilitation clinics with live patient presentation. A self-developed questionnaire assessed four domains: overall satisfaction, data security, technical setup, and the rehabilitation process. Each item was rated using a 5-point Likert scale. Participation was voluntary and anonymous. RESULTS: A total of 39 participants completed the survey, with 23 respondents (59%) from a rehabilitation institute and 14 (41%) from an acute care hospital. The majority were physicians (n = 32, 82.1%), with the most common age group being 50–60 years (n = 12, 30.8%). Overall satisfaction with the telemedicine process was high, with 31 participants rating their experience as either 4 or 5 (very satisfied). Only one respo […]

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Patterns of surgical management of hollow organ injuries in severe trauma.

Schurr LA, de Schultz MJ, Kupke P, Thiedemann C, Kaltofen T, Alt V, Schlitt HJ, Popp D

In the setting of severe trauma, abdominal injuries are common and potentially life-threatening. Such injuries predominantly affect parenchymal organs, particularly the liver and spleen. Hollow organ injuries occur less frequently but nevertheless require timely and accurate diagnosis as well as appropriate surgical management to avoid severe complications, including peritonitis and sepsis. In daily clinical practice, different surgical strategies are employed for the management of hollow organ injuries. According to overall injury severity and abdominal injury patterns, treatment may involve either definitive single-stage surgery or staged procedures incorporating damage control principles. This study aimed to retrospectively evaluate the incidence and patterns of surgical management of hollow organ injuries in severely injured patients and to describe postoperative course and complications associated with different treatment strategies. Between 2006 and 2020, 1794 patients aged ≥ 16 years with an Injury Severity Score (ISS) ≥ 16 were treated at University Medical Center Regensburg, a level 1 trauma center. Among these, 57 patients sustained transmural hollow organ and/or mesenteric injuries requiring surgical intervention. Definitive surgical management during the initial operation was performed in 20 patients, whereas 37 patients underwent staged management with at least one […]

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Influence of elastic stable intramedullary nailing on fracture reduction of paediatric diaphyseal forearm fractures.

Pukalski Y, Auroi J, Yanev P, Rashkov M, Schmidutz F, Zderic I, Sprecher C, Richards RG et al.

BACKGROUND: Diaphyseal fractures of both forearm bones (both-bone forearm fractures) are common pediatric injuries. When surgical treatment is indicated, elastic stable intramedullary nailing is widely used due to its minimally invasive nature and favorable outcomes. Nail design may vary – straight, pre-contoured, or a combination of both – however, the influence of these configurations on fracture reduction has not been clearly explored. The aim of this study was to investigate postoperative radiological outcomes after stabilization of both-bone forearm fractures with either two non-contoured nails, two contoured nails, or one non-contoured and one contoured nail in human cadaveric bones. METHODS: Twelve human cadaveric forearms from 6 adult donors were used and standardized AO PCCF22-D/4.1 transverse diaphyseal fractures of the radius and ulna were created. Titanium elastic nails (TENs) were inserted retrograde in the radius and anterograde in the ulna creating 3 groups with either 2 straight TENs (Group 1), 1 straight and 1 curved TEN (Group 2), or 2 curved TENs (Group 3). Anteroposterior and lateral radiographs of each intact and instrumented specimen were taken in supination. Parameters of interest included total bone length, maximal radial bow (MRB), and location of the maximal radial bow (LMRB). RESULTS: Both total bone lengths remained nearly unchanged after ETN instrum […]

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The future of trauma care in Germany 2030: challenges, opportunities, and strategic directions.

Spering C, Lemm M, Finke S, Wrobel M, Haering A, Franke A, Sprengel K, Seemann R et al.

BACKGROUND: Trauma care in Germany is a critical component of public healthcare, encompassing preclinical to rehabilitative phases of patient management. Demographic shifts, workforce shortages, and increasing patient complexity challenge the sustainability of trauma care. This study aims to analyze current trauma care structures, forecast future demands, and propose strategies to secure high-quality trauma care in Germany by 2030. METHODS: A mixed-methods approach was employed, combining a quantitative analysis of national hospital and ICD-coded diagnostic data (2010–2019) to model infrastructure, analyze patient volumes, and forecast demand to 2030 under different scenarios. This was supplemented with an online survey of 752 trauma surgeons nationwide to assess perceptions of future challenges and professional satisfaction, and a structured expert panel with 23 experts to qualitatively interpret the data and develop strategic recommendations. RESULTS: Survey respondents anticipate a significant rise in age-related trauma cases and a worsening shortage of both medical and non-medical staff. Hospital infrastructure is unevenly distributed, with particular accessibility issues in rural and eastern regions of Germany. Forecasts predict increasing trauma cases in older adults and a reduction in hospital beds, especially in rural areas. The data analysis identified critical areas f […]

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Long-term outcomes in physical function and quality of life after traumatic thoracolumbar A3/A4 fractures: a comparison of conservative versus surgical management.

Sienema AS, Reininga IHF, Hoekstra J

PURPOSE: To provide insight into the recovery rates of patient-reported outcome measures (PROMs) in terms of physical functioning and health-related quality of life (QoL) in patients with an A3/A4 thoracolumbar vertebral fracture, compared to the general population. Additionally, differences in outcomes between conservatively and surgically treated patients were assessed. METHODS: A retrospective cohort study with cross-sectional follow-up including patients with thoracolumbar A3/A4 vertebral fractures in a level 1 trauma center between 2010 and 2020. SMFA-NL was used to evaluate physical functioning, and EQ-5D was used to assess QoL. Outcomes were compared with normative data from the Dutch population. Patient-reported outcomes and complication rates were reported for each treatment type. Recovery was defined as reaching the lower limit of the 95% confidence interval of the normative data in all outcome measures. RESULTS: PROMs were available for 98 (37%) of the eligible patients with a median follow-up of 4.5 (IQR = 5.6) years. No significant differences in physical functioning or QoL were found between conservatively and surgically treated patients. The following non-recovery rates were found in the conservatively treated patients: physical functioning = 56–74%, QoL = 48% and in the surgically treated patients: physical functioning = 77–80%, QoL = 45%. Surgically treated pat […]

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The pararectus approach in acetabular fracture fixation: evidence of reproducibility and learnability.

Schaefer RO, Ivanova S, Leibold CS, Egli RJ, Bonel H, Keel MJB, Bastian JD

PURPOSE: The Pararectus approach is an anterior intrapelvic exposure introduced by Keel et al. for the management of anterior acetabular fractures. While its safety and effectiveness have been demonstrated, evidence regarding the learnability of this technically demanding method remains under discussion. This study compares the first fifty consecutive eligible cases treated by the approach's inventor (group A) with the first fifty independent cases performed by a surgeon without prior acetabular subspecialization (group B).

METHODS: Consecutive patients representing the first 50 Pararectus cases for displaced acetabular fractures in group A (2009-2013) and group B (2017-2021) were included. Baseline demographics were comparable except for age (64 vs. 73 years, $p = 0.001$) and trauma mechanism (high-energy trauma more frequent in group A, 46% vs. 12%; $p < 0.001$). Group A was treated using manually contoured reconstruction and buttress plates, whereas group B was managed with a standardized traction-table setup and a precontoured suprapectineal plate system ($p < 0.001$). Operative time, intraoperative blood loss, intra- and postoperative complications, CT-based reduction quality, reoperations ≤ 30 days, and conversion to total hip arthroplasty (THA) ≤ 24 months were analyzed.

RESULTS: The analyzed outcomes demonstrated comparable operative parameters between both groups: (i) Oper […]

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Is there a difference in plate positioning for clavicle shaft fractures? A retrospective cohort study of 355 cases.

Hinkelmann MA, Schibler D, Frei F, Allemann F, Pape HC, Barthel C

PURPOSE: Plate osteosynthesis is a common procedure for clavicle shaft fractures. The plate can be placed in an anterior, anterior-superior, or superior position. This study aims to evaluate surgical treatment options regarding different plate positions for clavicle shaft fractures and to compare them regarding patient and perioperative characteristics, frequency of use, appearance of callus, complication rates and implant removal. METHODS: A retrospective cohort study of patients treated with plate osteosynthesis after clavicle shaft fracture between January 2016 and December 2024 at a Swiss level 1 trauma center was performed. Patients aged ≥ 16 years treated with anterior, anterior-superior or superior plate placement and availability of clinical follow-up data with X-ray were included. Presence of callus was subdivided into early (≤ 6 weeks), intermediate (> 6 to ≤ 26 weeks), and late (> 26 weeks). RESULTS: A total of 355 cases with a median of 41 years (range 17 to 90 years) (81.7%, n = 290 males and 18.3%, n = 65 females) were analyzed. The most commonly used plate position was anterior (92.4%, n = 328), followed by superior (5.6%, n = 20) and anterior-superior (2%, n = 7). Plate placement had no significant influence on operation time, callus formation, implant removal, or refracture after implant removal. Complications occurred in 6.2% (n = 22) of cases with a trend tow […]

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Decreasing trend in prophylactic inferior vena cava filter usage: a retrospective analysis of Indications, patient profiles, and complications at a level 1 trauma center in the United States.

Barbosa Abrantes V, Navarro SM, MacArthur TA, Spears GM, Cirillo-Penn NC, Ballinger BA, Davis RW, Park MS

PURPOSE: To analyze trends in the use of prophylactic inferior vena cava (pIVC) filters at a level 1 Trauma Center in the United States and evaluate indications, patient characteristics, and complications. METHODS: A retrospective cohort study was conducted on adult trauma patients who received pIVC filters between January 2011 and October 2023. Demographic and clinical variables were collected. Indications, complication rates, and venous thromboembolism (VTE) outcomes were assessed. Temporal trends were analyzed using the Mann-Kendall trend test. Logistic regression analysis was used to identify predictors of VTE. RESULTS: Among 300 patients in the cohort, there was a significant decline in pIVC filter use (Tau – 0.846, $p < 0.001$). Symptomatic VTE occurred in 23% of patients, and 13 (4%) experienced filter-related complications. Filters remained in place > 90 days in 81% of patients. Increased age (OR 1.17 per 10-year increase, $p = 0.038$), BMI (OR 1.26 per 5-point increase, $p = 0.019$), and male sex (OR 2.31, $p = 0.026$) were independently associated with VTE. CONCLUSION: Prophylactic IVC filter use in trauma patients has declined significantly in recent years, consistent with guideline-driven recommendations. Despite more selective use, VTE and complication rates remain substantial, emphasizing the importance of careful patient selection and structured retrieval follow-up syste […]

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Blunt traumatic hollow viscus and mesenteric injuries: a comparison of the currently available scoring systems.

Granieri S, Altomare M, Milanesi I, Di Sabato ME, Costacurta S, Adami G, Casale C, Ghilardi G et al.

BACKGROUND: Blunt traumatic hollow viscus and mesenteric injury (THVMI) remains a diagnostic challenge. In 2020, our group proposed the Niguarda Score, a simple tool to predict clinically relevant THVMI based on six binary CT findings. This study aimed to evaluate the Niguarda Score and compare its performance with established CT-based and combined THVMI scoring systems. METHODS: We retrospectively analyzed the data of all consecutive adult patients undergoing contrast-enhanced abdominal CT after blunt trauma at the Niguarda trauma center from 2019 to 2021. All variables constituting the Niguarda, Faget, BIPS, BBMI, and RAPTOR scores were gathered in an electronic dataset. The Z-score was not included due to its complex grading system for abdominal tenderness. Discrimination (Area Under the Curve - AUC), and pairwise AUC comparisons through the DeLong test were performed to test the Niguarda score against the Faget, BIPS, BBMI, and RAPTOR scores. RESULTS: Over 3 years, 268 consecutive patients respecting the inclusion criteria were selected. Thirty-two patients (11.9%) underwent surgical exploration, and 11 had surgically relevant THVMI (4.1%). Ninety-four patients were intubated on the field; for them, data regarding abdominal pain/tenderness were not available. In the whole cohort, the Niguarda Scoring model achieved an AUC of 0.87 (95%CI: 0.69–1.00). Other THVMI scores AUCs […]

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Pediatric metacarpal and metatarsal fracture in pediatric emergency department.

Dönmez E, Arslan A, Guleryuz OD, Caglar AA

AIM: This retrospective study analyzes the epidemiological characteristics, trauma mechanisms, and anatomical distribution of metacarpal and metatarsal fractures in pediatric patients presenting to a pediatric emergency department (PED) due to hand and foot trauma. METHOD: We retrospectively evaluated 278 patients diagnosed with metacarpal or metatarsal fractures who presented to the PED between January 2019 and May 2024. Patient data, including age, gender, trauma mechanism, fracture location and type, and treatment methods, were analyzed using SPSS 22.0. A p-value of < 0.05 was considered statistically significant. RESULTS: Among the 278 patients under 18 years old, 63.3% had metacarpal fractures and 36.7% had metatarsal fractures. The majority were male (80.2%) and ranged in age from 10 to 15. Falls were the most common trauma mechanism. The fifth metacarpal and metatarsal bones were the most frequently fractured sites. Most fractures were managed conservatively with resting splints; only one open fracture required surgical intervention. Violence-related fractures were more common in adolescents older than 15 years, while falls predominated in younger children. In multivariable logistic regression, metacarpal fractures were independently associated with male sex, spring season, swelling at presentation, violence-related mechanisms, and trauma occurring in school, garden/stre […]

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Factors affecting the anatomical localization of proximal femoral fractures in geriatric patients.

Y■ld■r■m E, Er■en Ö, Tüzün HY, Kara M

PURPOSE: This study aims to investigate the relationship between proximal femoral geometry and the anatomical localization of proximal femoral fractures in geriatric patients. The analysis was based on morphometric measurements obtained from the contralateral, uninjured hip. METHODS: Between January 2017 and December 2024, 864 patients (549 females, 315 males; mean age: 80.09 ± 7.65 years) with proximal femoral fractures were retrospectively analyzed. Fractures were classified as subcapital, transcervical, basicervical, intertrochanteric, or subtrochanteric according to standardized radiographic criteria. Morphometric parameters measured on anteroposterior pelvic radiographs of the contralateral hip included hip axis length (HAL), femoral neck axial length (FNAL), femoral neck and head diameters, femoral shaft diameter, caput-collum-diaphyseal (CCD) angle, medial femoral offset, and center-edge (CE) angle. Absolute and proportional values (relative to HAL and FNAL) were compared among fracture types. RESULTS: Extracapsular fractures were more frequent (58.4%) than intracapsular ones (41.6%), with intertrochanteric fractures being the most common subtype (48.6%). Femoral shaft diameter, femoral shaft diameter/HAL ratio, and CE angle were all significantly greater in extracapsular fractures (p = 0.04, p = 0.003, p = 0.002). Among fracture subtypes, femoral neck diameter was highe […]

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Pelvic fracture related hematoma - emergency treatment and bleeding control.

Möller S, Dillinger D, Hempe S, Bieler D, Metzger F, Liodakis E, Sehmisch S, Schweigkofler U

PURPOSE: Unstable pelvic fractures present significant challenges in trauma care due to the risk of massive haemorrhage from arterial, venous, and osseous sources. Rapid identification of the bleeding origin is critical for targeted intervention. This multicentre retrospective study evaluated the use of the Pelvic Vascular Injury Score (P-VIS) and CT-based volumetric assessment (Kothari's method) for early detection and classification of pelvic vascular injury (PVI) in severely injured patients.

METHODS: Data from January 2021 to December 2022 were collected from three German Level I Trauma Centres. Adult patients (18-75 years) with Tile type B or C pelvic ring injuries, ISS ≥ 16, haemodynamic instability (SBP ≤ 100 mmHg), and solid intrapelvic hematomas ≥ 100 ml on CT were included. Hematoma size was estimated using Kothari's formula and validated with planimetric volumetry for volumes ≥ 500 ml. Clinical data (including bloodwork, pelvic stabilization method, need for blood transfusion, haemorrhage therapy, mortality etc.) was acquired and compared. P-VIS was applied pre- or peri-hospital to predict likelihood of PVI.

RESULTS: Among 229 eligible patients, 56 exhibited pelvic haemorrhage (median 233 ml). Significant bleeding ≥ 500 ml occurred in 1.7% of cases and correlated strongly with arterial injury (probability 45%) and transfusion requirement (risk ratio 4.7). Arterial […]

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Emergency trauma admissions in the oldest-old: short- and long-term mortality and the role of frailty in a Turkish national cohort of centenarians.

Güner M, ■ahbat Y, Bayram S, Ceylan S, Bozku■ R, Ata N, Cankurtaran M, Birinci ■

INTRODUCTION: This study aimed to evaluate the impact of orthopedic trauma on mortality among centenarians presenting to the emergency department, to examine the relationship between frailty and longevity, and to provide insights into post-trauma survival in long-lived individuals, as well as the role of frailty mechanisms. METHODS: Data were retrieved from the National Personal Health Record System (NPHRS) of Türkiye. The study included patients aged ≥ 100 years who presented to emergency departments with trauma between January 2016 and July 2024 across all levels of healthcare institutions. Frailty was assessed using the Canadian Institute for Health Information Hospital Frailty Risk Measure (CIHI-HFRM) and 5-factor modified frailty index (mFI-5) based on ICD-10 codes. Survival status was determined retrospectively from national death registry. RESULTS: A total of 1,180 patients were included, with a mean age of 101.3 ± 3.9 years. Hip fractures were the most frequent presentation, accounting for more than 40% of all cases. Overall, 906 patients (76.8%) died, with a median survival of 7.2 months (range: 0.1–51.1 months). Higher frailty scores according to the CIHI-HFRM were paradoxically associated with reduced mortality across follow-up intervals: 30-day (HR = 0.026, 95% CI: 0.006–0.118, p < 0.001), 90-day (HR = 0.091, 95% CI: 0.031–0.271, p < 0.001), and one-year mortality ([…]

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Efficacy of AriClot™ hemostatic powder versus standard pressure dressing for bleeding control in civilian penetrating trauma: A randomized controlled trial.

Abdolrahimzadeh Fard H, Paydar S, Hosseini H, Eftekharian HR, Naseri M, Hosseini SMH, Fazel S, Hosaini A et al.

BACKGROUND: Effective bleeding control is critical in trauma care to reduce morbidity and mortality. AriClot™ is a novel modified absorbable starch-based hemostatic powder (sodium carboxymethyl starch) that rapidly adsorbs fluid to concentrate clotting factors and form a gel barrier. This study aimed to compare the efficacy of AriClot™ hemostatic powder with standard cotton gauze in controlling bleeding in trauma patients to address a gap in the literature where RCTs on hemostatic powders for trauma are scarce. METHODS: A randomized controlled trial was conducted at Rajaee Emergency and Trauma Center, Shiraz, Iran, from June to September 2025. Fifty patients with traumatic wounds were randomized to receive either AriClot hemostatic powder (n = 27) or cotton gauze (n = 23). One patient in the control group was lost to follow-up, leaving 49 patients for analysis (27 AriClot, 22 cotton gauze). The primary outcome was bleeding control success rate at 3 and 6 min, with secondary outcomes including rebleeding events and investigator opinions on treatment efficacy and early side effects. Success in bleeding control was defined as no visible bleeding from the wound, assessed by outcome assessors using visual inspection. RESULTS: AriClot demonstrated significantly higher success rates in bleeding control at both 3 min (81.08% vs. 31.03%, p < 0.0001) and 6 min (89.19% vs. 51.72%, p = 0.0 […]

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Initial TEG 6s parameters for risk stratification of hemorrhage-related outcomes in massively transfused trauma patients.

Seo D, Kwon J, Heo I, Jung K

PURPOSE: While thromboelastography (TEG) and rotational thromboelastometry (ROTEM) allow rapid coagulation assessment, the prognostic value of arrival TEG 6s remains unclear. We examined the association between initial TEG 6s findings and in-hospital mortality and early transfusion requirements in trauma patients undergoing massive transfusion, excluding severe traumatic brain injury to better assess hemorrhage-related outcomes. METHODS: This retrospective study included trauma patients who received massive transfusion and TEG 6s testing upon arrival at a Level I trauma center in South Korea. Patients with transfers, incomplete records, or head/neck AIS ≥ 3 were excluded. TEG 6s parameters were classified using published thresholds. Multivariable logistic and linear regression analyses evaluated associations with in-hospital mortality and RBC transfusion volumes at 4 and 24 h. RESULTS: Among 194 patients, 54.1% had abnormal TEG 6s results. These patients received higher RBC transfusion volumes at 4 and 24 h, and had higher in-hospital mortality. Abnormal TEG 6s findings were associated with higher in-hospital mortality in the multivariable analysis (adjusted OR 3.505, p = 0.050) and increased transfusion volume at 4 h (B = 1.686, p = 0.023) and 24 h (B = 2.313, p = 0.042). CONCLUSION: Initial TEG 6s abnormalities were associated with in-hospital mortality and early RBC transfus […]

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Ballistic limb trauma: lessons from a decade of french civilian and military experience.

Ghabi A, Meric V, Micicoi G, Tannyeres P, Chateaux L, Odent JB, Mathieu L, de Geofroy B

BACKGROUND: Ballistic limb injuries are increasingly encountered in both civilian and military settings, yet robust collaborative cohorts remain limited. METHODS: We conducted a retrospective, multicenter study across six French trauma centers (three civilian, three military) from 2014 to 2022, including consecutive patients aged ≥ 16 years with ballistic limb injuries. Data collected included demographics, context (civilian vs. combat/terror), projectile type and velocity, lesion characteristics, and early management. The primary outcome was early surgical site infection (SSI, < 30 days). Secondary outcomes were late and persistent infection, reoperations, fracture union, and infection-free survival. Univariate tests and multivariable logistic regression identified predictors of early SSI; Kaplan–Meier analysis assessed infection-free survival after initial infection. RESULTS: Among 254 patients (mean age 33.3 ± 12.7 years), 238 (93.7%) were male and 203 (79.9%) were civilian cases. Bullets caused 178 injuries (70.1%), pellets 47 (18.5%), and fragments 29 (11.4%); 43 injuries (16.9%) were high-velocity. Lower limbs were affected in 127 cases (50.0%), and 136 patients (53.5%) had open fractures, including 33 (13.0%) Gustilo III.

Damage-control orthopedics was applied in 98 patients (38.6%), and immediate skin closure in 153 (60.2%). Early SSI occurred in 42 patients (16.5%), la […]

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Time to surgery and preoperative posterior tilt predict reoperation after femoral neck system fixation in older adults with nondisplaced femoral neck fractures: a retrospective cohort study.

Sasaki Y, Miyamoto T, Takahashi M, Kawamoto T, Makino T, Kasahara K

PURPOSE: Nondisplaced femoral neck fractures (FNFs) in older adults are typically managed with internal fixation; however, fixation failure remains clinically significant. The Femoral Neck System (FNS) provides angular and rotational stability, but evidence in this population is limited. This study evaluated reoperation rates after FNS fixation and identified factors associated with reoperation.

METHODS: This single-center retrospective cohort study included 103 patients aged ≥ 60 years with nondisplaced (Garden 1 and 2) FNFs treated with FNS during 2018-2024, all with ≥ 1-year follow-up. The primary outcome was reoperation for any cause. Clinical variables included age, sex, body mass index, American Society of Anesthesiologists Physical Status, and time to surgery. Radiographic parameters-Garden alignment index, posterior tilt (PT), and tip-apex distance-were assessed on preoperative and immediate postoperative radiographs. Receiver operating characteristic (ROC) analysis identified cutoffs, which were applied in multivariable logistic regression.

RESULTS: Reoperation occurred in 14/103 patients (13.6%). Compared with the non-reoperation group, the reoperation group had longer time to surgery and greater preoperative and postoperative PT. ROC analysis yielded cutoffs of 46.4 h (time to surgery) and 13.5° (preoperative PT). Multivariable logistic regression showed that time […]

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Beyond facial injury: prognostic utility of maxillofacial trauma scores in the emergency department.

Boztepe EK, Sönmez BM, Akçay G

PURPOSE: To evaluate the prognostic utility of two widely used maxillofacial trauma scoring systems (MFTSs) -the Facial Injury Severity Score (FISS) and Maxillofacial Injury Severity Score (MFISS)- in adult multiple trauma patients in the emergency department (ED). METHODS: This prospective observational study was conducted in the ED of a tertiary care hospital between 01.08.2020 and 01.08.2021. A total of 124 MFT patients aged ≥ 18 years with at least one concomitant injury to another organ system were included. FISS and MFISS were calculated for each patient. The associations between scores and coexisting injuries, intensive care unit (ICU) admission, and mortality were analyzed. Predictive performance for mortality was assessed using receiver operating characteristic (ROC) curve analysis. RESULTS: Higher FISS scores were significantly associated with intracranial injuries, including subarachnoid hemorrhage (SAH), cranial fractures, and pneumocephalus (all $p < 0.01$), as well as with pneumothorax, rib fractures, ICU admission, and mortality (all $p \leq 0.004$). MFISS scores were significantly higher in patients with SAH, while a negative association was observed between MFISS and abdominal injuries ($p = 0.043$). ROC analysis demonstrated good predictive performance for mortality, with areas under the curve of 0.820 for FISS and 0.758 for MFISS (both $p < 0.01$). CONCLUSIONS: FISS and […]

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Low-profile 2.7 mm VA-LCP plates vs. 3.5 mm LCP plates for midshaft clavicle osteosynthesis: a retrospective comparison.

Brun D, Lecoultre Y, van de Wall BJM, Pretz F, Geuss S, Babst R, Link BC, Beeres FJP

PURPOSE: Implant related irritation and subsequent removal are one of the main disadvantages of clavicle plate osteosynthesis. 2.7 mm VA-LCP clavicle plates with a lower profile were introduced in recent years, with the aim to improve fit and reduce plate prominence. This study aims to investigate whether this leads to a lower implant removal rate compared to the older 3.5 mm LCP clavicle plates. METHODS: All patients that underwent single-plate osteosynthesis for midshaft clavicle fractures between 2019 and 2024 in a Swiss level I trauma-center were included. Patients were categorized into two treatment groups: DePuy Synthes 3.5 mm LCP clavicle plates or DePuy Synthes 2.7 mm VA-LCP clavicle plates. Treatment groups were compared regarding healing, complications, and removal rate. RESULTS: 101 patients were included, 54 with a 2.7 mm VA-LCP plate, and 47 with a 3.5 mm LCP plate. There were no significant differences in implant removal due to hardware irritation (44% versus 49%), overall reintervention rate (46.3% versus 53.2%, p = 0.49), infection (0% versus 4.3% p = 0.21), and refractures (1.9% versus 6.4%, p=0.34). All fractures healed in both groups. CONCLUSION: Newer 2.7 mm VA-LCP clavicle single plating shows similar results in terms of complication and healing rates compared to older 3.5 mm LCP plating. Despite the design aiming for reduced plate-prominence, it does not d […]

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Risk factors affecting the outcome of upper cervical spine injuries in the elderly - role of operative vs. conservative treatment.

Perl M, Hauck S, Weiß T, Wagner S, Knoedler L, Bailey J, Gonschorek O, Perl M

STUDY DESIGN: Retrospective Cohort Study. OBJECTIVES: To evaluate risk factors for adverse outcome and complications of elderly patients with upper cervical spine fracture following conservative or surgical treatment. METHODS: 141 cases of elderly patients (≥ 65 years) admitted between 01/2013 and 12/2015 with upper cervical spine injuries were retrospectively analyzed through our prospectively collected database. Median follow-up was 113.5 days (IQR: 21–393). RESULTS: With a collective median age of 79, operatively treated patients (41%, 58/141) were younger compared to conservatively treated (76 vs. 83 years, p < 0.001). Ground level falls were the most common injury mechanism (55% vs. 54%, p = 0.91) while Anderson type II (29.1%) and combined odontoid/atlas fractures (28.2%) were the most frequent injury patterns. Complications were significantly increased in operated patients (67% vs. 49%, p = 0.035) when compared to those conservatively treated. Neurological and rheumatic comorbidities, as well prolonged operation time, had a significant association to postoperative delirium. The duration of surgery (47 vs. 59 min, p = 0.014) and the risk for secondary dislocation (22% vs. 60%, p = 0.086) were reduced through posterior C1-2 screw fixation, as compared to anterior direct osteosynthesis. CONCLUSION: Considerations regarding appropriate treatment of upper cervical spine injur […]

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Anatomical distribution and treatment for fractures and dislocations of the proximal and middle phalanges: a multicentre snapshot study of 461 patients.

de Haas LEM, Salentijn DA, Hendriks TCC, Schep NWL, van Heijl M, Snapshot Hand Fractures Collaborative Study Group

PURPOSE: To evaluate the anatomical distribution of fractures and dislocations of the proximal and middle phalanges, the current treatment modalities, and potential variations in treatment approaches. METHODS: This multi-centre cross-sectional study included patients presenting with single fractures of the proximal or middle phalanges and dislocations (excluding the thumb) over a three-month period in 2020. Data collection included demographic details, injury location and fracture pattern, and treatment modalities. Treatment categories included non-operative management (e.g., functional treatment allowing digit mobilisation, immobilisation, and immobilisation duration), surgical interventions, and post-operative care (functional treatment or immobilisation and immobilization duration). RESULTS: A total of 461 patients were included, with 395 (86%) managed non-operatively and 66 (14%) undergoing surgical intervention. The most frequently encountered injuries were proximal interphalangeal joint (PIPJ) avulsion fractures (137/461, 30%), followed by proximal phalanx shaft fractures (132/461, 29%), and dorsal PIPJ dislocations (57/461, 12%). Among patients with PIPJ palmar plate avulsion fractures managed non-operatively, 72% (88/123) were treated with functional treatment in the emergency department (ED), increasing to 94% after the initial follow-up. For proximal phalanx shaft fra […]

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The potential impact of changing the Norwegian trauma Registry inclusion criteria for patients not admitted with trauma team activation: Identifying patient groups to be excluded and considering potential risks.

Nordberg S, Holm KT, Weber C, Røise O

PURPOSE: One of the inclusion criteria of the Norwegian Trauma Registry (NTR) is a New Injury Severity Score (NISS) > 12 for patients not admitted with trauma team activation (TTA). Based on registry data and clinical experience, patients with NISS of 13 or 14 who are not admitted with TTA often have low-energy falls as the primary injury mechanism. Identifying and registering these patients requires considerable effort. However, it is unknown what effects exclusion of patients with a NISS of 13 or 14 not admitted with TTA will have on the risk and safety of this group of trauma patients. METHODS: Registry data from a cohort of patients admitted to Norwegian trauma hospitals during 2017–2021 with NISS 13 or 14 who were not met with TTA (group 1) were compared to a cohort of patients with the same NISS scores met with TTA (group 2), and the total registry population, respectively (group 3). RESULTS: Group 1 (n = 815) consisted mainly of elderly females (58%), with low-energy falls (73%). The group had a significantly higher proportion of injuries in the lower extremity (61%) and underwent fewer medical interventions than the two comparison groups. CONCLUSION AND IMPLICATION: Increasing the criteria would result in the exclusion of a patient group mainly consisting of elderly women injured with low-energy trauma and hence it is out of the primary scope of a national trauma regist […]

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Health-Related Social Needs Are Associated With Worse Physical Function, Pain, and Mobility in Hip and Knee Osteoarthritis Patients at Presentation.

Winter AD, Hayashi CL, Túa-Vargas LA, Chau N, Gibson DH, Rubin LE, Molloy IB, Wiznia DH

BACKGROUND: Health-related social needs (HRSNs) are mandatory reporting measures per the Centers for Medicare & Medicaid Services. We sought to assess the prevalence of HRSNs and associations with clinical presentation among preoperative hip and knee osteoarthritis patients.

METHODS: This prospective single-institution cross-sectional study assessed HRSNs (living situation, food, transportation, utilities, safety, financial strain, substance use, and mental health); Patient-Reported Outcomes Measurement Information System (PROMIS) Physical Function, Mobility, Pain Interference scores; and demographics (sex, race, education, income, household size, marital status) via in-person interviews in an academic arthroplasty practice. Kellgren-Lawrence Osteoarthritis Score, Charlson Comorbidity Index, and insurance were also determined. Odds of HRSNs by demographics were analyzed using logistic regression and relationship between HRSNs and PROMIS scores was determined by weighted least square linear regression.

RESULTS: The study included 253 patients. Prevalences of each HRSN were 15% for living situation, 13% for food insecurity, 11% for transportation, 4% for utilities, 0.4% for safety, 58% for mental health, 31% for substance use, and 27% for financial strain. Black patients and patients on Medicaid had higher HRSN prevalences. PROMIS scores were significantly worse in patients wit […]

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Striving for LGBTQI+ Health Equity in Arthroplasty.

Ross LA, Bellamy JL, Scott CE

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Social and Demographic Health Disparities in Knee Osteoarthritis and Total Knee Arthroplasty.

Dash AS, Ghanem OA, Tyler WK, Hung CT, Chahine NO, Sarpong NO

There has been increased awareness around the interplay between patients' social and demographic background and health outcomes. The influence of social determinants of health on patients outcomes has recently been recognized in orthopaedic literature. As knee osteoarthritis (OA) is one of the leading causes of disability in the world and the procedural volume of total knee arthroplasty (TKA) is rising with increased access and an aging population, better appreciation for these factors is imperative. This review seeks to examine the effect of sociodemographic factors on OA care and disease burden; and TKA outcomes. Our goal is to provide the arthroplasty surgeon community with a better understanding of how inequities in care and research impact our patients' OA and TKA outcomes. Previously published research has focused mainly on race and ethnicity, but the literature is limited, with what is available showing that there are disparate presentations and outcomes in both OA and TKA based on different social determinants of health. There is substantial work that will be required to better understand and address nonmodifiable risk factors and optimize surgery outcomes across all demographics.

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Barriers to Care for Total Joint Arthroplasty Patients With Medicaid Insurance: A Survey Study of the Surgeon Perspective.

Bloise C, Hall L, Sisco-Wise L, Randell T, Leonardi C, Cohen-Rosenblum A

BACKGROUND: The purpose of this study was to assess the demographic characteristics of orthopaedic surgeons in Louisiana taking care of patients with Medicaid insurance and to assess any perceived barriers to care, with a focus on total joint arthroplasty (TJA).

METHODS: An electronic survey was distributed to all practicing surgeon members of the Louisiana Orthopaedic Association with questions about practice type, subspecialty, demographics, estimated percentage of Medicaid patient volume, and perceived barriers to care. Responses were collected using a secure database, and statistical analysis was performed.

RESULTS: The 114 respondents were mostly male (95.6%), white (87.7%), and in private practice (61.4%). Most respondents (73.8%) did not accept Medicaid insurance (22.8%) or took care of <25% patients with Medicaid (50.9%). Fifty-eight respondents (50.9%) reported performing TJA. The majority of surgeons performing TJA (76.6% total hip arthroplasty [THA], 93.7% total knee arthroplasty [TKA]) either did not accept Medicaid insurance (36.2% THA, 47.3% TKA) or performed less than 25% of their TJA volume on patients with Medicaid insurance (40.4% THA, 36.4% TKA). The most commonly reported perceived barriers to care were lower physician and facility reimbursement, and difficulty referring to ancillary services. Surgeons in academic practice were significantly more likely to […]

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The Impact of Social Drivers of Health on Patient Portal Utilization for Total Joint Arthroplasty Patients: A Qualitative Study.

Antonioli SS, Alpert Z, Mishra S, Onakomaiya D, Vallurupalli N, Bosco JA, McLaurin T, Lajam C

BACKGROUND: Despite similar rates of osteoarthritis, minority populations undergo fewer hip and knee arthroplasties and have more complications. More than 90% of US hospitals have certified electronic health records, yet only 40% of patients utilize electronic patient portals (EPPs), with lower rates across some demographics. Adverse quality metrics and lower patient-reported outcome survey completion rates were noted for patients with inactive portals. Activation rates for EPPs can be lower among underrepresented groups, perpetuating existing disparities in access to care. Barriers to EPP activation must be identified to design interventions that improve portal utilization and, therefore, outcomes. We designed this study as there are no published reports analyzing factors which impede EPP utilization by total joint patients.

METHODS: In this IRB-exempt qualitative study, sixty-six arthroplasty patients were interviewed using a questionnaire designed to reveal reasons for EPP nonusage. Demographic factors including language, age, sex, insurance type, and zip code were collected from the electronic health records. Dedoose, a qualitative research tool, was used to analyze data and abstract trends from interview notes and transcripts.

RESULTS: We found arthroplasty patients' demographics and social drivers of health influenced utilization of EPPs. Older adults struggled with dig […]

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Association Between Regional Social Vulnerability and Race-Based Differences in Total Knee Arthroplasty Use Among Medicare Beneficiaries.

Cruse JJ, Schloemann DT, Danielson EC, Ricciardi BF, Balkissoon R, Franklin PD, Thirukumaran CP

BACKGROUND: There is inequity in total knee arthroplasty (TKA) use between racial groups. The social vulnerability index (SVI) may characterize mechanisms underlying disparity, which is essential for identifying targeted interventions. We examined the relationship of the overall SVI and its themes with TKA use rates for Black and White Medicare beneficiaries and the difference in TKA utilization between these groups.

METHODS: We used 2013-2017 Medicare data to calculate annual TKA use rates per 1000 beneficiaries in each hospital referral region (HRR). Multivariable mixed-effects linear regressions estimated the relationship between social vulnerability and TKA use.

RESULTS: The mean (standard deviation) TKA rates were 9.4 (1.7) and 5.9 (2.2) per 1000 White and Black beneficiaries, respectively. HRRs with highest social vulnerability (quartile 4) had lower TKA rates for White (estimate for quartile 4: -1.7, 95% confidence interval [CI]: -2.2 to -1.1, $P < .001$) and Black beneficiaries (quartile 4: -1.7, 95% CI: -2.3 to -1.0, $P < .001$) compared to HRRs with the least vulnerability. The SVI's minority status and language theme was linked with wider rate differences between White and Black beneficiaries (quartile 4: 0.6, 95% CI: 0.0-1.2, $P = .046$).

CONCLUSIONS: Worse overall SVI was related to lower TKA use rates for both White and Black Medicare beneficiaries. Areas with the hi […]

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Bridging the Empathy Gap: Why Health Equity Must Matter to All of Us.

Tucker KK

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Sociodemographic and Clinical Factors Associated With Response to 1-Year Postarthroplasty Patient-Reported Outcome Measures.

Sylla F, Stern BZ, Burla M, Suleiman LI, Franklin PD

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Rural vs Urban Outcomes in Total Knee and Hip Arthroplasty: A Study of Patient-Reported Outcomes and Web-Based Tool Efficacy.

Taylor JM, Dipane MV, Sassoon AA

BACKGROUND: Rural patients in the United States often face higher tobacco use, obesity rates, and lower health literacy, impacting arthroplasty outcomes. This study investigates whether rural residency is directly associated with poorer patient-reported outcomes (PROs) following total knee (TKA) and total hip arthroplasty (THA). It also evaluates the effectiveness of web-based PRO tools across urban and rural populations. Understanding these relationships is crucial for addressing disparities in care and improving surveillance for rural populations.

METHODS: From May 2021 to April 2023, TKA and THA patients at our institution were categorized based on their home zip codes, using the 2020 Census Bureau binary urban-rural classification. Web-based PROs were administered via text or email and assessed using the Knee Injury and Osteoarthritis Outcome Score, Joint Replacement; Hip Injury and Osteoarthritis Outcomes Score, Joint Replacement; University of California, Los Angeles Activity Score; Patient-Reported Outcomes Measurement Information System Global Physical and Mental Health Scores; and Forgotten Joint Scores. PROs were recorded preoperatively and at 6 weeks, 3 months, and 1 year postoperatively. Statistical analysis encompassed chi-square and independent samples t-tests.

RESULTS: Among 781 THA and 1094 TKA cases, survey response rates (71.9%-83.8%) were consistent across […]

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Health-Related Social Needs Associated With Worse Patient-Reported Outcomes and Increased Adverse Events Following Total Joint Arthroplasty.

Hayashi CL, Winter AD, Túa-Vargas LA, Chau N, Kalia A, Gibson DH, Rubin LE, Molloy IB et al.

BACKGROUND: Health-related social needs (HRSNs) are the individual-level adverse social conditions that negatively impact a person's health. This study characterizes the association of HRSNs with patient-reported outcomes and adverse events following total hip (THA) and knee arthroplasty (TKA).

METHODS: This single-institution cross-sectional study utilized the Centers for Medicare & Medicaid Services HRSN Screening Tool. In-person interviews captured HRSNs, patient-reported outcome measures, and demographic data from postoperative THA and TKA patients in an academic arthroplasty practice. Charlson Comorbidity Index, American Society of Anesthesiologists scores, discharge data, and 90-day complications were collected via chart review. Regression analysis was used to determine associations between HRSNs and outcomes.

RESULTS: Among 190 patients, food insecurity had a significant association with reoperation (odds ratio (OR) = 5.78, 95% confidence interval 1.44-23.2, $P = .013$). Patients with food and transportation HRSNs had significantly worse postoperative physical function, pain, and mobility (all $P < .05$). Black patients had significantly higher odds of visiting the emergency department (OR = 2.15, 95% CI 1.10-4.20, $P = .025$) or being readmitted (OR = 2.70, 95% confidence interval 1.11-6.58, $P = .029$) within 90 days postoperation compared to White patients.

CONCLUSIONS: Fo […]

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Impact of Rurality on Total Joint Arthroplasty Access and Outcomes: A Systematic Review.

Agrodnia SL, Call CM, Lachance AD, McDonald JC, Rana AJ, McGrory BJ

BACKGROUND: Total joint arthroplasty (TJA) is one of the most frequent orthopaedic surgeries in the United States; however, disparities in utilization and outcomes based on race, sex, and socioeconomic level have been well documented. The impact of rural geographic location, a characteristic that may impact access to TJA care and postoperative outcomes, remains understudied. This systematic review investigated associations between rural location and several metrics in total hip arthroplasty and total knee arthroplasty to investigate whether disparities are present based on rural geographic location.

METHODS: In November 2024, PubMed, Web of Science, Scopus, EMBASE, and Cochrane Review databases were queried. Ten studies investigating TJA utilization and outcomes that included patients in a rural location were included. Study quality was assessed using a modified Newcastle-Ottawa Scale.

RESULTS: Ten articles were included in the final analysis. Differences in complications, readmission rates, and length of stay due to rural location of patient and/or hospital were reported. Results indicate urban TJA patients more frequently discharge to rehab. Patient-reported outcome measures following TJA do not appear different between rural and urban patient groups.

CONCLUSIONS: Rural geographic location is associated with different TJA utilization and outcomes and may point to dispariti […]

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Patient Social Determinants of Health Are Associated With Hip Dysfunction and Osteoarthritis Outcome Scores and Outcomes After Total Hip Arthroplasty.

Yacoub G, Janney CA, Wang Z, Meremikwu R, Caird MS, Kheir MM

BACKGROUND: Disparities in access and outcomes persist for total hip arthroplasty (THA). The Hip Dysfunction and Osteoarthritis Outcome Score (HOOS) is a validated patient-reported outcomes and measure and a proxy for quality of care. We examined whether patient social determinants of health were associated with HOOS scores and outcomes.

METHODS: We retrospectively identified 2938 THAs at a single institution from 2015 to 2023. Multivariate analyses were performed to determine if associations existed between patient factors (race, sex, marital status, age, insurance carrier, among others) and HOOS scores (preoperative, 6-week, and 1-year postoperative) or outcomes (length of stay [LOS], discharge disposition, or perioperative complications).

RESULTS: Lower preoperative HOOS scores were associated with being younger, unmarried, Black, higher body mass index, higher Elixhauser score, government insurance, posterior approach, and female ($P \leq 0.01$). Lower 6-week and 1-year postoperative HOOS scores were associated with being Black and higher Elixhauser scores ($P \leq .05$). Decreased HOOS score improvement at 1 year was associated with older age ($P < .01$). Longer LOS and discharge to rehabilitation facility were associated with being older, unmarried, black, female, higher Elixhauser score, and posterior approach ($P < .01$).

CONCLUSIONS: Several social determinants of health were ass […]

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Arthroplasty Surgeons Taking Care of the Highest Social Risk Populations Are More Often Osteopathic Trained, Less Experienced, and More Likely to be Penalized by the Merit Based Incentive Payment System.

Holle AM, Gill VS, Lin E, Bingham JS, Spangehl MJ, Clarke HD

BACKGROUND: The purpose of this study was to examine how hip and knee arthroplasty surgeon Merit-based Incentive Payment System (MIPS) performance, surgeon demographics, practice characteristics, and patient population varied based on the social risk of their caseload in 2017 and 2021.

METHODS: Multiple databases published by the Centers for Medicare and Medicaid Services were combined and utilized to examine all US hip and knee arthroplasty surgeons. Surgeons were placed into quintiles of social risk based on the proportion of dual-eligible Medicare-Medicaid patients in their patient population. Demographics, Distressed Community Index scores, patient population information, and MIPS performance were compared between quintiles in years 2017 and 2021.

RESULTS: In 2017, arthroplasty surgeons with the highest social risk caseloads scored lower on MIPS (70.0 vs 73.5, $P = .012$) and were more likely to receive a negative payment adjustment (odds ratio: 1.64; 95% confidence interval: 1.01-2.68, $P = .046$) compared to surgeons with the lowest social risk caseload. In 2021, arthroplasty surgeons with the highest social risk caseloads scored higher on MIPS (86.2 vs 82.4, $P < .001$), but remained more likely to receive a negative payment adjustment (odds ratio: 3.98; 95% confidence interval: 1.63-10.53, $P = .003$) compared to those with low social risk caseloads.

CONCLUSIONS: These findi […]

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